



**GREEN
CLIMATE
FUND**

Meeting of the Board

15 – 18 July 2024

Songdo, Incheon, Republic of Korea

Provisional agenda item 10

GCF/B.39/02/Add.12/Rev.01

15 July 2024

Consideration of funding proposals – Addendum XII

Funding proposal package for FP237

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "E-Motion : E-Mobility and Low Carbon Transportation";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

It is noted that gender documentation has been updated.

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Funding Proposal

Project/Programme title:	E-Motion : E-Mobility and Low Carbon Transportation
Country(ies):	Argentina, Brazil, Colombia, Costa Rica, Dominican Republic, Mexico, Peru
Accredited Entity:	French Development Agency (AFD)
Date of first submission:	2021/03/17
Date of current submission	12/06/2024
Version number	V14



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

List of Abbreviations and Acronyms

AFD	French Development Agency
AMA	Accreditation Master Agreement
AML	Anti Money Laundering
ANEEL	Agencia Nacional de Energia Elétrica
ASI	Avoid-Shift-Improve
BAU	Business as Usual
BEB	Battery Electric Bus
BEV	Battery Electric Vehicle
BNCR	Banco Nacional de Costa Rica
BNDES	Banco Nacional de Desenvolvimento Econômico e Social
CAF	Development Bank of Latin America
CAPEX	Capital Expenditure
CB	Capacity Building
CNG	Compressed Natural Gas
CFN	Corporación Financiera Nacional
EASI	Electrify-Avoid-Shift-Improve
EPM	Empresa de servicios públicos de Medellín y Colombia
ESP	Environmental and Social Policy
EV	Electric Vehicle
FA	Financial Assistance
FAA	Funded Activity Agreement
FIRR	Financial Internal Rate of Return
GAP	Gender Action Plan
GDP	Gross Domestic Product
GEF	Global Environment Facility
GHG	Greenhouse Gases
GIZ	Deutsche Gesellschaft für Internationale Zusammenarbeit
HEV	Hybrid Electric Vehicle
ICE	Instituto Costarricense de Electricidad
ICES	Internal combustion engines
IEA	International Energy Agency
IFRS	International Financial Reporting Standards
ILO	International Labor Organization
KfW	Kreditanstalt für Wiederaufbau
LCV	Light Commercial Vehicle
NAFIN	Nacional Financiera
NDC	Nationally Determined Contribution
OECD	Organisation for Economic Co-operation and Development
OEM	Original Equipment Manufacturer
OPEX	Operational Expenditures
PHEV	Plug-in Hybrid Electric Vehicle
PPP	Public-Private Partnership
PROPARCO	Société de Promotion et Participation à la Coopération économique
PT	Public Transport
PTO	Public Transport Operator
SCC	Social Cost of Carbon
SDG	Sustainable Development Goals
SPV	Special Purpose Vehicle
TA	Technical Assistance
TCO	Total Cost of Ownership



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TTW	Tank to Wheel
WACC	Weighted Average Capital Cost
WHO	World Health Organization
WTT	Well to Tank
WTW	Well to Wheel

A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Programme	A.2. Public or private sector	Public
A.3. Request for Proposals (RFP)	Not applicable		
A.4. Result area(s)		GCF contribution	Co-financers' contribution ¹
	Mitigation total	Enter number %	Enter number %
	☐ Energy generation and access	Enter number %	Enter number %
	☒ Low-emission transport	100 %	100 %
	☐ Buildings, cities, industries and appliances	Enter number %	Enter number %
	☐ Forestry and land use	Enter number %	Enter number %
	Adaptation total	Enter number %	Enter number %
	☐ Most vulnerable people and communities	Enter number %	Enter number %
	☐ Health and well-being, and food and water security	Enter number %	Enter number %
	☐ Infrastructure and built environment	Enter number %	Enter number %
	☐ Ecosystems and ecosystem services	Enter number %	Enter number %
	A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	4,334,051	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)
A.7. Total financing (GCF + co-finance ²) ³	210,981,104 Euros 103,895,500 USD	A.9. Project size	Large (Over USD 250 million)
A.8. Total GCF funding requested	78,333,415 Euros 40,264,000 USD		

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² Refer to the Policy of Co-financing of the GCF.

³ Co-financing funds could be in EUR, USD or local currency

A.10. Financial instrument(s) requested for the GCF funding	<i>Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</i> ☒ Grant 22,333,415 Euros ☐ Equity <u>Enter number</u> ☒ Loan 56,000,000 Euros ☐ Results-based payment <u>Enter number</u> 40,264,000 USD ☐ Guarantee <u>Enter number</u>		
A.11. Implementation period	7 years	A.12. Total lifespan	25 years
A.13. Expected date of AE internal approval	12/1/2024	A.14. ESS category	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☒ In some countries ⁴
A.19. Complementarity and coherence	Yes ☒ No ☐		

⁴ Country Programs of many of the countries concerned by the Program are not yet finalized or published. The Program is included in the country Program of Mexico and not included in the country program of Colombia. Brazil is currently updating its country program of 2018 and all other countries have not yet finalized the country program.. All countries have either developed strategies, targets, roadmaps or have included in their NDCs concretely e-mobility. The Program intervenes in countries which are interested and where either the private or the public sector co-finance the project.

A.20. Executing Entity information

- **Agence Française de Développement (AFD)** is a financing company and a public industrial and commercial institution, acting as financier, advisor, partner and implementer to mobilise finance and expertise for development projects in emerging and developing countries. It is the main implementing agency for France's official development assistance to developing countries and overseas territories. Focusing on climate, biodiversity, peace, education, urban development, health and governance, AFD teams carry out more than 4,000 projects in France's overseas departments and territories and in 115 other countries. AFD is both the accredited entity and the executing entity.
- **PROPARCO** (société de Promotion et Participation à la Coopération économique). PROPARCO is a subsidiary of AFD, hence acts as agent in the name and on behalf of AFD, focused on private sector development, and provides funding (loans, guarantees, equity and quasi-equity) and support (technical assistance and expertise) to both businesses and financial intermediaries across Africa, Asia, Latin America and the Middle-East. The financing of infrastructure projects represents more than 30% of PROPARCO's activities in these countries. PROPARCO will implement the financing activities and TA of the Programme where Project Owners (PO) belong to private sector. PROPARCO's head office is located in Paris.
- **Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)**. GIZ is the German Agency for International Cooperation. It is a public-benefit federal enterprise. GIZ's sole shareholder is the Federal Republic of Germany, represented in Germany by the Federal Ministry for Economic Cooperation and Development (BMZ) and the Federal Ministry of Finance (BMF). GIZ is headquartered in Bonn and Eschborn but operates from about 120 countries around the globe. 70% of its 20,000 employees are national staff.

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

Climate Context

Transport contributes almost one-quarter of the current global energy-related greenhouse gas (GHG) emissions and is growing faster than any other energy end-use sector. Limiting the global temperature increase to below 2 degrees Celsius requires changing this transport emissions trajectory, which involves the development of an integrated electromobility ecosystem encompassing various transport modes, coupled with the low-carbon production of electricity and hydrogen (Paris Declaration on Electro-Mobility and Climate Change & Call to Action, 2015). To achieve this goal International Energy Agency (IEA) modelling says EVs need to represent 35% of global sales in 2030.

Avoid-Shift-Improve (ASI) has been a long-standing framework for classifying mitigation interventions in the transport sector. A new dimension needs to be added to this framework: EASI: Electrify-Avoid-Shift-Improve. Electrification is a game-changer and a **must** if carbon neutrality shall be achieved. Shift and improve measures without electrification are not capable to reach carbon neutrality in the transport sector. Electrification coupled with a fossil-free grid creates this opportunity. Many Latin American countries have recognized this fact and have started to develop a regulatory framework to go toward this direction by introducing Electric Vehicle (EV) tax incentives, EV targets, and favourable regulatory systems e.g. for e-buses (see Section B1 and Annex 2 Report 1). However, maximum efficiency is still required to reduce electricity demand and thus cost and challenges in the energy system. Additionally, other challenges in the sector, such as congestion, safety, access, health can be addressed through the use of more efficient

modes of transport, such as buses, and through non-motorised alternatives. Developing an electrified public transport as the backbone of urban transport is therefore key to a sustainable mobility system.

Social and Environmental Context

Air pollution is a large problem in most Latin American cities with levels affecting seriously human health. According to the latest urban air quality database, 98% of cities in low- and middle-income countries with more than 100,000 inhabitants do not meet World Health Organization (WHO) air quality guidelines. The WHO estimates that annually 4.2 million deaths result due to excessive exposure to fine particulate matter from ambient air pollution equivalent to 7.6% of all deaths. The transport sector is a major source of air pollution.

The COVID'19 pandemic has created an unprecedented challenge for many Latin American countries, from the health and economic perspectives, but it is also opening new opportunities for an accelerated sustainable transformation of its energy, transport, and urban landscapes while contributing to economic recovery. A recent report prepared by the International Labor Organization (ILO) shows that fostering of electric mobility can have a significant positive job impact (ILO, 2020).

Country Conditions and Ownership

Latin America has strong enabling conditions in terms of clean transport and e-mobility. The region has one of the cleanest electricity generation portfolios in the world with the median of Program countries having a carbon grid factor of 0.252 kgCO_{2e}/kWh. The adoption of an e-mobility strategy would allow the countries to move towards a decarbonisation pathway. The updated Nationally Determined Contributions (NDCs) of Program countries all include measures to reduce transport emissions. The majority of countries have included in the NDCs electric mobility as a core mitigation measure to reduce transport emissions and have formulated concrete electrification targets through regulations, policies or EV roadmaps.

The main motivating forces to foster e-mobility in the countries included in the Program are (i) reduce GHG emissions; (ii) improve air quality and reduce health problems associated with air pollution; (iii) reduce dependency on imported fossil fuels and exposure to external price shocks; (iv) achieve positive economic impacts related to reduced health costs, reduced fuel import bill and job creation; (v) Participate in the new growth industry of EV and battery manufacturing; (iv) increase the attractiveness of public transportation vis-a-vis private motorized transport.

This Program has been developed in close coordination with CAF whose FP195 has been approved by the GCF at B34 and which implements similar activities in Uruguay, Paraguay and Panama (private and public sector). The two programs will be closely coordinated by CAF and AFD and shall be implemented under the same label "E-Motion".

Intervention Strategy

The Programs goal is to accelerate Electric Vehicle (EV) deployment and enable a large-scale regional transition towards electro-mobility in Latin America through financial and technical assistance. The Program implements interventions to kick-start EV mass deployment significantly earlier than under a Business as Usual (BAU) scenario by reducing the risk profile of investments and by comprehensive technical assistance. The key strategic value of the Program is that it functions as market accelerator enabling a far faster uptake of e-mobility than under a BAU scenario avoiding a lock-in of long-lived assets in fossil technology. The Program fills the gap between initial pilots already existing in the region and long-term targets. These interventions are made in a time where e-mobility is commercially not yet viable and thus require initial investment support -like is the case in all countries which have a significant uptake of e-mobility.

The Program focus is on electrification of public transport buses. Electrification of buses is not a stand-alone measure but an important component to modernize and to upgrade the public transport system of

countries. New electric buses position the public transport system as modern, environmentally friendly and attractive. The investment in electric buses is linked with investments to improve the public transport system and non-motorized transport (NMT) in urban areas as well as transport demand management measures which result in increasing the attractiveness of low carbon forms of mobility. This results in a mode shift towards zero or ultra-low mobility systems. The Program directly modernizes public transport systems through establishing new business models, which increase the financial stability and economic efficiency of public transport systems e.g. through separation of bus ownership and bus operations and indirectly by realizing e-bus investments in cities which have programs to foster NMT and public transport financed nationally and/or internationally i.e. the complementarity of e-bus investments to such efforts is a selection criteria for investment projects of the Program. As example medium sized Colombian cities which are targeted by this Program have formulated urban mobility plans with the aim of modernizing and upgrading public transport whilst also fostering NMT and components of transit-oriented development.

Program activities improve the grid resilience and the resilience of urban infrastructure. All investments in grid infrastructure and bus depot up-gradings will be made climate resilient. Used batteries from EVs have a second-life application where they can serve as storage units and thereby (i) increase the resilience and independence of the grid at critical spots such as hospitals or schools; (ii) increase the grid resilience and reduce costs of renewable grids by increasing storage capacity capable of taking up electricity during times of over-supply; (iii) act as peak shavers in urban settings.

The program will also benefit from the activities implemented by GIZ and AFD in all program countries under the framework of the EUROCLIMA+ initiative whose purpose is to develop national policies for urban mobility and integrated multi-modal planning at the city level, in order to accelerate the transition of Latin American cities towards sustainable urban mobility with low carbon emissions. As a result, Euroclima+ will contribute to create the favorable conditions for the development of e-mobility in the framework of integrated and sustainable urban mobility plans. The TA of the FP will be limited to activities which are complementary to the ones already realized within EUROCLIMA. The Budgetary credit lines financed by AFD and supporting a high level political dialogue on transport decarbonization in Costa Rica and Dominican Republic also ensure that the projects implemented in these countries through E-Motion will have a key impact in public transport restructuring and decarbonization.

Barriers to EV Deployment and Program Approach to Overcome Barriers

Main barriers to EV deployment are (i) 2-3x higher Capital Expenditure (CAPEX) of EVs resulting in higher capital requirements, higher finance and insurance costs and higher risk exposure due to higher debt levels. The incremental CAPEX of e-buses result in incremental capital required from the investor and in additional finance costs compared to a fossil unit; (ii) lack of financial profitability for EVs in involved countries with long payback times; (iii) higher risk of EVs compared to fossil vehicles due to unknown performance and cost levels and potential income losses; (iv) governance structures and business models not conducive to EV deployment, especially in the public transport sector; (v) limited know-how on optimal EV ecosystem design, maintenance & operations; (vi) EV targets have been established by many countries but concrete and tangible support policies which are conducive to EV uptake are partially missing. All involved countries have a large interest in promoting e-mobility and various have also established fiscal incentives which are however targeted primarily to passenger cars. In all countries e-bus projects can be realized without requiring a change of the existing policy framework. The intervention strategy is to use concrete implementation projects to foster policies, regulations and frameworks conducive to a widespread adoption of e-buses. This can best be done by having concrete workable implementation cases e.g. of concession contracts or business models such as separation of bus ownership and bus operations conducive to e-buses. Such approaches together with high quality e-bus system design enables e-buses to be commercially competitive to fossil units. Only once it is apparent that e-buses can operate in a reliable manner and with comparable total costs to diesel buses will governments, city authorities, transport operators and other stakeholders be willing to establish and actually implement policy targets such as a ban of fossil buses. Stand-alone policy recommendations

without the proof of concept have limited impact. Stakeholders consulted by the Program have repeatedly stressed the point that investment projects involving significant fleets (not pilot projects) need to be implemented prior to taking critical regulatory decisions which affect public transport services. The proof of concept is thus critical and the sequencing of starting with investment projects which are then used as concept approach accompanied and followed by policy advice is therefore core to this Program. Not all counties experience the same magnitude of barriers and in some, e.g. Colombia, at least in some cities innovative business models are already under trial. The experience of Colombia as well as Chile (not part of this Program) is however, that the business models and financial structuring for large cities such as Bogota or Santiago cannot be transferred to medium and smaller cities where for example private capital has much less interest to participate due to offering smaller markets, financially fragile transport operators and weaker local city authorities including limitations in legal structures such as concession contracts. Charging infrastructure, which is a core barrier for passenger car deployment, is not listed as barrier due to the Program focus on electric buses which do not rely on a public charging network but have proprietary chargers either at the bus depot or along/close to their routes.

The Program tackles these barriers and offers a comprehensive e-mobility package including financial and technical assistance thereby improving the policy framework and the ecosystem and removing barriers that enable the mass deployment of EVs. The barrier of higher initial CAPEX and higher finance costs is tackled through concessional loans and through technical assistance (TA) which promote new business models with lower costs and risks for public transport (PT) operators and through an optimal e-bus system design which reduces costs and operational risks. The barriers of lack of financial profitability and high risks are overcome through concessional loans and TA which assists cities in designing optimal e-bus systems, which has a knowledge exchange platform and allows to learn from experiences of other cities in Latin America, and through training and capacity building of staff. The policy and regulatory work done with TA is also critical to increase profitability of e-bus investments e.g. through establishing concession contract lengths in line with the (longer) lifespan of e-buses or through tender documents which favour the participation of e-buses based on their positive environmental and health impact. As mentioned the policy and regulatory work builds upon the proof of concept investment projects and uses these to promote successful and workable policies based on concrete results. Stakeholders from the government and private sector are willing to establish and implement policies such as targets for EVs or ban of fossil vehicles but it needs to be demonstrated locally that this will not affect adversely the public transport system including its costs.

Program Components

For each country a market assessment has been performed analyzing (i) the current financial viability per vehicle category in terms of total cost of ownership, profitability, cash-flow and debt structure; (ii) financial instruments and support structures to overcome viability barriers; (iii) market readiness of EVs. Interventions are designed to accelerate market readiness and to foster widespread commercial deployment of EVs without further financial assistance.

Financial assistance is targeted to around 4 e-mobility projects in different countries focusing on e-bus systems and PT upgrading. The selection is based on the criteria established in section B3. E-mobility system investments include vehicles, charging infrastructure, grid upgrades and bus depot upgrades and are linked to urban public transport and NMT improvements. At maximum an individual country receives 35% of the total GCF funding. All countries receive technical assistance to enhance their e-mobility framework and policies. Investment projects can be executed in any of the participating country and are selected based on high quality proposals put forward on a running base to the AE. The available funds give an indication of investments in minimum 4 projects – for calculation and programming purposes the most promising and advanced projects have been used. In practice however, this might have variations and no country is included from proposing investment projects. During Program design city authorities, financial institutions and transport operators were consulted on upcoming investment projects for bus replacement or expansion of buse fleets and where interest would exist to purchase electric units if

technically and financially viable. The individual projects are listed in the country assessments included in Annex 3. For facility of review a seosrate excel sheet has been created listing all pre-identified investment projects (see Annex 3: List Investment Projects).

Technical Assistance will focus mainly on achieving Output 1.1: Establishment of a sustainable mobility ecosystem encouraging electrification and a shift to public transport. Work will be done to overcome the barriers associated with this output through the implementation of a set of activities that were specifically selected and subsequently structured according to a differentiation of scope at regional, national or local levels. Following TA activities are implemented in Argentina, Brazil, Colombia, Costa Rica, Dominican Republic, Mexico and Peru: (i) establishment of an e-mobility conducive ecosystem on the local, national and regional level by providing TA to create a policy and business framework conducive for massive deployment of EVs on a local level (project design, business model design and development, city EV policies, training etc.), national level (sectoral electrification roadmaps, support of national enabling policies for EV deployment, advice on battery re-usage, recycling and disposal) and a regional level (capacity building, knowledge materials, outreach and dissemination); (ii) Project sourcing, structuring and development; (iii) Program management. The Program has in addition a comprehensive and ambitious gender action plan targeting to improve gender equality and reduce sexual harassment in public transport.

Program Impacts

The Program supports a paradigm change to low-emission transportation systems. Impacts to be achieved are (i) GHG reduction: EVs have significantly lower lifecycle GHG emissions than fossil units in all countries included in the Program and mode shift triggered through program investments and technical assistance from modes with high levels of carbon emisisions per passenger-km to low carbon transport modes; (ii) improved air quality; (iii) reduced fossil fuel import dependency; (iv) increased energy efficiency: EVs are up to 4x more energy efficient than fossil vehicles.

Program key figures are:

- Total Program volume 210.98 MEUR and 103.89 MUSD
- GCF investment 56 MEUR and 40.264 MUSD of concessional loan and 22.3 MEUR grant for TA
- Co-finance ratio 62%
- Direct mitigation impact over lifetime of assets 4,334,051 MtCO_{2e}
- Direct impact on reduced air pollutants and energy saved
- Indirect large GHG impact due to accelerated uptake of EVs
- Economic internal rate of 16% of the total Program

Program Sustainability

E-mobility is still in a learning stage in Latin America. It therefore requires at this stage FA and TA. However, due to the efforts of the Program this is only required for a transitional period. By 2025 the cost of e-buses is expected to be at par with fossil units. However, the experience is that even if total cost of ownership of e-buses is below those of fossil untis, this does not automatically derive in operators purchasing such units. The initial CAPEX including charing infrastructure remains factor 2-3 higher thus requiring also more equity of investors and risks associated with e-buses (including risks of adequate maintenance) prevent investors from purchasing such units as long as insufficient concrete implementation experience is available in the country which clearly show that e-buses have comparable reliability and availability rates as fossil units, that actual operational costs are as expected (perceived risk of much higher than promoised energy consumption of e-buses) and that maintenance and repairs can be managed (this is also related in many countries to concerns over purchasing buses from countries with no local maintenance structure in place). The investments in projects and the TA allow to overcome the barriers of profitability and high upfront CAPEX through successfully implementing new business models

and a regulatory and policy environment which favour E-mobility investments. Interchange of experience and knowledge management instruments allow to learn from experiences and to structure in an optimal manner new investment projects thereby reducing costs and risks. Thus market forces of decreasing prices work hand in hand with Program activities which create an enabling environment with lower risk and higher profitability for follow-up investors in e-mobility, which will not longer require further technical or financial support.

Program Replicability and Transformational Impact

The potential market for e-mobility is very large in Latin America. However, especially secondary cities are facing financial and technical barriers towards adopting e-mobility. Whilst pilot projects have been successful in reducing some barriers they have not resulted in widespread adoption. The approach of the Program to invest in large-scale application combined with TA to foster new business models and with knowledge management instruments which allow for optimal e-mobility system design, learning from experience, training and policy regulations which favour EVs allows to combine all ingredients which allow for a transformational impact towards mass deployment of commercial e-mobility in Latin America. A modelling of EV deployment with and without Program was realized for different vehicle sectors and is included in details in reports 3 per country attached under Annex 3 of this FP and also available in excel version under File per country.

Program Pipeline

The initial indicative Program pipeline includes 4 investment projects (among which 50% in secondary cities) to be realized in 2023/2024 covering the total available FA. The full list of initially identified potential investment projects is for 9.6 billion USD i.e. around 20 times higher than the investment size of the proposed Program. This is a clear indication of the demand for such investments. Many investments are for fleet renovation over various years where a first batch could be financed with future funding being based on local banks. Telling is also the fact that many of the projects initially listed in 2021/2022 with the planned implementation 2023 could not be implemented due to lack of adequate finance. Other operators e.g. in Peru have decided to purchase instead fossil buses. The program will select the investment projects based on fulfilment of all Program investment criteria, on being projects with a high replication potential and demand driven i.e. first-come first-served. The Technical Assistance activities of the Program concerning the regional, national and local level will be implemented mainly by GIZ, and AFD will implement Technical Assistance for Projects feasibility studies and to structure projects from an operational, financial and legal perspective.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Climate Change Problem and the Relevance of Sustainable Transport (incl. Electric Mobility)

Transport contributes almost one-quarter of the current global energy-related GHG emissions and is growing faster than any other energy end-use sector. Latin America already accounts for 10% of global GHG emissions as of today and, transportation related GHG emissions from many countries in Latin America represent a higher proportion than the global average (16% excluding LULUCF), namely Argentina (13%), Brazil (18%), Colombia (16%), Costa Rica (37%), Dominican Republic (18%), Mexico (23%), and Peru (25%)⁵. Current transport demand per capita in developing and emerging economies is far lower than in Organisation for Economic Co-operation and Development (OECD) countries but is expected to increase at a much faster rate in the next decades due to rising incomes and development of infrastructure⁶. GHG emissions from transport are anticipated to rise from today's levels by nearly 20% by 2030 and close to 50% by year 2050 unless major action is undertaken. Limiting the global temperature increase to below 2 degrees Celsius requires changing this transport emissions trajectory, which involves the development of an integrated electromobility ecosystem encompassing various transport modes, coupled with the low-carbon production of electricity (Paris Declaration on Electro-Mobility and Climate Change & Call to Action, 2015). The Paris Declaration on Electro-Mobility calls for the deployment of electric vehicles (EVs) compatible with a 20% share of all road transport vehicles in 2030. To achieve this goal modelling performed by the IEA suggests that EVs need to represent 35% of global sales by 2030⁷.

Under a Business as Usual scenario World Data Lab calculations come to 13GtCO₂ of transport emissions by 2050 which is 67% more than 2022 and reflects a 1.9% annual growth rate of transport emissions⁸. Taking into account that all policies announced by today of governments worldwide are actually implemented GHG emissions of the transport sector would continue growing albeit at a slower pace (see figure below).

Figure 1: Projected World Transport GHG Emissions under a Stated Policy Scenario



Source: IEA

⁵ World Resources Institute, Climate Watch, 2018.

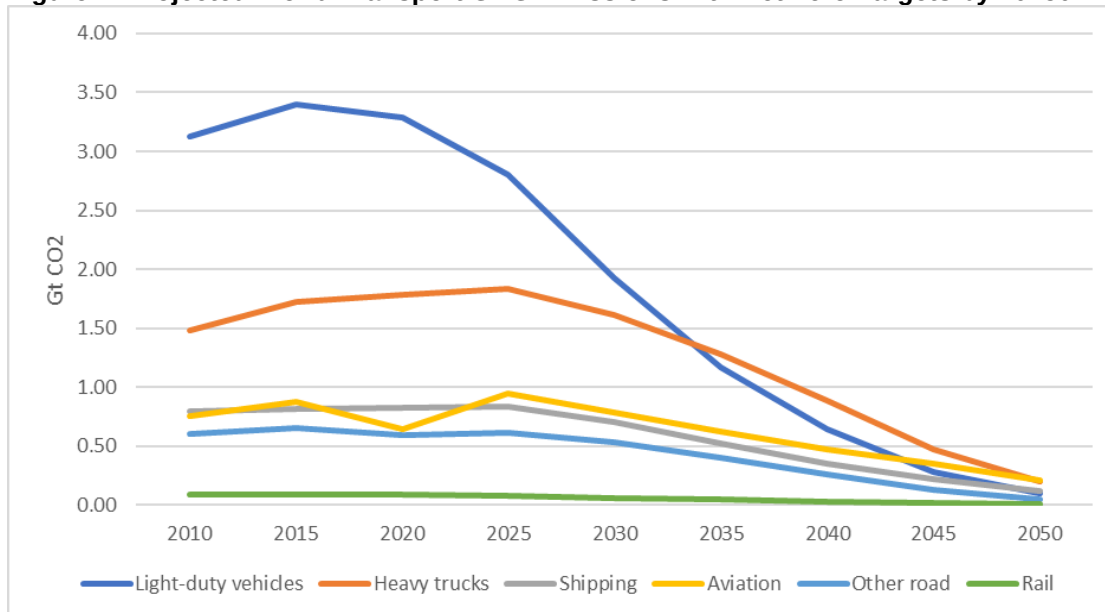
⁶ https://www.ipcc.ch/site/assets/uploads/2018/02/ipcc_wg3_ar5_chapter8.pdf

⁷ This includes BEVs, fuel cell and plug-in hybrid electric vehicles (PHEVs); [paris-electro-mobility-declaration.pdf](https://www.paris-electro-mobility-declaration.org/) ([windows.net](https://www.windows.net))

⁸ [The World of Tomorrow in Data - World Data Lab](https://www.worlddata.org/)

To achieve targets of decarbonization massive transformational changes are still required in the transport sector (see figure below).

Figure 2: Projected World Transport GHG Emissions with Net Zero Targets by 2050



Source: IEA

Avoid-Shift-Improve (ASI) has been a long-standing framework for classifying mitigation interventions in the transport sector. Electrification is part of the Improve dimension and a paradigm change and not just hybrid vehicles, eco-drive, increased vehicle occupation rates etc. i.e. typical efficiency measures. Electrification is a game-changer and a must if carbon neutrality shall be achieved⁹. No shift measures and no improve measures will be capable to reach carbon neutrality in the transport sector. Electrification coupled with a fossil-free grid creates this opportunity. Not surprisingly do carbon neutrality plans involve a fully electrified vehicle fleet. Already today electric passenger cars can result in less GHG emissions per passenger-kilometer than diesel buses, even including a life-cycle approach.

Electrification of public transport buses is an important component to modernize and to upgrade the public transport system of countries. This results in new and attractive buses which incentivize passengers to use public transportation. E-mobility is a game changer as new technology and higher quality enables cities to rethink and plan their entire public transport system.

The impact caused on modernization of public transport systems through establishing new business models which increase the financial stability and economic efficiency of public transport operators and public transport systems e.g. through separation of bus ownership and bus operations and the introduction of new investors with deep pockets to the sector. Both movements are spearheaded by the electrification of public transportation due to the different financial structure of e-buses with a high upfront investment and a long recovery time of investments. This shift of business models and development of public-private partnership incited by e-mobility can be clearly observed in Chile and in Colombia, the 2 countries in Latin America which have already embarked on electrifying their public transport fleet and both of which have within this process changed their business models how to deliver public transport services. In both cases large capital-heavy investors have entered the market (electric utilities, private capital funds and bus manufacturers) which results in public transport operations being financially more solid (in both countries

⁹ Avoid in theory also has the potential of zero GHG emissions— however, avoiding 100% of motorized transport is unrealistic

bankruptcy of traditional bus operators has been a major problem) and with better and lower cost access to capital.

The change of the business model is directly linked to e-mobility as (i) new investors are competitive with traditional bus operators with e-mobility due to the high amount of capital involved (ii) the interest of new investors is linked to electric mobility due to their business background; and (iii) e-mobility will not float with the traditional business model as operators do not have sufficient capital, to high finance costs and vested interests with the repair and spare parts market which is a major business for fossil buses but not for e-buses. This disruptive change of the business model solidifies public transport operations, reduces in the long-term its costs (due to lower financial costs) and allows for upgrading of the service offer with the separation of ownership and operations and investments not only in new buses but also in improving the network, fare integration, and increased usage of electronic means. All of these measures can be established better by a transport authority if transport operators are just operators and not system managers. This results in e-mobility being a game-changer not only in terms of fostering low-carbon vehicles but also in reorganization and modernization of the public transportation system increasing its attractiveness and thereby resulting in an important mode-shift towards public transport compared to a business as usual situation.

Program activities will also improve the grid resilience and the resilience of urban infrastructure. Used batteries from EVs have a second-life application where they can serve as storage units and thereby (i) increase the resilience and independence of the grid at critical spots such as hospitals or schools; (ii) increase the grid resilience and reduce costs of renewable grids by increasing storage capacity capable of taking up electricity during times of over-supply: this feature is especially important for grids moving towards high rates of (supply unpredictable) renewables; (iii) increase grid resilience through acting as grid shavers, especially in urban settings, thus reducing peak impacts.

Electrification is happening within all vehicle segments. However, long-range vehicles such as inter-urban buses or long-haul trucks are only at the start of electrification. Hydrogen technology might be, at least initially, a more appropriate technology solution for these vehicle segments. Battery Electric Vehicles (BEVs) are the most competitive and also have the largest environmental impact in urban areas. Hybrid and plug-in hybrid vehicle technologies have been outgrown by the fast development of battery technology and only offer a limited GHG impact. For commercial vehicle applications at least (buses, light-commercial vehicles) manufacturers have switched towards full electric vehicles. Hybrids as well as plug-in hybrids are therefore not considered in this Program (see for further details Benchmark Report in Annex 22).

The Program focuses on bus investment projects. Urban buses are the vehicle segment closest to being at cost-par with fossil units, have a high mileage and are used in urban areas where most people live. Replacing such units thus has a high GHG as well as air quality and health impact. Fostering commercial vehicles also avoids a negative social impact which could result from subsidizing private passenger cars.

Accelerating Transformation towards Low-Carbon Mobility

By 2022, there were about 18 million full electric cars on the worlds' roads¹⁰. 6 countries have nearly 90% of all electric cars in operation worldwide¹¹. EV deployment is still at an infant stage in Latin American countries and lags far behind Asia or Europe. None of the participating countries have reached 15,000 EVs yet. Under a BAU Scenario prices of EV's are dropping reaching cost parity or full commercial viability in 5-15 years depending on the vehicle category and the country¹². End 2022 700,000 battery electric buses were operating worldwide of which 670,000 in PR China¹³. In Latin America in total (including trolleybuses)

¹⁰ [Global EV Data Explorer – Data Tools - IEA](#)

¹¹ PR China, USA, Norway, France, Germany, and the UK

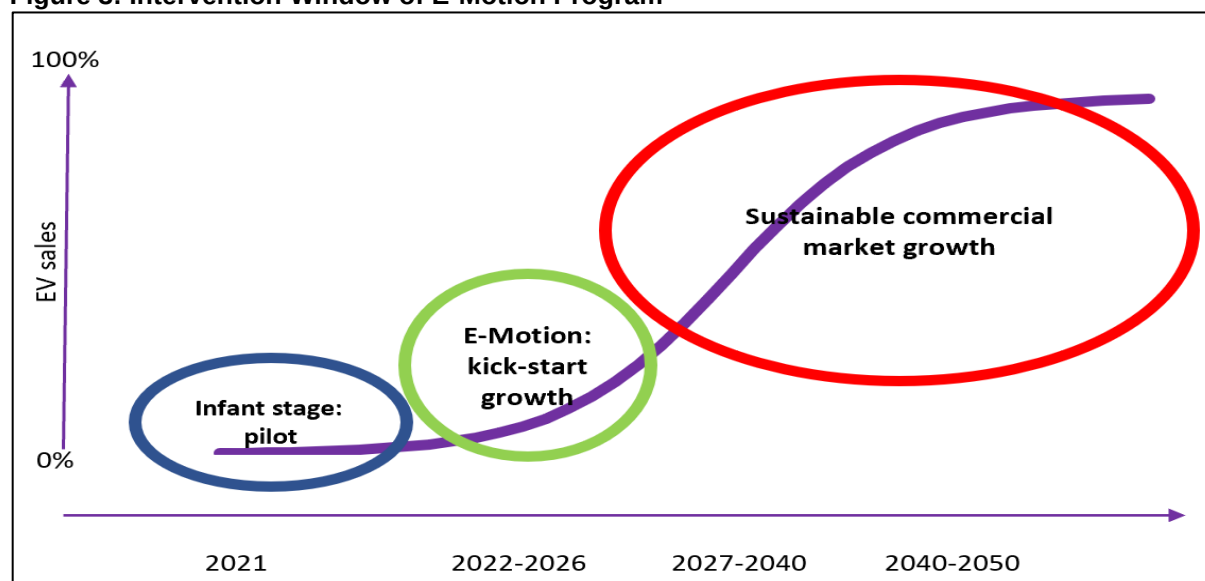
¹² See e.g. [Electric vehicle trends | Deloitte Insights](#) or [McKinsey Electric Vehicle Index: Electric Vehicle Trends | McKinsey](#); See for an analysis per country per Program targeted vehicle segment Annex 2

¹³ [Global EV Data Explorer – Data Tools - IEA](#)

some 5,100 electric buses were operating early 2024¹⁴. large fleets of electric buses are operating in Chile (in Santiago specifically with 2,000 out of in total 2,043 electric buses) and in Colombia (in Bogota specifically with 1,486 of in total 1,590 units). The other program countries had early 2024 the following number of battery electric buses (excluding trolleybuses): Brazil 142 buses; Mexico 129 buses; Costa Rica 4 buses; Peru 8 buses; Argentina: 22 buses; Dominican Republic: 2 buses

The Programs core function is to accelerate EV deployment in involved countries through financial and technical assistance (energy transition in transport). The Program ensures an at scale transition towards e-mobility and closes the gap between the current pilot stage and commercial large-scale sustained growth. The focus is on commercial EVs and therein on buses. The mass deployment is feasible if successful business models are established, if the government is willing to cover the viability gap and/or if creative financial structuring is done as is shown e.g. by Chile or Bogota. This linked with the experience gained in mass applications which reduces the performance risk of vehicles as well as decreasing costs of e-bus systems allows to ensure a mass replication of the interventions.

Figure 3: Intervention Window of E-Motion Program

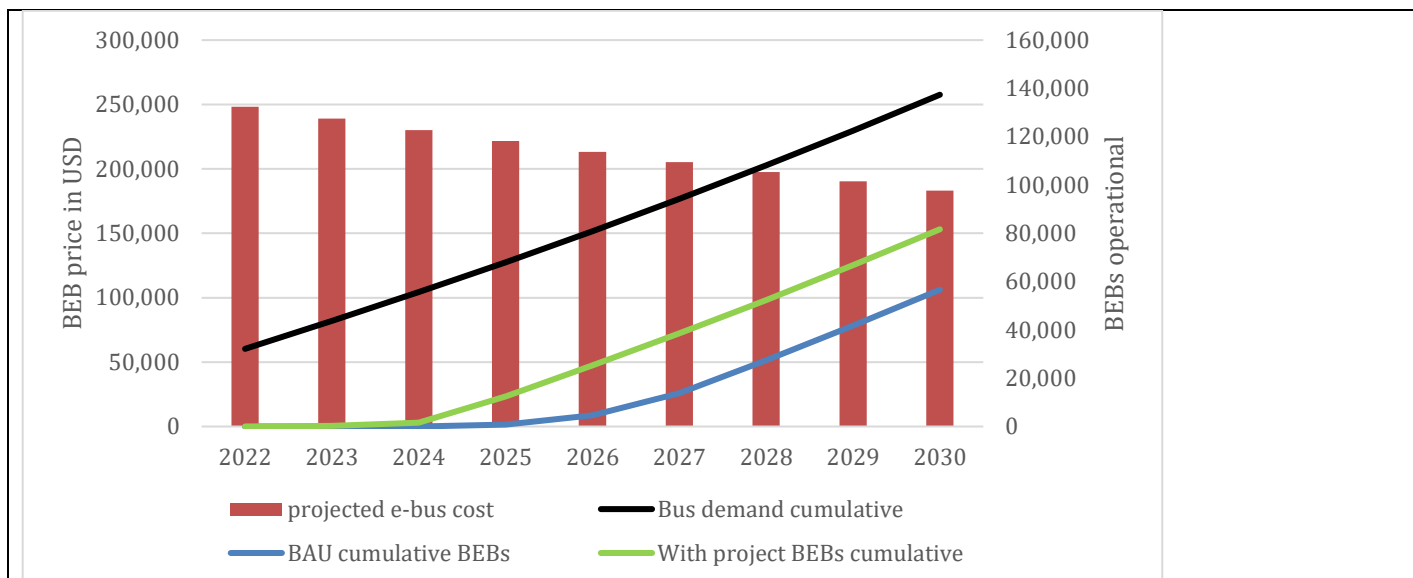


The Program has a strong focus on electrification, especially in its financial component. It implements the required interventions to kick-start mass deployment of electric vehicles significantly earlier than with a BAU scenario by reducing the risk profile of investments and by comprehensive technical assistance. The Program is focused in countries and vehicle segments where this gap can be closed i.e. where conditions for promoting on a larger scale e-mobility are met and where simultaneously commercially sustainable e-mobility deployment is feasible within 7 years by kick-starting the process and by eliminating barriers towards commercial uptake. This is the reason why the Program does not have the same sequencing and activities as well as support levels in all countries. The Program is thus not about pilots and initial technical assistance, which already exist in the region, but about accelerating market-based commercial EV deployment.

Per country modelling has been performed to determine the projected EV deployment with and without Program intervention. This is detailed in the reports 3 of each country included in Annex 3 as well as the corresponding country excel files also included under Annex 3. The following graph shows the modelled e-bus deployment with and without program for Mexico.

Figure 4: Projected deployment of e-buses with Program and BAU for Mexico

¹⁴ [E-BUS RADAR - E-BUS RADAR \(ebusradar.org\)](https://ebusradar.org)



Source: Annex 3; Report File 3 Mexico and File 3 Mexico excel; Grutter Consulting

Strengthening Public Transport

Electrification is an enabler for improving not only vehicles but increasing services quality and working with partners on policies for incentives to shift to public transport and increasing ridership. Introduction of EVs is complemented by technical assistance on the mobility transition and by investments in improving PT systems. Projects with e-buses are located in cities which want to upgrade their urban mobility system and modernize public transport to also increase its mode share.

Electric mobility represents a lever to shift the underlying paradigm that private vehicles are the preferred mode of transport. Introducing a large number of electric vehicles into the fleet requires transport operators and cities to fundamentally redesign and improve the overall public transportation system. Accelerating market-based commercial EV deployment across Latin America cannot be guaranteed by electric buses solely achieving cost-parity with rival technologies. Perceptions of risk, operational practices, business models, user choices and how local governments plan for integrated, sustainable urban transport systems, including non-motorised transport (NMT), also must change.

Most existing EV bus electrification initiatives focus only on the exchange of vehicles and the provision of the required charging infrastructure and training of those operating the new systems. Often only a few vehicles running on one or two lines are replaced. Initiating a larger-scale replacement of vehicles offers the unique opportunity to also re-think the overall mobility system to create an attractive alternative to private vehicles, utilising the exchange of vehicles to trigger a systemic change towards a comprehensive mobility environment centred around public transport and non-motorised and low emission modes.

Figure 5: Demand stimulation for public transport involving electric buses

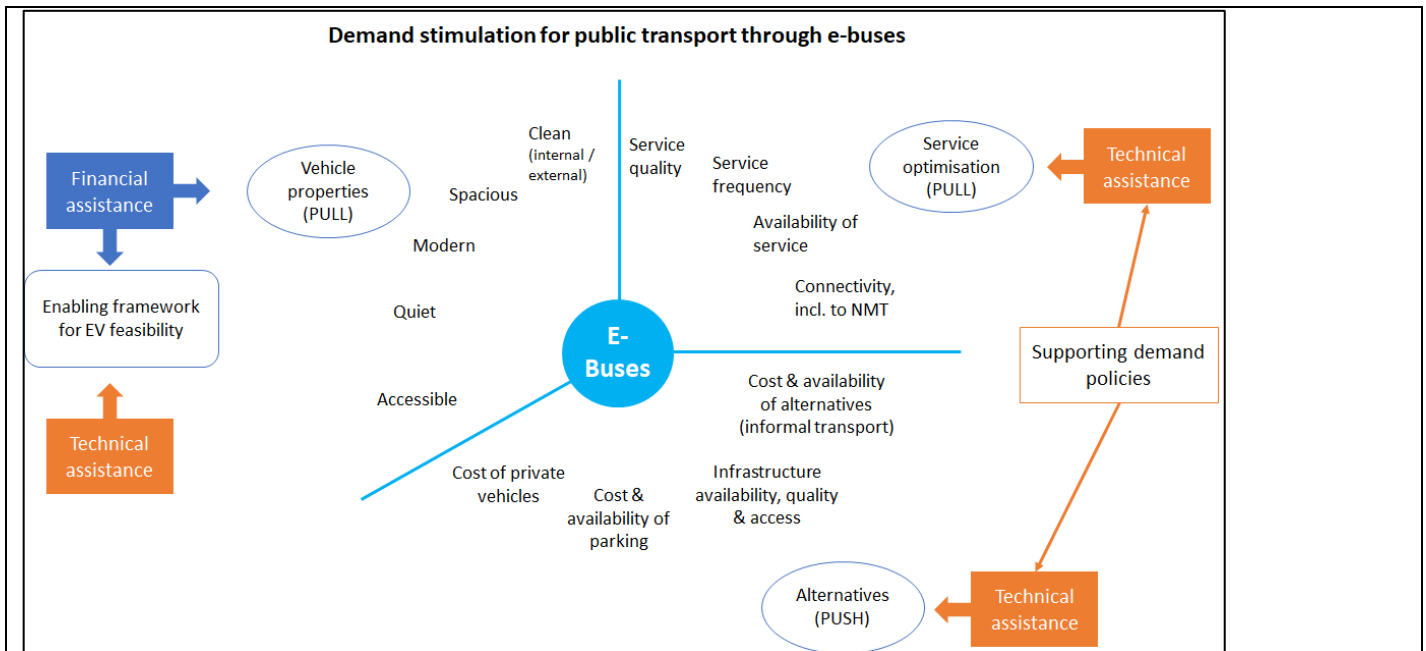


Figure 2 illustrates the push and pull elements that enable a shift towards more use of public transport and, if well-connected, to walking and cycling. The goal is to enhance efficiency of public transport systems and make the sustainable options more attractive than personal motorised transport to counter the past trend in declining public transport ridership. Shifting to electric buses can contribute to this by providing more attractive vehicles. The programme will provide financial assistance to de-risk financial products and accelerate the market take-up of electric vehicles. Technical assistance will support the development of conducive policy frameworks, as well as the identification of innovative business models and contractual arrangements that safeguard the financial and operational viability of public transport services.

The upgrade of the vehicle fleet is an opportunity for reforming the transport systems in cities. Electric mobility offers an unique opportunity to re-structure the PT sector, introduce new PPP and business models based on separation of asset ownership and operations, favour the entry of new financially stronger players, and all this leading to the structuring of a modern, attractive and integrated offer to potential users of PT. This is why the technical assistance component goes beyond electrification and complements the direct impact of fleet renewal with indirect results in terms of shift from high-carbon to low-carbon modes. Customers might also appreciate an EV more due to less noise or due to a positive image and therefore ridership of electric buses could increase resulting in a positive mode shift towards EVs and therefore additional ERs. A report found for example that e-buses could attract 1.9% additional ridership compared to diesel units (Currie, 2018). More important is however the impact caused on reorganization and modernization of public transport systems through establishing new business models which increase the financial stability and economic efficiency of public transport operators and public transport systems. This results in e-mobility being a game -changer not only in terms of fostering low-carbon vehicles but also in modernization of the public transportation system increasing its attractiveness.

The E-Motion program will also have a significant impact on the development of sustainable and low carbon mobility in the region. The Program is implemented in countries and cities supported by the EUROCLIMA+ program, implemented by AFD and GIZ, and that establishes all the enabling conditions to do this shift towards sustainable transport in the region through the identification of mitigation measures for the effects of climate change in the transport sector. Indeed, EUROCLIMA+ Program finances 2 Latin American national governments National Urban Mobility Plan (NUMP), 6 Sustainable Urban Mobility Plans (SUMP) approved by local governments fostering multimodal participatory integrated mobility planning and 3 pilot projects. The Budgetary credit lines financed by AFD and supporting a high level political dialogue on transport decarbonization in Costa Rica and Dominican Republic also ensure that the projects

implemented in these countries through E-Motion will have a key impact in public transport restructuring and decarbonization. The TA will be limited to activities which are complementary to the ones already realized within EUROCLIMA.

Local Context

Latin America has strong enabling conditions in terms of clean transport and e-mobility. The region has one of the cleanest electricity generation portfolios in the world (high hydroelectric production and growing deployment of wind and solar power plants). Therefore, the adoption of an e-mobility strategy would allow the sector to move towards a decarbonisation pathway. The table below shows the carbon intensity of electricity generated of Program countries. With these grid factors EVs can reduce GHG emissions on a well-to-wheel base by 70-90% compared to fossil units¹⁵.

Table 1: Carbon Grid Factor of Program Countries

Country	Carbon Grid Factor Electricity in kgCO ₂ /kWh
Argentina	0.288
Brazil	0.150
Colombia	0.208
Costa Rica	0.039
Dominican Republic	0.426
Mexico	0.359
Peru	0.252
Median	0.252

Source: IFI Version 3.2, 2022

Latin America is also well known for having developed innovative bus-based transportation systems (especially in Colombia and Chile) and cities like Bogotá, Medellín and Curitiba have become international benchmarks for innovation in this area. Finally, Latin America has a positive local manufacturing context: three large vehicle producers operate in the region (Argentina, Brazil and Mexico) and four countries have large lithium reserves (Argentina, Bolivia, Peru and Chile) – essential material to build EV's batteries. The main motivating forces to foster e-mobility in the countries included in the Program are: (i) Reduce GHG emissions; (ii) Improve air quality and reduce health problems associated with air pollution; (iii) Reduce dependency on imported fossil fuels and exposure to external price shocks; (iv) Re-structure and modernize the public transport system to increase its attractiveness and mode share; (v) Achieve positive economic impacts related to reduced health costs, reduced fuel import bill and job creation; (vi) Participate in the new growth industry of EV and battery manufacturing;

Air pollution is a large problem in most Latin American cities with levels affecting seriously human health. According to the latest urban air quality database, 98% of cities in low- and middle-income countries with more than 100,000 inhabitants do not meet WHO air quality guidelines¹⁶. The WHO estimates that annually 4.2 million deaths result due to excessive exposure to fine particulate matter equivalent to 7.6% of all deaths¹⁷. The transport sector is thereby a major source of air pollution. The most vulnerable population is disproportionately affected by air pollution, as they tend to live and work closer to its sources¹⁸. Children and the elderly are particularly vulnerable. Recent studies show that women are more affected by poor air quality than men¹⁹.

¹⁵ See Annex 23

¹⁶ <https://www.who.int/airpollution/data/cities-2016/en/>

¹⁷ <https://www.who.int/gho/phe/en/>

¹⁸ Mitchell G. and Dorling D. 2003. An environmental justice analysis of British air quality, *Environment and Planning*, vol. 35, 909-929

¹⁹ Clougherty J. 2010. A Growing Role for Gender Analysis in Air Pollution Epidemiology, *Environmental Health Perspectives*, 118(2), 167-176

The COVID'19 pandemic has created an unprecedented challenge for many Latin American countries, from the health and economic perspectives, but it is also opening new opportunities for an accelerated sustainable transformation of its energy, transport, and urban landscapes while contributing to economic recovery. A recent report prepared by the ILO shows that fostering of electric mobility can have a significant positive job impact primarily due to the induced impact of savings of consumers on fossil fuels and vehicle maintenance resulting in increased spending on goods with a high income elasticity which tend to be labour intensive service-goods (ILO, 2020).

AFD has conducted a feasibility study of the E-Motion Program including a diagnostic and market study of all countries involved (see Annex 2). Core information on each country given below is complemented with information on the countries NDC and transport sector in Section D5.

Argentina

Area	Population	GDP/Capita	Vehicle Fleet	Diesel price ²⁰	Electricity price ²¹
2,780,400 km ²	46 million	13,700USD	19 million	1.09 USD/l	0.08 USD/kWh

Sources: see Annex 2; data population and current GDP for 2022; energy prices January 2024

Argentina is one of the largest economies in Latin America. It is also, along with Brazil, Mexico and Colombia, one of the main producers of automobiles and auto parts in Latin America. Lithium reserves represent a unique opportunity for the transition towards electric mobility. The National Action Plan for Transport and Climate Change represents the set of initiatives that Argentina has planned to do to reduce GHG emissions in the transport sector. It is based on strategic intervention axes and mitigation and adaptation measures. Argentina launched the development of its National Electric Mobility Strategy in May 2018 with the support of UN Environment. Argentina is currently formulating specific legislation on electric vehicles, establishing the conditions for the installation or operation of charging centres, and at the same time, exploring possible options for the local development of the electric mobility industry. Various pilots are running with electric buses in Argentina.

Brazil

Area	Population	GDP/Capita	Vehicle Fleet	Diesel price	Electricity price
8,511,965 km ²	215 million	8,900 USD	98 million	1.19 USD/l	0.14 USD/kWh

Sources: see Annex 2; data population and current GDP for 2022; energy prices February 2024

The Brazilian automotive industry is among the ten most important in the world. The government promotes the local sales of EVs by providing manufacturers with "super credits". The NDC indicates that the transportation sector shall promote efficiency measures, improve urban public transportation infrastructure and biofuels to achieve mitigation targets. Brazil has already established various tax incentives for the import and purchase of EVs. A National Strategy of Electric Mobility is under development as part of the NDC strategy of the country and 21 bills are being discussed in parliament (tax reductions for fuel efficiency, stringent rules for manufacturers, import fare reductions).

Colombia

Area	Population	GDP/Capita	Vehicle Fleet	Diesel price	Electricity price
1,141,748 km ²	52 million	6,600 USD	8.5 million ²²	0.61 USD/l	0.17 USD/kWh

Sources: see Annex 2; data population and current GDP for 2022; energy prices end 2020

²² Registered vehicles 14 million; operating 8.5 million

Colombia has an automotive industry dedicated mainly to vehicle assembly, auto parts production and motorcycle assembly. The NDC contemplates for the transportation sector amongst others to achieve 600,000 registered electric taxis, buses, light commercial vehicles including small trucks and official vehicles. The Government has developed the National Strategy for Electric Mobility, which aims to promote the electrification of the transportation sector. Bogota has purchased as of end 2020 nearly 1,500 electric buses, whilst Cali and Medellin have pilot fleets of e-buses. However, all e-buses purchased to the moment are 9-12m units not operating on bus-only lanes. Initial pilots of e-taxis have also been deployed. Law 1964 requires that all buses purchased from 2035 onwards be electric. Colombia has established a good business model for e-buses. However, this model still requires a subsidy from the municipality (16% additional lifespan cost expressed in 16% higher cost per km paid to e-bus operators). Large municipalities such as Bogota are willing to do this and have the financial means. Smaller municipalities have different bus operation systems (no BRTs) and do not have these financial means. Also e-buses purchased to the moment are only for 8-12m feeder buses. For high-frequency trunk routes operated with 18-26m buses the e-bus technology would need to be different e.g. based on ultra-fast charging which requires a different business set-up due to large investments required in electric infrastructure for charging with different ownership structures (investments at bus stops and not bus depots). The program therefore focuses for Colombia on these areas and will not finance standard 8-12m feeder buses for cities with BRT systems.

Costa Rica

Area	Population	GDP/Capita	Vehicle Fleet	Diesel price	Electricity price
51,100 km ²	5 million	13,300 USD	1.5 million	1.19 USD/l	0.22USD/kWh

Sources: see Annex 2; data population and current GDP for 2022; energy prices end 2020

Costa Rica reaffirmed its aspiration of becoming a Carbon Neutral economy and aims for a decarbonized economy with net-zero emissions in 2050. The updated NDC of Costa Rica has concrete 2030 e-mobility targets for public transport, passenger cars and fleets (8% of the vehicle stock). For other vehicle categories such as motorcycles targets and measures shall also be developed to migrate towards EVs. Costa Rica has published a national plan for electric transport which includes concrete steps towards electrification of vehicles and has approved 2018 the law on incentives and promotion of electric transportation which includes targets for EV penetration, the establishment of a public charging infrastructure as well as important tax incentives for private EVs.

Dominican Republic

Area	Population	GDP/Capita	Vehicle Fleet	Diesel price	Electricity price
48,400 km ²	11 million	10,200 USD	4 million	1.00 USD/l	0.19 USD/kWh

Sources: see Annex 2; data population and current GDP for 2022; energy prices end 2020

The updated NDC of the Dominican Republic proposes as mitigation actions the electrification of buses, taxis, school buses, minibuses and "conchos" and the creation of policies to encourage the transition to electric and hybrid mobility for private usage. The country has realized a National Strategic Plan for Electric Mobility with specific EV targets for 2030 (30% of official vehicles and public buses, 10% of private vehicles) and for 2050 (100% EVs for official vehicles and public buses and 70% of all private vehicles shall be electric).

Mexico

Area	Population	GDP/Capita	Vehicle Fleet	Diesel price	Electricity price
5,144,300 km ²	128 million	11,500 USD	50 million	1.40 USD/l	0.22 USD/kWh

Sources: see Annex 2; data population and current GDP for 2022; energy prices end 2020

The automotive industry has a 3% share of the national GDP. Various manufacturers have announced intentions to manufacture e-buses in the country and one European and one Chinese manufacturer are already producing or assembling e-buses in the country. The Climate Change Strategy to 2050 published in 2016 established that within 10 years the use of EV in public transport shall be common. The National Electric Mobility Strategy Vision 2030, has as goal of having 10 urban areas with electric mobility as well

as a 5% share of total sales of new electric or hybrid vehicles by 2030. The Management Program to Improve Air Quality in the State of Mexico 2018-2030 proposes to prioritize the implementation of electric transportation and public transport as well as to implement an electric taxi program. Mexico City also published the Strategy for Electromobility of Mexico City 2018 - 2030, in which it proposes that 20% of the fleet will be electric in 2030 and 80% of the cabs will be hybrid or electric for the same year. Metrobus in the capital city has an initial fleet of 18m full electric buses it is using on one of its BRT routes and has the intentions of expanding the fleet but eventually choosing a different charging technology.

Peru

Area	Population	GDP/Capita	Vehicle Fleet	Diesel price	Electricity price
256,300 km ²	34 million	7,100 USD	2.4 million	1.01 USD/l	0.17 USD/kWh

Sources: see Annex 2; data population and current GDP for 2022; energy prices end 2020; CNG price: 0.29 USD/kg

Peru has very low levels of Compressed Natural Gas (CNG) prices due to having abundant national gas reserves. EVs cannot compete with these prices due also to the fact that the electricity price is relatively high. Thus the Program will not realize activities in areas connected currently or in the near future to the gas pipeline system. However, various cities such as Arequipa or Cuzco are and will not be connected in the medium future to the pipeline system and depend on diesel buses thereby suffering from high pollution levels. In these cases EVs make commercial sense. Therefore the Program will focus on these areas.

In none of the program countries diesel fuel is as of February 2024 being subsidized. Prices for diesel are, with exception of Colombia, over 1 USD per litre being virtually identical to the diesel price paid in the USA. Only Colombia had a lower price with 0.61 USD/l but also in this case this does not involve a fuel subsidy. Not all external costs are included in the fuel price, which is the case for **all** countries worldwide. In this sense all countries currently subsidize fossil fuels due to not including external costs fully (health, environmental costs etc). The current fossil fuel prices make the usage of electric buses in general profitable from a total cost of ownership perspective for all involved countries. However, payback times are long and risks potentially high. Risks are related to reliability of the technology – this is also linked to usage of buses basically from China with limited experience of operators in Latin America with Chinese buses and with partially negative experience with such suppliers including negative experience of post-sales service which for e-buses is critical to maintain acceptable bus availability ratios. Risks are also related to the lifespan of the battery and actual electricity usage as well as risks related to non-fulfilment of daily operational requirements due to range issues. Range issues are also related to the choice of the battery technology and the charging system – optimal choices require expert knowledge not widely available in Latin America. Many of these risk factors cannot be addressed well contractually or with guarantees. The risk appetite of public transport operators on the other hand is limited as they do not want to jeopardize their core business and revenue source namely transporting passengers. Therefore, technical assistance for high quality system design, advisory services in contract formulation, business models which allow for reducing risks and exposure and concessional finance are critical to initialise large fleets of e-buses. Summarized (i) fuels are not subsidized and (ii) the current fuel prices are adequate from a financial viewpoint and make investments in e-buses in general profitable but not sufficient to overcome the risks associated with the technology.

Barriers addressed by the Program

1.Limited commercial attractiveness of e-mobility investments:

- A core barrier to EV deployment is the 2-3x higher CAPEX of EVs including associated infrastructure. The higher CAPEX results in higher capital requirements, higher finance and insurance costs and higher risk exposure due to higher debt levels. Higher CAPEX can be partially or fully recovered with lower Operational Expenditures (OPEX), primarily lower energy and maintenance costs and a higher lifespan of EVs. However, the profitability of EVs is only in the long-term and insecure. In the majority of involved countries the differential investment in an e-bus

is not recovered during its lifespan, due also to having to re-invest in batteries and chargers during the asset lifetime. Municipalities and transport operators in developing countries therefore prefer to invest scarce capital resources in a larger number of new fossil vehicles although electric units have potentially lower total costs of ownerships (TCO). This barrier is addressed by the Program with concessional loans reducing financing costs, with the promotion of business models which reduce capital requirement of bus operators and which reduce finance costs and with TA which allows to identify the most convenient e-bus system approach which allows to reduce upfront investment and operational risks.

- The low financial profitability of EV investments reflected in low FIRR and very long payback times, constitutes an important barrier that guides investment decisions related to electromobility.
- Another barrier related to commercial attractiveness of EVs is the higher risk of EVs compared to fossil vehicles as the technology performance level in the respective country is unknown, the long-term profitability of EVs depends on long-term energy price development and cities and operators lack know-how in EV system design, maintenance and operations. This results in a high-risk profile of EVs. This barrier is primarily addressed by the Program through TA in the optimal design and management of EV systems and the information and capacity building efforts which allows to learn from experiences of already implemented projects.

2. Business models and policies non-conducive to sustainable urban mobility and EV deployment:

- This barrier is prevalent in various countries as governance structures and business models are not conducive to EV deployment. Innovative and localized business models that redefine the roles and the risk allocation among involved private and public actors are required. Local regulations, contract structures, payment systems and fare structures are frequently not favourable for e-buses (e.g. concession periods which are too short to recover the high investment in e-buses). Adjusting such structures can be a game-changer for e-mobility allowing to apply innovative business models adapted to the eco-system of e-mobility. This barrier is addressed by the Program through TA designing and promoting e-mobility appropriate business models and through outreach showcasing successful business models such as those established in Santiago de Chile or in Bogota.
- Non-conducive policies: Whilst many countries have established general targets or policies these have not yet trickled down to concrete and tangible support policies such as a reform of the public transport sector which increases the demand for attractive public transport, improve the framework conditions and allow for private sector investment and participation to reach scale. This barrier is also tackled with TA, in a regional manner establishing know-how platforms and interchange of information.

3. Limited know-how and experience with e-mobility systems and lack of experience in the design of effective interventions:

This barrier refers to a lack of know-how on optimal EV ecosystem design (identification of type of EVs to use, fast/slow/opportunity charging etc.), maintenance & operations, as well as on used battery management policies including options of 2nd-life usage, recycling and disposal. This barrier is addressed through TA of the Program including studies on best practices in different countries, assistance in the design of appropriate regulations and policies and with outreach activities to ensure that experiences learnt in countries are also taken into consideration in other countries. GIZ has for example worked in these different areas in various countries already and can use that experience and know-how applying it to other countries and regions.

The Program tackles these barriers and offers a comprehensive e-mobility package including financial and technical assistance thereby improving the policy framework and the ecosystem and removing barriers which allows to kick-start mass deployment of EVs. It further tackles the link between increasing service quality (PULL) and encouraging shift from cars to public transport (PUSH). The activities and instruments

used by the Program to remove these barriers are detailed in section B3. Currently many cities and countries have pilots deployed with mixed success. Pilot projects have the large drawback of resulting very expensive (bus prices are much higher if less than 50 units are purchased and the infrastructure cost gets very expensive) and with multiple problems related to maintenance, repair and problem solving. Manufacturers will not deploy appropriate after-sales services for few units. The step between pilots and full-blown implementation has not been made by transport operators as latter requires a high-quality technical design and a large investment which is considered as too risky by operators and by conventional finance institutions. As example Costa Rica has already made 2 pilots of 2 and then 3 buses – the first 2 decades ago and the second with GIZ funds finished 3 years ago. The last pilot had large-scale technical assistance developing business models etc. but no financial instrument and no transport operator nor the government hopped on. The same is true e.g. for pilots in Peru, Argentina or the Dominican Republic and even for countries like Mexico or Brazil where no large fleets of e-buses are operating. The project overcomes this first mover problem with concessional finance and targeted high quality technical assistance to ensure high quality buses, charging infrastructure, an optimal combination of charging system and buses. It also builds on an adequate business model distributing risks and purchase contracts which include multi-year service, maintenance and repair plus training and capacity building to be provided by the manufacturer.

Related projects/interventions

Multiple actors are engaged in promotion of e-mobility. AFD group and GIZ are important actors in e-mobility in various Program countries. The E-Motion Program is implemented by AFD in 7 countries of the region (Argentina, Brazil, Colombia, Costa Rica, Peru, Mexico and Dominican Republic).. The AFD FP and the GCF approved CAF FP (FP 195, approved at B34, no overlapping countries) are part of the same E-Motion program and approach aiming to promote e-mobility in the region.

UN Environment is implementing the GCF Readiness Program “Advancing a regional approach to e-mobility in Latin America” in thirteen countries in the region. The Program focuses on enabling strategies and policies to scale up electric mobility. The Global Environment Facility (GEF) is also implementing through UNEP a Global E-Mobility Program at the world level including Costa Rica. The individual TAs per country have been designed so that a complementarity exists with on-going and planned initiatives of other actors.

The Inter-American Development Bank (IDB) is also an active e-mobility player in various countries e.g. financing a pilot fleet of e-buses in the Dominican Republic. IDB also has an approved GCF e-mobility program (FP189) for the Latin American region, with complementarities with the E-Motion Program. The IDB program also includes Colombia and Costa Rica whilst all other countries covered by this Program are not included in the IDB program. The IDB program is broader in scope including also pilot hydrogen projects as well as electric vessels with potential electric vessel projects having been identified for Costa Rica as well as for Colombia. The three programs of CAF, IDA and AFD will collectively have the potential to define a transformational change in the transportation system of the Region.

Projects/programmes in the region implemented by intergovernmental organizations and multilateral development banks focusing on the introduction of electric mobility include:

1. GIZ – Changing Transport²³: Supporting partners in defining low carbon transport strategies, implementing climate actions to reduce emissions and harnessing sustainable transport benefits through the Transfer project in Colombia, the PROMOB-E platform in Brazil, the Mi-Transporte Program in Costa Rica and other initiatives in all the countries concerned by the Program. GIZ is also the organization in charge of delivering the TA activities of the e-mobility Program.

²³ <https://www.changing-transport.org/about/>

2. Euroclima+ program is a program implemented by AFD and GIZ with activities in the Program countries Argentina, Brazil, Colombia, Costa Rica, Mexico and Peru. Its objective is to develop national policies for urban mobility and integrated multi-modal planning at the city level to accelerate the transition of Latin American cities toward sustainable urban mobility low in carbon emissions. As a result, EUROCLIMA+ contributes to create the favourable conditions of the development of e-mobility in the framework of integrated and sustainable urban mobility plans. This is therefore an important link to achieve the structural change of the urban transportation system towards low carbon transport and links up e-mobility promotion with low carbon urban development. More specifically, with a total allocation of 10 million euros, Euroclima+ has the following goals: (i) Support the National low carbon Urban Mobility Policy in Ecuador and promotion of the National strategy of Electric Urban Mobility in Uruguay; (ii) Support the elaboration of the following 7 Sustainable Urban Mobility Plan (SUMP): Regional Plan for Mobility and Logistics of Bajada Santista (Brazil), Mobility and Sustainable Urban Transport Plan for the Provincial Municipality of Arequipa (Peru), Sustainable Urban Mobility Plan for Córdoba (Argentina), improvement of the Integral Plan for Sustainable Urban Mobility of the Guadalajara Metropolitan Area (Mexico); (iii) Support the implementation of 3 pilot projects: Mobility Observatory to improve the performance of public transport in Teresina (Brazil), Pilot Project on soft mobility in Cordoba (Argentina) and development of bike parkings in BRT stations in Puebla (Mexico); (iv) A regional platform/community of practice is strengthening exchanges and capacities of cities and national Latin American governments for NUMPs and SUMPs.

The E-Motion Program will leverage on ongoing programs of GIZ and Euroclima+ specifically through the following aspects:

- TRANSfer Colombia, being a "project preparation facility", designed a national program, including the structuring of a financial fund that could allow for major private investments, but also national guarantees and subsidies. Furthermore, an MRV system was developed, training curricula for business models, technical operation and planning were developed and a government mechanism was established, that E-Motion can build on.
- MiTransporte: the project implemented an electric bus pilot, including data collection, which enabled that local bus operators gained interest and trust in the new technology. E-Motion can build on this to accelerate the deployment. The project furthermore reduced barriers for the uptake of e-buses (e.g. preferential tariff for charging) through a National steering committee on e-mobility, has trained different actors (drivers, maintenance personnel, firefighters, electricians, etc.), implemented a pilot project with commercial light duty EVs with the Institute of Technology Costa Rica and companies, evaluated operator models and financing mechanisms for e-buses that can be implemented under E-Motion and has established an eco-system and a plant for battery recycling with a private company (develoPPP FORTECH)
- PROMOTE: This project created the framework for electric mobility in Brazil by promoting a multi-stakeholder dialogue in the National Platform for Electric Mobility (PNME), the introduction of standards and regulations for charging infrastructure and the support of pilot projects in urban public transport and urban logistics.
- Euroclima+: The NUMP in Uruguay will set a long-term vision regarding sustainable mobility that will enable different sectors to engage. The NUMP will generate capacity at the city level to develop mobility plans, those plans expect to be financed in private public partnership. Consideration for future financing models are welcome. Uruguay is building a working group that will set the agenda for fleet renewal as for energy production. Considering their almost 100% renewable electric grid, they are exploring Hydrogen and Bio-fuels. Governance capacity is suitable to pilot energy transition as to work on innovative multi-actors networks to accelerate the transition and to outreach lessons learned to other countries.

- TUMI E-Bus Mission: This project is supporting 10 Deep Dive Cities in Brazil, Colombia and Mexico to set ambitious targets for e-bus deployment, and to elaborate and implement a work plan for tendering electric buses.
- 3. UNEP - Global Programme to Support Countries with the Shift to Electric Mobility²⁴: The aim of the Global Programme (covering Antigua & Barbuda, Chile, Costa Rica, Ecuador, Grenada, Jamaica, Peru and St. Lucia in LAC) is to support low and middle-income countries in overcoming these barriers through a combination of technical assistance and investment. Technical assistance on the Country Child Project level includes awareness raising campaigns, capacity building and the development of instructional structures to support the introduction of electric mobility. It furthermore includes laying the ground for large-scale market introduction of electric mobility through 1) The development of policies, including fiscal, regulatory and local measures; 2) The establishment of adequate business models and finance mechanisms; and 3) The development of plans and studies to ensure environmental sustainability. The investment component of the Country Child Projects will be used for electric mobility demonstration projects to allow for the creation of local experience to de-risk the technology and to attract investors to upscale electric mobility in the respective countries.
- 4. UNEP GCF Readiness Programme on E-Mobility in Latin America: The UNEP Regional Office in Panama currently implements a GCF Readiness Programme on E-Mobility with the title “Advancing a regional approach to e-mobility in Latin America” including the Program countries Argentina and Costa Rica. The project aims at building capacity, development of e-mobility policies and business models, identification of strategies to finance the shift to electric mobility in the region and will be implemented combining a national with a regional approach, whereby close collaboration is sought for with the GEF Global E-Mobility Programme.

UNEP - MOVE Electric Mobility in Latin America²⁵: In 2016, UNEP and Mario Molina Centre Chile launched a regional platform to accelerate deployment of electric mobility in Latin America. Since then, MOVE has provided (1) capacity building to public technical officers from almost 20 countries in the region through online webinars and face-to-face meetings, (2) knowledge creation on the uptake of electric mobility in Latin America and the Caribbean, (3) technical assistance, through the elaboration of National Electric Mobility Strategies in the Program countries Colombia and Argentina and (4) resource mobilization to enable countries to transition to electric mobility – with a special focus on electric road mass transit.

The ongoing and planned project / programmes are addressing the current inexistence of socially and economically viable alternatives to fossil fuel road transport. However, despite those ongoing initiatives, barriers to the broad adoption of EVs continue to exist in many countries. This is related to a significant extent to the projects focus only on technical assistance lacking financial assistance instruments and not allowing to implement actually new business models. For an accelerated shift of the transport sector towards e-mobility, overcoming the investment barriers will be key in the Program countries. Gap finance and financial incentives are still missing to a larger extent. The E-Motion Program will contribute significantly in closing the funding gap of higher up-front cost, while providing business model support and complementary TA elements for commercial fleets and general enabling activities, which will help accelerating the transformation towards low-carbon mobility in the respective countries.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

IF low carbon urban mobility systems are implemented and an enabling environment for e-mobility can be established **THEN** the population will have access to affordable, attractive and convenient urban transport

²⁴ <https://www.thegef.org/project/global-programme-support-countries-shift-electric-mobility>

²⁵ <https://movelatam.org>

and investors will be interested in pouring resources into EV projects which helps the involved countries to meet their GHG reduction targets **BECAUSE** transportation systems will have a more gender balanced and comprehensive mobility offer using low carbon EVs.

The potential **paradigm shift** of the Program in terms of **scale** is a shift from the current reliance on fossil-fuel powered transportation towards electrified low-emission transportation means. The Programs technical and financial assistance in the design and implementation of business models in line with the specific needs of electric mobility are decisive factors to foster this paradigm shift. The development and implementation of business models which tackle the barrier of high upfront investments pulls in other investors and allows for massification at scale in a commercial manner. In terms of **replicability** the Programs investments in electrified urban transport combined with the market trend of strongly decreasing prices of EVs allow for solutions to be replicated across other cities in Program countries. This is enabled through knowledge management and outreach instruments and the implementation of at-scale implementation projects which reduce decisively the risk levels for follow-up investors. The paradigm shift is **sustainable** due to the promotion of government policies favouring investments in EVs and resilient infrastructure combined with a market where EVs will achieve price-parity with fossil units within this decade. The interventions make cities more liveable resulting also in a behavioural change towards public transport. The Programs technical assistance in the design and promotion of policies which enable a commercial mass-uptake of e-mobility and the interaction with the private sector in the implementation of projects are critical for achieving this target. The main contribution of the GCF is that this paradigm shift towards low-emission electric mobility happens faster and at a much larger scale than under a BAU scenario. It is expected that commercial investors will crowd in to the e-bus area once the framework conditions (e.g. guarantees, concession contracts) and the performance risks of e-buses are manageable. This has been the case in Santiago de Chile as well as Bogota. A coalition of investors announced end 2021 the intention to invest 1 billion in e-buses in Latin America²⁶. The Programs TA creates conditions which are favourable to actually materialize such investments and the investment component of the Program reduces the risks to investors and operators. The Program is thus critical to unlock the private investment potential.

The expected **outcomes** of the Program are (1) improved public and private sector capacity and a more favourable enabling policy framework for low carbon transport; (2) reduced GHG emissions from the transport sector due to a shift from private to public modes of transport and due to a shift from fossil to electric means of transportation resulting in 4.33 MtCO_{2e} decreased GHG emissions and reduced energy usage estimated at 48,343 TJ of energy saved due to EVs being on average factor 3 times more energy efficient than fossil units and PT being more energy efficient per passenger-km than other modes of transport

Expected **co-benefit outcomes** are (i) Reduced economic & social cost of air pollution due to an improved air quality resulting in environmental co-benefits of 41 tons less PM_{2.5} and 4,842 tons lower NO_x emissions; (ii) Increased use of public transport resulting in lower costs for mobility as economic benefit and gender co-benefits due to implementing measures which favour a safe usage of public transport by women; (iii) increased climate resilience of urban transport infrastructure.

Outputs of the Program are (1.1) Establishment of a sustainable mobility ecosystem encouraging electrification and shift to public transport. (2.1) E-mobility is combined with PT measures to foster mode shift to low carbon transport. This is made to strengthen attractiveness and convenience of PT including measures such as exclusive bus lanes, improved pedestrianization and public transport accessibility, transit-oriented development, fostering of electric micro-mobility and gender sensitive interventions realized in 2 countries. (2.2.) Electric bus fleets are deployed with innovative business models to enable their mass application in 4 countries.

²⁶ [New \\$1 billion dollar commitment from investors to deliver zero-emission buses in Latin America - C40 Cities](#)

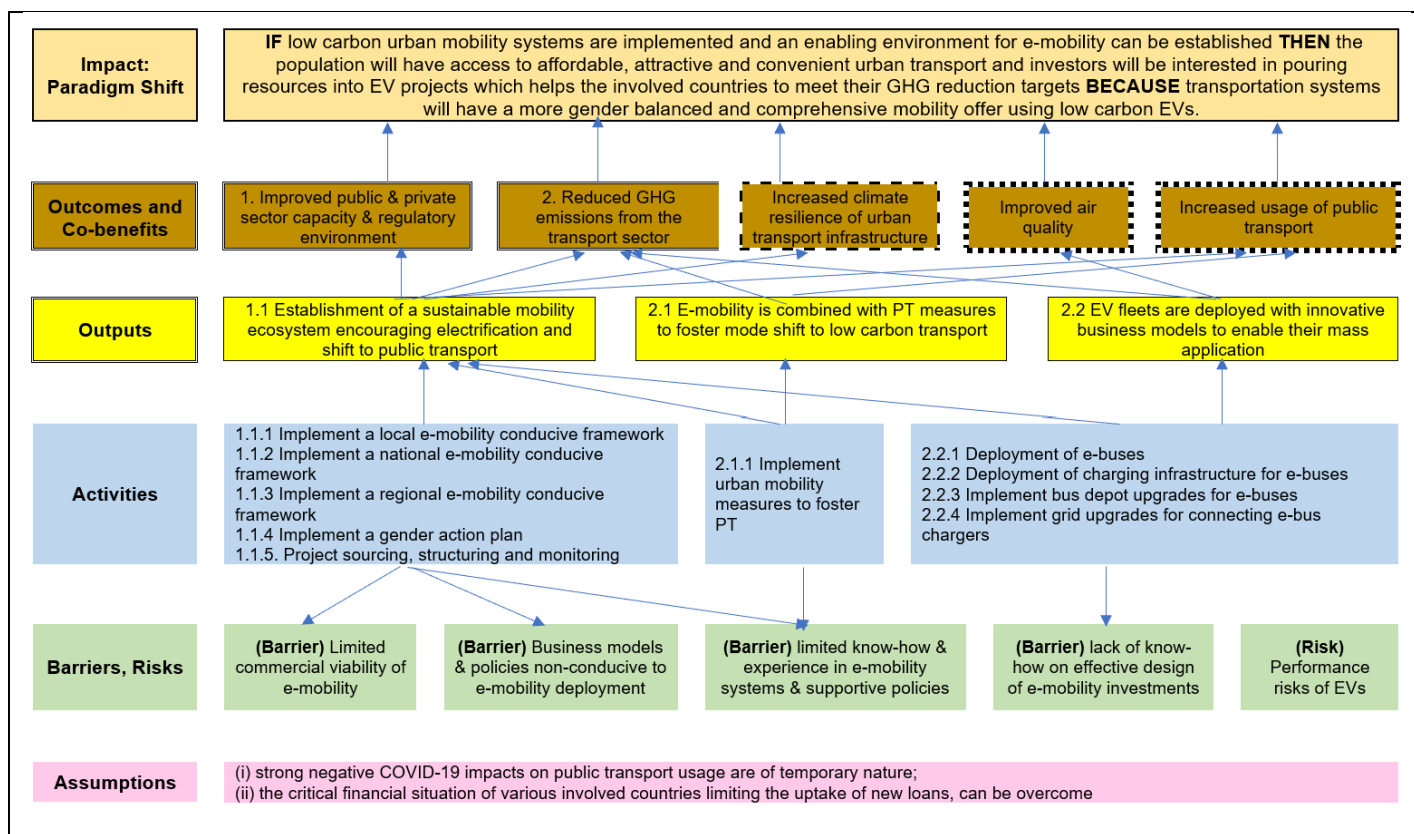
Activities related to the Establishment of a sustainable mobility ecosystem encouraging electrification and shift to public transport are (1.1.1) A sustainable mobility local ecosystem encouraging electrification and shift established with activities related to the local environment such as business model design and development, local policy advice, advice on concession contracts for bus services (e.g. concerning concession length as this is important for e-bus deployment), capacity building and training of operators, safety staff, maintenance staff etc.; (1.1.2) A sustainable mobility national ecosystem encouraging electrification and shift established with activities such as design and divulgation of sectoral roadmaps, support of national enabling policies for EV deployment, advice on battery re-usage, recycling and disposal and capacity building at a national level e.g. in public financial institutions for the generation of financing mechanisms and in insurance companies; design of business models to facilitate the scaling up of e-buses deployment in medium size cities; (1.1.3) A sustainable mobility regional ecosystem encouraging electrification and shift established which is core for learning and experience exchange between Latin American countries with activities such as the realization of different knowledge materials including publications, webinars, benchmark and best-practice studies, preparation of capacity building guidelines and curricula, dialogue with EV suppliers and outreach events including trainings and workshops; (1.1.4.) Implementation of the Gender Action Plan; (1.1.5) Project sourcing, structuring, design and monitoring.

Activity related to the integration of e-mobility with PT: (2.1.1) Implement urban mobility measures such as pedestrianization, bus lanes & route restructuring.

Activities related to the deployment of EV fleets linked with innovative business models are (2.2.1) deployment of 850 electric buses; (2.2.2) Charging infrastructure for e-buses installed; (2.2.3.) Bus depot upgrades realized for e-buses; (2.2.4) Upgrade for grid connections for e-buses realized Finance covers the entire e-bus system of buses, charging infrastructure, grid connection and bus depot upgrades to accommodate e-buses. This needs to be designed individually per project as the optimal technology choice (e.g. overnight charging, fast/opportunity/at boarding station ultra-fast charging) is dependent on route and operational characteristics as well as local costs.

The **barriers** identified and which shall be overcome with this Program are (i) limited commercial viability of e-bus investments related to high upfront costs, low or negative profitability, long payback times and possible loss of income if assets cannot perform at the equal level as fossil units e.g., due to range limitations; (ii) Business models and policies which are not conducive to a mass EV deployment including e.g. atomized public transport operators with limited capital access, contracts for public transport services which have a too short duration to be viable for EVs, policies favouring investments in roads and private means of transport or transport policies not taking into account gender aspects. This barrier is linked to limited experience with and knowledge on innovative business models, E-mobility system designs and on effective policy instruments (iii) limited know-how and experience in e-mobility systems and supportive policies; and (iv) lack of know-how on the effective design of e-mobility investments. The identified major **risk** is technological, operational and performance risks of EVs linked also with a limited availability of skilled maintenance staff.

Core **assumptions** are (i) strong negative COVID-19 impacts on public transport usage are of temporary nature; it is expected that system will recover gradually from the strong demand reduction experienced during COVID-19; (ii) the critical financial situation of various involved countries limiting the uptake of new loans, can be overcome.



B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

[illegible]

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1 Improved air quality	☒	☐	☒	☐	☐	☐
Co-benefit 2 Increased usage of PT and NMT	☒	☐	☒	☐	☐	☐
Co-benefit 3 Adaptation to climate change	☐	☒	☐	☐	☒	☐

Adaptation co-benefit: This program's main focus is on mitigation of climate change impacts and reduction of GHG emissions through fleet replacement. As such and due to a necessary investment prioritisation, adaptation to climate change impacts was not selected as a core infrastructure component of the program and considered as a co-benefit only. Measures related to adaptation will be addressed through the large technical assistance included in the program. This TA will look at aspects such as resilience of the local grid and charging infrastructure to flooding and other climatic events, promotion of non-motorised transport (NMT) along and around mass-transit corridors, improvement of drainage capacity through the selection of porous materials for NMT infrastructure, inclusion of bioswales, etc.

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Program Description & Intervention Strategy

The program aims to stimulate the operation of more electric buses and increase their ridership. It plans to achieve this by creating an enabling environment for electric mobility, embedded in a policy and system environment that enhances the efficiency and attractiveness of public transport and through this make electric buses financially viable.

The Program has a comprehensive approach and includes implementation-oriented activities such as targeted policy advice for an integrated sustainable urban mobility and transport demand management, business model development and capacity building to ensure a favourable e-mobility environment combined with financial instruments to ensure deployment of large-scale fleets. The establishment and operation of large-scale EV fleets works as proof of concept of the commercial viability of EVs and fosters a mass replication without further concessional finance. The Program can reduce risks through the investment interventions combined with the implementation of business models appropriate for EVs (e.g., asset and operation separation; leasing and PPP schemes) and measures that increase ridership, thus enhancing the financial viability of public transport systems.

For each country a market assessment (see for details section below) has been performed analysing (i) the current financial viability of commercial EVs in terms of total cost of ownership, profitability, payback period, cash-flow and debt structure; (ii) required financial instruments to overcome current financial viability barriers; (iii) market readiness of EVs in the specific country based on energy and vehicle prices and future expected EV costs. Interventions are designed to accelerate market readiness and to foster widespread commercial deployment of EVs without further financial assistance. Therefore interventions in each country are targeted to vehicle segments where (i) existing conditions are favourable for EVs (in terms of policies, energy prices, interest of investors); (ii) limited financial support can make a decisive

change and (iii) market conditions allow in the medium term (by end of the Program) for a sustained uptake of EVs based on commercial conditions.

A power sector assessment has been made in detail per country. This includes power availability, electricity generation, and expected grid factors based on renewable energy (RE) trends²⁷. All countries have trends of increasing electricity demand and supply and an increase in RE shares resulting in a decrease of the carbon grid factor. Commercial EVs will only result in marginal incremental electricity usage (<5% even with the high growth EV scenario by 2030 – see Annex 2 market studies). Especially for commercial vehicles the issue is much more on localized grids and e.g. supply of sufficient electricity and updated grid connections and substations for bus depots. At design stage various options how, when and where buses can be charged are assessed to find localized solutions per investment project.

The Program is for commercial EVs and charging infrastructure related to the commercial EVs only. Private passenger cars and motorcycles are not financed. Commercial vehicles result in a much larger impact in terms of reduced GHG emissions and improved air quality and have significantly lower GHG abatement costs than private vehicles. With a small number of vehicles and a limited investment a large impact and a high visibility can be achieved. Concessional finance for private vehicles is considered to be socially inequitable and could result in a mode-shift from public to private transportation. Commercial EVs targeted are electric buses which have attracted the most attention from stakeholders in the involved countries.

The Program only includes 100% BEVs. Hybrid electric vehicles (HEVs) and plug-in hybrid electric vehicles (PHEV) are not included. HEVs and PHEVs were an intermediate technology as long as BEVs were not yet technologically mature (see for details Annex 22). Hybrid buses or taxis can recover their incremental investment but only have a limited environmental impact of around 20% of emission reductions. A review of >50,000 plug-in hybrid buses deployed in PR China revealed that the overwhelming share of plug-in hybrid buses are never re-charged and thus only result in 20% energy and GHG savings (ADB, 2018). Exceptional cases where such buses, can make sense are in countries or cities with electricity supply problems and frequent prolonged black-outs²⁸ or on very demanding routes. PHEV taxis are not a commercially attractive option. The battery size of PHEVs is small resulting in an electric driving range of only 30-50km. Most PHEVs models can only use AC slow chargers, which means charging of more than 1 hour to re-charge their already small batteries. A recent study of ICCT revealed that commercial PHEVs only run 20% of real-world driving on electricity (ICCT, 2020). In addition, PHEVs are more complex and expensive to maintain due to having both electric and internal combustion powertrain. The limited operational savings are insufficient to recover incremental investments of PHEVs compared to fossil or hybrid units. Hydrogen or fuel-cell vehicles are not included in the Program. They use on a life-cycle base 3-4x more electricity than BEVs due to highly energy intensive hydrogen production. They therefore result in most countries in an increase of GHG emissions compared to fossil units, except if hydrogen supply is exclusively based on electrolysis using to an overwhelming extent renewable energy. In such cases, hydrogen can be of interest especially for long-haul vehicles or large vessels. However, this is not the scope of this Program.

The Program focus is on electrification of commercial vehicles, especially public transport buses. As outlined in Section B.1, electrification of buses is however not a stand-alone measure but an important component to modernize and to upgrade the public transport system of countries. New electric buses position the public transport system as modern, environmentally friendly and attractive. The investment in electric buses (PULL) is linked in 2 cities (potentially in Colombia and Brazil) with investments to improve the public transport system as well as non-motorized transport (NMT) in urban areas which result in increasing the attractiveness of low carbon forms of mobility. Together with transport demand management policies (PUSH) this results in a mode shift towards zero or ultra-low mobility systems. The

²⁷ See Annex 2

²⁸ This is the reason why plug-in hybrid buses and not full electric buses were purchased for the BRT Peshawar.

Program directly modernizes public transport systems through establishing new business models, which increase the financial stability and economic efficiency of public transport systems e.g. through separation of bus ownership and bus operations. Indirectly various urban mobility projects fostering NMT and public transport financed nationally and internationally are realized in the cities where investments in e-buses take place (the complementarity of e-bus investments to such efforts is a selection criteria for investment projects of the Program). As example medium sized Colombian cities which are targeted by this Program have formulated urban mobility plans (SUMP) with the aim of modernizing and upgrading public transport whilst also fostering NMT and components of transit-oriented development.

To achieve the program goal of mass deployment of EV and increase share of sustainable transport, an intervention logic has been designed that combines TA and FA efforts to promote the implementation of large-scale EV projects and contribute to the efficiency and attractiveness of public transport for both users and investors, generating an important mode-shift,. The combination of TA and FA will allow to orient the available resources in projects structured under innovative business models and provoke the necessary transformation of the sector by working on strengthening the current regulatory and normative framework, the creation of necessary promotional public policy instruments, the development technical capabilities and other aspects that will allow addressing the barriers that the development of electro-mobility faces in the region (See Section B.2 Theory of change. Activities).

There are many ongoing activities to foster electrification in the transport sector in Latin America. However, most either concentrate on defining policy frameworks, without practical application and often with a focus on passenger vehicles. Projects aiming to replace conventional buses with electric alternatives only cover a small number of vehicles out of the total fleet in the specific urban context and do not address the wider system sustainability.

The program is designed to bridge this gap and combine local and national framework creation with large-scale implementation in model cities. Operating in medium-sized cities and converting a large share of the existing diesel-based fleet, supported by the technical assistance will enable the envisaged system change. This will not only allow to demonstrate the functionality of technologies, but also how business models and policy frameworks can work hand-in-hand with technology to deliver environmentally and commercially sustainable systems.

- a. Linking policy design with on-the-ground large-scale implementation also allows to ensure that
- b. Policy frameworks address actual challenges faced by cities in introducing electric bus systems;
- c. The investment in vehicles and infrastructure is not hampered by existing policy frameworks;
- d. Business models don't need to be exclusively structured around existing policy, but can influence policy design to enable more favourable concepts.

The Program will benefit from activities implemented by AFD and GIZ in the Program countries under the Euroclima+ initiative, whose objective is to develop national policies for urban mobility and integrated multi-modal planning at the city level to accelerate the transition of Latin American cities toward sustainable urban mobility low in carbon emissions. As a result, EUROCLIMA+ will contribute to create favourable conditions of the development of e-mobility in the framework of integrated and sustainable urban mobility plans.

Market Assessment

A market viability assessment has been made for each for urban buses, taxis and Light Commercial Vehicles (LCVs) in each country based on local prices, costs and operating conditions (See Annex 2). This reflects the current market situation for EVs which is then expanded, based on projected future selling prices of EVs until 2030 (again per vehicle segment and country) to assess commercial viability of vehicles without financial support. Financial parameters used for the assessment are primarily TCOs, the FIRR

compared to the weighted average capital cost (WACC) of the transport sector, the dynamic payback, cash-flow and debt levels and required equity investment levels (see Annex 2). Whilst results differ between countries and vehicle segments core take-aways are:

- Non-discounted TCOs of e-buses as well as e-taxis are in most countries lower or at-par with fossil units. However, this depends on various critical assumptions including life-time of EVs versus fossil vehicles, maintenance costs, energy usage and energy costs and battery replacement costs. A critical part is thereby the bus availability rate which depends on the reliability of the vehicle itself, maintenance and availability of qualified maintenance staff and repairs respectively availability of spare parts. This has been a critical issue with e-buses, especially with pilot fleets, with availability rates being as low as 60% whilst diesel buses have, well maintained, 95%. This results in huge cost increases as well as difficulties to comply with the time-table and service offer. The reason for low bus availability rates is lack of experience with the technology, low quality bus suppliers, lack of adequate maintenance and trained staff and lack of spare parts. Issues related with poor after-sales service of some providers have resulted in many transport operators being sceptical about changing bus providers and bus technology. These barriers or problems can be overcome with adequate technical specifications, contracts including guarantees, maintenance and service contracts requiring on-site manufacturer staff and training. However, this is not feasible in financial terms for small or pilot fleets and most public transport operators are not willing to bet on a new technology with fleets of 100 or more buses at one go. They thus wait to see what other operators are doing which shows the relevance of having established in the country initial large fleets of e-buses to prove their commercial viability.
- The differential FIRR (profitability of the incremental investment in an EV versus a fossil vehicle due to operational expenditure savings) is in some cases below the benchmark WACC which means the investment is not profitable and in others is insufficient to warrant for the extra risks involved in changing the propulsion technology..
- The dynamic payback time of EVs is long and in general exceeds the concession periods of transport operators and in some cases also exceeds the asset lifetime.
- EVs require 2-3x more investment capital, 3-4x higher equity investment²⁹ and higher loan volumes resulting in a balance sheet laden with debt and high consequential risks. Most investors prefer to purchase more fossil vehicles thus renewing their fleet more rapidly or increasing revenues instead of investing in costly EVs.
- EV investments result in prolonged periods of cumulative negative cash-flow which can result in liquidity problems of investors.
- The lack of large fleets results in scepticism concerning the reliability and performance of EVs and a lack of know-how on EV maintenance. Also, spare parts are scarce in the country without deployment of large fleets. No party wants to be the first-mover and take the risk, although many acknowledge that this might be an interesting investment.
- Regulations are often not conducive for EV investment e.g. the length of concession contracts is too short for long-term investments or concession payment systems are not conducive for other ownership structures.

Future market development has been modelled based on expected future price development of EVs per sector per country based on international trend forecasts of EV cost parity and battery prices (see Annexes 2 and 23 for a description of the modelling approach).

Business models in many countries are not conducive for EV deployment. New business models (see for more details section below) based on separation of asset ownership and operations or leasing models also

²⁹ This rate is higher even than the total investment capital as FIs in general only finance a percentage of the vehicle cost and not the associated charging infrastructure, grid connection or bus depot upgrades which are considered non-marketable assets.

require risk instruments to be in place e.g. fare systems managed by third parties with automated debiting to the vehicle asset company to ensure payments, vehicle transfer agreements in case of operator default, vehicle insurance systems or bankable concession contracts.

Financial Support Instruments

The market assessment identified that without initial support commercial EVs are not yet viable in Program countries. Concessional finance is required to promote EVs. No investment grants are given. Concessional loan conditions of GCF for public sector projects and private sector projects are 0.75% interest rate with a maximum loan tenure of 20 years.

The Program will give loans in Euro or USD. The maximum USD requested loans from the GCF under this Program will be 44.264 MUSD, and the maximum EUR requested loans from the GCF will be 56 MEUR.

In absence of concessional loans investment in fleets will not take place. Initial financial support for large-scale fleet deployment of electric buses, taxis as well as LCVs is critical and the international experience in this area is clear: No country worldwide has been able to deploy on a large-scale e-bus fleets without initial financial support (see Benchmark report in Annex 22).

Table 2: Investment Grants for E-Bus Purchase

Country	Incentive
China	100% of incremental bus cost subsidized; chargers and grid connection up to 100% subsidized; electricity price initial 3-5 years subsidized; decreasing rates since Program start in 2009
India	80-100% of incremental bus cost subsidized in Fame II; chargers subsidized. FAME I 2015-2019 had 20 percentage points higher subsidy levels than FAME II
Germany	80% of the total incremental cost of e-buses and 40% of all other costs for charging infrastructure, grid connection, bus depot upgrades subsidized plus coverage of 100% of costs for the establishment of new maintenance centers
Switzerland	80-100% of incremental costs of buses and 100% of charging infrastructure and grid connection costs subsidized
UK	75% of incremental bus and charging infrastructure costs subsidized under Phase II (since 2019); under Phase I (2016-2019) 90% of incremental costs were financed
Poland	100% of incremental cost of entire e-bus system subsidized
USA	90% of incremental cost of entire e-bus system subsidized
Colombia	100% of incremental CAPEX of entire e-bus system was paid in Medellin and Phase I in Bogota. 100% of incremental total lifetime cost of e-bus system (CAPEX plus OPEX) subsidized in Phase III in Bogota (16% incremental cost compared to fossil units) ³⁰
Chile	100% of incremental cost of entire e-bus system for Phase I (200 buses) subsidized with monthly installments during 10 years; phase II longer concession periods for e-buses compared to conventional units (+40%) and additional points in tenders
Uruguay	100% of the incremental cost of e-bus plus chargers is paid in monthly installments during 7 years; the government subsidy for e-buses equals the government savings on subsidies saved on diesel fuel

Source: Grutter Consulting (2020), Benchmark Report (Annex 22)

Countries consistently subsidized 80-100% of the entire e-bus system incremental cost for the initial phase. Countries decreased their subsidy rates significantly within few years (e.g. China, India, UK, Chile). The importance of upfront investment subsidies – independent of TCOs – to kick-start deployment of e-buses is thus clear. It also shows however that once a large fleet is established subsidies can be reduced gradually and even be eliminated due to more competitive prices of e-buses combined with the experience of mass operations of e-buses. Concrete examples are³¹:

³⁰ Phase II had no subsidies for e-buses and all contracts were awarded to fossil bus systems as e-bus offers were non-competitive

³¹ See Benchmark Report in Annex 22

- In PR China, the average subsidy level per 12m urban e-bus decreased within 5 years from 150,000 USD per bus to 30,000 USD per bus and is expected to be fully eliminated by 2022³².
- In India, under FAME I 2017-2019, e-buses were subsidized with 60% of incremental costs. This was reduced to 40% under FAME II 2020-2021 (40% of a lower absolute value meaning absolute subsidies were reduced by more than 50% within 2 years).
- In the UK, the 1st scheme for low carbon buses (until 2017) subsidized 90% of the incremental cost and the 2nd scheme, since 2018, 75% of the incremental cost of e-buses (including charging infrastructure).
- Chile subsidized for the first lot of 200 electric buses 100% of incremental costs. The next lots received no more investment gap subsidies but some additional benefits for e-buses relative to fossil units in tenders which is translated into higher payments per distance driven or per service unit for e-buses (TransMilenio, Colombia analysed the tenders of Santiago and estimates the subsidy to amount to more than 16% of total costs per km³³).
- Uruguay subsidized for the first lot of 30 buses 100% of the incremental cost, through a National law that was intended to be up to 150 buses for the whole country.

Business Models

Two components stand out that can improve in a decisive manner the attractiveness and commercial viability of commercial e-bus projects: (i) bulk purchase –that will be promoted through the TA-, and (ii) separation of asset ownership and operations/management. Bulk purchase is important not only due to significant price discounts but also as this allows to optimize the e-mobility system in terms of charging infrastructure and bus types allowing e.g. to use fast-charged buses with common usage of chargers (up to 12 buses per charger). Separation of asset ownership & management is critical due to high investment costs of buses which require strong financial players with access to cheap long-term finance. Private bus operators typically have a weak balance sheet and are financially not preferred clients of FIs. Business models with asset separation have been successfully implemented in Chile with e-buses involving initially electric utilities and also bus manufacturers as well as private investment companies. To thrive such business models also need the regulatory framework defining payment schemes, procedures in case of bankruptcies of parties, clear long-term concession contracts etc. Separation of ownership and management can be done with public led delivery systems (public asset companies) or with private led delivery vehicles (special purpose vehicles within a private public partnership PPP). This needs to be adjusted and structured per country based on national and even local requirements, priorities and conditions. It is also essential to work on standardized bidding documents to acquire EVs in order to strengthen the ability to replicate this business model in Latin American cities. One of the main features of the Program will thus be to design, establish and implement such business models which can work as delivery vehicles for widespread commercial replication of e-buses without further need of subsidies.

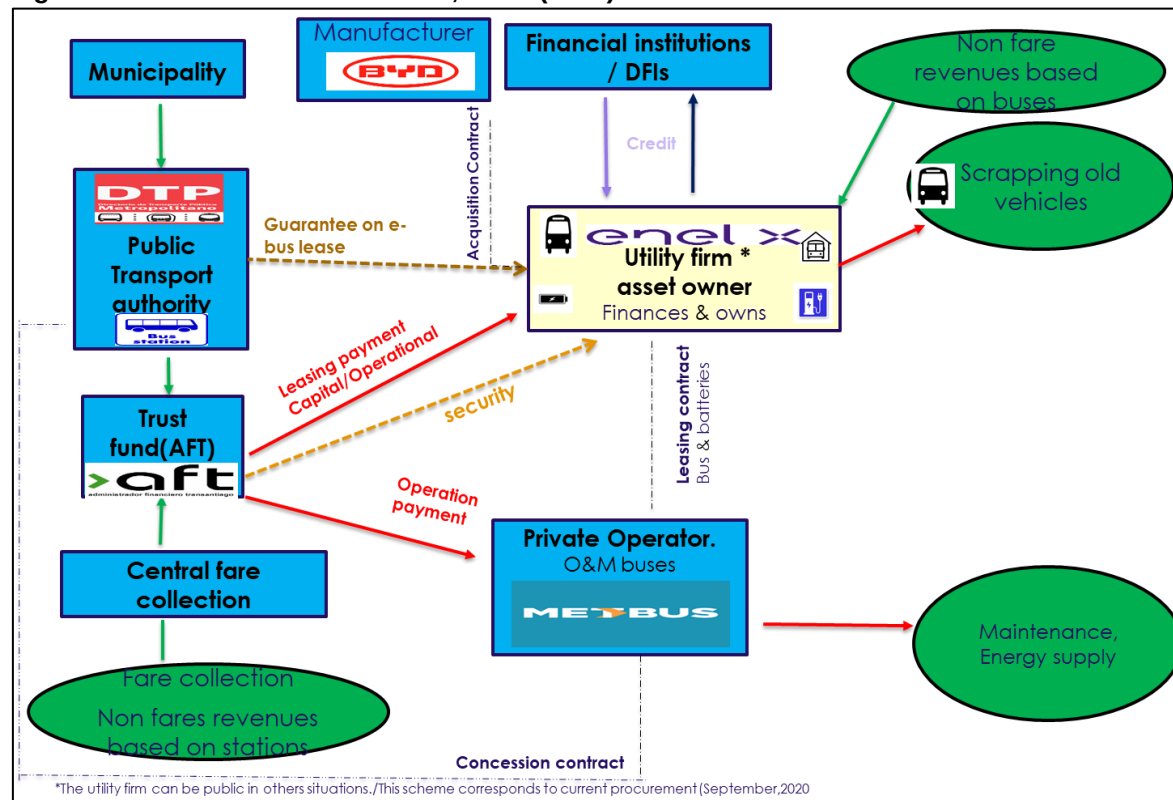
This shift of business models incited by e-mobility can be clearly observed in Chile and in Colombia, where in both cases large capital-heavy investors have entered the market (electric utilities, private capital funds and bus manufacturers) which results in public transport operations being financially more solid (in both countries bankruptcy of traditional bus operators has been a major problem), more efficient and with better and lower cost access to capital. New business models promoted by the Program separate asset ownership and asset management and allow for large investment firms with solid balance sheets and access to cheaper finance to enter the game. This has been the case for example in Chile or Colombia. This results in e-mobility being a game-changer not only in terms of fostering low-carbon vehicles but also in modernization of the public transportation system increasing its attractiveness and thereby resulting in an important mode-shift towards public transport compared to a business as usual situation. Unbundling

³² Source: Grutter Consulting; contract with ADB for monitoring of e-bus fleets in 18 cities with 60,000 buses financed by leasing funds with support by ADB

³³ Grutter Consulting, personal communication with TransMilenio

ownership and operation can also include the provision of power, chargers and bus depots. Leasing by fleet owners can be limited to the vehicle or include also maintenance and energy. Leasing can effectively reduce the barrier of upfront investment. E-buses can have lower TCOs and leasing models can bring this advantage to bus operators. Various models are used separated in the type of leasing, the type of leasing company or components leased.

Figure 6: Business Model e-Buses, Chile (2020)



Source: AFD; dotted lines contracts; fixed lines: fund flows

Examples of Chile and Colombia in e-Bus Promotion

Chile and Colombia have been particularly successful in promoting e-buses. Why that? What was the game-changer?

Chile

The case of Chile is especially interesting due to the introduction of new business models separating bus ownership and bus operations. The first batch of 200 electric buses was based on investments of electric utilities leasing e-buses to operators, in order to boost their core business (energy sales and the installation of charging infrastructure). Critical for the first batch of e-buses was not only the investment of electric utilities but also a government subsidy for the entire incremental investment cost of e-buses versus fossil units as well as changes in concession contracts reducing risks for investors in vehicle assets and ensuring payments from vehicle operators. In subsequent tenders ownership of e-buses is mixed with some offers based on JVs of bus operators with bus suppliers and others being JVs of bus operators with investment firms. Electric utilities are no longer engaged or have teamed up with capital investment firms as they do not consider vehicle investments to be a part of their core business (e.g. AMP Capital and NEoT Capital have entered as investors). The government no longer pays for incremental investment costs but still offers other advantages for e-buses such as longer concession periods and additional points in the bidding process representing effectively a subsidy to e-buses. Critical take-aways from the Chilean case are: (i) separation of ownership and operations can be an effective instrument to promote e-buses and can bring in new financially strong players into the industry; (ii) for kick-starting the process investment subsidies are critical; (iii) e-buses still require incentives to be competitive with fossil buses; (iv) required subsidies can be reduced significantly after an initial fleet of e-buses and after having established a working business model.

Colombia

Large Colombian cities already run successfully since many years BRT systems with private operators and a public system manager paying companies based on distance driven cum service delivery targets. E-buses have been introduced for secondary routes based on local e-bus targets. The initial batch of 500 e-buses (2019) was based on a tender which only allowed for the provision of e-buses. A subsequent second tender which allowed for any technology resulted in all winning offers to be fossil buses, thereby clearly showing that e-buses are not yet commercially competitive (end 2019). The third tender (2020) included significant additional points for e-buses as well as longer concession periods. This made electric buses competitive with fossil units but also resulted in 16% higher per-km costs of buses which is being paid by the municipality. The experience of Colombia reveals two important points: (i) e-buses can be inserted into public transport operations and will be offered by private operators if concession contracts are sufficiently attractive and if e-buses receive benefits compared to fossil units; (ii) without financial incentives e-buses are commercially not yet competitive. However, some large cities such as Bogota are willing to pay for these incremental costs due to e-buses improving air quality and reducing GHG emissions.

The importance of these business models for e-buses relies in their ability to reduce upfront CAPEX barriers which are larger than for fossil vehicles. Difficulties of leasing models for e-buses are the long-payback period resulting in long lease terms required to make e-buses competitive. This again results in difficulties as concession contracts might be for a shorter duration than the payback time for e-buses and question concerning the actual lifespan of buses as well as the re-sale value of a leased e-bus arise (there is only a very thin market currently for used e-buses). Without proper estimate of the re-sale value of used e-buses and the actual lifespan a determination of the lease fee is complex. Long payback periods and a high CAPEX also result in high finance costs and the importance of low interest rates. Asset separation can allow for new, financially stronger companies, to enter the market and act as SPV leasing vehicles as they can have access to less costly financing sources. Supporting these business models will add to the additionality of GCF resources by de-risking the sector and providing incentives to new actors to invest in the sector. The Program will work with municipalities, the national government, transport operators and companies willing to invest in urban transport schemes in concession contracts, regulations and the structuring of business models conducive for e-bus deployment. The delivery scheme for component 3 of the Program (deployment of EV fleets) can thereby be public or private led (see figures below).

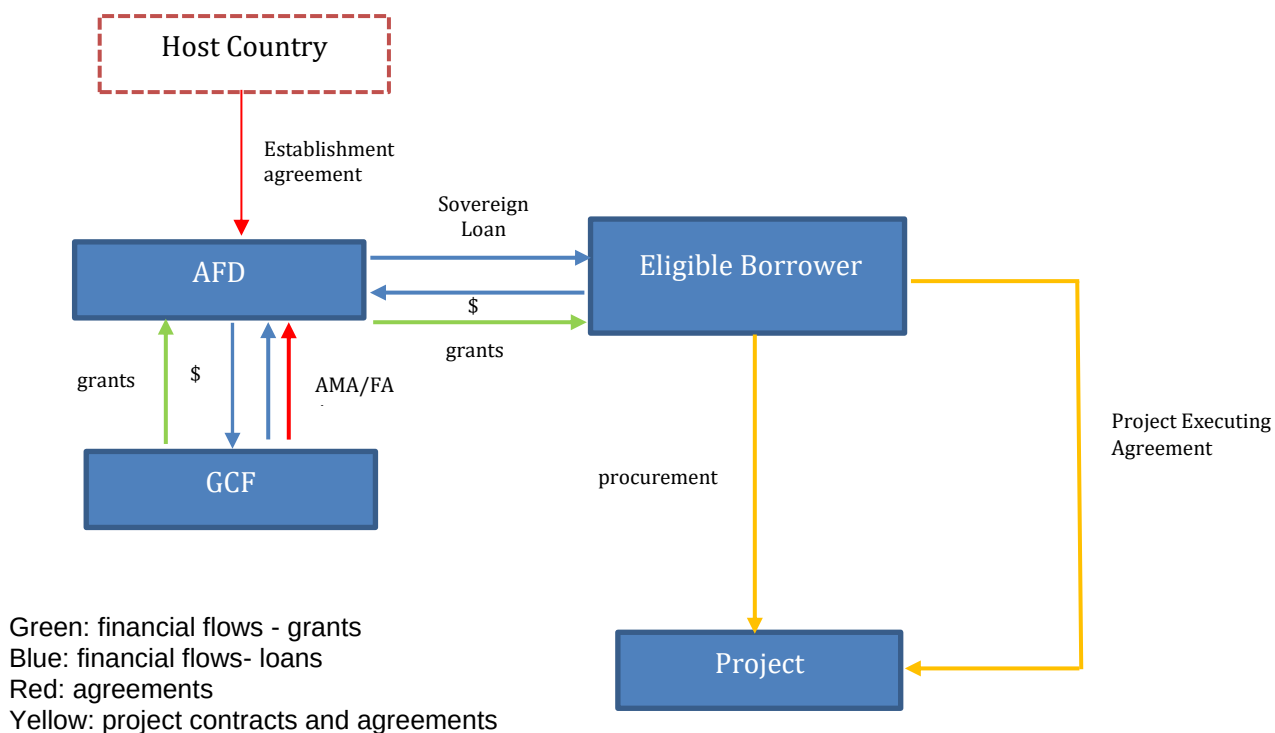
Public-Sector Led Delivery

In the case of Sovereign Investment Loans or Grants to directly finance Projects under this Funding Proposal, the AFD eligible borrower, acting as EE (EE 1st level) or a designated EE (EE 2nd level) (if different from the eligible borrower) will directly execute all project activities following AFD rules including, detailed specifications and prepare bidding documents, manage procurement processes under AFD rules, perform financial management and reporting of results.

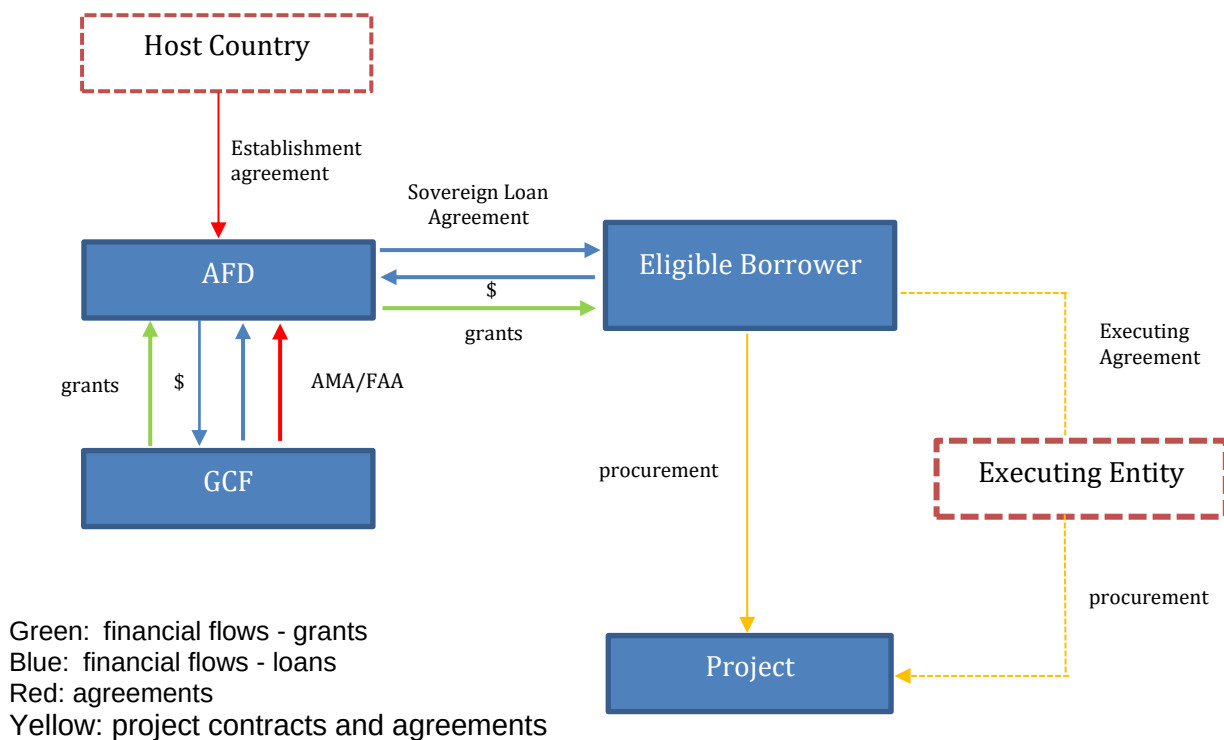
Accordingly, the most simple execution arrangement corresponds to the structure presented in the Figure below. In this structure, Subsidiary Agreements (Sovereign Investment Loans and Grants) are entered into between the AFD and an AFD eligible sovereign borrower. The AFD eligible sovereign borrower will act either as EE or transfer proceeds to a separate EE (in this case a project execution agreement is normally required). The EE will be in charge of the execution of the Project and will carry-out any procurement. AFD as AE will have an agreement with each EE, as required pursuant to the AMA.

Figure 7: Public Sector-led Delivery for Urban Buses

Eligible sovereign borrower acting as EE:

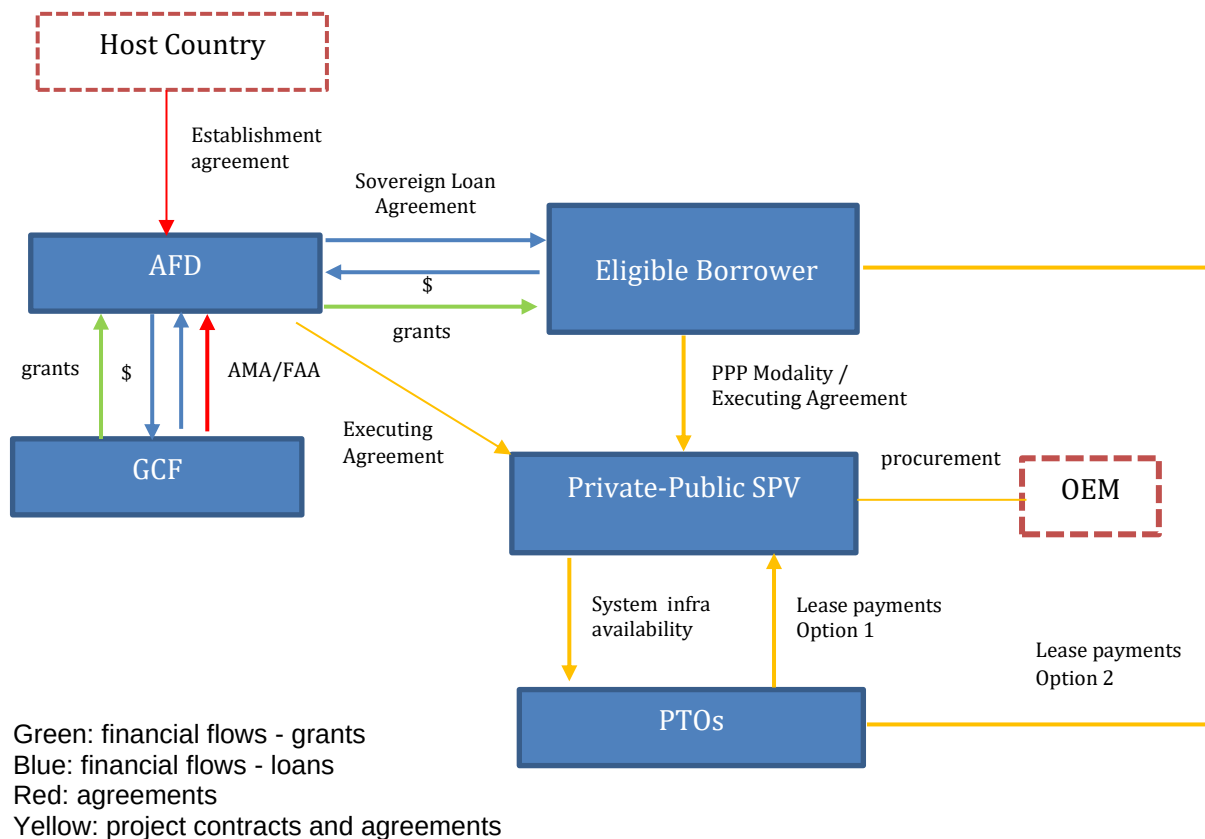


Eligible sovereign borrower transfers proceeds to a separate EE:



The role of the Public Third Party (PTP) could be undertaken, *mutatis mutandis*, by a public-private entity, as illustrated in the following figure:

Figure 8: Public-Private Sector-led Indirect Delivery for Urban Buses

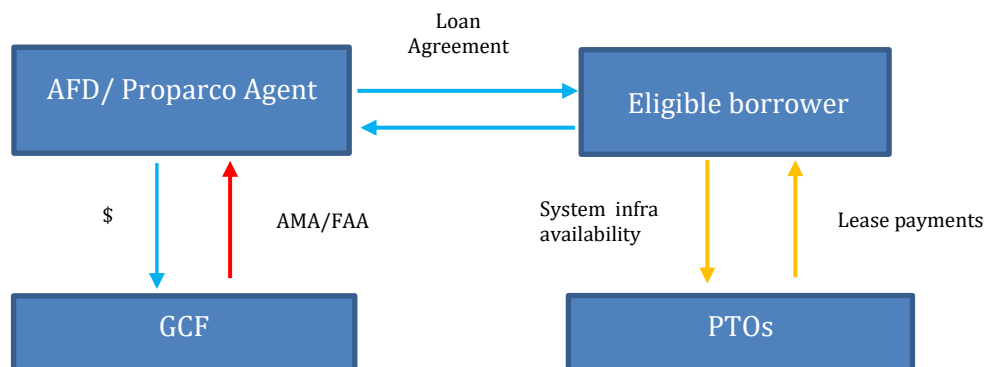


EE: Executing Entity; OEM: Original Equipment Manufacturer; PTO: Public Transport Operator; SPV: Special Purpose Vehicle; PPP: Public-Private Partnership blue loans; green: grants; orange: related to OEM system purchase

The indicative structure presented in the figure above includes a private-public special vehicle (SPV) -instead of a PTP- as an EE of funds being channeled through an AFD eligible borrower. The SPV would typically be selected following a competitive process and its relationship with the corresponding AFD eligible borrower (or its corresponding national or subnational government agency) would be established under the terms of a PPP arrangement. The SPV would be responsible for delivering and financing the BEB fleet and charging infrastructure assets. To this effect, the SPV may: (i) structure and raise financing from selected financiers and investors; (ii) procure the buses and charging equipment assets from an OEM; (iii) ensure the availability to PTOs of buses and charging equipment; (iv) provide maintenance training and additional spare parts inventory. OEMs can also be involved directly in the SPV or team up with SPVs for tenders. The financing modality for PTOs will in general be through leasing which can include the vehicles plus charging infrastructure only or it can also include service and maintenance.

Private Sector Led Delivery

Figure 9: Private Sector-led Delivery for Urban Buses



Green: financial flows - grants

Blue: financial flows - loans

Red: agreements

Yellow: project contracts and agreements

PTO: Public Transport Operator

In case of a private sector led delivery the GCF loan is a public sector loan (e.g. to the MOF) with sovereign guarantee or either a loan without sovereign guarantee directly to the private SPV or a public sector loan to the municipality without sovereign guarantee which on-lends it to the private SPV or directly to a PTO based e.g. on a leasing arrangement

OEMs can also be involved directly in the SPV or team up with SPVs for tenders as has been the case e.g. in Colombia or Chile. Also in Brazil, the PROMOB project has established a Working Group on Electric Buses within National Platform for Electric Mobility (PNME). Private and public national banks will be co-financier of many projects of the program e.g. as PPPs and public development banks can also be financed by the program (e.g. national banks like FDN in Colombia or NAFIN in Mexico can finance e-bus projects). In some countries, e.g. Costa Rica or Colombia, national banks are already involved or have established lending facilities for EVs which are however not being used by clients due to lack of attractiveness (example credit line for purchase of electric passenger cars, taxis, buses or LCVs established by various banks in Costa Rica which is not being used with exception of passenger cars as the purchase of EVs is not profitable even with slightly more attractive loan conditions). Therefore, to ensure an active participation of national development banks in the Program and allow them to gain experience and de-risk the sector, the Program will propose technical assistance and financial assistance- which are key to ensure that the credit line proposed by these banks are disbursed.

Objectives, Impacts and Outcomes

The Program supports a paradigm change to low-emission transportation systems. The Program shall overcome the barriers identified to kick-start the mass deployment of commercial EVs and increase ridership in public transport. The following impacts shall be achieved:

- GHG reduction: EVs have significantly lower lifecycle GHG emissions than fossil units in all countries included in the Program. Public transport mode shift results in additional GHG reductions.

- Improved air quality: EVs have no combustion emissions and are a crucial instrument to achieve clean air in cities. Commercial vehicles are a major source of PM_{2.5} and NO_x emissions in cities. Replacing fossil diesel powered commercial vehicles with EVs improves air quality significantly. Public transport mode shift results in additional reductions of pollutants.
- Reduced energy dependency: EVs use domestic resources and reduce reliance on imported fossil fuels. They thereby also increase the resilience of the country's economy to external oil price shocks.
- Increased energy efficiency: EVs are up to 4x more energy efficient than fossil vehicles. In addition, public transport is more energy efficient than private motorized modes. Electricity consumption even if pursuing an ambitious EV penetration level such as proposed by the EV30@30 target supported by the International Energy Agency (IEA) is marginal compared to national production levels – however, localized grid problems need to be addressed. Public transport mode shift results in additional energy efficiency impacts.

The expected GCF fund-level impacts are reduced GHG emissions through increased access to low-emission transportation resulting in direct emissions reductions of 4,334,051 tCO_{2e} over the asset lifetime of investments co-financed by the Program excluding benefits from mode-shift towards public transportation which are induced by the Program and the introduction of innovative business models in public transportation but are considered an indirect and non-quantified Program impact³⁴. The potential impact of de-risking EV investments has been calculated in the country reports. Investments are previewed in 4 countries only whilst TA is delivered in all countries. The GHG reduction calculations are only based on the 4 countries with investment projects. Investment projects can be executed in any of the participating country and are selected based on high quality proposals put forward on a running base to the AE. The available funds give an indication of investments in minimum 4 countries with 1 project each – for calculation and programming purposes the most promising and advanced projects have been used. In practice however, this might have variations and no country is included from proposing investment projects.

Technical Assistance

To enable the envisaged system change, technical assistance is an essential element of the program, supporting and enabling the financial assistance. The exact activities and themes addressed in each city and country will depend on the existing frameworks and an initial gap analysis. Gaps will be based on the analysis already conducted during the preparation of the program, updated at the start of the program to reflect recent developments.

All activities developing strategies, policies or concrete system design will be conducted using a multi-stakeholder approach supported by technical inputs generated within the program. Relevant stakeholders will be determined in close collaboration with the responsible government agencies and will be involved through workshops, dialogues and other forms of interaction.

Program Components/Outputs

The Program is structured in 3 components:

- Component 1: Establishment of a sustainable mobility ecosystem encouraging electrification and shift to public transport (TA)
- Component 2: Investive measures for PT mode shift (FA)
- Component 3: Deployment of EV fleets (FA)

³⁴ Multiple other factors difficult to separate from the Program activity influence the impact on mode shares

Additionally Program Management is put as separate activity.

Component 1: Establishment of a sustainable mobility ecosystem encouraging electrification and a shift to public transport

The outcome of this component is (1) Improved public & private sector capacity & regulatory environment for e-mobility.

Component 1 is linked with Output (1.1.) Establishment of a sustainable mobility ecosystem encouraging electrification and shift to public transport.

The activities of component 1 are (1.1.1) Implement a local e-mobility conducive framework; (1.1.2) Implement a national e-mobility conducive framework; (1.1.3) Implement a regional e-mobility conducive framework; (1.1.4) Interventions realized with a gender perspective.

This component is TA only and realizes the TA activities for components 2 and 3 which are FA only.

This component provides technical assistance to create a policy and business framework conducive for sustainable mobility and massive deployment of EVs. Includes a set of activities aimed at overcoming the barriers faced by the development of sustainable mobility in general and multiplication of electromobility projects in particular.

Experience has shown that the introduction of new technologies can only follow a sequential path of applying TA and then FA, when the barriers that the market faces are only partially explained by financing aspects. Only under these scenarios is it understood to adopt an intervention strategy that starts with TA to create the required enabling conditions (development of normative and regulatory framework, development of the required technical capacities, develop of required public policy instruments, etc.). This work paves the way to the entry of the necessary resources that will allow the implementation of projects.

In the case of the market of EV for public transport, the main barriers that were identified are associated with financial aspects (high investment costs, problems of access to financing, risks associated with a high debt exposure, etc.). This is caused by atomized ownership structure, financially weak transport operators, often high levels of subsidies and lack of an integrated attractive offer for clients, resulting in declining use of public transport and a steadily increasing motorization rate. The Corona pandemic has further decreased the attractiveness of public transport. Therefore the required strategy must be comprehensive, contemplating simultaneously both the generation of the enabling conditions that this market requires and the effective implementation of projects that will drive the expected development. The market needs to see projects implemented and learn from them.

Through the development of details feasibility studies in each of the proposed country, extensive information has been obtained to know and understand the electromobility market in the region. The starting situation of the transport sector in general and the electro-mobility market in particular was described and made it possible to understand the main barriers that the electromobility faces in the region (Section B.2 Theory of change. Risks and barriers). Based on this information, the TA component was structured,.

Activity 1.1.1 Implement a local e-mobility conducive framework. TA supports the FA components by working with the cities³⁵ in which FA activities will take place to create the conducive environment for sustainable mobility based on electric buses, as described above. This includes required local policies and frameworks for electric vehicle deployment and support to cities to develop the system in a gender sensitive manner and catering to a diverse set of users. It further supports cities to optimise public transport systems, including route planning, depot design and location, and city planning, including the

³⁵ Indicative 4 cities; the activity 1.1.1 is realized in all cities where investment projects take place

integration with active mobility. TA targets all stakeholder groups involved in public transport planning and operation in the selected cities, as well as utilities and grid operators.

Sub-Activities will support cities to:

- Develop or, as appropriate, revise sustainable urban mobility plans (SUMP) in selected cities to provide the strategic context for policies and investments, incorporating opportunities through the electrification of bus systems;
 - Better understand available technology options;
 - Develop or adjust demand policies, including planning, system optimisation for public transport and integration with walking and cycling, and instruments discouraging private vehicles use;
 - Develop locally appropriate business models, financial structuring and concession contracting;
 - Identify national policies and frameworks that are required to enable cities to implement electricity-based, sustainable public transport systems. This will directly feed into component 1.2 and support activities at the national level.
 - Enhance capacity: Technical training in the operation, management and maintenance of EV buses and charging infrastructure to public and private service providers of the selected cities. Enabling south-south learning on gender-responsive planning and design of mobility systems and the design and implementation of gender responsive transport projects and planning at the city level.
- Identifying and disseminating lessons learned and experiences from the implementation of projects.

Activity 1.1.2 Establish a national e-mobility conducive framework. TA supports all seven countries in developing a national EV ecosystem conducive for electrification of buses in cities throughout the country in the context of an overall sustainable urban transport system that aims to enhance demand for public transport in a gender sensitive way. This means including gender from the perspective of users and as parts of the supply chain, for example the role of women as drivers, technicians and planners. TA targets transport and energy ministries and other institutions potentially involved in or responsible for urban planning and energy in the respective countries.

Sub-Activities will support city and national stakeholders to:

- Develop or, as appropriate, revise national urban mobility programmes (NUMPs), sectoral electrification strategies or related sections of national transport strategies;
- Develop or adjust policies, regulations and guidelines that will support the deployment of electric buses within the entire country, such as minimum requirements for electric bus fleets, procurement guidelines or contractual regulations impacting concession contract design. This also includes policies that foster the integration of electric buses into the energy system, for example through standardisation of equipment or flexible electricity pricing;
- Develop or adjust policies, regulations and business models that address the re-use, recycling and disposal of batteries at the end of their useful lifetime in vehicles, especially those that support grid stabilisation and resilience through second-life use as grid-integrated storage;
- For countries without a financial assistance component: The experiences of countries with FA interventions shall be used also in other countries localizing them based on country and city circumstances.
- Enhance capacity: Transfer of knowledge on available technology options for government officials and national stakeholders. Development of general training courses and curricula for vocational training for technicians and drivers with a gender inclusion perspective. Assess implications of e-mobility at project investment level and at a national level on jobs and skills required to enable a just transition. This entails for investment projects to identify areas of job losses (e.g. vehicle maintenance) and identify new job opportunities and skill sets required including the development of a transition program. On a national level this includes working with relevant government entities, including Ministries of Education on approaches to capture benefits of new jobs and opportunities

given by e-mobility through having flexible workforces and skills. This shall be integrated for example in e-mobility roadmaps or as stand-alone roadmaps for just transition in e-mobility.

Roadmaps are an important strategic tool to support the implementation of long-term goals and underpin these with intermediate targets and action areas. They can be developed at different levels (regional, national & city) and can address the overall sector or individual transport segments. For individual countries, they ideally build on an integrated national transport strategy or National Urban Mobility Policy or Programme (NUMP), that outlines the overall vision for the sector and empower cities to implement sustainable mobility planning. But they can also be specific to urban transport, as for example in city-specific Sustainable Urban Mobility Plans (SUMP). Some countries and cities also prefer to develop specific electrification roadmaps. In any case, roadmaps provide guidance for policy design and can provide clarity and predictability for investors, IF they are underpinned with the actual implementation of policy and it can be demonstrated that envisaged goals are practically feasible. The program design aims to ensure that these crucial conditions are met and that actual policies are implemented that address actual challenges faced on the ground while demonstrating technical and commercial feasibility in the context of improved urban mobility systems. GIZ has developed detailed tools and processes to develop national and local urban mobility plans (available at <https://changing-transport.org/toolkits/>) which will support the program activities. A range of countries and cities are currently in the process of developing NUMPs and SUMP, some with support from GIZ and AFD. To provide an example, the English summary of the SUMP of Santo Domingo in the Dominican Republic can be found here:

https://www.mobiliseyourcity.net/sites/default/files/2021-03/Santo%20Domingo%20SUMP%20Summary_final.pdf

Activity 1.1.3 Establish a regional e-mobility conducive framework: Capacity-building, knowledge sharing and experience exchange are conducted at the regional level. Activities include exchange and training between countries as well as between cities. The program will utilise existing platforms and institutions where available to disseminate information and experiences generated within the program.

Sub-Activities include:

- Dialogue with EV suppliers on the strengthening of regional equipment and support industries. If at the moment local manufacturers mainly focus on assembly of e-vehicles, this sub-activity will also look at initiating a platform for increased transfer of know-how and footprint of local manufacturers in the region;
- Dissemination of knowledge materials developed in the city and national components;
- Sharing of training courses and curricula developed in other components;
- General outreach activities.

Through its capacity-building activities, the program will be in a unique position to also identify potential candidates for further financial and technical support, which may be provided by consortium partners or external funders.

Activity 1.1.4 Implement a Gender Action Plan: The technical assistance component will work to ensure that:

- The design and revision of roadmaps and policies includes a gender perspective and adequately considers the gender equality gaps and opportunities for enhancing the participation of women in related jobs;
- The design of transport projects and investments considers gender equality;
- Gender-disaggregated data on urban transport is collected to enhance the knowledge base;
- Protocols against sexual harassment are promoted through communication campaigns.

Component 2: Investive measures for PT mode shift

The outcome of this component is (2) Reduced GHG emissions from the transport sector

Component 2 is linked with Output (2.1.) E-mobility is combined with PT measures to foster mode shift to low carbon transport. Component 2 is only FA.

The activities of Component 2 are (2.1.1) Implement urban mobility measures to foster PT in 2 countries. Urban mobility measures are implemented by the city. They are measures to increase the attractiveness of public and NMT and can include demand as well as supply measures. Declining PT shares are a fact in many countries. One of the reasons is the lack of attractiveness and convenience of PT. The program will work in the involved cities to increase PT attractiveness through measures such as bus only lanes thereby reducing travel time and making PT faster than private means of transport, route restructuring, tariff and mode integration, transport policies which favour in cities PT and de-incentivize private means of transport e.g., through parking policies. E-bus projects are embedded in urban transport projects which include measures such as bus route restructuring, exclusive bus lanes, integrated fare systems, or inter-modal integration to increase the attractiveness and usage rate of PT. Projects are also integrated with measures to increase the usage of NMT including segregated cycling lanes, increased pedestrianization and improved accessibility, and improved last-mile mobility services including (electric) micro-mobility (see component 1). Common urban spaces and new urban mobility services facilitate the independence and mobility of all residents and are key to promote sustainable, competitive cities while increasing the quality of life of citizens taking into account the different mobility patterns in women and men.

For the advancement of low-carbon sustainable mobility, this component will finance (i) the allocation and improvement of supporting urban spaces and infrastructure for public e-transport; (ii) the creation of urban mobility plans and policies; (iii) public transport improvements. The Program will prioritize solutions that integrate into the urban fabric, are compatible and conducive to modes of sustainable mobility guaranteeing universal access. This can include the infrastructure connecting public transport systems to alternative modes of urban mobility (electric and non-motorized) such as cycling lanes, pedestrian streets, etc., and/or providing the infrastructure conditions to facilitate these alternative modes of urban mobility as bicycle parking, park and ride, waiting areas, charging areas etc.

This component will only be implemented by AFD and not by PROPARCO.

Component 3: Deployment of EV fleets

The outcome of this component is (2) Reduced GHG emissions from the transport sector.

Component 3 is linked to the Output (2.2) E EV fleets are deployed with innovative business models to enable their mass application.

The activities of Component 3 are (2.2.1) Deployment of e-buses; (2.2.2) Deployment of charging infrastructure for e-buses; (2.2.3) Implement bus depot upgrades for e-buses; (2.2.4) Implement grid upgrades for connecting e-bus chargers.

This Component is FA only.

This component addresses the barrier of limited commercial attractiveness to invest in e-buses (barriers initial CAPEX barrier, profitability barrier and performance risk barrier).

It is expected that this component will be applied in 4 countries with investment projects (Brazil, Colombia, Peru and Mexico) with an indicative number of 850 e-buses deployed of different sizes. The goal is to identify, design and implement localized innovative business models in public transport which modernize and increase the attractiveness of public transport and which deploy large-scale fleets of electric buses to showcase their technical and commercial viability to operators. Technology-wise this includes different e-bus technologies (slow-, fast-, opportunity and ultra-fast charged buses), different bus sizes (8m to 26m buses) and different operational structures (mixed traffic buses as well as units operating on bus-only routes). The Program will not finance pilot projects but mass application linked with innovative business models. Buses are operated by municipal transport operators or private parties (see above for pre-identified business models).

E-bus systems always include 4 components: (i) the vehicle itself which accounts for around 70% of system costs; (ii) the charging infrastructure; (iii) grid upgrades to accommodate the charging infrastructure especially at bus depots but also, if on-route charging or opportunity charging is used, in the areas of chargers (this involves the installation e.g. of mini-substations) and (iv) bus depot upgrades which include e.g. roofs over chargers and increased space or layout changes to the bus depot as buses cannot be stacked equally if they are charged with plug-in. Charger network types and numbers depend on each city and the final feasibility study. This can be e.g. ultra fast charging at stations whilst passengers board, end of route opportunity charging, intermediate fast charging or overnight charging. This changes the number of chargers, the charging system, the output power of chargers and also results in different bus depot layout changes as well as different grid impact and grid upgrades. The feasibility studies realized per city will define the optimal e-bus system configuration per city. The system design is critical as bus features including battery types and battery size needs to be optimized with the charging infrastructure and with financial and technical criteria concerning the grid to not create too much grid stress. There is no one-solution fits all for e-buses and an assessment needs to be made per city based on operational conditions, bus size and characteristic, grid conditions, climatic conditions etc. Batteries will need to be changed after 8-10 years (guarantees are today available for 8 years with 80% SOH of batteries and in some cases and some battery types even longer). However, this investment will not be that critical in 8 years from now due to falling battery prices and is in the range of a complete engine overhaul and bus refurbishment.

Table 3: Indicative Projects to be Financed (MUSD)³⁶

Country, city	# e-buses	Total CAPEX	GCF loan	AFD loan	National finance	GHG impact
Brazil, Florianopolis ³⁷	aprox. 300 e-buses	113 MUSD	37 MUSD	64 MUSD	12 MUSD	904,000 tCO _{2e}
Colombia, secondary cities ³⁸	aprox. 280 e-buses	114 MUSD	35 MUSD	68 MUSD	11 MUSD	2,986,000 tCO _{2e}
Mexico, Sinaloa	200 1e-buses	63 MUSD	25 MUSD	32 MUSD	6 MUSD	348,000 tCO _{2e}
Peru, Arequipa	aprox. 80 e-buses	29 MUSD	15 MUSD	13 MUSD	1 MUSD	97,000 tCO _{2e}

³⁶ Any country can propose investment projects – the listed projects are those which are currently the most advanced

³⁷ Includes PT investments

³⁸ Includes PT investments

Projects can also be financed in other countries of the Program i.e. all countries participating have access to finance and can forward their projects to enable the highest quality ones to be selected. Examples of other projects discussed include:

- Argentina: Replacement of up to 300 diesel buses with electric units in Buenos Aires;
- Argentina: Replacement of up to 30 diesel buses with electric units in Tucuman;
- Costa Rica: Replacement of up to 200 diesel buses with electric units in the larger urban area of San Jose linked up with the LRT project;
- Dominican Republic: Replacement of up to 150 articulated and standard diesel buses with electric units in Santo Domingo and modal integration with the metro lines; For a full list of discussed investment projects totalling an investment volume of nearly 10 billion USD see File 3 “Pipeline Investment Projects”.

Program Management

Activity 1.1.5. Project sourcing, structuring and monitoring. This includes technical, financial and institutional advisory and the full feasibility of the project including economic, environmental, social, gender and legal. This includes the definition of a project contractual scheme, ensuring a proper risk allocation amongst key stakeholders and assuring the bankability of the project and securing a proper allocation of GCF concessional resources. This support is key as innovative business models are often needed to bridge the cost gap between EVs and internal combustion engines (ICEs). Project preparation can also support the verification at the suppliers’ of the conformity of e-buses with the bidding documents and test developments. Projects sourced and designed under this activity will be financed within components 2 and 3 and result in the Outputs (2.1) and (2.2.) Project monitoring is also part of this activity.

This activity will contribute to a proper structuring of the Projects, an adequate risk allocation and bankability and to the deployment of innovative business models.

This activity will be implemented by AFD only.

Project Selection Criteria

All investments must be compliant with the GCF investment framework and follow the principles, terms and conditions agreed by AFD group and the co-financers, including project eligibility criteria, governance, implementation arrangements, environmental and social management systems, gender consideration, monitoring & evaluation and verification. The Program assists in identifying climate change related projects and maximizing their GHG mitigation benefits and contributes to the bankability of projects by introducing catalytic funding. The Program applies to all projects a standard GHG accounting methodology to determine GHG benefits going beyond the baseline.

Project Eligibility Criteria

Eligible projects must fulfil the following minimum criteria:

1. All investment projects financed under Components 2 or 3 must reduce GHG emissions relative to the baseline and have a positive sustainable development impact. The approach to be used is determined in Annex 22a and is based primarily on UNFCCC methodologies registered under the Clean Development Mechanism (CDM);
2. All investment projects financed under Components 2 must be linked to transport demand policies encouraging shift to public transport;
3. All investment projects financed under Components 2 or 3 must demonstrate a transformational potential and increase ridership;

4. All investment projects financed under Components 2 or 3 must comply with economic internal rate of return of minimum 12%;
5. All investment projects financed under Components 2 or 3 must be in compliance with relevant national and local laws and regulations and have country ownership;
6. Investments in e-buses financed under Component 3 are realized in cities which are planning to or in the process of modernizing and upgrading their public transport system including initiatives to foster NMT and electric micro-mobility;
7. All investment projects financed under Components 2 or 3 shall have a gender action plan taking into account guidelines developed in Annex 8;
8. All investment projects financed under Component 2 will have a climate risk and vulnerability assessment to identify potential risks to transport infrastructure financed under the Program and to incorporate actions to mitigate climate risks if required;
9. All investment projects financed under Components 2 or 3 shall strive to generate employment and local economic development, and therefore shall demonstrate a maximised share of local content;
10. Only full-electric battery-electric vehicles are eligible for financing under Component 3. Hybrid trolleybuses (trolleybuses with battery) can also be financed if they prove to be a more cost-efficient option than usage of battery-electric buses considering also infrastructure replacement and maintenance costs;
11. No projects with less than 30 EVs are financed under Component 3;
12. The focus is on financing e-bus deployment in intermediate cities under Component 3. 50% of GCF FA resources shall be invested in intermediate cities. Cities which are not eligible under this criteria are: Colombia: Bogota, Barranquilla, Bucaramanga, Cali, Cartagena, Medellin, Pereira; Brasil: Belo Horizonte, Brasilia, Curitiba, Fortaleza, Manaus, Recife, Rio de Janeiro, Salvador, Sao Paulo; Peru: Lima; Mexico: Mexico City;
13. All investment projects financed under Components 2 or 3 have a co-finance level (non-GCF funds) of minimum 40%. Minimum co-financing rate of 60% over total Program (all components);
14. In minimum 4 countries (indicative Brazil, Colombia, Mexico, and Peru) minimum 1 investment project with e-buses under Component 3 and in minimum 2 countries (indicative Brazil and Colombia) investments in upgrading PT and/or NMT infrastructure under Component 2 will be realized;
15. No investment grants are given for e-bus investments under Component 3;
16. The target is to finance one e-bus investment project which will be realized by the private sector or as a PPP (indicative in Mexico) under Component 3; and
17. The interest rate to the private sector cannot be lower than the one for the public sector.

Not more than 35% of GCF funds shall be invested in one specific country. Investment projects will be determined on a “first come, first served” basis, depending on their level of maturity at the time of availability of GCF proceeds.

EVs are purchased instead of new conventional vehicles i.e. the old unit is at the end of its lifespan and not retired or phased out prior normal replacement. The Project Owner (PO) has the option at that moment to purchase a fossil or an electric bus. In both cases (baseline fossil as well as project case electric) a new unit enters and the old unit is also the same in both cases. There is thus no increased number of old units with the project case versus the baseline case. Therefore, the project in no case results, with or without scrapping, in additional emissions due to the purchase of NEW vehicles but only accounts for emission reductions due to purchase of a vehicle with a different drive technology. The intervention of the Program is that instead of a new fossil vehicle a new electric vehicle is purchased. What happens with the old vehicle is the same in the project case or in the baseline case. The difference is not the old vehicle but the BAU new vehicle (fossil) against the Program new vehicle (electric). All emission impacts and financial calculations are also a comparison of a new fossil against a new electric unit. The Program does not promote earlier replacement or scrapping of vehicles. Scrapping Programs in any case have limited merits

as for example in the case of buses old municipal buses could still be used to replace even older rural or private transport buses with low mileage. The Program does not change the number of vehicles but the vehicle technology. Instead of new diesel buses (BAU) the Program will support new electric buses. What happens with the old buses being replaced by new diesel or electric units is the same in the baseline and project case. Therefore to be very clear: the Program does not add vehicles. In absence of the Program the exactly same amount of vehicles would be purchased, just fossil instead of electric units. See for further details chapter 6.2. of Annex 22.

Finance Instruments

The Program uses loans and grants. Grants are only used for technical assistance in component 1 channelled mainly through GIZ, the implementing agency for TA (34 M€ grants of which 21 M€ from the GCF) and for project sourcing and development managed by AFD (6.5 M€ grant of which 1.1 M€ from the GCF). The investment components 2 and 3 are purely loan financed.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

GIZ has a strong technical expertise in the development of large TA programs and in the sustainable mobility sector. GIZ finances together with AFD the National Urban Mobility Policy (NUMPs) of Ecuador and Uruguay (Euroclima+ program). Moreover, GIZ is particularly active in the promotion of e-Mobility in the region and supports the design and implementation of e-Mobility policies through several grant-funded projects: support to introduce specific criteria for electric buses within the renewal fleet program REFROTA, integration of bus systems and electrification in Florianopolis (FELICITY program with EIB) and technical assistance to strengthen conditions to consolidate electrical mobility in Brazil (PROMOB-E program), development of a National E-Bus Promotion Program in Colombia, and support to the operationalization of e-buses in Costa Rica (MiTransporte program). Through the Cities Finance Facility (CFF) program, GIZ is also developing two bankable projects for new electric buses on 2 routes in Quito and providing assistance for new electric bus corridor and bicycle lanes in Mexico City.

The current regional Program aims at completing these multiple initiatives, which are currently being implemented across Latin America, with a common ambitious financing and technical assistance proposal specifically targeted towards the commercial EV segments, which are now able to reach the scale and become commercially viable in the short to medium term.

Implementation Arrangement

E-Motion will be implemented jointly by the following institutions:

- AFD, as accredited entity, executing entity and co-financier for the Financial assistance targeting public sector project owners (Components 2 and 3), for the technical assistance dedicated to Projects' preparation, and for the gender action plan activities related to the investments projects.
- PROPARCO, as executing entity and co-financier for the Financial Assistance targeting private sector project owners;
- GIZ, as executing entity for technical assistance activities;

AFD project teams will be responsible for the identification, appraisal, implementation and evaluation of eligible investments in the public sector. The projects' appraisal process will follow AFD's procedures. In particular, task team leaders will carry out due diligence and appraisals on each project in accordance with AFD Group standards and procedures.

PROPARCO will be in charge of E-Motion implementation within the private sector. As per AFD projects, PROPARCO project teams will be responsible for the identification, appraisal, implementation and evaluation of eligible investments in the private sector and appraisal process will follow PROPARCO's procedures, which are in line with AFD Group standards and involve AFD at decision taking stage.

GIZ will be responsible for the implementation of activities 1.1 (majoritarily³⁹), 1.2 and 1.3, and 1.4 (i) to (v) within the Technical Assistance Facility. GIZ will ensure the implementation of activities using a combination of own technical staff based in respective partner countries as well as independent consultants. GIZ will report to the E-Motion Steering Committee and to the AFD Programme Manager. AFD and PROPARCO will have the responsibility to apply GCF proceeds in the framework of the financed sub-projects.

Contractual arrangements

The GCF and AFD will sign a Funded Activity Agreement (FAA) and all GCF funds, whether reimbursable or non-reimbursable funds, will be transferred from the GCF to AFD according to the provisions set in the FAA. The GCF loan proceeds will be provided to AFD in the form of a sub-participation, and the GCF grant proceeds will be provided in the form of a grant to AFD.

AFD will sign a grant subsidiary agreement with GIZ and will channel GCF Technical Assistance funds to GIZ for the purpose the implementation of the majority of the activities 1.1, 1.2 and 1.3 and 1.4 (i) to (v) of component 1;

AFD will sign a subsidiary agreement with PROPARCO, which will define the fiduciary and implementation arrangements between AFD and PROPARCO and PROPARCO will have a mandate to act on the behalf of AFD, with AFD being the lender. A grant agreement might be signed to delegate part of TA project preparation funds to PROPARCO;

Financing agreement between AFD or PROPARCO and the beneficiaries

Eligible projects within AFD's scope (sovereign and non-sovereign public sector) and PROPARCO's scope (private sector) will benefit from financial support extended by AFD and PROPARCO through the Financial Assistance on a project-by-project basis. For this purpose, PROPARCO and AFD will sign financing agreements (grant agreement and loan agreement) with the respective implementing bodies (developers, ministries, etc.). Financing agreements will describe the provisions and modalities for disbursing and using the funds for the purpose of implementing the projects, and will be developed and negotiated according to AFD Group templates and procedures.

A detailed financial assessment will be realized by AFD and PROPARCO in the framework of the appraisal process of the project to ensure that the borrowers answer to the required criteria to sign a Credit Facility Agreement with AFD and PROPARCO (risk profile, revenues etc.).

Service contracts between AFD and consultants

AFD will hire consulting companies to implement the TA dedicated to the preparation of the projects and to support the PMU activities. In order to implement the Projects' preparation Facility of E-Motion, a call for tenders will be launched to hire a pool of consulting firms that shall be mobilised to meet the specific needs

³⁹ In order to secure consistency with the FA preparation process, and to maximize synergies between TA and FA, AFD might implement a part of activities 1.1(iv), 1.1(v) and 1.1(vii), as per the logframe.

of each project. This will allow flexibility in the TA activities and speed in the mobilisation of consultants for the projects.

AFD will also hire external consultants as part of the Program Management Unit. For the Project Preparation AFD intends to mobilize 2 MEUR grant to contribute to the Project's preparation. If approved, the funds should be committed progressively during the Program's commitment period.

For the Technical Assistance Component, a co-financing by the German Government (of up to 1 MEUR) is intended to be included in Component 1 of E-Motion and subject to a separate bilateral contractual arrangement between GIZ and the Federal Ministry for the Environment, Nature Protection and Nuclear Safety. The co-financing will enable GIZ to support the program management unit and link to other German Government funded projects.

In addition, the following TA activities, already committed and being implemented between 2020 and 2026, are part of the Funded Activities of the Program:

- Sustainable Urban Mobility support program for Dominican Republic through the Caribbean Investment Facility (CIF) comprising technical assistance activities implemented by AFD and whose purpose is notably to support the implementation of the National Urban Mobility Policy, the ongoing reform of urban transport sector, and the implementation of the Santo Domingo sustainable mobility plan including e-mobility development. The amount dedicated to activities facilitating the development of e-mobility is estimated at 4.6 MEUR.
- Euroclima+ technical assistance activities, financed by the EU and implemented by AFD and GIZ, whose purpose is to develop national policies for urban mobility and integrated multi-modal planning at the city level, in order to accelerate the transition of Latin American cities towards sustainable urban mobility with low carbon emissions. Euroclima+ will contribute to create the favourable conditions of the development of e-mobility in the framework of integrated and sustainable urban mobility plans. The amount of the Euroclima+ activities dedicated to the E-Motion countries is 8 MEUR.

For the Program Management Unit, AFD and GIZ will mobilize 4.3M EUR (65% of the total cost of the PMU) with an in-kind contribution of staff of each institution. 35% of the PMU costs will be financed by the GCF.

Governance and Program Management

An E-Motion **Steering Committee** (SC) will be constituted for the whole E-Motion Program. It will meet on a quarterly basis and be responsible for making the strategic decisions required for the execution of the programme. Its members are representatives of the executing entities (AFD, GIZ, and PROPARCO). The SC shall notably review and approve programme annual budget and work plans, discuss implementation issues and identify solutions, and ensure coordination and communication between executing entities.

The following additional committees will be constituted to ensure a proper implementation of the Program:

- A Technical Assistance coordination committee to ensure the consistence of the TA activities implemented at local, national and regional level. Its members are representatives of the executing entities (AFD, GIZ, and PROPARCO).
- For each sub-program (driven by AFD):
 - An investment committee to pre-validate Concessional GCF loans proposed to be mobilized on the Financed Projects. Its members are representatives will be PROPARCO, GIZ and AFD.
 - A Project Preparation Committee to coordinate and validate the mobilization of the TA dedicated to Projects' preparation. Its members are representatives will be PROPARCO, GIZ and AFD.

Considering the large geographical scope targeted by E-Motion and the diversity of financial and technical assistance tools managed by the Program, it is essential to have a strong Program management Unit (PMU) to ensure the coordination between the partners and a proper implementation of the Program.

The PMU staff will be composed of professionals from AFD group and GIZ that will be in charge of handling the respective components of the Program. The PMU will be chaired by an AFD Task Team Leader, part time dedicated to the Program and based in Latin America. A Task Team Leader from GIZ will also be dedicated part time to the Program Management Unit.

In addition, the PMU will be composed of external consultants: an e-mobility expert, an Environmental, Social and Gender expert, a communication Expert and a Project Management support officer. AFD will be in charge of contracting the PMU Consultant team.

The PMU will be in charge of:

- The general coordination and management of the Program, which will include the animation of the governance bodies of the Program, the technical coordination and animation, and the administrative and financial follow-up for structuring;
- The management of the Technical Assistance for the Projects' preparation;
- The preparation of the reporting and monitoring reports due to the GCF and the supervision of the evaluation process;
- The implementation and the follow up the E-Motion communication plan, supporting the organization and implementing of events, seminars and conference related to the program and preparing all the communication support (web content, press release etc.);
- The implementation and maintenance of a knowledge and records management system.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Rationale for and Additionality of GCF Funding

The Program introduces a concessional financing mechanism that addresses barriers to the uptake of innovative public transport business models with EVs and supports a transformative shift to low carbon transportation. Currently fossil vehicles dominate the market and the fleet renewal. In all involved countries no large fleets of commercial EVs operate. The paradigm shift towards electric mobility not only entails new vehicles but a new fuelling/charging infrastructure, changes in vehicle maintenance and operations and frequently also requires changes in the business model. Actors in the commercial vehicle sector fear such disruptive changes as it can affect their core business. Governments on the other hand are keen on promoting e-mobility as this can reduce emissions and dependence on (often imported) fossil fuels whilst also creating long-term sustainable jobs. They therefore often announce political targets or objectives to promote e-mobility coupled with stand-alone EV pilots. The missing link is however to kick-start EV massification with a first batch of at-scale implementation projects. Such initiatives are dependent on concessional funding to make them financially and from a risk perspective feasible. These projects are the foundation for meaningful, practical, realistic and action-oriented roadmaps and for developing policies which are cost-effective and reduce barriers towards e-mobility and encourage shift to public transport. Stand-alone pilots or technical assistance has proven to be insufficient to kick-start mass deployment of EVs. GCF concessional financing is critical to ensure this comprehensive e-mobility approach – in absence of concessional GCF finance, such projects will not materialize in the short or medium term.

The major financial barriers for the potential beneficiaries which prevent alternative financing options are (i) the low financial profitability of EV investments reflected in a low FIRR and long payback times (ii) the high risk profile of EV investments due to performance risks, asset risks (e.g. unclear resale value of assets), and replacement investments (batteries) and (iii) high capital and investor's capital demand resulting in high debt and risk exposure levels of the investing entity. Commercial funding is thus not willing to enter

this market. Alternative funding vehicles with a higher risk appetite are potentially interested but are critical of the lack of profitability and/or the high risk profile. A good example for this case is Chile where traditional banks were reluctant to finance commercial EVs and private electric utilities entered the market with a leasing scheme for buses thus creating a new business approach (see Benchmark report Annex 22). However, also these entities were only willing to kick-start the process when the government recognized and paid for the incremental investment cost of e-buses plus guaranteed payment of the leasing fees i.e. without concessional finance and risk mitigation instruments the deal would not have taken place.

The main barriers why investors, cities or PTOs do not want to purchase e-buses is the lack of profitability and the higher upfront CAPEX resulting in higher financial costs. Guarantees do not resolve this problem and will increase the total cost. Potential investors as well as FIs are only interested in guarantees if these incur no cost. They however also mention that the guarantee will change the decision to purchase or not EVs as guarantees do not tackle the problem of profitability and high upfront costs. Only very high guarantee levels (100%) would result in slightly lower interest rates. For some concession contracts in certain countries guarantees could be useful. However, no specific case could be identified where the guarantee instrument would be relevant. Highly concessional finance is critical for deployment of initial commercial EV fleets. Concessional finance is not required in replication projects which happen at a later stage as (i) EV market prices decrease: EV prices are decreasing in all vehicle segments due to decreasing battery prices and increasing competition and supply of EVs. TCO alone is not a sufficient argument to move for EVs if the upfront incremental investment is still significantly higher and the risks are still perceived to be higher. However, decreasing vehicle costs are a very important ingredient to make EVs commercially viable without need for further financial assistance. (ii) Reduced risks and costs for 2nd and 3rd projects due to showcasing of successful implementation projects and business models.

Risks which are reduced by large fleet applications are (i) vehicle reliability and vehicle performance risks; (ii) improved information and therefore lower risk on assessing the re-sale value of used EVs important for introducing business models based on leasing; (iii) less risk concerning maintenance & repair capabilities and availability of spare parts including secondary spare parts markets; (iv) reduced risk on the workability of new business models in practice. The Program assists in this process by designing and implementing business models and by enabling with concessional finance the uptake of large fleets which are critical to reduce these risks. (v) Policies which are more conducive for e-mobility and restrict fossil vehicle usage -if it is an objective set by the country. This allows EVs to be more competitive. Lower costs, reduced risks, and improved framework conditions all create favourable conditions for investors to pour capital into this market as profit rates improve and risks get down. This will allow for widespread replication based on commercial terms in 5-10 years if the country has gained experience with initial fleet deployment. The concessional finance is required as kick-off. The market development and experience of countries shows that at the end of the Program such support is no longer required.

Concessional finance is not only needed to compensate the higher upfront costs of electric vehicles but also to reduce and share the risk of investing in this type of technology. Indeed, there are several barriers beyond 'just higher upfront costs: associated with this, there are higher capital requirement, higher equity requirement and higher interest burden and guarantees (also due to uncertain resale value) with usually limited debt framework. Moreover, the construction of charging infrastructure and depot infrastructure needs to be considered as a sunk cost. Finally, there is a higher risk in comparison: TCO calculations and savings potentials are based on assumptions on electricity price development and usage, development of diesel prices, maintenance costs and operational readiness of e-vehicles, lifetime of batteries, operational risks due to lower flexibility (e.g. e-buses cannot be used on all routes).

Conventional financial institutions are also not willing to enter the market due to the novelty of the technology. A pilot project will not dispel their concerns as these are associated basically to the lifespan of the technology including batteries, associated replacement costs, the quality and reliability of vehicle providers (basically Chinese bus manufacturers which are not all well known in the market), resale value of vehicles, and potential savings in terms of maintenance and repair costs. Financial institutions perceive

these risks and are thus reluctant to finance EVs. The establishment of new technologies and new business models also entails significant additional transaction costs. Concessional GCF finance including grants can provide for technical assistance to design and structure e-mobility investments with a reduced performance risk whilst the financial instruments of the Program reduce risk exposure and increase profitability of the investment. Increasing the ridership of public transport strengthens operators and transport providers and reduces the risk with e-mobility investments.

The multi-country approach is critical for a transformative and accelerated shift towards e-mobility, which is again required to achieve the target of limiting the global temperature increase to below 2 degrees Celsius. Multi-country approaches are critical for learning and experience exchange between Latin American countries with activities such as the realization of different knowledge materials including publications, webinars, benchmark and best-practice studies, preparation of capacity building guidelines and outreach events including trainings and workshops. The interchange of experience is included in component 1.3 and is realized within the program. Through participation of various countries with a similar context the learning effect can be much larger helping to get to replication. Therefore, outreach and learning from experiences of other countries is critical. If the program is realized only in one country, the replication will be basically on a national level. National programs are always important but experiences from other countries can be brought in much easier if the Program covers multiple countries. A multi-country programmatic approach is also necessary given the scale of climate change challenges facing countries and the critical need to deliver transformative change at scale as quickly as possible as vehicle replacement rates are low thus resulting in long lock-in times of investments. The GCF involvement is critical to support low-carbon transport investments across multiple countries and vehicle sectors. With a multi-country programmatic approach, resources can be used efficiently to highest priority investment projects – this cannot be achieved with a piecemeal, project-by-project approach. Individual projects are in most cases too small to warrant a Funding Proposal to the GCF with all transaction costs involved. AFD would thus refrain from financing electric mobility in most cases and not integrate it into its transport investments. The Program will also capitalize on synergies and lessons learned across countries and vehicle sectors and can effectively ensure transfer of knowledge and capacity building between stakeholders.

Concessionality of GCF Funding

GCF concessional finance including grants are used for TA and for FA. GCF assistance enables linking finance with technical and policy assistance. Investments alone are insufficient to achieve the transformation in climate action needed by the transport sector. Strategic sectoral planning and policy reform are critical aspects to enable the transformation towards electric mobility. The GCF, with the goal of supporting paradigm shifts in climate action, is critical for sufficient funding of the policy and technical assistance component of the Program. The Program involves private transport operators, national FIs, and potentially SPVs of PPPs for vehicle ownership. The private involvement is stimulated by involving in projects national FIs and by having PPPs as the major delivery agent for projects.

PTOs are in most involved countries private companies. FIs are included in the national finance part. Also SPVs for private led delivery in a separation of ownership and operation of assets are private companies. In some countries however public sector led SPVs might be more appropriate depending on country circumstances.

FA concessional finance are critical to reduce upfront CAPEX, to make investments commercially viable and to compensate for risks from initiating climate investments in a new technology and in new business models. GCF's ability to offer concessional terms in investments enables prospective clients to invest in e-mobility which would otherwise not be able to do with AFD finance alone. The benchmark report (Annex 22) clearly shows that all countries with large fleets of commercial EVs (>100 units) have used, at least initially, financial support instruments in the form of grants to kick-start mass EV deployment. Incremental investment costs of e-buses was grant financed by countries by 80-100%. The magnitude of subsidies can be reduced with adequate policies and regulations (e.g. concession periods) and business models per

vehicle sector. The experience of countries is also that once a large fleet is established subsidies can be reduced gradually and even be eliminated within 5-10 years due to more competitive prices of EVs combined with the experience of mass operations. This reduces for investors in EVs primarily performance risks related to energy usage and costs, battery replacement and maintenance costs plus reduced income risks related basically to the availability of a fast charging infrastructure. GCF concessional finance is critical for sustainable market uptake linking initial pilot projects with commercial EV viability by inserting the missing part of initial large scale fleets combined with business models, policy advice and technical designs which reduce risks related to EV investments and allow to close the gap between political aspirations and commercial EV uptake.

Long tenures and low interest rates are critical for the affordability of such loans as the fiscal situation of countries and of operators has tightened in LAC drastically with COVID-19. Longer tenure and low interest rates reduce the debt service levels significantly allowing countries to take up new loans without creating significant additional fiscal imbalance. Also, the public transport sector has been hit hard by COVID 19 with additional subsidies required by government to maintain service levels. Without such subsidies the public transport system will decline in importance due to bankruptcy of multiple private operators. Concessional loans with long tenure and low interest rate increase the fiscal space of governments to provide subsidies to the public transport sector which increases the sustainability of the services and thus of the program. They are thus absolutely critical for Program success.

Grants are used for technical assistance such as new policies or business models, battery regulations, workshops, trainings etc. They result in no revenues. Therefore reimbursable systems will not work. TA is considered as absolutely essential for the paradigm shift. Only with changing policies, business models and with trained and informed people will this happen. Without this effort one-time projects and interventions will occur which will not result in a transformational change.

The Program will benefit from **technical assistance** that will be made available to accompany each project and will ultimately serve to enhance the viability of the projects. The TA package has been designed for each country individually based on the FA interventions, a demand analysis and an assessment of other actors in this area in the same country and other regional readiness programs to avoid duplication of effort. TA will establish a sustainable mobility ecosystem encouraging electrification and shift on a local, national and regional level with activities structured from the local to the regional level. The Program will provide direct technical assistance to projects that will potentially receive financial assistance. TA includes pre- and full feasibility assessment, legal, institutional, financial and technical advisory as well as advisory on supportive transport demand policies as well as possible business models and financial structuring. This includes charging networks and the interface between vehicle, charging, and the grid. TA includes also:

- Capacity building, training and monitoring: TA is used to ensure delivery of adequate training from equipment suppliers or third parties for EV and charger operations and maintenance, safety trainings and capacity building in optimal management of EV fleets. This activity also includes performance monitoring of EVs in technical (energy usage, emissions) and financial terms (operational expenditures, vehicle availability rates), which is core to reduce the perceived performance risk of EVs and to create a solid base of actual costs and benefits of EVs in the country.
- Support of enabling public policies for shift to public transport and EV deployment: Technical support will be provided to the relevant national and local authorities in the countries and cities where the Program will support investments in order to set-up, improve and/or enforce enabling public policies and legal, regulatory, fiscal and/or normative frameworks for e-mobility. The Program intends to establish in all countries where not yet available roadmaps for electrification of public transport and roadmaps for taxis and urban freight. Sectoral and specific roadmaps are a tool to steer climate investments and initiatives beyond the Program's scope and will help to inform

and/or further detail the countries long term mobility strategies and their NDCs. Policy advice is also performed for battery handling, re-usage, re-cycling and disposal. TA in policy also includes advice on concession contract issues, electricity pricing (e.g. demand charge for public charging networks).

- Development and advisory services on optimal business models and financial structuring which relate to the peculiarities of EVs. This is targeted for public transport to identify business models e.g. for atomized urban transport providers, models which allow for increased influx of private capital and which lower financing costs and risks for operators, leasing models, inclusion of electric utilities etc.
- Inclusion of gender aspects in all projects to be financed as well as on a national and regional level.
- Knowledge management: The Program will develop knowledge products and share experiences and cases through channels such as webinars, electronic reports, case-studies, and workshops to disseminate experiences and build a base for replication thereby accelerating EV uptake on a regional scale. The Program will not create own platforms or channels for information dissemination but use existing channels such as the one created by UNEP/MOVE. An interchange of experience shall also take place where experiences can be shared from countries more advanced in specific issues, e.g. Colombia on bus systems.

B.6. Exit strategy (max. 500 words, approximately 1 page)

Exit Strategy

Repayments under concessional loans provided with GCF funds under the Program will be flown through the AFD-GCF Special Account in accordance with repayment schedules set forth in AFD or Proparco's loan agreement. It is envisaged that tenors will not exceed 15 years. All loans will be monitored by AFD group. GCF resources will be reflowed back to the GCF in accordance with the terms of the FAA.

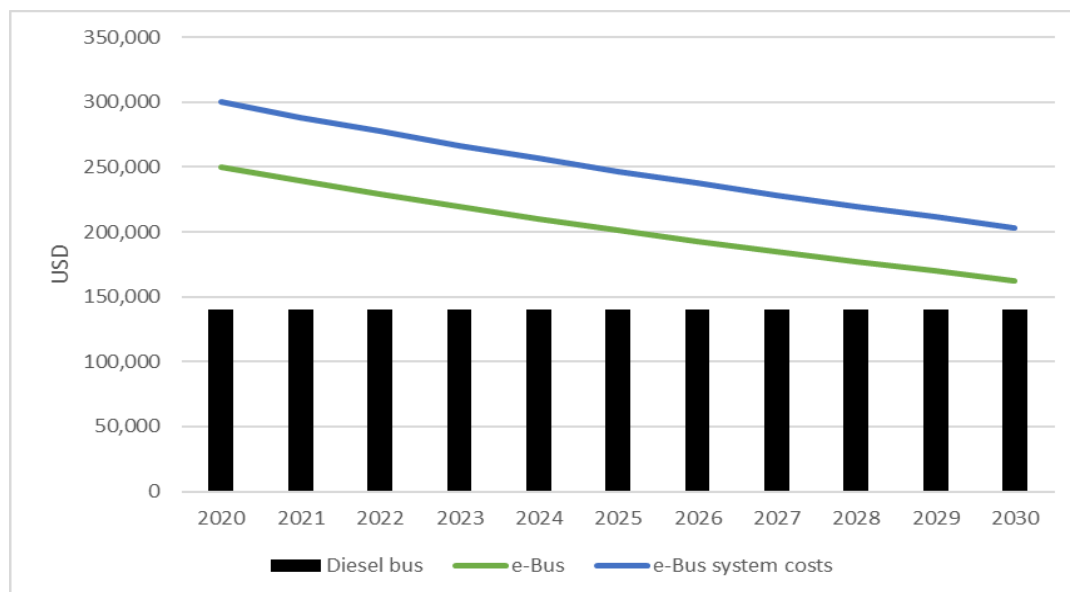
Program Sustainability

At the end of the Program (based on an implementation period⁴⁰ of 7 years this would be 2032) there are operational large fleets of e-buses in all Program countries. This forms the base for a large scale commercial replication without grants. The projections of dropping E-bus prices point to cost based parity (TCO) around 2025 and strongly decreasing upfront price tags. The market analysis performed per country shows that a significant share of investors would then be willing to purchase electric buses without subsidies given that experience is available in the country with an initial large-scale fleet which conforms core assumptions concerning energy usage, maintenance cost, vehicle reliability, and system technology choice (types of chargers, charging method and site, battery type and battery size on buses).

Whilst the investment cost is still expected to be higher for an e-bus system the significantly lower OPEX (basically energy plus maintenance cost) make the TCOs significantly lower than for fossil bus systems (in many countries the TCOs are already today lower but not sufficient i.e. the payback period is still long and sometimes longer than the concession period of PT operators thus not making the investment financially attractive under a risk viewpoint).

⁴⁰ Implementation period refers to the period of development of the TA and contractualization and disbursement of the credit lines.

Figure 11: Projected Electric and Diesel Investment Cost per Urban Bus (real constant USD of 2020)



Source: Grutter Consulting

It is expected that by 2030 e-buses will be close to sticker price parity with diesel units, whilst total system costs per e-bus are still expected to be around 40-50% higher than of a fossil unit. However, the significantly lower OPEX of e-buses together with the technology and market maturity will make purchases of e-bus the standard, without necessity of further concessional finance. TCO costs will be much lower than of diesel buses and payback periods of incremental e-bus investments under this investment price trend will be less than 5 years. This commercial uptake will only be realized if the investment risks of e-buses are reduced. This is possible through the establishment of new business models and through operations of initial large fleets which allow for performance monitoring thereby reducing uncertainty for follow-investors. Countries which originally subsidized 80-100% of the entire e-bus system incremental cost for the initial phase and have purchased or subsidized large fleets have been able to decrease their subsidy rates significantly within few years. The importance of significant upfront investment subsidies – independent of TCOs – to kick-start deployment of e-buses is thus clear. It also shows however that once a large fleet is established subsidies can be reduced gradually and even be eliminated with 5-10 years due to more competitive prices of e-buses combined with the experience of mass operations of e-buses.

Concrete examples are:

- In PR China the average subsidy level per 12m urban e-bus decreased within 5 years from 150,000 USD to 30,000 USD per bus and it is expected that it will be eliminated fully by 2022⁴¹.
- In India under FAME I 2017-2019 e-buses were subsidized with 60% of incremental costs. This was reduced to 40% under FAME II 2020-2021 (40% of a lower absolute value meaning absolute subsidies were reduced by more than 50% within 2 years).
- In the UK the 1st scheme for low carbon buses (until 2017) subsidized 90% of the incremental cost and the 2nd scheme since 2018 75% of the incremental cost of e-buses (including charging infrastructure).
- Chile subsidized for the first lot of 200 electric buses 100% of incremental costs. The next lots received no more investment gap subsidies but some additional benefits for e-buses relative to fossil units in tenders which is translated into higher payments per distance driven or per service

⁴¹ Source: Grutter Consulting; contract with ADB for monitoring of e-bus fleets in 18 cities with 60,000 buses financed by leasing funds with support by ADB

unit for e-buses (TransMilenio, Colombia analysed the tenders of Santiago and estimates the subsidy to amount to more than 16% of total costs per km⁴²).

The establishment of a conducive EV framework, local business models which match the EV situation (e.g. separation of ownership and usage of assets) as well as reduced risks of operating EVs due to the experience gained with large fleets all make it feasible that at the end of the Program implementation period EVs are taken up massively and at a far larger scale than in absence of the Program.

Based on the market studies performed the Program expects that a commercial uptake of EVs without further support from the GCF is feasible in the targeted countries at the Program end. This is feasible due to the intervention of the Program accelerating in a decisive manner the commercial uptake of EVs and fast-forwarding their financial feasibility. The sustainability is given by (i) improved framework conditions where the Program intervenes with different policy actions including roadmaps; (ii) attractive business models which are implemented with the projects and which provide investors with confidence of the attractiveness of investing in this market; (iii) reduced risk for investors due to clarity on performance of buses with the projects implemented and monitored by the program plus training of mechanics and support staff thus reducing the risk of stranded assets for investors and (iv) the market development which results in reduced costs of e-buses and system components resulting in increased profitability. Thus the combination of factors which create better framework conditions, clarity on investment models combined with reduced investment risks by knowing better the product and not being a first mover and increased profitability due to reduced CAPEX all work hand-in-hand to create a sustainable future for e-mobility investments.

It is expected that commercial investors will crowd in to the e-bus area once the framework conditions (e.g. guarantees, concession contracts) and the performance risks of e-buses are manageable. This has been the case in Santiago de Chile as well as Bogota. A coalition of investors announced end 2021 the intention to invest 1 billion in e-buses in Latin America⁴³. The Programs TA creates conditions which are favourable to actually materialize such investments and the investment component of the Program reduces the risks to investors and operators. The Program is thus critical to unlock the private investment potential. For commercial investors to crowd in the TA is crucial to establish a conducive framework including e.g. concession contracts and business models as well as clear government policies. Just as important is that large-scale fleet deployment with such business models take place as proof of concept and to reduce performance risks for operators and financiers. Both TA as well as concessional finance for FA are however not required anymore in the future once the framework conditions are established and once the risks are reduced as lower costs of EVs make these profitable already in the short to medium term under the mentioned conditions of an adequate e-mobility environment.

The technical assistance supports long-term sustainability through a number of crucial factors:

Roadmaps at the national or city level are developed and are underpinned with actual policies that strengthen the EV ecosystem and make public transport more attractive. These frameworks will support additional investment in other cities by providing the right incentives, reducing risk through enhanced attractiveness of buses, and reduce uncertainty for investors. Coupled with the demonstration effect of large-scale deployment of new technology and business models in the selected four cities, this will enhance the commercial feasibility of future investments.

Countries within the program have very different starting points. Those with already existing roadmaps will serve as role models implementing all components at both national and local level. Others will focus on developing their roadmaps and maybe implement some policies mostly at the local level, where components 2 and 3 are implemented. For countries without actual implementation components, the policy

⁴² Grutter Consulting, personal communication with TransMilenio

⁴³ [New \\$1 billion dollar commitment from investors to deliver zero-emission buses in Latin America - C40 Cities](#)

frameworks will be essential to trigger enhanced bus electrification and increased use of public transport. Experiences from on-the-ground implementation in other countries can help shape better policies and enable a more ambitious strategy.

Enhanced understanding of business models in the specific context of Latin America, including for second-life usage of batteries, combined with a proof-of-concept on the ground will reduce the risk of investors and motivate additional action.

Capacity-building activities ensure that there is a broader understanding of technology options and their suitability to different Latin American contexts at the national and city level in all countries covered by the program. The development and dissemination of experiences with business models as well as financing strategies will further add to the commercial attractiveness of EV buses after the program period. The development of general training and curricula for vocational training of drivers and technicians will support the availability of skilled personnel for enhanced deployment of electric buses in the region.

In addition and under the TA component of the Program, specific studies will be undertaken on e-vehicles batteries' replacement, disposal and recycling issues, to ensure environmental sustainability of this new business model (refer to Annex on Technical Assistance, where this is further detailed in activity 1.2).

C. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	78.333415 million euro (€) 40.264 million USD (\$)			million euro (€) million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	56 million euro (€) 40.264 million USD (\$)	20 years	5 years	0.75 %		
(ii) Subordinated Loans	Enter amount	Enter years	Enter years	Enter %		
(iii) Equity	Enter amount			Enter % equity return		
(iv) Guarantees	Enter amount	Enter years				
(v) Reimbursable grants	Enter amount					
(vi) Grants	22.333415 million euro (€)					
(vii) Results-based payments	Enter amount					
(b) Co-financing information ⁴⁴	Total amount			Currency		
	Enter amount			Options		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
AFD, KfW	Senior Loans	85.5 63.6315	million euro (€) million USD (\$)	20 years 5 years	Enter%	pari passu
AFD, KfW, GIZ/BMUV	Grant	18.314335	million euro (€)	Enter years Enter years	Enter%	Options
national finance	Public: grant, in-kind Private: Equity, loan	25.833333	million euro (€)	Enter years Enter years	Enter%	Options
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options

⁴⁴ Co-finnancing amounts could be in USD, EUR or local currency

(c) Total financing (c) = (a)+(b)		Amount					Currency			
		<u>210.981104</u> <u>103.8955</u>					<u>million euro (€)</u> <u>million USD (\$)</u>			
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)		Notes: 1. National finance: dependent on project if grant (public sector led projects), equity capital, loans (private sector led implementations) 2. The maximum USD that will be requested by AFD from the GCF is 40.264 MUSD								
C.2. Financing by component										
Comp onent	Outp ut	Indi cati ve cost milli on euro (€)	Indica tive cost million USD (\$)	GCF financing			Co-financing			
				Amo unt millio n euro (€)	Am oun t milli on US D (\$)	Financi al Instru ment	Amount million euro (€)	Amount million USD (\$)	Financi al Instru ment	Name of Institutio ns
<u>Compo nent 1</u> <u>Establi shmen t of a</u> <u>sustain able</u> <u>mobilit y</u> <u>ecosys tem</u> <u>encour aging</u> <u>electrifi cation</u> <u>and</u> <u>shift to</u> <u>public</u> <u>transp ort</u>	<u>Outp ut</u> <u>1.1</u> <u>activi ties</u> <u>1.1.</u> <u>to</u> <u>1.4</u>	<u>34</u>	<u>0</u>	<u>21</u>	<u>0</u>	<u>Grants</u>	<u>13⁴⁵</u>	<u>0</u>	<u>Grants</u>	<u>AFD,KfW</u> <u>, GIZ</u>
<u>Compo nent 2</u>	<u>Outp ut 2</u>	<u>27</u>	<u>15.8</u>	<u>8.5</u>	<u>6.1</u>	<u>Loans</u>	<u>13.5</u>	<u>9.7</u>	<u>Senior loans</u>	<u>AFD, kFW</u>
							<u>5</u>		<u>Grants</u>	<u>National</u>

⁴⁵ Funds already committed and being implemented between 2020 and 2026

<u>Investive measures for PT mode shift</u>										
<u>Component 3 Establishment of innovative public transport business models deploying e-buses</u>	<u>Output 2</u>	<u>143</u>	<u>88</u>	<u>47.5</u>	<u>34.1</u>	<u>Loans</u>	<u>75</u>	<u>53.9</u>	<u>Senior loans</u>	<u>AFD, KfW, CAF and PROPARCO</u>
							<u>21</u>		<u>Grants</u>	<u>National</u>
<u>PMC</u>	<u>Output 1.1 Activity 1.5</u>	<u>7</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>Grants</u>	<u>6</u>	<u>0</u>	<u>Grants</u>	<u>AFD</u>
Indicative total cost		<u>211</u>	<u>103.8</u>	<u>78</u>	<u>40.2</u>		<u>133.5</u>	<u>63.6</u>		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☒ No ☐

Capacity Building

Capacity Building is delivered under Component 1. This includes specifically:

- Project specific training during design and planning stage specifically the transfer of skills for optimal technology choice, design of charging infrastructure, preparation of tender documents, design of relevant business models, financing plans, risk sharing mechanisms.
- Project specific training and capacity building per implemented project on technology related issues such as maintenance of EVs and charging infrastructure, ecological driving skills with EVs, optimal vehicle dispatch, planning and design as well as technology choice of EVs,

safety and risk avoidance with EVs, battery management and disposal policies and regulations etc.

- Capacity building and policy advice specifically for the establishment of sub-sectoral roadmaps for electric mobility in the fields of public transportation, taxis and LCVs.
- Capacity building with knowledge products and their dissemination based on the best global practices and the experiences in the different projects. This is realized through different products and channels to reach a wider audience in Latin America.
- Capacity building on gender mainstreaming
- Training and CB of service providers including maintenance and repair staff for buses as well as charging infrastructure to minimize the risk of low vehicle availability rates.

The purchase contracts with equipment providers (buses, chargers) will include specific training and CB courses which the equipment providers must realize for operators personnel including drivers, maintenance staff, service staff and transport operators. The procurement contracts will also include on-site maintenance services to be provided by the equipment provider for buses and chargers during minimum 2 years in collaboration with the staff of the equipment owner cum operator.

Technology Transfer

Technology Transfer be facilitated through identification and design of appropriate e-mobility technologies and projects and their subsequent funding. Technology providers such as bus and charging equipment providers are tied with purchasers into 2–3-year service provision contracts where the supplier must provide for on-ground expert staff to train local maintenance and repair staff. At the same time this extended service contract includes guarantees on vehicle availability and spare parts availability which ensure smooth operations and high vehicle availability rates. The supplier must therefore ensure that the vehicle technology is understood by the local operators and staff in charge and ensures this through training programs which are also specified in the bidding documents for suppliers.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

Area of emission reductions are (i) impact of replacing fossil with electric vehicles and (ii) increased ridership in public transport.

Impact of E-Buses

GHG emission calculations have been made per for potential investment projects for e-buses per country. Whilst baseline emissions per vehicle category are comparable for different countries (variations are due to vehicle mileage, fuel type used and to a lesser extent due to different specific fuel consumption values) the carbon grid factor per country influences the emissions of the EVs and therefore the emission reductions.

The GHG impact is determined based on a well-to-wheel approach. This includes for EVs the carbon emissions from the grid including production, as well as transmission and distribution losses. With the average grid factors of the countries included in the Program electric vehicles targeted by the report reduce GHG emissions by 70-90%. The methodology report on GHG reductions of the Program also includes a life-cycle or cradle-to-grave approach to determine emission reductions to assess the magnitude of leakage (see Annex 23). Inclusion of upstream emissions (vehicle manufacturing including batteries) only changes the picture marginally. For the grid factor the IFI database last version 3.1 of January 2022 is taken for electricity consumption. The assumed lifetime of measures is 15 years in accordance with e-bus lifespan with 1x battery replacement.

Impact of Public Transport Measures

This includes GHG reductions from Program investments in public transport measures to increase its patronage. The methodology used for calculation GHG reduction is based on the UNFCCC methodology ACM0016 and the UNFCCC methodological tool "Baseline emissions for modal shift measures in urban passenger transport". GHG reductions are due to additional passengers which in absence of using the bus would have used other modes of transport with on average higher emission factors. Estimations of the incremental patronage of PT systems is thereby based on monitored results of already implemented projects. Measures which increase the PT usage include infrastructure investments in bus lanes and bus stations, smart ticketing, improved connectivity and intramodality, modern electric bus fleets and preference for PT systems in urban settings. The emission factor of private means of transport is per passenger-kilometre factor 2x higher than for PT. The assumed lifetime of measures is 25 years in accordance with infrastructure lifespan.

The total projected GHG direct impact of the Program is estimated at 4,334,051 tCO_{2e} over the lifetime of the project based on the initial pipeline of projects⁴⁶. This includes only the impact of the 4 countries. Actual Program emission reductions will depend on which projects are actually implemented and will be monitored by the Program.

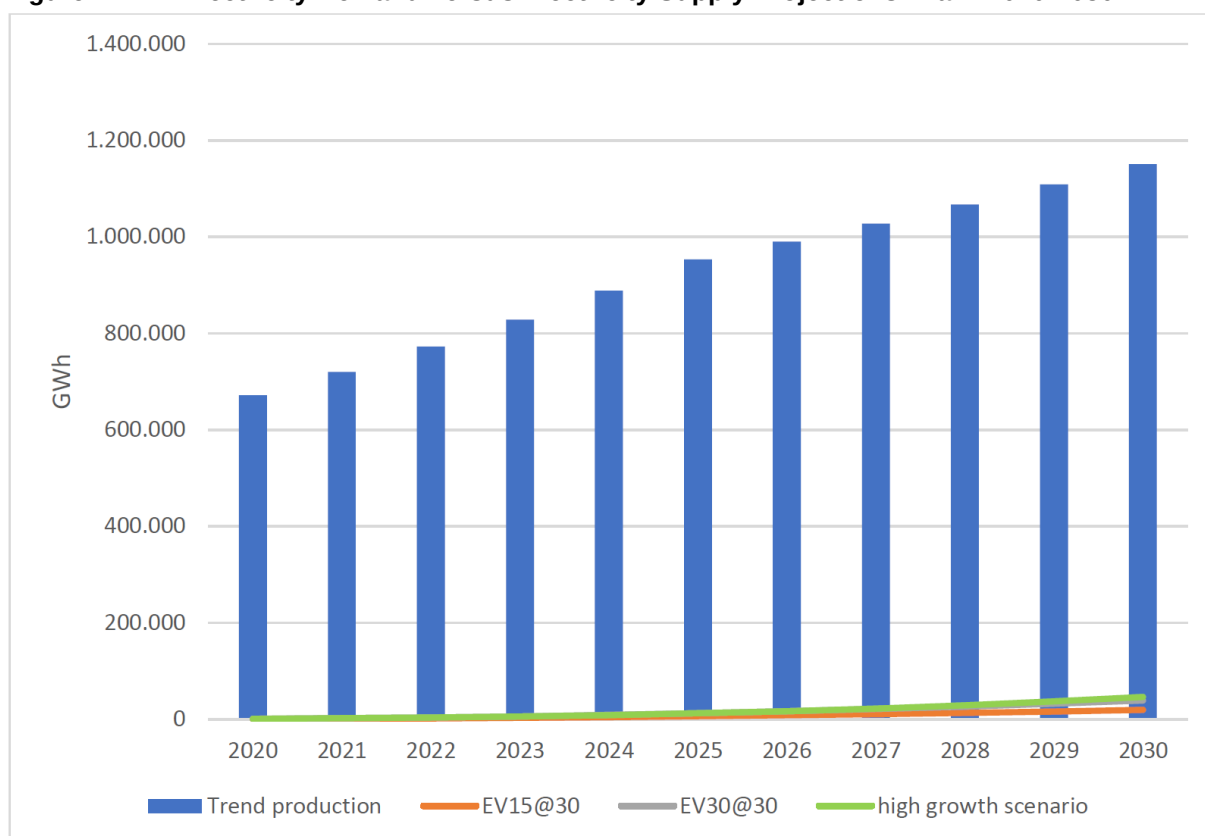
The important aspect of the Program is however that it creates the conditions for an accelerated uptake of EVs. This has been modelled based on an investor-risk model which per country and vehicle sector calculates the annual FIRR of investing in EVs (the FIRR increases over time due to decreasing relative EV prices), the investment risk level and the target WACC⁴⁷.

⁴⁶ Expected impact per country: Brazil 440,000 tCO₂; Colombia 285,000 tCO₂; Mexico 195,000 tCO₂ and Peru 95,000 tCO₂ (rounded values)

⁴⁷ See for a detailed description Annex 23

Market assessments made in the different countries show that electricity demand for EVs even with a high penetration rate as targeted by the IEA scenario EV30@30 (30% of new registration of vehicles are electric in 2030) represent 1-digit figures of demand and less than the annual growth rate experienced in the countries (see figure below as example from one country). In no country the electricity demand by 2030 is more than 5% of the current supply even if the highest growth strategy of EVs of IEA would be achieved. This is due to the fact that vehicle turnover rates are low i.e. the change of the vehicle stock takes long (this is also true in other countries e.g. in Norway nearly 90% of vehicles sold in 2022 were electric but the total vehicle stock of EVs is still below 30%). The limiting factor by 2030 is in general not electricity supply or new demand created by EVs. The problems which grids must face are of limited localized power supply, limited additional capacity at substations and peak power demand. Therefore design of adequate charging methods for e-buses is for example critical to avoid localized problems and to avoid increasing peak demands.

Figure 12: EV Electricity Demand versus Electricity Supply Projections Brazil 2020-2030



Source: AFD, Country diagnostic report Brazil

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

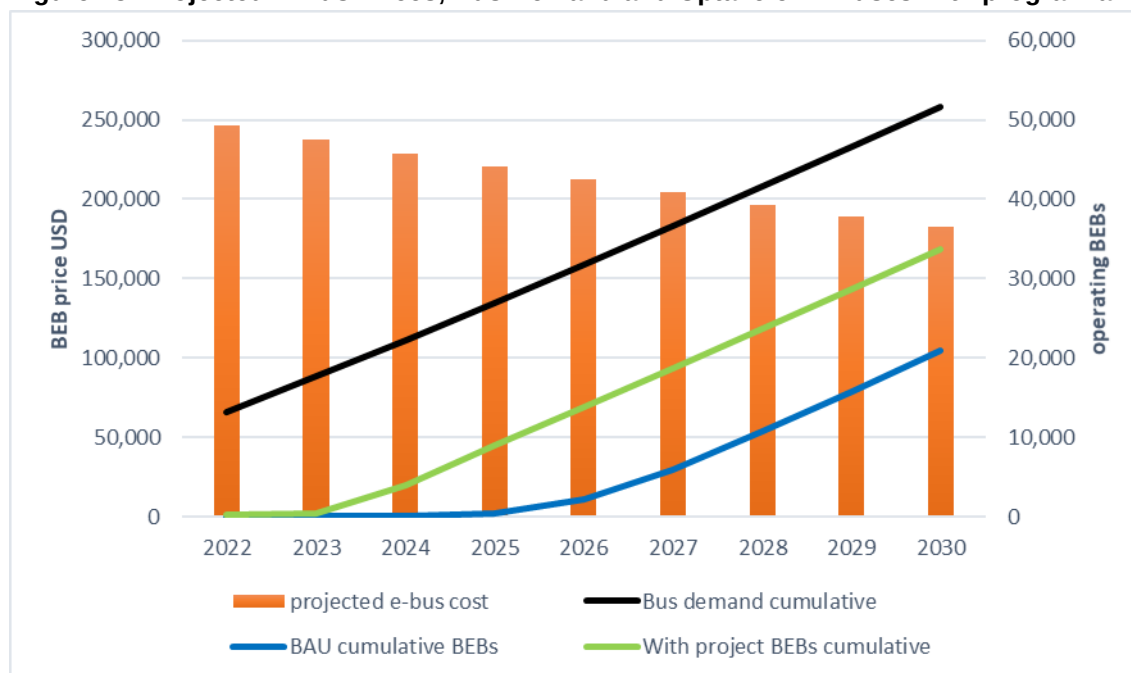
The **paradigm shift** of the proposed Program is achieved by having as long-term outcome a EV conducive ecosystem which results in investors purchasing EVs on a commercial base and a modal share of sustainable transport increased. Barriers towards mass EV deployment have been successfully eliminated or significantly reduced through Program interventions including FA and TA. Accelerated EV investments take place due to experiences made, business models established, reduced entry barriers, lower risks and costs and new market players. The transformation of public transportation into an attractive and efficient system is achieved by increasing services quality and working with partners on policies for incentives to shift to public transport and increasing ridership. Both approaches, the energy transition in transport and the mobility transition, together lead to a paradigm change towards sustainable mobility. The transformative shift is possible through the uptake of commercial EVs and the improved ecosystem for EVs

resultant of the concrete interventions as well as of policy assistance and capacity building. The transportation sector is transformed to a sustainable low-carbon system with a dominance of public transportation and NMT. Multiple countries in the region are establishing Green Recovery Plans due to the COVID-19 crisis as a vehicle to transform the society by investing in profitable infrastructure which creates economic, environmental and social benefits, short-term jobs and a long-term greening of the economy. Electrified and sustainable public transportation systems are thereby a core feature. Countries are interested in shifting to e'mobility ' this is expressed in updated NDCs as well as in activities on'going in all countries in the realm of e'mobility. What lacks is clarity on policies and an enabling environment for EVs, appropriate business models especially for e'buses and experience with initial large fleets, especially also in Colombia outside the large cities with established mass transit systems. The Program activities can remove these barriers and can showcase successful and commercially attractive implementation of large fleets. This combined with the falling costs of EVs of all categories allows to fast'start EV deployment on a mass scale and move from a fossil'based transportation system to an electric one. The willingness of the public sector is there - the interest of the private sector is also manifest if the implementation has manageable risks and an acceptable profit rate – what can bring these components to spur e-mobility deployment is the design and implemetnaiton of adequate policies, framnework conditions, business models and performance experience of large fleets.

In absence of a program E-mobility will also be the norm in Latin America as in the rest of the world – just at a later stage. The key issue is about accelerating mass transformation at a faster pace and at an earlier stage. An e-mobility program in this context is not different than any other GHG mitigation project – the question is not so muchi IF it will happen but if we can accelerate with GCF investments WHEN and to what EXTENT it will happen. The figure below shows a modelling made for Colombia with and without Program showing the projected deployment of EVs. The same modelling was made for all involved countries for e-buses, LCVs and taxis. The results including the model assumptions and modelling approach are detailed in the country reports per country (country reports number 3 (E-Motion Country Intervention Strategies) and the respective Excel Files per country included under Annex 3.

The initial pipeline developed in 2021/2022 and updated thereafter showed that the majority of investment plans to purchase e-buses have not been fulfilled. The stakeholders contacted mention the lack of concessional finance and the prevailing risks as core reason. The program country which has progressed strongly in e-buses is Colombia but focusing to an overwhelming extent on Bogota plus to a much smaller extent in Cali and Medellin. Investments in smaller cities, targeted by this Program, have not taken place. The need for the Program is thus obvious and the delay in approving this Program is one of the reasons why Latin American cities have not deployed more e-buses resulting in retarding dearly needed initial investments in this technology to reduce with speed GHG emissions.

Figure 13: Projected E-Bus Prices, Bus Demand and Uptake of E-Buses with program and under BAU



Source: AFD (2021), E-Motion Country Intervention Strategy Colombia

The **potential for scaling up and replication** is ensured at the national and the international level through capacity created in the E-mobility ecosystem, proven cases of successful business models, improved profitability and reduced risks of EV investments and knowledge products. The experience of various countries shows that once fleets of EVs are operational and not only pilot projects, the technology is taken up quickly if the business environment is conducive. The projects financed under this Program function as trigger projects demonstrating the commercial viability of EVs. Performance risks are reduced greatly and business models have been introduced which make the uptake of EVs on a purely commercial base viable. This is linked with decreasing investment costs of EVs reaching cost-parity to fossil units in the medium term. Scaling up and replication is also enabled through TA at policy level including the development of sectoral roadmaps, design and establishment of appropriate policy incentives and knowledge management instruments as well as capacity building.

It is expected that commercial investors will crowd in to the e-bus area once the framework conditions (e.g. guarantees, concession contracts) and the performance risks of e-buses are manageable. This has been the case in Santiago de Chile as well as Bogota. A coalition of investors announced end 2021 the intention to invest 1 billion in e-buses in Latin America⁴⁸. The Programs TA creates conditions which are favourable to actually materialize such investments and the investment component of the Program reduces the risks to investors and operators. The Program is thus critical to unlock the private investment potential. For commercial investors to crowd in the TA is crucial to establish a conducive framework including e.g. concession contracts and business models as well as clear government policies. Just as important is that large-scale fleet deployment with such business models take place as proof of concept and to reduce performance risks for operators and financiers. Both TA as well as concessional finance for FA are however not required anymore in the future once the framework conditions are established and once the risks are reduced as lower costs of EVs make these profitable already in the short to medium term under the mentioned conditions of an adequate e-mobility environment.

Potential for knowledge sharing and learning: The TA Component includes capacity building, training and monitoring. Technical assistance is used to ensure delivery of adequate training from equipment

⁴⁸ [New \\$1 billion dollar commitment from investors to deliver zero-emission buses in Latin America - C40 Cities](#)

suppliers or third parties for EV and charger operations and maintenance, safety trainings and capacity building in optimal management of EV fleets. Training materials, reports and knowledge products elaborated under this header in the different countries will be shared through multiple channels under the knowledge management activity. This will be realized at a national level thus removing one of the barriers to more widespread deployment of EVs in each country and also on an international level by sharing documents and experiences between countries and also with countries outside the Program. The Program will also develop professional knowledge products and share experiences and cases through channels such as webinars, electronic reports, case-studies, and workshops to disseminate experiences and build a base for replication. The Program will use existing channels within GIZ, AFD, CAF and KfW as well as those of other platforms for information dissemination.

Contribution to the creation of an enabling framework: The TA Component includes the support of enabling policies for EV deployment: Technical support will be provided to the relevant national and local authorities in order to set-up, improve and/or enforce enabling public policies and legal, regulatory, fiscal and/or normative frameworks for e-mobility. One of the main instruments used thereby are sector roadmaps for electrification of public transport, roadmaps for electrification of taxis, and roadmaps for electrification of LCVs. Whilst various countries have general EV roadmaps or targets these remain at a macro-level. Sectoral roadmaps combined with investment projects perform a base for more specific targets, intervention instruments and actions allowing to take steps towards actual implementation. Sectoral and specific roadmaps are a tool to steer climate investments and initiatives beyond the Program's scope and will help to inform and/or further detail the countries long term mobility strategies and their Nationally Determined Contributions.

Contribution to the regulatory framework and policies: The TA Component includes activities such as roadmaps (see above) which outline an agreed-upon set of pragmatic and realistic measures to achieve sustainable mobility targets. Regulatory activities realized by the Program will tackle transport demand policies but also include aspects such as battery management and disposal regulations, charging infrastructure standards, pricing policies for EV charging, public transport regulatory aspects including aspects such as concession contracts (concession length, vehicle turnover guarantees), separation of vehicle ownership and vehicle operations, tariff structure etc. Technical assistance in this area is linked up with investment projects making it more credible and pragmatic. Roadmaps are developed in collaboration with all stakeholders and have clear short and medium term targets, means to achieve these targets to be realistic and implementable and to go beyond a vision statement. An example of a recent and clear roadmap which helps to accelerate a transition is the Transit zero emission vehicle roadmap of the State of Colorado, USA which outlines the pathway for full electrification of buses⁴⁹. An interesting roadmap for the region is also from Chile⁵⁰.

Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans: Transport contributes almost one-quarter of the current global energy-related GHG emissions and is growing faster than any other energy end-use sector. Limiting the global temperature increase to below 2 degrees Celsius requires changing this transport emissions trajectory. The Paris Declaration on Electro-Mobility and Climate Change & Call to Action, calls for the deployment of EVs compatible with a 20% share of all road transport vehicles in 2030. All countries have however expressed their keen interest in embarking on this transformational shift towards e-mobility. Without significant GHG reductions in the transportation sector countries will not be able to achieve their decarbonization plans. Fostering e-mobility and kick-starting EV deployment is thus consistent with relevant national climate -resilient development pathways.

⁴⁹ [colorado-transit-zev-roadmap-2021-11.pdf \(codot.gov\)](https://colorado-transit-zev-roadmap-2021-11.pdf)

⁵⁰ [Chile estrategia electromovilidad 27dic.pdf](#)

D.3. Sustainable development (max. 500 words, approximately 1 page)

Sustainable Development Goal alignment

The Program has as main target to reduce GHG emissions. It contributes significantly to sustainable development goals (SDG) 3 (“good health and well-being”), SDG goal 7 (“affordable and clean energy”), SDG goal 9 (“industry, innovation and infrastructure”), SDG goal 11 (“sustainable cities and communities”), and SDG 13 (“climate action”).

Environmental co-benefits

Major environmental co-benefits are reduced emissions of pollutants and reduced noise emissions. The major concern for air pollution in the cities is PM_{2.5} and NO_x emissions. The impact of the Program on these pollutants has been quantified based on combustion emissions of new fossil baseline vehicles (compliant with vehicle emission standards of the country) using the EU COPERT model i.e. this is a conservative approach as emission reductions are not based on comparing the old replaced with the new EV but on comparing a new (BAU) fossil with a new EV. For the methodology used, the database as well as calculations see Annex 23. The following table shows the projected reduction of pollutants of the Program.

Table 4: Projected Lifetime Pollutant Reductions (in tons lifetime)

Projected tons of PM _{2.5} reduced	Projected tons of NO _x reduced
41	4,842

Note: Total Program impact calculated with the same approach as for GHG reduction

Source: Annex 23

Social co-benefits including health impacts

The major social benefit is improved air quality (plus reduced noise). Air pollution is a major problem in most Latin American cities with levels affecting seriously human health. According to the latest urban air quality database, 98% of cities in low- and middle-income countries with more than 100,000 inhabitants do not meet WHO air quality guidelines⁵¹. The WHO estimates that annually 4.2 million deaths result due to excessive exposure to fine particulate matter equivalent to 7.6% of all deaths⁵². The poor are disproportionately affected by air pollution as they tend to live and work closer to its sources⁵³. Children and the elderly are particularly vulnerable. Recent studies show that women are more affected by poor air quality than men⁵⁴. The transportation sector is thereby a major source of urban air pollution. Commercial fossil vehicles are a major source of air pollution in cities. Although they do not represent a majority of vehicles they are extensively used (large mileage), have high emission factors and are basically diesel powered. The diagnostic reports realized in all Program countries shows that the targeted commercial vehicles contribute in general more to air pollution than private passenger cars or motorcycles (see Annex 2). Replacing such vehicles with EVs thus has a significant impact on air quality and health of people.

The COVID'19 pandemic has created an unprecedented challenge for many Latin American countries, from the health and economic perspectives, but it is also opening new opportunities for an accelerated sustainable transformation of its energy, transport, and urban landscapes while contributing to economic recovery. A recent report prepared by the ILO shows that fostering of electric mobility can have a significant positive job impact primarily due to the induced impact of savings of consumers on petrol and maintenance resulting in increased spending on goods with a high income elasticity which tend to be

⁵¹ <https://www.who.int/airpollution/data/cities-2016/en/>

⁵² <https://www.who.int/gho/phe/en/>

⁵³ (Mitchell, 2013)

⁵⁴ (Clougherty, 2010)

labour intensive service-goods (ILO, 2020). A report of McKinsey also reveals that with the COVID pandemic the interest in EVs has risen amongst customers due to the fact seemingly that the pandemic has raised the awareness among people of the negative impact of fossil transport modes being suddenly being able to experience clean air during lockdowns⁵⁵.

Economic co-benefits

The monetary value of reduced air emissions (GHG and air pollutants) has been calculated (see Annex 23 for the methodology, dataset and results). The economic cost of pollution is calculated by assigning a monetary value to emissions of PM_{2.5} and NO_x for each country based on local levels of pollution at the ground level and the impact on health and costs caused by this type of pollution per country. This is based on the exposure of the population to contamination and how increased pollution increases mortality risks using the World Health Organization's dose response functions to concentration. The greater risk of mortality or, more precisely, the cost of premature death is valued economically on the basis of stated preference studies as performed by the OECD.

EVs are more energy efficient than fossil vehicles. PT is more energy efficient than usage of private vehicles. The energy savings resulting from the Program are estimated at 48,343 TJ over the lifetime of vehicles.

The TA will include plans for a just transition to capture the potential positive benefits in terms of new jobs being created with e-mobility whilst enabling people which loose jobs e.g. in the maintenance area to find new jobs.

Gender-sensitive development impact

Gender issues and development impact will be described in the gender-disaggregated targets in projects' targets. Each Program will be required to meet AFD's policy on gender equality. Individual projects will also be guided by the gender action plan (GAP) as included in Annex 8. The GAP includes actions and targets (i) on improved access for women to public transportation; (ii) improved sex-disaggregated data collection on urban transport; (iii) Improved women's access to jobs; (iv) interventions including awareness raising and infrastructure to reduce sexual harassment in public transport; and (v) trainings and workshops to disseminate best practices for reducing sexual harassment in public transport.

On a cross-sectoral level the GAP has as target to integrate gender considerations in transport sector activities. This includes (i) the usage of a gender lens when designing the roadmap for different EV segments and EV support policies; (ii) Increased awareness about gender equality gaps and opportunities in transport through capacity buildings and training; and (iii) Increased knowledge in how to include gender equality in Urban Transport projects through specific training materials.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

⁵⁵ [Mobility investments in the next normal | McKinsey](#)

The market study (Annex 2) identified the following barriers hindering an take-up of EVs in the countries involved:

- High upfront investment;
- Lack of profitability of EV investments;
- High risk of EV investments (financial and technical risks);
- Lack of know-how on appropriate system design, business models, financial models and technical know-how;
- First-mover problem as no commercial EV fleets yet operate in the country;
- Lack of an enabling (policy) framework for sustainable mobility and EV deployment.

AFD has also realized market costing studies on the different vehicle segments for different price ranges and countries (see Annex 2). The commercial attractiveness of EVs versus fossil vehicles has been assessed based on parameters such as the Total Cost of Ownership (TCO), FIRR, capital and equity requirements, cash flow and risks. This again serves as base to structure the financial instruments required to resolve the barriers to commercial viability of EVs in the selected vehicle segments. The conclusions from the market study that upfront financial support is required combined with policy advice to create a favourable e-mobility ecosystem is supported by the benchmark study (Annex 22) which shows that countries which have followed this path are succeeding in deploying on a mass scale EVs whilst others remain in the pilot stage.

Contrary to renewable energy generation that is increasingly cost competitive, EVs are not yet commercially viable and remain largely underfunded. The market and benchmark studies show that, if unaddressed, commercial EV deployment on a mass scale will take place in a delayed manner. This will delay the transformation towards low carbon transport systems in the Program countries.

The EV sectors targeted have (i) limited access to commercial funding, (ii) early-mover costs and disadvantages including high performance risks (iii) lack of sufficient commercial viability including profitability and acceptable risk levels. Therefore financiers are not in a position to fund such activities with commercial loans at the current stage.

Governments and transport operators also lack capacity and know-how on the appropriate technology choices and the most appropriate and conducive policies and business models to foster in a cost-effective manner EV deployment. This is clearly evident from public statements in favour of electric mobility which is however not followed up with concrete actions and implementations.

D.5. Country ownership (max. 500 words, approximately 1 page)

Section B1 already includes a short overview of countries including the carbon grid factor, population and vehicle data, energy prices and major EV policies. This section therefore focuses on the climate policies of the involved countries. The majority of countries have included in the NDCs electric mobility as a core mitigation measure to reduce transport emissions and have formulated concrete electrification targets through regulations, policies or EV strategies. The project supports the countries in implementing the following measures and targets:

In the framework of the elaboration of the E-Motion Concept Note until today, the French Development Agency and its partners (Proparco, CAF, KfW, GIZ) have hold meetings with the NDAs of the countries concerned by the Program and key actors of the e-mobility sector to ensure that the FP answers to the needs of the countries (See Annex 7). These actors include development banks, private banks, municipalities and regional states, power utilities, transport operators etc. The following table gives an overview of consulted stakeholders per country and the identified potential investment projects per country. With all the listed stakeholders virtual and in many cases physical meetings were organized. With many stakeholders multiple meetings were realized. The Program also made per country virtual kick-off meetings with stakeholders, a large amount of individual meetings and a discussion of final results. The FP, the

planned instruments and activities are the result of the discussions with more than 100 stakeholders of the countries involved.

Table 5: Overview stakeholder meetings

Country	Stakeholder meetings	Identified Investment Projects
Argentina ⁵⁶	Individual meetings with 23 stakeholders including municipal and city governments; national and local transport authorities; national and local power companies; financial institutions; other programs in the area of e-mobility	Multiple individual meetings with 17 potential owners of investment projects in various cities for urban buses, taxis, light commercial vehicles incl. discussion of financial terms they have negotiated so far
Brazil ⁵⁷	Individual meetings with 23 stakeholders including national ministries (economics, science & technology, environment), provincial and city governments; national and local transport authorities; national and local power companies; financial institutions incl. BNDES; other programs in the area of e-mobility	Multiple individual meetings with 9 municipalities for investment projects for urban buses incl. large cities such as Sao Paulo and smaller cities such as Santa Teresina or Niteroi incl. discussion of financial terms they have negotiated so far
Colombia ⁵⁸	Individual meetings with 31 stakeholders including national ministries (UPME, transport, national planning department, environment, energy), provincial and city governments; national and local transport authorities; national and local power companies; financial institutions incl. Findeter; other programs in the area of e-mobility	Multiple individual meetings with 6 potential owners of investment projects for urban buses in 13 cities (Findeter with investments in 13 cities), taxi and Light Commercial Vehicle projects plus detailed discussions of all expansion plans of TransMilenio for Bogota
Costa Rica ⁵⁹	Individual meetings with 14 stakeholders including national ministries (environment & energy, presidency, regulatory transport authority, transport); financial institutions incl. various banks and leasing companies; other programs in the area of e-mobility	Investment projects had already been identified with previous studies in the field of electric vessels, urban electric buses through leasing schemes and light commercial vehicles e.g. with the Costa Rican postal system or private operators.
Dominican Republic ⁶⁰	Individual meetings with 24 stakeholders, including national ministries (economics and planning, environment, energy), national and local transport authorities and associations; power companies; financial institutions; transport companies; other programs in the area of e-mobility	Multiple individual meetings with 15 potential owners of investment projects for urban and interurban buses, taxis and light commercial vehicles including current financial conditions for investments
Mexico ⁶¹	Individual meetings with 22 stakeholders including national ministries (e.g. INECC, ministry environment, ministry of finance, Protram), provincial and city governments; national and local transport authorities; national and local power companies incl. ENGIE or CFE; financial institutions incl. Banobras and NAFIN; other programs in the area of e-mobility	Multiple individual meetings with 9 potential owners of investment projects for urban buses and taxis plus detailed discussions of all expansion plans of Metrobus for Mexico City. Projects identified are in Mexico city as well as other cities such as Guadalajara, Hermosillo or Culiacan
Peru ⁶²	Individual meetings with 18 stakeholders including national and city transport authorities; power	Multiple individual meetings with 11 potential owners of investment projects for urban buses and meetings with 2 banks for

⁵⁶ Annex 7 File 9

⁵⁷ Annex 7 File 10

⁵⁸ Annex 7 File 11

⁵⁹ Annex 7 File 12

⁶⁰ Annex 7 File 13

⁶¹ Annex 7 File 15

⁶² Annex 7 File 16

	companies; financial institutions; other programs in the area of e-mobility	electric mobility credit lines (acceso and cofide)
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Argentina

Argentina's total GHG emissions are estimated at 361 MtCO_{2e} in 2020 (excluding LUCF) of which the transport sector 39 MtCO_{2e}.⁶³ In its NDC (Nationally Determined Contribution, 2015) Argentine commits to an absolute net emission target of 359 million tCO_{2e} by 2030. The updated NDC of 10/2021 includes that electric mobility is a core mitigation measure in the transport sector. Argentina launched the development of its National Electric Mobility Strategy in May 2018 with the support of UN Environment. Argentina is currently formulating specific legislation on electric vehicles, establishing the conditions for the installation or operation of charging centers, and at the same time, exploring possible options for the local development of the electric mobility industry. Decree 51/2018 establishes a tariff reduction for the import of electric buses up to a certain level. In terms of electric vehicle charging infrastructure, provision 283/2019 regulates the service of electric charging at fuelling stations and defines safety specifications on the installation and registration of charging stations.

Brazil

Brazil's total GHG emissions in 2020 were 1.06 GtCO_{2e} (excluding :LUCF) and transport emissions 190 MtCO_{2e}.⁶⁴ In its updated NDC of 11/2023 Brazil commits to reduce GHG emissions by 53% by 2030 compared to the year 2005 with the goal of achieving climate neutrality by the year 2050. The NDC indicates that the transportation sector shall promote efficiency measures and improve urban public transportation infrastructure to achieve mitigation targets. Brazil has already established various tax incentives for the import and purchase of EVs. There are 21 bills pending in the Chamber of Deputies on electric mobility that seek greater incentives for electric vehicles (Promobe, 2019). Resolution 97/2015 reduces the import tariff from 35% to 0% for electric vehicles and fuel cells. Senate Bill No. 454/2017 which seeks to prohibit the sale of internal combustion vehicles by 2060, is currently under discussion in the Federal Senate.

Colombia

Colombia's GHG emissions for 2020 are estimated at 187 MtCO_{2e} (excluding LUCF) Transportation emissions for the same year are 28 MtCO_{2e}.⁶⁵ Colombia's NDC Update estimates that according to the reference scenario for 2030 emissions would reach 346 MtCO_{2eq}. Within the mitigation goals Colombia commits to emit a maximum of 169 MtCO_{2e} in 2030 (equivalent to a 51% reduction of emissions). The NDC contemplates for the transportation sector amongst others to achieve 600,000 registered electric taxis, buses, light commercial vehicles including small trucks and official vehicles. The Electric Mobility Law has managed to provide for measures in public transportation services such as compliance with a minimum quota of 30% of EVs in new acquisitions or contracts, taking into account the commercial offer in Colombia. According to the same law, the goals for the incorporation of EVs in the acquisition of the fleet of zero-emission mass transportation systems must follow the scheme of minimum proportions of 10% in 2025, 20% in 2027, 40% in 2029, 60% in 2031, 80% in 2033 and 100% in 2035 (Congress of Colombia, 2019). As a complement to the Law, the National Government has developed the National Strategy for Electric Mobility (ENME), which aims to promote the electrification of the transportation sector. In addition to the above, the National Energy Plan (PEN) 2020-2050 presents projections for the incorporation of EVs,

⁶³ Climatewatch data; in 2020 COVID-19 resulted in large GHG reductions of the transport sector of more than 20% compared to pre-COVID whilst total GHG emissions dropped by only 4%.

⁶⁴ Climatewatch data; in 2020 COVID-19 resulted GHG reductions of the transport sector of 5% compared to pre-COVID whilst total GHG emissions dropped by only 1%.

⁶⁵ Climatewatch data; in 2020 COVID-19 resulted GHG reductions of the transport sector of 14% compared to pre-COVID whilst total GHG emissions remained stable.

under the scenario of meeting the GHG reduction commitments (20% by 2030) (UPME, 2019). Projections for 2030 include 630,000 electric motorcycles, 370,000 electric light vehicles, 40,000 e-taxis and 20,000 electric urban freight vehicles.

Costa Rica

Total GHG emissions of Costa Rica are estimated at 10.9 million tCO_{2e} in 2019 with land transport being responsible for more than 50% of emissions. Emissions under a BAU scenario are expected to increase by 45% by 2050. The updated NDC of Costa Rica includes as target net emissions of 9.1 MtCO_{2e} by 2030 and holds on to the net zero target by 2050. Greening the transportation sector is key to achieving these targets. Electrifying mobility is considered as essential and a national priority. The updated NDC has quantified 2030 electric mobility targets for public transport, passenger cars and fleets (8% of the vehicle stock). For other vehicle areas e.g. motorcycles targets and measures shall be developed to migrate towards EVs. Costa Rica has also developed a national plan for electric transport which includes concrete steps towards electrification of vehicles and has approved 2018 the law on incentives and promotion of electric transportation which includes targets for EV penetration, the establishment of a public charging infrastructure as well as important tax incentives for private EVs.

Dominican Republic

The estimated GHG emissions of the Dominican Republic in 2020 were 37 million tCO_{2e} (excluding LUCF) of which 7.2 million tCO_{2e} from the transport sector⁶⁶. The updated version of the Nationally Determined Contribution NDC-RD 2020 (Gobierno de la República Dominicana, 2020) has as target a 27% reduction in GHG emissions in relation to a Business as Usual (BAU) scenario by 2030. In terms of electric mobility, the NDC-RD proposes (i) Electrification of the fleet of diesel buses; (ii) Renewal of public transportation vehicles, such as cabs and "conchos" with electric and hybrid vehicles; (iii) Introduction of electric buses for school transportation service; (iv) Regulation and creation of policies to encourage the transition to electric and hybrid mobility for private usage. In 2020 the Dominican Republic has adopted a National Strategic Plan for Electric Mobility (INTRANS, 2020) developed by INTRANS in collaboration with the IDB. It provides for a short, medium and long term transformation of the transportation sector. The plan is aligned with both the UN Sustainable Development Goals (SDGs) 2030 and the Sustainable Development Plan 2030 of the Dominican Republic (Ministerio de Economía, Planificación y Desarrollo, 2012) and aims at all modes of road transportation. Targets of the plan are 30% of official vehicles and public buses shall be electric by 2030, whilst the target for the private sector is 10%. 14,000 public chargers shall be operational by 2030. By 2050 the respective goals are 100% EVs for official vehicles and public buses and 70% of all private vehicles shall be electric. For the freight sector the target for 2050 is 50% of all units to be electric.

Mexico

The GHG emissions of Mexico for 2020 are estimated at 592 MtCO_{2e} (excluding LUCF) with the transport sector contributing 109 MtCO₂⁶⁷. The baseline scenario for 2030 projects GHG emissions of 973 MtCO_{2e} of which 266 MtCO_{2e} from the transport sector (Gobierno de México, 2015). The Government of Mexico committed in its NDC to reduce its emissions by 22% by 2030. For the transport sector the target is set at 18%, reaching 218 MtCO₂ by 2030. The *Climate Change Strategy to 2050* published in 2016 established that within 10 years the use of EVs in public transport shall be common and within 40 years in all types of transport (SEMARNAT-INECC, 2016). In addition to the national instruments, the federal states

⁶⁶ Climatewatch data; in 2020 COVID-19 resulted GHG reductions of the transport sector of 9% compared to pre-COVID whilst total GHG emissions dropped by 7%.

⁶⁷ Climatewatch data; in 2020 COVID-19 resulted GHG reductions of the transport sector of nearly 30% compared to pre-COVID whilst total GHG emissions dropped by only 10%.

incorporate within their Programs strategies to incorporate electric mobility. As example the *Management Program to Improve Air Quality in the State of Mexico 2018-2030* proposes to prioritize the implementation of electric transportation and public systems over those that use fossil fuels, to provide the necessary infrastructure for electric vehicles, and to implement an electric cab Program, among others (Secretaria del Medio Ambiente, 2018). Mexico City published the *Strategy for Electromobility of Mexico City 2018 - 2030*, in which it proposes to promote electric public transportation with incentives and support for both public and private operators, with the goal that 20% of the fleet will be electric in 2030 and 80% of the cabs will be hybrid or electric for the same year.

Peru

The GHG emissions of Peru are estimated at 90 MtCO_{2e} for 2020 (excluding LUCF) of which the transport sector 20.5 MtCO₂⁶⁸. The Peruvian State committed in December 2020 that its net greenhouse gas emissions would not exceed 208.8 MtCO₂ in 2030 (unconditional target). In addition, the Peruvian government considers that GHG emissions could reach a maximum level of 179.0 MtCO₂ depending on the availability of international external financing and the existence of favourable conditions (conditional target). Adding both targets, unconditional and conditional, a total reduction commitment of 40% would be reached. This shall limit GHG emissions to a maximum of 209 million tons in 2030, and in case of international support, the limit is fixed at 179 million tons (NDC, 2020). The technical background document to determine the NDC target proposes the entry of around 7,000 electric buses and 170,000 electric light-duty vehicles by 2030 (measure ECM30) (Peru Ministerio de Ambiente, 2018). Initiatives will be limited for investment projects in areas without access to gas pipelines (as long as EVs are not competitive with CNG vehicles) with exception of TA activities which are also realized on a national level. The Competitiveness and Productivity Plan 2019 -2030 that includes (i) a regulatory package for the promotion of electric vehicles and their supply infrastructure by the end of 2019 (ii) technical standards for charging stations by mid-2025.

Regional Initiatives

The GEF Global E-Mobility Program implemented by the UN Environment Program (UNEP) in partnership with the International Energy Agency (IEA) is active in various targeted countries. Its focus is on technical assistance and pilot projects i.e. it complements very well the proposed Program. Complementarity will be ensured through close coordination with the existing programs and initiatives. The proposed Program is focused on investment projects, demand management and market transformation whilst existing initiatives are more for creating readiness for electric mobility in general.

Engagement with civil society

In each country AFD has performed country diagnostics and market studies study with multiple interviews and meetings realized with all major stakeholders including national and local governmental bodies, energy utilities, transport operators and civil society. Consultations have also been realized with stakeholders during the formulation of the Concept Note and the Funding Proposal (see Annex 7 for details).

D.6. Efficiency and effectiveness (max` 500 words, approximately 1 page)

The Program introduces a concessional financing mechanism that addresses barriers to the uptake of commercial EVs and supports a transformative shift to low carbon transportation. GCF concessional

⁶⁸ Climatewatch data; in 2020 COVID-19 resulted GHG reductions of the transport sector of 22% compared to pre-COVID whilst total GHG emissions dropped by only 10%.

financing is critical – in absence of concessional GCF finance, such projects will not materialize in the short or medium term. The benchmark report (Annex 22) clearly shows that all countries with large fleets of commercial EVs (>100 units) have used, at least initially, financial support instruments in the form of grants to kick-start EV deployment. The market studies realized in all countries also show that EVs are currently commercially not viable (with different degrees of non-viability). The market studies however also reveal that in the medium term due to price decreases, a commercial uptake of EVs is viable. Countries can take advantage of this circumstance from an early stage if they invest early-on in fleets of EVs to reduce prevalent risk factors associated with EV deployment.

62% of the total investment is co-financed (leverage ratio of 1.6).

The major financial barriers for the potential beneficiaries which prevent alternative financing options are (i) the low financial profitability of EV investments reflected in low FIRR and very long payback times (ii) the high risk profile of EV investments due to performance risks, asset risks (e.g. re-sale value of assets), and replacement investments (batteries) and (iii) high capital and investor's capital demand resulting in high debt and risk exposure levels of the investing entity. Commercial funding is thus not willing to enter this market. Conventional financial institutions are also not willing to enter the market due to the novelty of the technology.

Projects involved in the proposed Program will not be dependent on a continuous flow of GCF funds. Rather, the Program aims to achieve substantial reduction in the need for concessional financing in future project. It is expected that, over the course of implementation of the Program, commercial e-mobility will gradually move towards commercial viability in countries which have kick-started the process. This process has clearly been validated in the market studies realized in all countries and the Program will only engage in vehicle segments and in countries or regions where this condition of mid-term (by end of Program) commercial viability is given and can be achieved with the interventions of the Program. The Program therefore acts clearly as accelerator and massively reduces GHG emissions which is fully in-line with the global climate targets and the urgency to achieve earliest possible a change in greening long-term assets. The use of GCF funds de-risks future investments and demonstrates the viability of e-mobility investments on a commercial scale. It closes the gap between initial stand-alone initial pilot ventures and mass market deployment.

The FIRR and EIRR of individual projects is determined during the project preparation and the due diligence phase. AFD will assess the financial soundness of each project.

The total Program volume is 309 MEUR with a requested GCF finance of 117 MEUR and a direct GHG reduction of 4,334,051 tCO₂ resulting in an effectiveness of the GCF investment of 27 EUR/tCO₂. GHG marginal abatement costs of transport projects tend to be higher than of other interventions as they are also not realized primarily due to the GHG impact. The value of 27 EUR/tCO₂ lower than subsidies given by countries with significant EV numbers for buses (see table below) which result in average costs per tCO₂ of 190 EUR.

Table 6: Average GHG Abatement Cost of Fostering E-Buses

Country	Subsidy per bus USD	Carbon factor (kg CO ₂ /kWh)	GHG reduction per bus lifespan tCO ₂	Cost per tCO ₂ in USD
China (8y bus usage; subsidy level 2019)	40,000	0.71	412	97
India (FAME II)	80,000	0.90	626	128
Germany (2019)	300,000	0.51	1,039	289
UK (2019 level)	230,000	0.25	1,307	176
Switzerland (only carbon finance; excl. Municipal subsidy, 2020)	200,000	0.04	1,521	132
Poland (2019)	300,000	0.90	630	476
USA (2019)	400,000	0.43	1,116	358
Chile initial 200 e-buses (2019)	180,000	0.42	1,130	159

Average				227
Source: Grutter Consulting; see Annex 23 Notes: 1. Subsidies include bus plus charger 2. Carbon grid factor based on IEA for 2019 grid factors Based on the project pipeline AFD has realized a financial and economic modelling (see for details in Annex 3). For EVs FIRR and EIRR are based on differential analysis i.e. the FIRR and EIRR expresses the profitability of the incremental investments of purchasing an electric versus a fossil vehicle. Only costs which are differential such as CAPEX, energy, maintenance, finance and insurance are included. All other costs such as staff or overhead are the same between the 2 technologies and are not included. Revenues are not included as they are the same between the 2 technologies. The FIRR and the EIRR of E-buses therefore does not express the profitability of operations overall for a transport operator but reflects if the operator makes more profits or less losses if purchasing an EV instead of a fossil unit. Without GCF finance the average FIRR for 4/2024 is at 1% which is below the average Weighted Average Capital Cost (WACC) of 8% whilst with GCF finance the FIRR is on average at 9% or above the WACC. The EIRR of e-buses is without GCF 8% and with the GCF 15%. The total Program EIRR is estimated at 18% including the measures of e-buses and public transport (in 2 countries) against the total Program cost including TA (see Annex 3). Efficiency considerations The mitigation cost is 27 Euro/tCO₂ which is in the medium to higher range of the GCF portfolio reflecting the higher average GHG abatament costs of transport interventions. Effectiveness considerations The economic internal rate of returns are superior to the social discount rate for all countries. Based on the financial analysis, the FIRR with be below the WACC without the GCF for all countries. The grant amount will be used for technical assistance measures that cannot be financed by the private sector while necessary to ensure the replication impact of the program, to increase the attractiveness of the public transport system and to ensure the intended shift towards e-mobility. Long-term financial viability The long-run viability of the project is based on proven business models and the decreasing technology cost in the future. The sensitivity analysis of the financial model shows that for if the CAPEX for e-buses drops by 10%, the FIRR is in most cases higher than the WACC <u>without</u> GCF. In summary, as the CAPEX of e-buses drops, the projects will be financially viable with attractive returns in all the countries.				

E. LOGICAL FRAMEWORK				
E.1. Project/Programme Focus				
☒ Reduced emissions (mitigation) ☐ Increased resilience (adaptation)				
E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)				
Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Transportation in Program countries is nearly to 100% fossil fuel based.	<u>Low</u>	The paradigm shift is a move from fossil to EVs. Operators get accustomed to managing EVs and business models get common which favour the usage of EVs	The Program intervention already reduces 4.33 MtCO ₂ . The shift to e-mobility goes in scale beyond the project as initial positive applications are scaled up within the city to encompass the entire fleet. Results obtained in certain areas of the city can then be scaled up to the entire city as stakeholders recognize profitability of managing EVs.
Replicability	Electric mobility is not yet used at scale and thus replicability is very limited to individual purchases.	<u>Low</u>	The combination of successful implementation of new business models and electric fleet operations combined with the market force of decreasing EV prices form the base of a fast and massive replication potential. Cities which adopt successfully e-mobility systems will be replicated within the country and	Knowledge management instruments re used to disseminate successful business models and performance results of EVs amongst stakeholders. Capacity created will also allow to design new systems at a high quality. Once the high initial barrier in electric mobility of uncertainty about performance, operational risks and financial cum business models to make them commercially viable is removed, the lower TCO is a core argument to achieve widespread

			can serve as lighthouses also for other countries.	replication. Due to the national TA provided by the Program replicability can also be achieved in other cities of the country and due to the regional TA efforts results can also be replicated outside the countries where investive measures take place.
Sustainability	The NDCs of countries increasingly include targets on EVs. However, targets are often not lined with a regulatory framework and an attractive business proposal thus not achieving the goals.	<u>Medium</u>	Successful business models and a policy framework which effectively encourages the adoption of e-mobility linked with decreasing market prices of such vehicles make efforts sustainable.	The program will work with local and national institutions in designing and implementing business models and policies which are effective in encouraging the adoption of EVs. Appropriate policies and business models together with the declining EV prices results in high levels of sustainability.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term ⁶⁹	Final ⁷⁰	
<u>MRA2 Low-emission transport</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Ex-ante and ex-post analyses (conducted by a 3rd party contractor) 1 ex-ante; annual reporting of results based on continuously	0	104,275 tCO _{2e}	End of Program: 789,072 tCO _{2e} Lifespan: 4,334,051 tCO _{2e}	No delays in implementation which would result in lower mid-term and final but not lifespan GHG reduction. Lifespan BEBs on 16 years and PT

⁶⁹ Cumulative year 3

⁷⁰ cumulative year 7

		monitored parameters for e-buses (distance, energy consumption) and for PT passengers (number of passengers transported)				infrastructure measures 25 years Annual average GHG reduction of 183,812 tons Source: Annex 22 FP Methodology: CDM AMS.III.C and ACM0016
	<u>Supplementary 1.5</u> <u>Improved low-emission vehicle fuel economy</u>	Ex-ante and ex-post analyses (conducted by a 3rd party contractor) 1 ex-ante; annual reporting of results based on continuously monitored parameters for e-buses (distance, energy consumption)	0	3,970 *10 ⁶ MJ energy saved	End of Program: 12,903*10 ⁶ MJ energy saved Liferspan: 48,343 *10 ⁶ MJ	Annual average 2,233*10 ⁶ MJ Calculated based on differential energy usage fossil bus and electric bus multiplied with activity indicator (distance driven). Fossil buses energy usage is calculated based on liters/km, fossil fuel density and NCV of diesel. Electric buses energy consumed is based on kWh/km and conversion factor to MJ

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 5:</u> <u>Degree to which GCF investments contribute to strengthening institutional and</u>	Regulatory measures and policies to promote EVs are in many countries weak,	<u>medium</u>	Regulatory measures and policies are in place which favour the purchase and usage	Component 1 of the Program is the establishment of a conducive e-mobility framework. The	<u>Multi-countries</u>

<u>regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	incomplete or not enforced		of EVs, especially of commercial units.	outputs of this component are the establishment of an e-mobility conducive local, national and regional ecosystem, and mainstreaming of increased climate resilience of urban transport infrastructure	
<u>Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation</u>	Baseline vehicles are fossil powered.	<u>low</u>	In all Program countries fleets of BEBs operate. The Program countries have a vibrant commercial EV market.	The Program finances the purchase of E-mobility systems and provides TA for their appropriate design, implementation and maintenance. The Program will monitor the number and share of e-buses (share of new bus sales and share of fleet) per country to assess the impact of the Program on an accelerated deployment of e-buses. The sales and cumulative share is compared to other countries and to historic trends	<u>Multi-countries</u>

				excluding Program interventions.	
<u>Core indicator 7:</u> <u>Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level</u>	Women share and participation in EV technology and PT low and PT suffers from declining market shares	<u>low</u>	The of emobility results in an attractive PT system which can contribute to reverse the trend of declining PT market shares.	The Program's work to build gender sensitive infrastructure will support market development. More broadly, the Program will promote the benefits of electric mobility and PT to the general public	<u>Multi-countries</u>
<u>Core indicator 8:</u> <u>Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	Limited awareness and know-how on effective business models & policies to promote EVs.	<u>low</u>	Various business models conducive to EV fleet deployment have been applied to countries and cities depending on local and national circumstances.	The Program's KM strategy will ensure that lessons learnt are shared with interested parties and stakeholders on a national, regional and international level.	<u>Multi-countries</u>

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1: Improved public and private						

sector capacity & regulatory environment						
Output 1 Establishment of a sustainable mobility ecosystem encouraging electrification and shift to public transport	E-mobility roadmaps	Publications	0	5 sectoral roadmaps	13 sectoral roadmaps 3 regional outreach events	Municipal and national leadership can delay processes
	Trainings and CB events realized	Training and CB event reports	0	1 event and 1 CB report per investment project; battery policy regulations in 3 countries	1 event and 1 CB report per investment project; battery policy regulations in 7 countries	Municipal and national leadership can delay processes
Outcome 2: Reduced GHG emissions from the transport sector						
Output 2.1: E-Mobility is combined with PT measures to shift to low carbon transport	Electric buses are integrated with PT measures	Reports from city authorities in cities with e-bus deployment providing evidence of PT measures taken	0	- 1 projects with e-buses and PT measures	- 2 projects with e-buses and PT measures	Municipal and national leadership can delay processes; Investment in PT measures under component 2 take place in 2 cities
Output 2.2 EV fleets are deployed with innovative business	e-bus fleets deployed	Vehicle inventories of operators	0 ⁷¹	- 572 e-buses - charging	- 849 e-buses - charging systems for e-	Municipal and national leadership can delay processes

⁷¹ This refers to project financed e-buses. It does not include pilot e-buses operating in all participant countries nor fleets of e-buses already operational in Colombia (primarily Bogota, Cali and Medellin) as this would not allow for a clear identification of actual results of the GCF Program due to the fact that non-project e-buses would then also have to be included in mid-term as well as final-term target. For informational purposes based on e-bus radar the current

models to enable their mass application				g systems for e-buses in 3 cities - bus depots upgraded for e-buses in 3 cities - grid upgrades for e-buses in 3 cities - 3 cities with e-buses - Report on business models implemented	buses in 4 cities - bus depots upgraded for e-buses in 4 cities - grid upgrades for e-buses in 4 cities - 4 cities with e-buses - report on business models implemented	
Project/programme co-benefit indicators						
Co-benefit 1: Improved air quality	Reduced amount of pollutants	3rd party report based on default values for baseline fossil vehicles replaced by EVs	0	- 3 t PM _{2.5} reduced - 322 t NO _x reduced	- 12 t PM _{2.5} reduced - 1,444 t NO _x reduced	Baseline fossil vehicle emissions based on EU COPERT model; impact based on assumptions of mileage of vehicles
Co-benefit 2: Increased usage of PT	Increase in number of PT users	PT operators and surveys	Constant PT ridership (0% growth)	- +20% PT users in projects in 1 cities	- +20% PT users in projects in 2 cities	Refers to increased ridership on PT due to measures taken.

number of e-buses outside the program is as of 2.2024: Argentina: 22, Brazil: 142, Colombia: 1,590, Costa Rica: 4, Dominican Republic: 2, Mexico: 129, Peru: 8

			Share of female users determined per project in the baseline	- constant or increasing share of female users	- constant or increasing share of female users	Share of femal users monitored with surveys
Co-benefit 3: Increased climate resilience of urban transport infrastructure	More climate resilient cities	Reports AFD and cities	0	2 cities with increased resilience	4 cities with increased resilience	Measured based on investments realized by cities in adaptation

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 1.1.1 Implement a local e-mobility conducive framework	Realize policy advice, training and CB on areas conducive to commercial EV deployment at the local level	1.1.1.1 Support the development or update of sustainable urban mobility plans (SUMPs) 1.1.1.2. Advisory on the integration of non-motorised transport 1.1.1.3. Support of enabling public policies for demand management policies 1.1.1.4. Advisory on optimal technology solutions for the local context 1.1.1.5. Development and advisory services on optimal business models, and financial structuring and concession contracts which relate to the particularities of EVs Advice on concession contracts conducive for improved transport services and e-bus deployment	1.1.1.1. Promoted in all investment projects 1.1.1.2. Promoted in all cities with investment projects 1.1.1.3. Promoted in all cities with investment projects 1.1.1.4. Training courses realized with each investment project 1.1.1.5. Per project invested at least 1 knowledge management product

		1.1.16 Support of enabling public policies for demand management policies and EV deployment & integration 1.1.17. Capacity building & training of EV operators of (drivers, maintenance staff), FIs, and safety staff 1.1.1.8. Realize knowledge materials (publications, webinars) on demand management policies and commercial EV deployment	
Activity 1.1.2. Implement a national e-mobility conducive framework	Realize policy advice, training & CB on areas conducive to commercial EV deployment at the national level	1.1.2.1. Advisory on national urban mobility programmes and sSectoral roadmaps for commercial EVs 1.1.2.2. Promote sustainable mobility demand management policies and regulations incl. electrification and shift 1.1.2.3. Promote electrification policies and regulations 1.1.2.4. Advice on battery re-usage, recycling and disposal policies 1.1.2.5. Realize knowledge materials (publications, webinars) on demand management policies and commercial EV deployment CB at national level 1.1.2.6. Develop training courses, guidelines and curricula for vocational training	1.1.2.1. 13 sectoral roadmaps 1.1.2.2. Policy advice in 7 countries 1.1.2.3. Battery policy regulation prepared in 7 countries 1.1.2.4. CB activities at the national level for government officials & other stakeholders at a national level

Activity 1.1.3. Implement a regional e-mobility conducive framework	Realization of KM products and policy dialogue on a national and regional level	1.1.3.1. Realize Disseminate knowledge materials (publications, webinars, benchmark studies, best practice studies) on sustainable mobility transformation and EV deployment 1.1.3.2. Dialogue with EV suppliers 1.1.3.3. Outreach eventsDisseminate training course guidelines and curricula	1.1.3.1. KM publications on different topics 1.1.3.2. Continuous dialogue 1.1.3.3. 3 regional outreach events
Activity 1.1.4. Implement a GAP	Implementation of GAP activities.	1.1.4.1. Usage of a gender perspectives including mainstreaming of the same when designing and reviewing roadmaps and policies for different EV segments . 1.1.4.2. Increased awareness about gender equality gaps and opportunities in EV sector 1.1.4.3. Increase knowledge on how to include gender perspective in Urban Transport projects via CB online 1.1.4.4. South-to-South learning from best practices in the region (e.g. “Bajale al acoso” Quito) implemented 1.1.4.5. Design of a communication campaigns about sexual harassment and protocols of attention in public transport	1.1.4.1 100% of policies and roadmaps supported by the Program are reviewed with a gender perspective 1.1.4.2.a 80% of capacity building activities of the facility include awareness on gender equality and urban transport 1.1.4.2.b 30 % of participants of the exchange platform attending trainings and capacity building activities on EV are women 1.1.4.2.c 1 regional network on women in e-mobility to share and exchange experiences, promote mentorship and produce knowledge, is supported and strengthened 1.1.4.3. 1 online module of training on Gender and Urban Transport is elaborated, with participation of at least 1 representative from the local public

		1.1.4.6. Include gender perspective in the investment plans 1.1.4.7. Improved sex disaggregation of data collected on urban transport 1.1.4.8. Improved women's access to jobs 1.1.4.9. Raise awareness on sexual harassment in public transport through capacity building 1.1.4.10 Promotion of gender equality at the work place	authority in each city where an investment of the program is foreseen 1.1.4.4 1 South-to-South learning visit to best practices in the region 1.1.4.5 1 Public communication campaigns about sexual harassment and protocols of attention in public transportation is developed 1.1.4.6. 100% of projects financed include a Gender Equality Strategy with a GAP which contains specific and measurable activities 1.1.4.7. 100% of urban passenger transport projects whose user's data systems collects and presents data disaggregated 1.1.4.8. 80% of projects with direct finance by the Program guarantee at least 30% of skilled jobs related directly to the project are reserved for women, with special attention to measures that can attract and retain women in the industry 1.1.4.9.a 100% of public transport projects financed by the Program include activities (communication campaign and workshops) aiming at preventing sexual harassment 1.1.4.9.b 100% of projects financed by the Program have reporting systems designed for
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			victims of sexual harassment in collaboration with local authorities 1.1.4.10.a One workshop on gender equality and work-life balance for each public transport project financed by the Program 1.1.4.10.b 100 % of public transport projects financed by the Program include annual surveys on work-life balance to their staff and sharing the results 1.1.4.10.c 100% of public transport projects financed by the Program with one focal point in charge of gender equality
Activity 2.1.1 Implement urban mobility measures to foster PT	Design and implement measures for PT enhancement	2.1.1.1 Feasibility report (problems, options, feasibility) 2.1.1.2 Financial structure and implementation structures 2.1.1.3 Tendering 2.1.1.4 Implementation	Measures in 2 cities
Activity 2.2.1 Deployment of e-buses	Design and implement e-bus systems bus component	2.2.1.1 Feasibility report 2.2.1.2 Financial structure and implementation structures 2.2.1.3 Tendering 2.2.1.4 Implementation	850 e-buses in 4 cities
Activity 2.2.2. Deployment of charging infrastructure	Design and implement e-bus systems charging component	2.2.2.1 Feasibility report in coordination with 2.2.1 2.2.2.2 Financial structure and implementation structures 2.2.2.3 Tendering together with 2.2.1	Charging system for 850 e-buses in 4 cities

		2.2.1.4 Implementation together with 2.2.1	
Activity 2.2.3. Implement bus depot upgrades for e-buses	Design and implement e-bus systems bus depot component	2.2.3.1 Feasibility report in cooperation with 2.2.1 and 2.2.2 2.2.3.2 Financial structure and implementation structures 2.2.3.3 Tendering 2.2.3.4 Implementation	Bus depot upgraded for 850 e-buses in 4 cities
Activity 2.2.4. Implement grid upgrades for connecting e-bus chargers	Design and implement e-bus systems grid component	2.2.4.1 Feasibility report in cooperation with 2.2.1 and 2.2.2 and 2.2.3 2.2.4.2 Financial structure and implementation structures 2.2.4.3 Tendering 2.2.4.4 Implementation	Grid system upgraded for charging system for 850 e-buses in 4 cities

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Project level

During the disbursement period, AFD Group will receive regular reports from the Beneficiaries, in order to track progress and identify potential issues as well as improvement opportunities. In this document, the Beneficiary will report on performance indicators, as well as qualitative information, as defined in the Financing Agreement signed with AFD Group.

AFD Group local agencies, supported by headquarters' portfolio teams in the case of Proparco, will ensure the day-to-day monitoring by overseeing the implementation of the projects, meeting with the Beneficiaries on a regular basis, ensuring coordination with TA, exchanging with local government, etc. Every semester, they will review projects' performance and assess any potential risks. The local agency staff will be able to check the validity and robustness of the projects' results thanks to their proximity, strong knowledge of the situation on the ground, and these regular exchanges and visits.

Project Task Team Leaders (based in AFD's Paris headquarters, or in the PMU offices in Latin America) will visit projects and meet the Project Owner on average once in a year. Following this mission, a project implementation report will be shared with all the project stakeholders. At Proparco, a third party technical advisor hired at project level usually handles these tasks. Project TTLs at AFD, and portfolio officers at Proparco, will be responsible for overseeing project implementation and for approving funding disbursements. On a yearly basis, all projects financed by AFD

will be assessed through an internal quality review process based on several indicators (technical implementation, disbursement, procurement, outcomes, etc.). Red flag projects that meet any kind of difficulties are subject to a specific monitoring process before returning to a regular implementation pathway.

At the end of the disbursement period, AFD will prepare, for each project financed by AFD, a project completion report summarizing the main events of the implementation phase, the results achieved, and the lessons learned. This report will also provide recommendations for future improvement.

During the repayment period, a biannual monitoring of the use of funds will be carried out by the local agencies and at corporate level by the middle office.

In addition, about a year after the drafting of the project completion report, the Evaluation department of AFD may conduct an ex-post evaluation of projects or appoint a local consultant to do so. In most cases, AFD prefers to select a local consulting firm based in the country where the project is located. This allows the Agency to take advantage of deep contextual knowledge and to help develop local evaluation skills.

The consultant will be tasked with evaluating the impact of the project (economic, environmental, etc.) and writing a report that will be shared with all relevant stakeholders (central and local governments, companies, NGOs, etc.) and could also be made public.

Program level

An Annual Performance Reports (APR) will be submitted to the GCF following reporting requirements under the AMA and FAAs, in parallel to performance and financial information at country and Borrowers level. In particular, the APR will include a narrative report on the implementation progress based on the logical framework, considerations on the ongoing performance of the Programme, updates on the indicators, and a report on ESS as well as gender. Proparco and GIZ, as Executing entities, will be required, through the subsidiary agreement, to provide periodic reporting, in line with GCF requirements, to indicate progress on fund utilization as well as development outcomes.

The APR will be submitted to the Secretariat on an annual basis for the period ending on 31 December within 60 days after the end of the relevant annual period. The first APR will be submitted following the end of the calendar year after the Parties have entered into the relevant FAA and the last APR will be submitted within six months of the end of the relevant reporting period.

The implementation reporting period of the Programme will start from the date of effectiveness of the FAAs until the Programme implementation end date. Financial reporting will continue until full repayment of the loans.

The mid-term and final independent evaluation of the Programme will be organized and managed by the Evaluation department of AFD, on the same basis as any project funded under the Programme.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Selected Risk Factor 1

Category	Category	Category
Credit	Credit	Credit
Description		

Inability of lender to service debt

Mitigation Measure(s)

AFD's comprehensive due diligence and technical support are designed to assess the borrower's ability to service the loan. All loans from the program will be aligned with AFD loans. As such the credit or loan default risk on repayment of program loans is low. To further ensure that underlying projects of the Program are financially sustainable themselves and don't create a further burden on the lender, all economic and financial analyses of each project shall be conducted in accordance with AFD's principles for the economic analysis of projects. The Program explicitly focus on developing financially bankable models at the project level itself.

Selected Risk Factor 2

Category	Probability	Impact
Technical and operational	Low	Medium
Description		

EVs do not perform as expected and the operator cannot realize standard operations.

Mitigation Measure(s)

AFD's comprehensive technical design and structuring of projects minimize this risk. All technologies have been proven at least in pilot cases in other countries.

Selected Risk Factor 3

Category	Probability	Impact
Other	Low	Medium
Description		

Beneficiaries' failure to comply with national regulations and/or AFD and GCF environmental, social policy, gender requirements and compliance with standards, policies and procedures.

Mitigation Measure(s)

Each project will be subject to AFD's Environmental and Social Management System as well as gender assessment. Each Project will be appropriately appraised and structured to meet AFD's and GCF's requirements. Environmental, Social and Gender Action Plans will be developed during project appraisal to achieve this. Implementation thereof will be covenanted through financing agreements. Its implementation will be monitored by AFD.

Selected Risk Factor 4		
Category	Probability	Impact
Other	Low	Medium
Description		
No or limited commercial uptake of EVs after initial investment uptake resulting in limited indirect Program impact. The declining EV price is an external factor. Whilst the magnitude and speed of EV price decline is disputed, the trend of reduced price differential between fossil and EVs is undisputed. Modelling has been done per country of the potential uptake of EV with Program and under BAU (see Reports 3 per country and the corresponding excel file included in Annex 3)		
Mitigation Measure(s)		
Successful EV investment projects and well implemented TA are core to reduce the investment risk levels of EVs. Investment risk levels are together with the EV price the 2 core factors which result in a commercial uptake of EVs. Good design and implementation of investment projects and of TA can thus minimize this risk		
Selected Risk Factor 5		
Category	Probability	Impact
Prohibited practices	Low	Medium
Description		
Risk of projects being involved in illicit practices. (money laundering, terrorist financing or other prohibited practices		
Mitigation Measures(s)		
AFD has its internal system of control. All investment projects realize a first level due diligence to ensure that this risk is mitigated. GCF financing will be provided as loans or grants to sovereign and sovereign-backed entities, to cities or then to private entities. As GCF funds will be provided alongside co-financing from AFD, AFD integrity principles and associated due diligence will apply at project level, which will also mitigate this risk. All projects shall comply with AFD's anticorruption policy and acknowledge that AFD reserves the right to investigate directly, or through its agents, any alleged corrupt, fraudulent, collusive or coercive practice relating to the projects; and recipients of the grant/loan shall cooperate with any such investigation and extend all necessary assistance for satisfactory completion of such investigation. As per AFD policies and procedures, prior to the approval of each specific project, AFD's fiduciary team will perform and/or update an institutional capacity assessment on the corresponding EE. This assessment includes overall technical capacity, adequacy of information systems, internal and external controls, and recommendations on any fiduciary risks identified. None of the project activities will be undertaken in any jurisdiction which is subject to or affected by United Nations Security Council. The AFD project procurement policies that will regulate the procurement activities of the program establishes that firms of a country or goods manufactured in a country may be excluded if by an act of compliance with a decision of the United Nations Security Council taken under Chapter VII of the Charter of the United Nations, the Borrower's country prohibits any import of goods from, or payments to, a particular country, person, or entity. Where the Borrower's country prohibits payments to a particular firm or for particular goods by such an act of compliance, that firm may be excluded.		

Selected Risk Factor 6		
Category	Probability	Impact
Other	Medium	Medium
Description		
Limited capacity or experience to design and implement e-mobility projects by beneficiaries		
Mitigation Measure(s)		
Technical assistance will focus on supporting entities to identify, design and structure e-mobility projects. Detailed legal and technical due diligence will be carried out by the AFD to develop robust projects. Support will be provided under the Program to assist in developing favourable policies for deployment of e-mobilities.		
Selected Risk Factor 7		
Category	Probability	Impact
Other	Low	Medium
Description		
battery supply chain risks		
Mitigation Measure(s)		
This is managed by the vehicle manufacturers and might be a concern for individual manufacturers but not the program itself. The program is targeted to procure about 850 buses over 2-3 years which is lower than the annual demand for e-buses of about 200k units. Moreover, one major e-bus supplier has sold about 35,000 buses in 2023. ^{72,73} Hence the risk of the e-bus supply shortage is low. Separately, battery chemistry is also an area that is rapidly evolving – in terms of technologies and input materials. It is expected that market forces will adjust.		
Selected Risk Factor 8		
Category	Probability	Impact
Other	Low	Low
Description		
Sustainability risk of vehicle maintenance and repairs if local capacity is low.		
Mitigation Measure(s)		
It is envisaged that the bus supply contracts shall come with a 3-year maintenance and service contract with guaranteed bus availability rates and with fixed training programs for the local mechanics. Tender documents shall include also lists of spare parts, equipment for repair and minimum training and capacity building courses. A trained chief mechanic of the supplier must be on-site during the 3-year contract. This is the reason why fleets of 50 buses plus are supported and not pilot fleets. These conditions will be reviewed and subject to further adjustment, pending detailed technical due diligence to be conducted for each sub-project.		

⁷² Fortune Business Insights. Electric Bus Market Size, Share & Growth Trends.
<https://www.fortunebusinessinsights.com/electric-bus-market-102021>

⁷³ [Yutong Bus Reflects on Landmark Year, Cumulative Global Sales of New Energy Buses Exceeded 175,000 Units - PR Newswire APAC \(prnasia.com\)](#)

Selected Risk Factor 8		
Category	Probability	Impact
Other	Low	Medium
Description		
Risk related to EV batteries during operations and end of life		
Mitigation Measure(s)		
The Technical Assistance component would conduct studies on this aspect on battery disposal/recycling. It also has training on safety aspects of batteries during operations. Depending on the outcome of the prevailing technologies, best practices, etc., the TA may consider recommendations that are likely to include, for example, (ii) drafting battery regulations in line with those approved. For example, inclusion of Extended Producer Responsibility (EPR) which obligates the manufacturer selling new and refurbished batteries or importing batteries to ensure environmentally sound management of batteries. This would prohibit the manufacturer and/or importer from disposing waste batteries in landfills and incinerators and obliges the manufacturer and/or importer to recycle, refurbish or re-purpose waste batteries. It may also include a tracking system for all batteries. The batteries have a guaranteed lifespan of 8 years at 80% State of Health(SOH) and thus significant quantities of batteries which warrant a recycling industry on a national scale will only be possible in around 10 years from now. In the meantime, the used batteries are likely to be re-purposed or re-exported to the manufacturing country for which the Basel convention has a provision. See for details section B3 TA and ESG Annex.		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

AFD has a comprehensive Environmental and Social Policy (ESP). All projects need to comply with these requirements. All projects undergo environmental and social appraisal both to help AFD decide if the project should be financed and, if so, the way in which environmental and social risks and impacts should be addressed in its planning, implementation and operation. The appraisal process also identifies opportunities for additional environmental or social benefits. AFD seeks that projects are designed, implemented, and monitored in compliance with its policies, applicable regulatory requirements and good international practice. The ESP sets out principles, rules, procedures and guidelines for conducting environmental and social due diligence of the potential projects. These procedures and guidelines also describe the process for developing measures to avoid and mitigate potential adverse impacts as well as opportunities to improve the environmental and social outcomes of the projects.

AFD is committed to the principles of transparency, accountability and stakeholder engagement, and promoting adoption and implementation of these principles by its clients. Proportionate to the nature and scale and environmental and social risks and impacts of the project, AFD requires its clients to disclose sufficient information about the risks and impacts arising from projects, engage with stakeholders in a meaningful, effective, inclusive and culturally appropriate manner and take into consideration the feedback provided through such engagement.

Each project under the Program will require a Stakeholder Engagement Plan, including a grievance mechanism. AFD will disclose on its website appropriate E&S information for each project under the Program.

AFD works very closely with governments, authorities and civil society to uphold the principle of country ownership via direct partnerships (regional offices). Previous to the realization of this Funding Proposal AFD has realized in all countries diagnostic and market assessment missions which have extensively discussed intervention options with the country and with stakeholders.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

A gender assessment has been realized for all countries as well as a gender action plan. This serves as guideline for each individual project to be funded under the Program (see Annex 8).

The last two decades brought significant advances for women in Latin American Countries. All of the countries in this Program ratified the Convention Belem do Pará, whose main achievement was to incorporate specific legislation to include “femicide” as a specific aggravated crime. Since its ratification in 1994, countries have been advancing to prevent, sanction, and eradicate violence against women. In general, for the Latin American and the Caribbean region, Gender Based Violence (GBV) is endemic and underreported, and before 2006 no country in Latin America established femicide as a crime, while by 2015 16 countries already did. Likewise, in many parts of the World, civil society has raised strongly against GBV and México and Perú have marched under the motto “Ni una menos” (Not one woman less).

The seven countries have included quotas of around 30 percent for political representation, and many countries have reached it at the national level yet at the local level, most lag behind. These countries have advanced enormously in maternal health, but inside every country, disparity persists geographically and by ethnic group or race. Parity in education is mostly reached in all countries at the primary and secondary levels, and in many cases, girls are outperforming boys; whereas in tertiary education attainment varies by country, and careers in STEM (Science, Technology, Engineering and Maths) remain predominantly

masculine. Occupational segregation persists across countries and women's careers concentrates in low paid jobs and informality, which moreover, have little and low benefits, resulting in lower earnings and well-being. Despite many laws, gender wage gaps persist across sectors and gender roles impede women's access to better economic opportunities since domestic responsibilities are mostly in their hands.

Education in girls is affected by adolescent pregnancy, which is still high and despite the downward regional tendency in fertility rates, it is only exceeded by Sub-Saharan Africa. Marriage and sex are illegal for minors in all these countries, yet teenage pregnancies and minors living in union occur. Plans, strategies, and entities (Ministries or agencies) have been given more power or have been recently created. These entities are not funded enough to carry on, enforce, and follow up with all the gender equality agenda.

The outlook to both keeping the progress and continue reducing the gaps in gender equality is challenging after the COVID-19 pandemic. UN Women and other international organizations fear that recent gains in gender equality might be lost. The COVID-19 pandemic has worsened women's situation and has aggravated differences amongst groups. In Brazil, the maternal death rate due to COVID-19 is twice as higher for black women than for white ones (UN Women, 2020). Gender based violence has increased and women in lockdowns are living with their aggressor under the same roof and can't access support networks. Only in Brazil, from March to April 2020 compared to the same period in 2019, there has been an increase of 22% in femicides and a decrease of 28% in reporting cases of rape. Complaints increased 27% and reports for physical aggression cases decreased by 26%⁷⁴. In Colombia the calls to the domestic violence helpline increased by 91%, and in Mexico by 36%. It is estimated that less than 40% of domestic violence survivors report it; adding to that, figures of actual gender-based violence are worse during COVID-19 because the fear of getting infected prevents women from seeking help in health services. Lack of critical resources and access to key reproductive and health services due to stay-at-home orders might have increased teenage pregnancies and maternal mortality. There is evidence that other forms of violence, such as sexual exploitation and abuse have increased in others health crises⁷⁵. Many women have abandoned the labor market to lead their household's dynamics during the pandemic, with severe lockdowns in most countries in the region and schools and workplaces closed, while access to decent work has deteriorated for all.

According to UN Habitat, two-thirds of the population will be living in cities by 2050 (UN Habitat, 2018). Together with pollution, overcrowding, lack of modal integration, are some of the problems that the increase of urban population is exacerbating. 46% of surveyed people in Peru responded that Transportation is the second most important problem affecting their quality of life (first is insecurity) (IADB, 2019). Women from the countries involved do not have limitations of mobility in the legal structures, but their agency is limited. Agency in mobility is the ability to make and act upon decisions related to one's mobility and make full use of public transport systems (WB, 2020). Lack of access to transportation and safety issues limits women's participation in the labor market and reduces their participation in the economy by 16.5%⁷⁶. Women turn away working opportunities depending on the traveling conditions to the work station (time spent, return timing, expenses). Besides providing access to education, health and jobs, the sector generates jobs itself but traditionally most opportunities are likely to remain in men's hands. Hence, when we use gender lens to analyze urban transportation, we discover that the conditions, priorities, and beliefs about urban transport infrastructure and services vary between women and men and, consequently, the decisions that they make about mobility are shaped by these conditions, priorities and beliefs. The lack of certain safety features in the design of the transport system can constrain a woman's decision to choose a certain mode of transportation, sometimes in ways that are inoculate to men, such as

⁷⁴ Addressing Violence against women under COVID-19 in Brazil, WB Ag 2020

⁷⁵ Gender dimensions of the COVID-19 Pandemic, WB April 2020

⁷⁶ World Employment and Social Outlook: Trends for women 2017. International Labour Office – Geneva: ILO, 2017

dark bus stops or overcrowded buses. When it comes to the design of the transport infrastructures, the realities of women, the elderly and people with disabilities are many times alike and need be taken into account when designing a transport operation. There is myriad of links between gender and urban transportation. There are gender differences in the purpose, time, mode of transportation, or perceptions on safety, that affect women's mobility agency and access to economic opportunities, and the differences comprised in this section are shared among the countries in the region.

The investments projects of the Program need to include a gender strategy with actions to increase women's mobility, promote safety and access to jobs. This Program can improve women's agency in mobility if mainstreaming gender equality in its projects becomes systematic. Gender-based violence is endemic in these countries and underreported; the COVID-19 pandemic is exacerbating the numbers. The urban transport system represents an environment where harassment and sexual abuses occur quotidianly in the regular trips women do every day. Incidents will happen in the e-buses, in the e-taxis, while waiting, walking to and from the stops or using charging stations; incidents will increase at night, in isolated vehicles and unattended places, but in crowded units as well. Women are not represented in the jobs generated by the construction and transport sector in the region. The pandemic complicates the return to work for those women who lost their jobs and those whose care responsibilities have increased; the Program is an opportunity for women in these eight countries to access the new jobs that will be generated. The investment projects for e-taxis adoption must learn from the segregated ride-hailing market which is growing in the region, and adapt to vulnerable groups without access to technology.

The Gender Action Plan guarantees that these general recommendations are transformed into specific activities at the Program and project levels, and that they are measured and adequately monitored, and adjusted when needed. The technical assistance will support the adopters implement a people centered design. Adding to the differentiated risks and vulnerabilities of men and women detailed in the gender assessment, the adopters will be guided on how to apply intersectionality enablers. Adding to the afro descendants and indigenous peoples present in many of these countries, intersecting identities of age, social economic status, social cultural context, education, disability among others⁷⁷ need to be considered. Together with universal design approach, accessibility and reasonable accommodations⁷⁸ will be factored in when electrifying transport systems. The technical assistance will guide the adopters to use tools that guarantee that women and vulnerable groups are represented during multi-stakeholders' consultations so that their perspectives can be factored into policies, road maps, trainings and campaigns. The adopters will be guided to use tools such as accessibility and reasonable accommodations in a way that maintain confidentiality and privacy to allow women and vulnerable groups to share their perspectives. An environment where women and vulnerable groups feel safe will allow the adopters to include these groups perspectives during design and implementation that might not be considered otherwise. Consultations in a safe environment will be particularly emphasized when designing and implementing sexual harassment programs and prevention campaigns. Multi stakeholder consultations should include women and vulnerable groups perspectives so that they are represented, and they can input into policy, road maps, training, and campaigns.

⁷⁷ UN Women, UNPRPD, and more (2021). Intersectionality resource guide and toolkit, An Intersectional approach to leave no one behind

⁷⁸ Universal design means the design of products, environments, programmes and services to be usable by all people, to the greatest extent possible, without the need for adaptation or specialized design. Universal design shall not exclude assistive devices for particular groups of persons with disabilities where this is needed.

Reasonable accommodation means necessary and appropriate modification and adjustments not imposing a disproportionate or undue burden, where needed in a particular case, to ensure to persons with disabilities the enjoyment or exercise on an equal basis with others of all human rights and fundamental freedoms

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Financial Management

The detailed provisions regarding financial management of the GCF resources by AFD will be described in the FAA. Reporting will be provided by AFD to the GCF specifying among others:

- The amounts already committed and disbursed by AFD to local partners, by country, by project and by instrument;
- The amounts already committed and disbursed by GIZ by country;
- The remaining amount on the different GCF accounts;
- A provisional disbursement schedule on a one-year rolling horizon;
- For the loan component, if any, a list of the incidents recorded during the repayment period of the loans provided to the borrowers.

Subsidiary agreements will be signed between AFD and PROPARCO, and AFD and GIZ.

For non-sovereign borrowers, a credit risk monitoring will be carried out by AFD through the assessment of the financial statements of borrowers, monitoring of financial covenants, etc. Audits will be performed as per AFD's procedures and as per the AMA.

Disbursement

At AFD level, disbursement will be related to:

- Loan disbursements on projects. Disbursement conditions will be defined on a project-by-project basis and will be described in the financing agreement signed between AFD and the borrower;
- Fund transfers to GIZ and PROPARCO whose disbursements will be reported to AFD;
- Payments of the consultants in charge of providing support for projects' preparation;

Procurement

AFD's Procurement rules are in line with international standards provided by the World Bank Group. The AFD Procurement guidelines will apply to the publicly owned final beneficiaries in charge of implementing the projects financed by the Program. For a detailed presentation of AFD Procurement rules, see the Procurement Guidelines for AFD Financed Contracts in Foreign Countries available on AFD's website.

For privately owned projects sponsors, PROPARCO will apply the procedures dedicated to the specificities of the PROPARCO's client in the private sector.

GIZ procurement and awarding rules apply for procurement within the TA component for which GIZ is the procuring agency.

AFD, PROPARCO and GIZ are accredited to the GCF; as such their procurement rules and procedures were assessed and evolutions are communicated on a yearly basis during the AEs' self-assessment exercise.

G.4. Disclosure of funding proposal

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence.

Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People’s Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project

- | | | |
|-------------------------------------|----------|--|
| ☐ | Annex 19 | Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity |
| ☐ | Annex 20 | First level AML/CFT (KYC) assessment |
| ☒ | Annex 21 | Operations manual (Operations and maintenance) |
| ☒ | Annex 22 | Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects) ⁷⁹ |
| ☒ | Annex 23 | Benchmark report |
| | Annex 24 | Annex Technical Assistance |

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

⁷⁹ Annex 22 is mandatory for mitigation and cross-cutting projects.



*Subsecretaría de Relaciones Financieras
Internacionales para el Desarrollo
Presidencia de la Nación*

To: The Green Climate Fund ("GCF")

Buenos Aires, 16 June 2022

Re: Funding proposal for the GCF by Agence Française de Développement regarding the Programme "E-Motion: E-Mobility and Low Carbon Transportation"

Mr. Henry González
Deputy Executive Director - Green Climate Fund - Korea

Dear Madam, Sir,

We refer to the programme titled "**E-Motion: E-Mobility and Low Carbon Transportation**" in Argentina as included in the funding proposal submitted by **Agence Française de Développement** to us on 31 May 2021.

The undersigned is the duly authorized representative of Argentina, Mr. Leandro A. Gorgal, Under-Secretary of International Financial Relations for Development, the National Designated Authority/focal point of Argentina.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

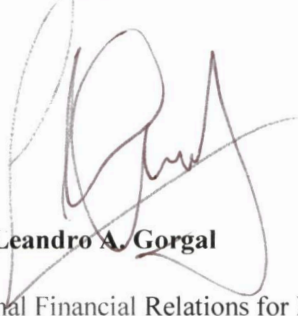
- (a) The government of Argentina has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Argentina's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Leandro A. Gorgal

Undersecretary of International Financial Relations for Development - NDA



MINISTRY OF FINANCE
Secretariat for International Affairs
National Designated Authority for the Green Climate Fund
Esplanada dos Ministérios, Bloco P, 2º andar, sala 219-B
CEP 70048-900 - Brasília/DF - (61) 3412-2296 - e-mail and.gcf@fazenda.gov.br

OFÍCIO SEI Nº 84829/2021/ME

Brasília, 14 de abril de 2021.

Mr. Yannick Glemarec
Deputy Executive Director
Green Climate Fund
G-Tower, 24-4 Songdo-dong, Yeonsu-gu
Incheon City, Republic of Korea

Re: Funding proposal for the GCF by AFD regarding the program "E-Motion: E-Mobility and Low Carbon Transportation"

Referência: If you answer this letter, please indicate the Process n.º 12120.100151/2021-12.

Dear Mr. Glemarec,

1. We refer to the programme entitled "E-Motion: E-Mobility and Low Carbon Transportation" as included in the funding proposal submitted by AFD to us on March 23rd, 2021.
2. The undersigned is the duly authorized representative of the Secretariat for International Economic Affairs, Ministry of Economy, the National Designated Authority for Brazil.
3. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.
4. By communicating our no-objection, it is implied that:
 - a) The government of Brazil has no-objection to the programme as included in the funding proposal;
 - b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Brazil; and
 - c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.
5. We also confirm that our national process for ascertaining no-objection to the programme, as included in the funding proposal, has been duly followed.
6. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.
7. We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Documento assinado eletronicamente

ERIVALDO ALFREDO GOMES

Secretary for International Economic Affairs

Ministry of Economy of Brazil



Documento assinado eletronicamente por **Erivaldo Alfredo Gomes, Secretário(a) de Assuntos Econômicos Internacionais**, em 19/04/2021, às 10:39, conforme horário oficial de Brasília, com fundamento no art. 6º, § 1º, do [Decreto nº 8.539, de 8 de outubro de 2015](#).



A autenticidade deste documento pode ser conferida no site https://sei.economia.gov.br/sei/controlador_externo.php?acao=documento_conferir&id_orgao_acesso_externo=0, informando o código verificador **14824701** e o código CRC **1B7850A6**.



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DNP
Departamento
Nacional de Planeación

Bogotá D.C., jueves, 22 de abril de 2021

DADS



Al responder cite este Nro.
20214500369981

THE GREEN CLIMATE FUND – GCF

Att. fundingproposal@gcfund.org

Subject: Funding proposal for the GCF by Agence Française de Développement – AFD
regarding E-Motion: E-Mobility and Low Carbon Transportation

Dear Madam, Sir,

We refer to the programme titled E-Motion: E-Mobility and Low Carbon Transportation in Colombia as included in the funding proposal submitted by Agence Française de Développement – AFD to us on March the 25th 2021.

The undersigned is the duly authorized representative of the National Planning Department of Colombia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Colombia has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Colombia;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

DNP

Firmado
digitalmente





El futuro
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DNP
Departamento
Nacional de Planeación

We acknowledge that this letter will be made publicly available on the GCF website.

Best Regards,

AMPARO GARCÍA MONTAÑA
General Territorial Deputy Director
National Planning Department

CC: cpolycarp@gcfund.org; ahaider@gcfund.org; hgiron@gcfund.org; sbasnyat@gcfund.org; cuevasj@afd.fr; andrer@afd.fr; countries@gcfund.org

Prepared: Carlos Eduardo Casallas
Revised: Carolina Díaz Giraldo; Juan Sebastián Rivera



DM-293-2022
July 21st, 2022

The Green Climate Fund (“GCF”)

**Re: Funding proposal for the GCF by French Development Agency (AFD)
regarding E-Motion : E-Mobility and Low Carbon Transportation**

Dear sirs

We refer to the programme titled *E-Motion : E-Mobility and Low Carbon Transportation* in Costa Rica as included in the funding proposal submitted by French Development Agency (AFD) to us on 6 April 2022.

The undersigned is the duly authorized representative of The Ministry of Environment and Energy, the National Designated Authority of Costa Rica.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Costa Rica has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Costa Rica;
- (c) In accordance with the GCF’s environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely



Franz Tattenbach
Minister of Environment and Energy

To: The Green Climate Fund ("GCF")

Santo Domingo, D.N.
April 5, 2021

Re: Funding proposal for the GCF by French Development Agency (AFD) regarding "E-Motion: E-Mobility and Low Carbon Transportation"

Dear Madam, Sir,

We refer to the programme titled "E-Motion: E-Mobility and Low Carbon Transportation" in Dominican Republic as included in the funding proposal submitted by French Development Agency (AFD) to us on March 2021.

The undersigned is the duly authorized representative of the Ministry of Environment and Natural Resources, the National Designated Authority of Dominican Republic.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Dominican Republic has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Dominican Republic;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Milagros De Camps
Deputy Minister of International Cooperation
Ministry of Environment and Natural Resources
Dominican Republic





To: The Green Climate Fund ("GCF")

Mexico City, 6 July 2022.

Re: Funding proposal for the GCF by French Development Agency (AFD) regarding E-Motion : E-Mobility and Low Carbon Transportation

Dear Madam, Sir,

We refer to the programme titled *E-Motion: E-Mobility and Low Carbon Transportation* in Mexico as included in the funding proposal submitted by French Development Agency (AFD) to us on 6 April 2022.

The undersigned is the duly authorized representative of The Ministry of Finance and Public Credit, the National Designated Authority of Mexico.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

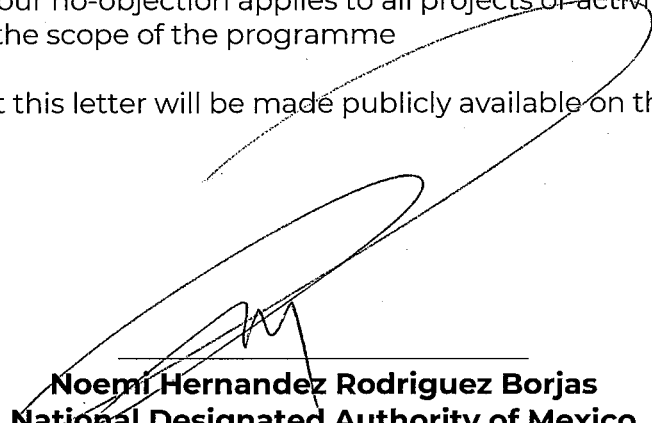
- (a) The government of Mexico has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Mexico;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


Noemi Hernandez Rodriguez Borjas
National Designated Authority of Mexico





HACIENDA

SECRETARÍA DE HACIENDA Y CRÉDITO PÚBLICO

**Subsecretaría de Hacienda
y Crédito Público
Unidad de Crédito Público**

Cc:

1. María del Carmen Bonilla Rodríguez, Head of the Public Credit Unit and International Affairs, Ministry of Finance and Public Credit of Mexico.
2. Laura Elisa Aguirre Tellez, General Director, Ministry of Finance and Public Credit of Mexico.





**MINISTERIO DE ECONOMÍA Y FINANZAS
DIRECCIÓN GENERAL DE ASUNTOS DE ECONOMÍA
INTERNACIONAL, COMPETENCIA Y PRODUCTIVIDAD**

**"DECENIO DE LA IGUALDAD DE OPORTUNIDADES PARA MUJERES Y HOMBRES"
"AÑO DEL BICENTENARIO DEL PERÚ: 200 AÑOS DE INDEPENDENCIA"**

To: The Green Climate Fund ("GCF")

Lima, July 6th, 2021

Re: Funding proposal for the GCF by the French Development Agency (AFD) regarding the E-Motion: E-Mobility and Low Carbon Transportation programme

Dear Madam, Sir,

We refer to the E-Motion: E-Mobility and Low Carbon Transportation programme in Peru as included in the funding proposal submitted by AFD to us on June 10th, 2021.

The undersigned is the duly authorized representative of the Ministry of Economy and Finance, the National Designated Authority of Peru.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Peru has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Peru;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

[DIGITAL SIGNATURE]
JOSÉ LA ROSA BASURCO
Director
General Directorate of International
Economic Affairs, Competition and
Productivity
Ministry of Economy and Finance of Peru



BICENTENARIO
PERÚ 2021

Esta es una copia auténtica imprimible de un documento electrónico archivado por el Ministerio de Economía y Finanzas, aplicando lo dispuesto por el Art. 25 del D.S. 070-2013-PCM y la Tercera Disposición Complementaria Final del D.S. 026-2016-PCM. Su autenticidad e integridad pueden ser contrastadas a través de la siguiente dirección web <https://apps4.mineco.gob.pe:443/st/r?ctrln=6f136663-6300-4f12-a75d-f2a93c2ab025-693842> ingresando el siguiente código de verificación EFHKEJFD

Sede Central
Jr. Junín N° 319, Lima 1
Tel. (511) 311-5930
www.mef.gob.pe

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	E-Motion: E-Mobility and Low Carbon Transportation
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	Agence Française pour le Développement (AFD)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Argentina, Brazil, Colombia, Costa Rica, Dominican Republic, Mexico, and Peru
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Friday, June 14, 2024
Language(s) of disclosure	English, Spanish, and Portuguese
Explanation on language	The languages chosen are either the official language of the target countries or a language understandable by the affected peoples.
Link to disclosure	English: https://www.afd.fr/en/ressources/annex-6-environmental-and-social-management-framework-e-motion-program Spanish: https://www.afd.fr/es/ressources/anexo-6-marco-de-gestion-ambiental-y-social-programa-e-motion Portuguese: https://www.afd.fr/pt/ressources/annex-6-environmental-and-social-management-framework-e-motion-program
Other link(s)	https://www.afd.fr/en/ressources-accueil?f%5B0%5D=resource_type%3A38
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the E-Motion Environmental and Social Management Framework.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, June 14, 2024
Language(s) of disclosure	English, Spanish, and Portuguese
Explanation on language	The languages chosen are either the official language of the target countries or a language understandable by the affected peoples.

Link to disclosure	English: https://www.afd.fr/en/ressources/annex-6-environmental-and-social-management-framework-e-motion-program Spanish: https://www.afd.fr/es/ressources/anexo-6-marco-de-gestion-ambiental-y-social-programa-e-motion Portuguese: https://www.afd.fr/pt/ressources/annex-6-environmental-and-social-management-framework-e-motion-program
Other link(s)	https://www.afd.fr/en/ressources-accueil?f%5B0%5D=resource_type%3A38
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the E-Motion Environmental and Social Management Framework.
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity's website	- An indicative Outline of Stakeholder Engagement Plan, Grievance Mechanism, and information disclosure - An indicative Outline for Indigenous Peoples Planning Framework (IPPF) / Friday, June 14, 2024
Language(s) of disclosure	English, Spanish, and Portuguese
Explanation on language	The languages chosen are either the official language of the countries and the language understandable by affected people.
Link to disclosure	English: https://www.afd.fr/en/ressources/annex-6-environmental-and-social-management-framework-e-motion-program Spanish: https://www.afd.fr/es/ressources/anexo-6-marco-de-gestion-ambiental-y-social-programa-e-motion Portuguese: https://www.afd.fr/pt/ressources/annex-6-environmental-and-social-management-framework-e-motion-program

Other link(s)	https://www.afd.fr/en/ressources-accueil?f%5B0%5D=resource_type%3A38
Remarks	An indicative Outline of Stakeholder Engagement Plan, Grievance Mechanism, and information disclosure, consistent with the requirements for a Category B project, is contained in the E-Motion Environmental and Social Management Framework (ESMF). An indicative outline for Indigenous Peoples Planning Framework (IPPF) is also included in the ESMF as an outline for developing Sociocultural Analysis (SCA) and Indigenous Peoples Plans (IPP).
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, June 14, 2024
Place	In the AFD agencies of each of the target countries where the programme is implemented, hard copies of the ESMF will be made available: Argentina : Avenida Carlos Pellegrini 1141 Piso 11 BUENOS AIRES Brazil : SCS Quadra 9, Lote C S/N, Bloco A, Torre C, Sala 1103 Edifício Parque Cidade Corporate 70308200 Brasilia Colombia : Avenida 82, n° 10-33, Of. 302 BOGOTÁ Costa Rica : Del Centro comercial Plaza del Sol 425m al Sur, Curridabat, SAN JOSE Dominican Republic : Avenida Gustavo Mejía Ricart No. 98 Esq. Abraham Lincoln Edificio Corporativo 2010, Local 201, Ens. Piantini BP 285-7 Mexico : Torre Omega, Campos Eliseos No 345, Piso 16 oficina 1500, Col. Chapultepec-Polanco, 11560 MÉXICO, D.F. Peru : Calle Rio de la Plata 167 Of. 502, San Isidro 27 Lima
Date of Board meeting in which the FP is intended to be considered	

Date of accredited entity's Board meeting	December 2024
Date of GCF's Board meeting	Monday, July 15, 2024

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP237

Proposal name:	E-Motion: E-Mobility and Low Carbon Transportation
Accredited entity:	Agence Française de Développement (AFD) (French Development Agency)
Countries:	Argentina, Brazil, Colombia, Costa Rica, Dominican Republic, Mexico, Peru
Programme size:	Large

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The programme consistently scores medium-high or high against GCF's investment criteria, and is consistent with GCF's aim of supporting a low-carbon transformational shift and innovation in low-carbon transport.	The risk of limited commercial uptake of electric vehicles after the programme investments is being mitigated in the project design, but it needs to be carefully monitored to potentially take corrective action.
GCF's concessional finance will accelerate an e-mobility market transition in Latin America. GCF and Agence Française de Développement will be providing essential financial assistance to (i) de-risk transactions; and (ii) boost the market in both the public and private sectors.	In order to assure strong co-financing and parallel financing to materialize and support the market transition, the entities need to continue the strong country engagement demonstrated through the funding proposal development phase.
The programme provides technical assistance to develop conducive policy frameworks and identify innovative business models to create synergies with other investment platforms in the region, which is crucial for e-mobility success in a region with a potential for rapid growth in electric vehicle uptake.	The programme needs to carefully balance public and private investments during implementation to react to developing market opportunities due to the rapidly evolving nature of the public/private e-mobility market.
The programme demonstrates strong country ownership by aligning with nationally determined contributions and low-emission, urban, and climate policies.	

2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity, and, if considered appropriate, subject to the conditions set out in Annex II of document GCF/B.39/02.

II. Summary of the Secretariat's assessment

2.1 Programme background

3. The programme objective is to accelerate electric vehicle (EV) deployment and enable a large-scale regional transition towards electric mobility in Latin America through financial and technical assistance. The programme will implement interventions to kick-start EV mass deployment significantly earlier than under a business-as-usual (BAU) scenario by reducing the risk profile of investments and utilizing comprehensive technical assistance.

4. The key strategic value of the programme is that it functions as a market accelerator, enabling a faster uptake of e-mobility than under a BAU scenario, thus avoiding a lock-in of long-lived path dependencies and allocations of assets into fossil fuel technology pathways. The programme fills the gap between initial pilots already existing in the region and long-term transformational targets. These interventions are made now, at a time when e-mobility is not yet commercially viable and thus initial investment support is required, as is the case in all countries that have a significant uptake of e-mobility.

5. Latin America has strong enabling conditions for clean transport and e-mobility. The region has one of the cleanest electricity generation portfolios in the world with the programme countries having a median carbon grid factor of 0.252 kilograms of carbon dioxide equivalent (CO₂eq)/kilowatt-hour, thus enabling e-mobility to have a significant greenhouse gas (GHG) mitigation impact. The adoption of e-mobility strategies further enables countries to move towards a decarbonization pathway. The updated nationally determined contributions (NDCs) of programme countries all include measures to reduce transport emissions. In their NDCs, the majority of countries have prioritized e-mobility as a core mitigation measure to reduce transport emissions and have formulated concrete electrification targets through regulations, policies or EV road maps, which are also supported through the present funding proposal.

6. The main motivating forces to foster e-mobility in the countries included in the programme are (i) reduce GHG emissions; (ii) improve air quality and reduce health problems associated with air pollution; (iii) reduce fossil fuel dependency on imported fuels and exposure to external price shocks; (iv) achieve positive economic impacts related to reduced health costs, reduced fuel import bills, and job creation; (v) participate in the new growth industry of EV and battery manufacturing; and (vi) increase the attractiveness of public transportation vis-à-vis private motorized transport.

7. The programme financial envelope is EUR 307.31 million (of which EUR 210.98 million and USD 103.90 million), utilizing loan and grant instruments. The GCF contribution to the total is 37.6 per cent (EUR 78.33 million and USD 40.26 million). This consists of EUR 56 million and USD 40.27 million in concessional loans and EUR 22.3 million in grants for technical assistance. The direct mitigation impact over the programme lifetime of 25 years is 4,334,051 million tonnes of carbon dioxide equivalent (MtCO₂eq).

8. GCF categorizes the programme as B (medium) for environmental and social safeguards.

2.2 Component-by-component analysis

9. The programme is structured in three components:

- (a) **Component 1:** Establishment of a sustainable mobility ecosystem encouraging electrification and shift to public transport (technical assistance);
- (b) **Component 2:** Investments into electric public transport mode shift (financial assistance); and
- (c) **Component 3:** Deployment of EV fleets (financial assistance).

Component 1: Establishment of a sustainable mobility ecosystem encouraging electrification and a shift to public transport (total cost: EUR 34 million; GCF cost: EUR 21 million in grants)

10. The outcome of component 1 is improved public and private sector capacity and an improved regulatory environment for e-mobility. Component 1 is linked with output 1.1: Establishment of a sustainable mobility ecosystem encouraging electrification and a shift to public transport.

- (a) The activities of Component 1 are: 1.1.1 – Implement a local e-mobility-conducive framework; 1.1.2 – Implement a national e-mobility-conducive framework; 1.1.3 – Implement a regional e-mobility conducive framework; and 1.1.4 – Realize interventions with a gender perspective;
- (b) Component 1 is technical assistance only.

Component 2: Investment measures for public transport mode shift (total cost: EUR 27 million and USD 15.8 million; GCF cost: EUR 8.5 million and USD 6.1 million in loans)

11. The outcome of component 2 is reduced GHG emissions from the transport sector. Component 2 is linked with output 2.1.: E-mobility is combined with public transport measures to foster mode shift to low-carbon transport.

12. The activities of component 2 are: 2.1.1 – Implement urban mobility measures to foster public transport in two countries.

13. This component is focused on lending and financing activities only.

Component 3: Deployment of EV fleets (total cost: EUR 143 million and USD 88 million; GCF cost: EUR 47.5 million and USD 34.1 million in loans)

14. The outcome of this component is reduced GHG emissions from the transport sector. Component 3 is linked to output 2: EV fleets are deployed with innovative business models to enable their mass application.

15. The activities of component 3 are deployment of e-buses; deployment of charging infrastructure for e-buses; implementing bus depot upgrades for e-buses; and implementing grid upgrades for connecting e-bus chargers.

16. This component is focused on lending and financing activities only.

Project management (total cost: EUR 7 million; GCF cost: EUR 1 million in grants)

17. Activity 1.1.5. refers to project sourcing, structuring and monitoring. This includes technical, financial and institutional advisory services and the full feasibility of the project including economic, environmental, social, gender and legal aspects. This includes defining a project contractual scheme, ensuring a proper risk allocation among key stakeholders, assuring the bankability of the project, and securing a proper allocation of GCF concessional resources. This support is key as innovative business models are often needed to bridge the cost gap between EVs and internal combustion engines. Project preparation can also support verification of the conformity of e-buses at the supplier end, including the bidding documents and test developments. Projects sourced and designed under this activity will be financed within components 2 and 3 and result in outputs 2.1 and 2.2. Project monitoring is also part of this activity, which will contribute to a proper structuring of the projects, an adequate risk allocation and bankability, and the deployment of innovative business models.

III. Assessment of performance against investment criteria

18. The programme consistently scores medium-high or high against GCF's investment criteria. It is also consistent with GCF's aim of supporting a low-carbon transformational shift and innovation, in this case in e-mobility and low-carbon transport. The programme is well

suited to the needs of the recipient countries. A detailed assessment of the programme's fit with GCF investment criteria is provided below.

3.1 Impact potential

Scale: Medium to high

19. The impact potential is expected to be 4,334,051 Mt CO₂eq of GHG emissions avoided. This is based on an initial conservative estimate related to an indicative pipeline of subprojects in four countries receiving investment support.
20. GHG emission calculations for e-buses have been made for potential e-bus investment projects per country. While baseline emissions per vehicle category are comparable for different countries (variations are due to vehicle mileage, fuel type used and, to a lesser extent, different specific fuel consumption values), the carbon grid factor per country influences the emissions of the EVs and therefore the emission reductions.
21. In addition, GHG emissions include calculations for public transport measures. This includes GHG reductions from measures to increase patronage. The methodology used for calculating GHG reduction is based on the United Nations Framework Convention on Climate Change (UNFCCC) methodology ACM0016 and the UNFCCC methodological tool "Baseline emissions for modal shift measures in urban passenger transport". GHG reductions are due to additional passengers who, in absence of the electric bus option, would have used other modes of transport with higher emission factors. Measures which increase public transport usage include infrastructure investments in bus lanes and bus stations, smart ticketing, improved connectivity and intramodality, modernized bus fleets, and related measures to promote public transport systems in urban settings.
22. The emission factor per passenger-kilometre of private means of transport is two times higher than for public transport. The assumed lifetime of the programme is 25 years in accordance with infrastructure lifespan.

3.2 Paradigm shift

Scale: High

23. The programme will lead to transformation of urban EV transport systems and penetration via enabling policies, infrastructure investment, and technology and knowledge transfer for sustainable EV market development across the region.
24. The programme has potential for upscaling and replication at the national and international level through capacity created in the e-mobility ecosystem, successful business models, improved profitability and reduced risks of EV investments and knowledge products.
25. It also has high potential for knowledge-sharing and learning through the technical assistance component at regional, national and local levels. In addition, the programme also to contribute to the creation of an enabling framework. Technical assistance includes the support of enabling policies for EV deployment. Technical support will aim to set up, improve and enforce enabling public policies and legal, regulatory, fiscal and normative frameworks for e-mobility.
26. The programme also aims to have an overall contribution to climate-resilient development pathways consistent with national climate change adaptation strategies and plans. It will lead to significant GHG reductions in the transportation sector. Without this contribution, the countries will not be able to achieve their decarbonization plans. Fostering e-mobility and kick-starting EV deployment is consistent with relevant national climate-resilient development pathways.

3.3 Sustainable development potential

Scale: Medium to high

27. The programme contributes to the following key Sustainable Development Goals (SDGs): SDG 3 (Good health and well-being), SDG 7 (Affordable and clean energy), SDG 9 (Industry, innovation and infrastructure), SDG 11 (Sustainable cities and communities), and SDG 13 (Climate action).

28. The main environmental co-benefits are reduced emissions of pollutants and reduced noise emissions. The major concern for air pollution in the cities is PM_{2.5} and NO_x emissions.

29. The major social co-benefit is improved air quality, plus reduced noise. Air pollution is a major problem in most Latin American cities, with levels seriously affecting human health. According to the latest urban air quality database, 98 per cent of cities in low- and middle-income countries with more than 100,000 inhabitants do not meet World Health Organization (WHO) air quality guidelines. The WHO estimates that annually 4.2 million deaths result due to excessive exposure to fine particulate matter, equivalent to 7.6 per cent of all deaths.

30. The programme also delivers economic co-benefits due to the monetary value of reduced air emissions (GHG and air pollutants). The economic cost of pollution is calculated by assigning a monetary value to emissions of fine particulate matter (PM_{2.5}) and nitrogen oxides for each country based on local levels of pollution at the ground level and the impact on health and costs caused by this type of pollution per country.

3.4 Needs of the recipient

Scale: High

31. GCF concessionally is expected to address financial and technical capacity barriers, de-risk and leverage public-private capital for the EV sector, support public finance flows and incentivize a shift to EV in urban contexts.

32. The majority of countries have included electric mobility in their NDCs as a core mitigation measure to reduce transport emissions and have formulated concrete electrification targets through regulations, policies and/or EV strategies. The programme's activities in each country are fully contextualized in the country plans, as further described in the funding proposal and the feasibility study.

33. The accredited entity (AE) Agence Française de Développement (AFD) has also followed a robust stakeholder consultation process, which includes the national designated authorities (NDAs) in each country as well as development banks, private banks, municipalities and regional states, power utilities, transport operators, etc.

34. The EV sectors in targeted countries have (i) limited access to commercial funding; (ii) early-mover costs and disadvantages including high performance risks; and (iii) lack of sufficient commercial viability including profitability and acceptable risk levels. Therefore, financiers are not in a position to fund such activities with commercial loans at the current stage. The programme countries depend on technical assistance that serves to enhance the viability of the subprojects and leads to long-term transformational impacts. The programme countries would not be able to source the required public funds on its own due to current fiscal realities (e.g. economic weakening, coronavirus disease 2019 (COVID-19) recovery, high inflation). The GCF grants are therefore essential for the successful implementation of the required technical assistance activities.

3.5 Country ownership:

Scale: Medium to high

35. No-objection letters have been issued by all programme countries.

36. The programme is well aligned with countries' policy objectives in low-emission transport, urban development, climate change commitments in NDCs and sectoral plans, which include concrete e-mobility targets. All countries have also established/are formulating EV policies and legislation. Some of the countries have established specific targets for the EV sector, to which this programme will contribute.

37. The design of the programme, from the concept note stage to funding proposal development, was based on a detailed stakeholder assessment, which included multiple meetings with the NDAs and stakeholders in the public and private sector relevant to the programme, as well as countries' diagnostics and market studies.

38. The EV sector in programme countries have (i) limited access to commercial funding; (ii) early-mover costs and disadvantages including high performance risks; and (iii) lack of sufficient commercial viability including profitability and acceptable risk levels. Therefore, financiers are not in a position to fund such activities with commercial loans at the current stage. Governments and transport operators also lack capacity and know-how on the appropriate technology choices and the most appropriate and conducive policies and business models to foster cost-effective EV deployment.

3.6 Efficiency and effectiveness

Scale: Medium to high

39. The total programme cost is EUR 210.98 million and USD 103.90 million, with a GCF contribution of EUR 78.33 million and USD 40.26 million, resulting in very efficient mitigation costs of EUR 26.7/tonne of carbon dioxide (tCO₂) and a leverage ratio of 1:1.66.

40. The economic internal rate of return (EIRR) is estimated at 18 per cent, including the measures for e-buses and public transport (in two countries), against the total programme cost, including technical assistance. The EIRR is above the social discount rates calculated by the Secretariat for the programme countries (between 5 to 8 per cent).

41. The analysis was also conducted on the average financial internal rate of return (FIRR) for e-buses without concessional GCF finance (1 per cent) and with GCF concessional finance (9 per cent). Without concessional finance, the FIRR is below the weighted average cost of capital (WACC) (8 per cent), while it is above the WACC with concessional finance.

42. The mitigation cost is EUR 27/tCO₂, which is in the medium to higher range of the GCF portfolio, reflecting the higher average GHG abatement costs of transport interventions.

43. The AE has provided a comparison with eight other countries, where costs ranged between USD 97/tCO₂ (China) to USD 476/tCO₂ (Poland), with a total average of USD 227/tCO₂ across the eight countries. The programme therefore has an efficient and effective cost structure.

44. The long-run viability of the programme is based on proven business models and the decreasing technology cost in the future. The sensitivity analysis of the financial model shows that if the capital expenditure for e-buses drops by 10 per cent, the FIRR is in most cases higher than the WACC without GCF. In summary, as the capital expenditure of e-buses drops, the subprojects will be financially viable with attractive returns in all the countries.

45. The AE's financial analysis shows that GCF concessional financing improves the FIRR of subprojects. In Brazil and Mexico, GCF provides a significant profitability enhancement by enabling the FIRR to improve above or close to the WACC of 8 per cent. In Colombia and Peru, however, even with GCF support, the subproject FIRR remains at 6 per cent, which is slightly below the target of 8 per cent. As AFD has presented an indicative pipeline of subprojects, the Secretariat does not have certainty of the exact amount of GCF resources to be allocated to each programme country. To mitigate the risk of country concentration, AFD has set a maximum of

35 per cent of the total resources per country. The financial analysis supports the need for concessionality in aggregate.

46. Grant instruments will be used for technical assistance measures that cannot be financed by the private sector, but are necessary to ensure the replication impact of the programme, increase the attractiveness of the public transport system and ensure the intended shift towards e-mobility.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

47. **Programme Background.** The programme is part of the regional efforts that aims to accelerate EV deployment and enable a large-scale transition towards electro-mobility in 7 countries in Latin America. The programme intends to invest the deployment of e-buses and upgrading the public transport systems. The AE indicates 4 investment projects in the programme pipeline, with focus on financing: (a) urban mobility measures such as pedestrianization and bus lanes and route restructuring, which include infrastructure connecting public transport systems to alternative modes of urban mobility; and (b) deployment of 850 e-buses, the related charging infrastructure, bus depot upgrades, and grid upgrades for connecting e-bus chargers. Among the key co-benefits of the programme includes the reduction of air pollution, noise levels and GHG emissions, and subsequently improving public health. Main social co-benefits include increased road safety as non-motorized transport and micro-mobility options tend to lower accident rates, lesser road congestion, and improved the quality of life for urban dwellers with more resilient urban transport infrastructure.

48. **Environmental and Social Risk Category and safeguards instrument.** The AE assigned the programme low to moderate environmental and social (E&S) risk category that is equivalent to Category B as per the GCF's E&S risk categorization. This is rationalised by financing activities with potential limited adverse environmental and/or social risks and impacts that are few, generally site-specific, largely reversible, and readily addressed through mitigation measures. No activities with potential significant adverse environmental and social risks and impacts that would fall under Category A will be supported under the programme. The Secretariat confirms the E&S risk category and that this is within the AE's risk accreditation level. The AE has provided an Environmental and Social Management Framework (ESMF) applicable to all the programme components to ensure safeguards compliance.

49. Compliance with GCF's Environmental and Social Safeguards (ESS) Standards. The paragraphs below describe the programme's compliance with GCF's ESS standards:

50. **ESS 1: Assessment and Management of Environmental and Social Risks and Impacts.** The ESMF provides an analysis of the potential E&S risks based on the characteristics of the programme, suggests potential mitigation measures focusing on EV deployment and the construction of charging infrastructure, grid connections and bus depot upgrades, and outlines the eligibility criteria, exclusion list, and the E&S screening checklist for subproject due diligence. All subprojects will be required to prepare an ESIA and ESMP. Where gaps and moderate risks are identified as a result of the AE's subproject E&S due diligence, an E&S commitment plan (ESCP) will be required and be included as part of financing agreement. All subprojects under the programme are required to comply with the applicable national and local E&S laws and regulations in the respective target countries as well as AE's E&S Safeguards that are aligned with the International Finance Corporation (IFC) Performance Standards on Environmental and Social Sustainability.

51. **ESS 2: Labor and Working Conditions.** The programme identifies potential risks and adverse impacts on occupational health and safety for workers involved during the construction, implementation and operations of EVs, charging stations, grid connection and bus depot upgrades. The ESMF triggers the AE's E&S safeguard on working conditions and considers the terms and conditions of employment, non-discrimination and equal opportunity, worker's organizations, labour protection and the prevention of child labour and forced labour, etc. as part of the E&S due diligence. The programme will exclude any activities involving the use of forced labour or child labour. The programme will promote the employability of local labour in the construction area with gender focus, encouraging hiring service providers locally.
52. **ESS 3: Resource Efficiency and Pollution Prevention.** The programme is expected of minor short-term impact of natural resources due to construction activities. ESMF is guided with the aim of avoiding and minimizing negative impacts on human health, biodiversity, and ecosystems through the prevention of pollution. Activities under the programme will potentially generate wastewater and hazardous materials as regards the use, replacement, reuse and recycling, and disposal of EV batteries (primarily lithium batteries and the peripheral components). Additional impacts can be attributed to small-scale construction activities of grid-connected charging stations and bus depot upgrading, which may generate noise, dust particles, solid and liquid waste, and soil and water contamination from and scrapping of out-of-use vehicles, such as oils and lubricants, lead-acid batteries, tires and scrap metals. The programme will exclude the production, trade, storage or transport of large volumes of products hazardous chemicals, or commercial scale use of hazardous chemicals, and provide technical assistance to the targeted countries and requires an integral EV Battery Waste Management Program must be developed together with the ESMP, which will need to comply with current or proposed new regulatory requirements established by each target country for the management, treatment and disposal of End-of Life EV Batteries. The programme envisions that necessary legal framework and capacities on EV battery management will be developed and deployed during the implementation of the programme. ESMF specifies that if the local legislation does not have specific regulations in this regard, the best international practices must be applied. Given the regulatory framework on the proper management of EV batteries in the targeting countries are yet to be developed, the AE is recommended to further elaborate the guideline on EV battery management and detail the best practice and technical standards to be followed at the subproject level.
53. **ESS 4: Community Health, Safety and Security.** The investment under the programme could potentially adversely affect the community health and safety. These could include infrastructure and equipment safety risks during the construction, maintenance and operation of the EV charging stations and upgrading related systems; increased accidentality as the low or non-existent noise of EVs are less noticeable during operation; potential for community exposure to hazardous materials and chemical substances related to EV battery disposal and scrapping conventional vehicles; and conflict risks between institutions, companies and community during the installation and cabling activities which may fluctuate the electric power service and cause power outage in the affected communities, therefore, the community may have power outage. The ESMF has provided measures to manage these risks and required further risk assessment will be conducted at subproject level. The AE is recommended to incorporate measures to avoid and minimize community exposure to hazardous materials in the EV battery waste management plan at subproject level.
54. **ESS 5: Land Acquisition and Involuntary Resettlement.** The programme will exclude subprojects that involves land acquisition and involuntary resettlement through the exclusion list. As part of the subproject due diligence process, the programme will consider minimizing resettlement through altering project design, compensation and benefits for affected persons, community engagement, physical and/or economic displacement, and collaboration with other responsible agencies or sub-national jurisdictions. The programme will construct small-scale

urban charging stations and new connecting power lines located on lands of public domain and property of municipal or state authorities. The programme will build e-bus charging stations on sites belonging to the bus companies.

55. **ESS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources.** The AE's does not consider any eligible projects will pose risks on any modification to protected areas, critical habitats and or sensitive areas or affect any modified, natural and critical habitats or legally protected and internationally recognized areas. Hence the ESMF does not trigger AE's safeguard in this regard. The Program will exclude projects that involve any modification to natural and critical habitats or legally protected and internationally recognized areas. As part of the subproject due diligence, the programme will exclude any operation leading to or requiring the destruction of critical habitat, or any forestry project which does not implement a plan for improvement and sustainable management.

56. **GCF Indigenous Peoples Policy and ESS 7: Indigenous Peoples.** All of the countries the project will work in have Indigenous Peoples. As the project activities focus on accelerating Electric Vehicle deployment and a large-scale regional transition towards electro-mobility in Latin America through financial and technical assistance, the management of batteries for electric vehicles or of the hazardous waste from disposal of batteries may affect Indigenous Peoples. Furthermore, there are no specific regulations or policies for the management of batteries for electric vehicles or of the hazardous waste from disposal of batteries in the countries that are part of the Programme. Accordingly, the AE has included an Indigenous Peoples Planning Framework (IPPF) as an annex to the ESMF. The contents of the IPPF need to meet the requirements of the Green Climate Fund Indigenous People Policy. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group is available to provide advice to the AE who is also encouraged to share emerging good practices and success stories with the Group.

57. **ESS 8: Cultural Heritage.** The ESMF establishes specialized requirements for subproject specific ESMP, when warranted, must contain measures to ensure the protection of the cultural heritage from loss or damage and support its preservation. Per the AE's assessment, eligibility projects under the program do not expect to affect properties and sites of archaeological, historical, cultural, artistic, and religious significance. The Program will exclude projects that affect cultural heritage. As part of the due diligence, the programme will confirm the identification and legal protected cultural heritage areas. The project will exclude Any activity leading to an irreversible modification or significant displacement of an element of culturally critical heritage or any activity leading to any risks and impacts on cultural heritage.

58. **Sexual Exploitation, Sexual Abuse, and Sexual Harassment (SEAH) safeguarding.** The AE's has identified potential SEAH risk related to the deployment of EVs in the public transport sector and in an low gender-based violence (GBV) and SEAH during the construction of small-scale EV infrastructure works. The AE will develop a SEAH action plan following if any questions will be identified. Potential mitigation measures include training transport operating companies to detect SEAH incidents, raising awareness and creating spaces free of gender-based violence (GBV) and SEAH, and supporting the establishment of an adequate regulatory framework for public transport operating companies to implement gender policies preventing SEAH and GBV. All subprojects will comply with the requirements established in the GCF regarding Sexual Exploitation, Sexual Abuse, and Sexual Harassment.

59. **Implementation arrangements.** The AE will monitor and supervise the environmental and social performance and compliance with contractual requirements of the specific subprojects financed by programme. A project management unit (PMU), composed of professionals from the AE and the Executing Entity and supported by an external E&S and Gender expert, will be established and will support early E&S screening and to ensure the day-to-day implementation and monitoring of the programme's impacts. The PMU's E&S specialists

will also review E&S due diligence, ensure E&S supervision of all subproject. The AE will also ensure E&S monitoring. Three steps are determined for the appraisal at the project level, being: identification, instruction and commitment; Early E&S screening verifications, requirements and categorizations carried out by the project owner and PMU specialists; elaboration of ESDD to be conducted under the responsibility of the project owner with review of the PMU's E&S specialist, elaboration of the ESCP having the project owner responsible for including E&S specifications from the ESMP into procurement documentation and contract works. Regarding the supervision component, the implementation of the ESMP and ESCP measures conducted by the project owner and PMU's E&S specialist, in coordination with AFD. The evaluation step is defined by an environmental and social ex-post evaluation conducted by AFD or external consultants.

60. **Stakeholder engagement and information disclosure.** The development of the programme has evolved based on consultations conducted with various stakeholders including national government representatives such as from National Designated Authorities, national companies in charge of electric energy, municipalities of the main cities in targeted countries, and main actors including those involved in capacity building and financing EV, charging infrastructure and in the battery value chain. The programme also provides for engagement of stakeholders during the implementation of the supported subprojects and capacity building activities through an ongoing process involving stakeholder analysis and planning, disclosure and dissemination of information, consultation and participation, grievance mechanism, and environmental and social performance reporting to affected communities. The ESMF provides an indicative outline of the stakeholder engagement plan required by all the subprojects. The AE will also disclose environmental and social documents at project level, in the same manner, and time frame as the safeguards documents, a summary of the activities, along with the environmental/social information.

61. **Grievance redress mechanism.** The programme provides access to grievance redress mechanisms at the AE, implementing partners and subproject levels. The AE will use its own E&S complaints mechanism to ensure the independent handling of complaints. The project owner will design a GRM to receive and facilitate concern or conflict resolution in relation with E&S risks and impacts, with the purpose of resolving concerns or conflicts through an understandable and transparent consultative process consistent with project and/or local context. A project-level grievance redress mechanism will also work in conjunction with AFD's Environmental and Social Complaints Mechanism. So far, the AE does not have a specific procedure to manage the risks related to GBV and SEAH. This is however covered by the operating procedures, which aims to identify, prevent or mitigate environmental and social risks and impacts and human rights violation that could result from AFD-funded activities. In particular, the AE relies on and applies the principle of the World Bank Environmental and Social Framework, which includes risk management measures related to SEAH. The AE is recommended to keep all relevant stakeholders informed once its corporate GRM is established and keep the ESMF and guidance to subproject GRMs updated.

4.2 Gender policy

62. The AE provided a gender assessment and action plan with the funding proposal and therefore **complies with the requirements of the GCF Gender Policy.**

63. The gender assessment (using secondary data) provided an analysis of gender equality and women's empowerment and how they link with the transport sector in the targeted countries of Argentina, Brazil, Colombia, Costa Rica, Dominican Republic, Mexico and Peru. All these countries have unique situations, and while the assessment has put a spotlight on most of their gender dynamics, country-level and transport sector-specific gender assessments are

required to comprehensively unpack these gender issues and tailor the countries' gender action plans (GAPs) in alignment with the gender/social issues and the funding proposal. The AE has committed (in the GAP) to conducting additional sector-specific and country-level gender assessments with their accompanying GAPs, one for each targeted country.

64. The gender assessment report also presents initiatives and good practices emerging from the targeted countries and region on gender equality and urban transportation. In general, for the Latin America and the Caribbean (LAC) region, GBV remains a challenge and is underreported. Before 2006, no country in the LAC region had established femicide as a crime, while by 2015, 16 countries had done so. Likewise, in many parts of the world, civil society has become strongly activated against GBV, and Argentina, Chile, Mexico, Peru and Uruguay have marched under the motto "Ni una menos" (Not one woman less).

65. While the programme countries have laws and regulations that foster women's access to economic opportunities, studies have shown that there are still constraints on women's emancipation in many areas, including: access to productive resources (jobs, equal pay/benefits, etc.); constraints on women's mobility (usage, fare schemes, time/commute factors); and safety (exposure to harassment and/or sexual violence and risks involved). The gender gaps and inequalities in the transport sector are enormous and require robust programming that is gender-responsive and potentially transformative in the long-run. The AE has put in place several activities to be implemented as outlined in the GAP. The implementation of the GAP will be led by the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ); it is important that the AE is clear on the modalities for implementation and ensures that the gender specialists to be recruited have practical gender experiences in the local countries. The AE is reminded that gender accountability is critical in achieving the gender results/outcomes presented in the funding proposal and annex 8 in particular.

66. The programme will also integrate gender components in the projects financed by the programme, and this will be done through the inclusion of gender considerations in the investment plans; improved sex disaggregation of data collection on urban transport; improved women's access to jobs; increased awareness on sexual harassment in public transport through capacity-building; promotion of gender equality at the workplace; and capacity-building for the staff of public transport projects on gender equality and work-life balance issues. The AE is required to ensure that the grievance mechanisms of the programme are accessible to both women and men.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

67. GCF is providing a senior loan of EUR 56 million and USD 40.27 million and a grant of EUR 22 million to accelerate EV deployment in seven countries in Latin America. The total co-financing is estimated at EUR 133.5 million and USD 63.6 million from different co-financiers such as AFD, Corporación Andina de Fomento (CAF), GIZ and KfW, and private and public investors in a form of grants, in-kind contributions, equity and loans. The exact allocation of each co-financier is yet to be determined, and the currency of the co-financing is indicative.

68. The programme allows for the provision of financing for both public and private sector entities. The level of support and a business model per country will be determined during the implementation period on a "first come, first served" basis, depending on the needs of the subproject in each country and their level of maturity at the time of availability of GCF proceeds. The GCF grant will only be used to provide technical assistance (component 1) that cannot be financed by the private sector. All the subprojects must be aligned with the eligibility criteria in the term sheet.

4.3.2. **Accredited entity/executing entity capability to execute the current programme (low risk)**

69. AFD is the AE and co-executing entity (co-EE) (components 2 and 3) for this programme. In addition, GIZ will be acting as co-EE for the programme. AFD will be responsible for the implementation of eligible investments in the public sector. Proparco will act as an AFD agent and be in charge of the implementation of financing activities and the technical assistance measures of the programme in cases where the project owners belong to the private sector. GIZ will be implementing some of the technical assistance activities (activity 1.1, 1.2, 1.3 and 1.4 (i) to (v)).

4.3.3. **Programme-specific execution risks (medium risk)**

70. **Delays in implementation and impact risk:** The programme is planning to implement different business models (e.g. public sector-led direct/indirect delivery, public/private sector-led indirect delivery) in multiple countries. The final financial structure and business model will be subject to due diligence by the AE and agreement with the borrower during the implementation. This great flexibility and complex implementation arrangement may pose a risk of delays in the programme. The current impact potential at the end of the programme (year 7) accounts for ~18 per cent of the total lifespan impact potential. This assumes that there is no delay in implementation. The success of the programme will depend on the ability of the AE/executing entity (EE) to generate the pipeline that aligns with the eligibility criteria and arrange the appropriate modality in a timely manner.

71. **Credit risk:** GCF will be exposed to the credit risk of borrowers. GCF loans may be provided to sovereign with sovereign guarantee, municipalities without sovereign guarantee, and to the private sector. The sovereign credit rating of the participating countries ranges from non-investment grade to investment grade (from Ca to Baa1 by Moody's). The projects are mainly targeted for intermediate cities, which may not always have sovereign guarantee. The credit risk of different borrowers varies depending on the countries and borrowers. GCF will rely on AFD's due diligence to assess the borrower's ability to service the loan. GCF and AFD funding will be provided in two separate tranches. Comfort may be drawn from the fact that the credit risk will be shared between GCF and AE on a pari passu basis and all the other unsecured and unsubordinated creditors. In addition, no more than 35 per cent of GCF funds will be invested in one specific country.

72. **Co-financing risk:** The total co-financing amount is indicative. The equity contribution from the government or sponsor and other co-financing are not confirmed. Instead, the project eligibility criteria in the term sheet sets a minimum co-financing of 40 per cent in each subproject and a minimum co-financing of 60 per cent at the overall programme level. The covenants include a specific co-financing ratio for GCF reimbursable funds in component 2 (1:1.9) and 3 (1:1.8) and GCF non-reimbursable funds for technical assistance (1.2:1).

73. **FX risks:** As the GCF reimbursable funds will be provided in hard currencies (in EUR and USD), the volatility of the local currencies in participating countries could potentially affect returns of subprojects, and consequently the borrowers' ability to repay. If the GCF proceeds approved by the Board become insufficient to complete the funded activities resulting from the exchange rate fluctuations, GCF shall not bear any responsibility of providing additional financing.

4.3.4. **Project viability and concessionality**

74. The programme enables a commercial uptake and mass deployment of EVs without further support from the GCF at the end of the programme in the targeted countries. It is

expected that commercial investors will be crowded in to the e-buses once the framework condition (e.g. concession contracts) and the performance risks of e-buses are manageable. However, the current term sheet does not have a split allocation between public and private sector projects. There is no specific target or indicators to track the likelihood of commercial uptake of EVs in the funding proposal.

75. The interest rates of the GCF senior loans are differentiated by the public and private borrower. While the pricing to the public sector is fixed, the pricing to the private sector will be determined by the AE/EE on a case-by-case basis with different spread. The AE/EE will ensure that it follows the minimum concessionality principle and contractually ensure that the GCF concessionality passes down to the final borrowers. The pricing to the private sector will not be lower than the pricing for the public sector.

4.3.5. Compliance risk (medium risk)

76. Following a thorough assessment of various risk factors outlined in the proposal, the Secretariat has determined a compliance risk rating of 'medium' for the project. Upon evaluation, we have identified several key points warranting attention and consideration.

77. First, regarding risk assessment, it is noted that due diligence of counterparts is contingent upon the project's structure. Given the current status of preparation of subprojects (pre-feasibility stage), the organizational structure remains fluid, posing challenges to conducting comprehensive due diligence at this stage. It is essential to differentiate between eligibility criteria and due diligence questions, ensuring that the latter focuses on implementation and organizational issues.

78. Second, concerns surrounding sanctions and prohibited practices have been addressed with assurances that project activities will not take place in jurisdictions affected by UN Sanctions List and that measures are in place to prevent involvement with sanctioned individuals or entities. Additionally, mechanisms are established to mitigate risks related to illicit practices through internal controls and adherence to anti-corruption policies.

79. Third, the importance of robust counterparty due diligence is emphasized, with adherence to accredited procedures and regular assessments of technical capacity and fiduciary risks. Similarly, the anti-money-laundering and countering the financing of terrorism policy ensures compliance with integrity principles and due diligence at the project level.

80. Additionally, stringent measures already in place to address E&S risks can also serve to address potential concerns regarding money-laundering, terrorist financing, or prohibited practices. GRMs, including stakeholder engagement plans and an established mechanism for handling complaints, underscore the commitment to transparency, accountability, and stakeholder engagement.

81. In summary, the Secretariat has conducted a review of the proposal in accordance with relevant Board-approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the Secretariat has determined a risk rating of 'medium' and has no objection to this proposal proceeding at this time.

Summary risk assessment	
Overall programme	Medium
Accredited entity/executing entity capability	Low
Programme-specific execution	Medium
GCF portfolio concentration	Low

Compliance	Medium
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4.3.6. GCF portfolio concentration risk (low risk)

82. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of results area and single proposal is not material.

4.4 Fiduciary

83. The AFD is both the AE and the EE for the programme. GIZ is also the EE for the project.

84. The programme will be implemented by three development banks that are members of the International Development Finance Club. These are AFD Group (including Proparco), CAF, KfW Development Bank, and a bilateral technical cooperation agency (GIZ). The roles of each are illustrated as follows: AFD, as AE, EE and co-financier for the financial assistance targeting public sector project owners (components 2 and 3), for the technical assistance dedicated to project preparation, and for the GAP activities related to the investment projects; Proparco, as an agent of AFD and co-financier for the financial assistance targeting private sector project owners; GIZ, as EE for technical assistance activities; and KfW Development Bank and CAF, as co-financiers for the financial assistance.

85. A funded activity agreement (FAA), subsidiary agreement and co-financing agreement will be signed with each relevant party. A financing agreement will also be made between AFD or Proparco and the beneficiaries, which will describe the provisions and modalities for disbursing and using the funds for the purpose of implementing the projects, and will be developed and negotiated according to AFD Group templates and procedures.

86. AFD will hire consulting companies to implement the technical assistance dedicated to the preparation of the projects and to support the Programme Management Unit activities. AFD will also hire external consultants as part of the Programme Management Unit.

87. The programme will have a steering committee with the EEs (AFD, CAF, GIZ, KfW Development Bank and Proparco) as representative members. The steering committee shall notably review and approve the programme's annual budget and work plans, discuss implementation issues and identify solutions, and ensure coordination and communication between EEs.

88. The Programme Management Unit will ensure coordination between the partners as well as proper implementation of the programme.

89. During the disbursement period, the AFD Group will receive regular reports from the beneficiaries in order to track progress and identify potential issues as well as improvement opportunities. In this document, the beneficiary will report on performance indicators, as well as qualitative information, as defined in the financing agreement signed with AFD Group. An annual performance report will be submitted to GCF following reporting requirements under the accreditation master agreement (AMA) and FAAs, in parallel with performance and financial information at country and borrower level.

90. Audits will be performed as per AFD's procedures and as per the AMA. AFD procurement will follow the procurement guidelines for AFD-financed contracts in foreign countries.

4.5 Results monitoring and reporting

91. The E-Motion programme by AFD is a low-emission transport programme with an expected emission reduction impact of 4.3 MtCO₂eq over a lifespan of 25 years. The overall programme targets seven countries in the LAC region; however, it is essential to note that the mitigation impact calculated for this programme is generated from only four countries.
92. The theory of change adequately explains how the results chain will cascade from the goal statement of the programme to the proposed activities and also identifies the linkages between outputs, barriers and risks. Articulating the goal statement and outcomes helps understand the intervention's climate impact and ancillary benefits (co-benefits).
93. The logical framework has been well designed per the Integrated Results Management Framework. The paradigm shift potential indicators and enabling environment indicators describe the current scenario well as well as the potential target scenario, which rests on the successful implementation of EV deployments through new business models and a conducive policy framework for market adoption.
94. Section E.3 of the logical framework has been designed with relevant details, including reliable means of verification and reporting on the appropriate core and supplementary indicators for the targeted results area. The annex on the mitigation impact calculation provides required information on the relevant clean development mechanism methodologies being applied to calculate the estimated impact. A separate monitoring manual has been submitted that explains the approach and how the various parameters related to EVs and public transport measures will be monitored. The monitoring and evaluation budget of the programme is within the range specified in the evaluation policy and provides the required details in the relevant annexes.

4.6 Legal assessment

95. The AMA was signed with the AE on 11 November 2017 and became effective on 16 January 2018. The AE's five-year accreditation term lapsed on 15 January 2023 but was extended by decision B.37/18, paragraph (q), until the earlier of three years from the date of lapse of the five-year accreditation term or the date on which a revised accreditation framework is adopted by the Board. The Board approved the re-accreditation of the AE pursuant to decision B.36/10, and the negotiation of the amended and restated AMA is ongoing.
96. The AE has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the programme.
97. The proposed programme will be implemented in seven countries: Argentina, Brazil, Colombia, Costa Rica, Dominican Republic, Mexico and Peru. GCF is not provided with privileges and immunities in these seven countries. This means that, among other things, GCF is not protected against litigation or expropriation in these countries, which risks need to be further assessed. The Secretariat has engaged with the relevant countries as follows:
 - (a) **Argentina:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of Argentina most recently on 7 February 2023. The latest communication was sent by the GCF Secretariat to the Government of Argentina on 16 March 2023;
 - (b) **Brazil:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of Brazil most recently on 1 February 2023. The latest communication was sent by the GCF Secretariat to the Government of Brazil on 22 February 2024;
 - (c) **Colombia:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of Colombia most recently on 8

May 2024. The latest communication was sent by the GCF Secretariat to the Government of Colombia on 8 May 2024;

- (d) **Costa Rica:** The GCF Secretariat and the Government of Costa Rica have an agreed version of the draft agreement on privileges and immunities. The latest communication was sent by the GCF Secretariat to the Government of Costa Rica on 5 March 2024;
- (e) **Dominican Republic:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Dominican Republic most recently on 31 March 2023. The latest communication was sent by the GCF Secretariat to the Government of the Dominican Republic on 31 March 2023;
- (f) **Mexico:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of Mexico most recently on 24 January 2023. The latest communication was sent by the GCF Secretariat to the Government of Mexico on 24 January 2023; and
- (g) **Peru:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of Peru most recently on 24 July 2023. The latest communication was sent by the GCF Secretariat to the Government of Peru on 24 July 2023.

98. The Heads of the Independent Redress Mechanism and the Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities.

99. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Submission by the AE to GCF of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 120 days after Board approval, confirming that the AE has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed programme;
- (b) Signature of the FAA in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the AE has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and
- (c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP237

Proposal name:	E-Motion: E-Mobility and Low Carbon Transportation
Accredited entity:	Agence Française de Développement (AFD) (French Development Agency)
Countries:	Argentina, Brazil, Colombia, Costa Rica, Dominican Republic, Mexico, Peru
Programme size:	Large

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. The programme intends to promote an integrated transformational shift to climate resilient zero emission mobility systems through investments in climate resilient urban transport infrastructure and large-scale deployment of electric bus (e-bus) systems combined with mode-shift measures to increase the performance and convenience of urban public and non-motorized transport in seven countries (Argentina, Brazil, Colombia, Costa Rica, Dominican Republic, Mexico, Peru). The programme will support countries undergoing shifts in sustainability, replicability, and scalability by investing in greening public transport and expanding non-motorized transport. The paradigm shift of the programme will be achieved by addressing early-stage market deployment risks to demonstrate proof-of-concept in the targeted Latin American markets.
2. The proposed programme's financial envelope is EUR 307.7 million, utilizing loan and grant instruments. The requested GCF contribution to the total financing volume is 37.6 per cent, EUR 115.6 million, consisting of EUR 93.3 million in concessional loans and a grant of EUR 22.3 million for technical assistance. The direct mitigation impact over the programme's lifetime of 10 years is expected to be 4,334,051 million tonnes of carbon dioxide equivalent (Mt CO₂ eq).
3. The funding proposal is a resubmission, and the independent Technical Advisory Panel (iTAP) commends the accredited entities (AEs) and the executing entities (EEs) involved for addressing all the previous iTAP assessment conditions in a thorough and satisfactory manner.

1.2 Impact potential

Scale: High

4. The programme objective is to accelerate electric vehicle (EV) deployment and enable a large-scale regional transition towards electric mobility (e-mobility) in Latin America through financial and technical assistance. The programme will implement interventions to kick-start EV mass deployment significantly earlier than under a business-as-usual scenario by reducing the risk profile of investments and by providing comprehensive technical assistance.
5. The key strategic value of the programme is that it is supposed to function as a market accelerator enabling a faster uptake of e-mobility than under a business-as-usual scenario avoiding a lock-in of long-lived path dependencies and assets into fossil-fuel technology pathways. The programme has the potential to fill the gap between initial pilots already implemented in the region and long-term transformational targets.

6. Strong enabling conditions for clean transport and e-mobility are in place in Latin America. The region has one of the cleanest electricity-generation portfolios in the world with the programme countries having a median carbon grid factor of 0.252 kg CO₂ eq/kWh, thus enabling e-mobility to have a significant greenhouse gas (GHG) mitigation impact. The adoption of e-mobility strategies further enables countries to move towards decarbonization pathways. The updated nationally determined contributions (NDCs) of programme countries all include measures to reduce transport emissions. In their NDCs, the majority of the programme countries have prioritized e-mobility as a core mitigation measure to reduce transport emissions, and have formulated concrete electrification targets through regulations, policies, or EV road maps, which are also supported through the present funding proposal.

7. The main motivating forces to foster e-mobility in the countries included in the programme are (i) reduction in GHG emissions; (ii) improvement of air quality and reduction in health problems associated with air pollution; (iii) reduction in fossil-fuel dependency on imported fuels and exposure to external price shocks; (iv) achievement of positive economic impacts related to reduced health costs, reduced fuel import bill, and job creation; (v) participation in the new growth industry of EV and EV battery manufacturing; and (vi) increase in the attractiveness of public transportation vis-à-vis private motorized transport.

8. The direct mitigation impact over the programme's lifetime of 10 years is expected to be at 4,334,051 Mt CO₂ eq in reduced GHG emissions, based on the indicative pipeline of projects in four of the seven countries. Annex 22 to the funding proposal sets out reasonable assumptions made, and latest methodologies used to calculate this total figure. However, given that the exact pipeline of investments and countries is unknown for this programme, only the potential impact can be evaluated. The iTAP is encouraged by two fundamental programme assumptions which are well-documented globally: (i) replacing diesel buses with EV buses will be an improvement in carbon avoided; and (ii) the proposal assumes that the current renewable generation of each country's grid will remain the same it is currently, which is a very conservative assumption, as globally the iTAP sees increased penetration of renewable energy. Finally, the panel is also encouraged by the detailed measures for monitoring the actual results shown in annex 11b to the funding proposal.

9. More specifically, the anticipated emission reductions are calculated by comparing a with-project scenario with a baseline scenario. Under the baseline scenario, in the target cities, bus fleets reaching the end of their lifetimes are replaced with diesel vehicles, whereas in the project scenario they are replaced with lower-emission e-buses (exact emissions depend on each country's grid emissions factor, all of which are low in the target markets). The e-bus impact is calculated for four investment projects in four cities.

10. Equally important in the calculations is a figure for modal shift to public transport in two projects in two countries, as new passengers are attracted by investments (through AFD co-finance to the programme in these countries) that make public transport more appealing, such as the establishment of bus lanes, integrated ticketing systems, etc. (such that trips that would have been made using on-average higher emission transport modes are replaced by lower-emission trips on public transport).

1.3 Paradigm shift potential

Scale: Medium

11. Most importantly, these investments, if successful and sustainable, would avoid lock-in of long-lived high-emission infrastructure in public services, providing additional benefits to society in terms of independence from fossil-fuel imports, clean air and health benefits, and the increased use of public transportation.

12. The programme aims to enable paradigm shift by shifting in an accelerated manner the transition from a fossil mobility sector to an electrified low to zero carbon mobility sector. The paradigm shift is achieved through the development of an EV-conducive ecosystem in each of the markets and target cities, and the de-risking of EV investments in e-bus systems.
13. However, based on the funding proposal, the programme design is very broad at this stage and the implementation aspect specifically remains largely undefined as no specific, fully prepared investment projects under the programme are being presented. There are many interesting conceptual implementation arrangements such as the three options for Public-Private Partnership (PPP) models. These models vary from purely public sector led implementation to mostly private sector led with the public entities overseeing. The difference in the three implementation options is significant in terms of impact and paradigm shift, for example, (i) the ability to crowd in private-sector finance and private-sector operators; (ii) the creation of an enabling environment; and (iii) the creation of scale and replicability. The stated goal of the programme is to “accelerate the market uptake of EVs by reducing risk factors for investors” – however it is unclear in both the theory of change and in its implementation *how this will be achieved: goal* depends entirely on which modality of public-private implementation will be agreed with the relevant countries, and how will risks be allocated between the public and the private. And if any risks are not allocated between the parties, how will they be mitigated or absorbed through higher cost of capital.
14. Secondly, AFD, as the AE, fails to present a cohesive strategy for leveraging co-investment from the local financial sector. Although the EV sector is in its early stages and this programme will certainly contribute to scaling up new models for investing in EV buses, the programme does not sufficiently address how to enable and mobilize local capital to invest in this sector long term. By some International Monetary Fund 2023 estimates, local savings in Latin America are about USD 1.1 trillion and if the programme (through technical assistance or co-investment) effectively channels savings into the development of public EV bus and ancillary infrastructure, it would contribute to long-term sustainable urban transportation and climate resilience beyond this one-off programme. Currently, the link between this programme financing a large-scale pilot in each city and the scaling up of EV buses is alluded to but not defined. Making that link stronger and part of the programme’s design would result in a stronger paradigm shift, as would developing a cohesive strategy for leveraging co-investment, from both public-sector and private-sector sources.
15. The iTAP strongly recommends that the AE design co-investments with private local players, where possible, for the long-term sustainability of the sector and market creation.

1.4 Sustainable development potential

Scale: High

16. The programme contributes to key Sustainable Development Goals (SDGs): SDG 3 (Good health and well-being), SDG 7 (Affordable and clean energy), SDG 9 (Industry, innovation and infrastructure), SDG 11 (Sustainable cities and communities), and SDG 13 (Climate action).
17. The main environmental co-benefits of the proposed programme are expected to be reduced emissions of pollutants and reduced noise emissions. The major concern is for air pollution in the cities through emissions of fine particulate matter and nitrogen oxides.
18. The major social co-benefit would be improved air quality (plus reduced noise). Air pollution is a major problem in most Latin American cities, with levels seriously affecting human health. According to the latest urban air quality database, 98 per cent of cities in low- and middle-income countries with more than 100,000 inhabitants do not meet World Health Organization air quality guidelines. The World Health Organization estimates that 4.2 million

deaths each year result from excessive exposure to fine particulate matter, equivalent to 7.6 per cent of all deaths.

19. The programme also has the potential to deliver economic co-benefits due to the monetary value of reduced air emissions (GHG and air pollutants). The economic cost of pollution is calculated by assigning a monetary value to emissions of fine particulate matter and nitrogen oxides for each country based on local levels of pollution at the ground level, and the impact on health and costs caused by this type of pollution in the programme countries.

1.5 Needs of the recipient

Scale: Medium

20. The majority of programme countries have included e-mobility in their NDCs as a core mitigation measure to reduce transport emissions and have formulated concrete electrification targets through regulations, policies, or EV strategies. The programme's activities in each country are fully contextualized in the country plans, as further described in the funding proposal and the feasibility study.

21. The accredited entity, AFD, has also followed a robust stakeholder consultation process, which includes the national designated authorities in each country as well as development banks, private banks, municipalities, and regional states, power utilities, transport operators, etc.

22. A detailed power-sector assessment has been made per country as separate report. See new report attached as additional report under annex 2. This includes power availability, electricity generation, and expected grid factors based on renewable-energy trends, which anchors the important paradigm-shift argument that electrification of transport systems in countries with high renewable energy will lead to lower emissions.

23. However, each country's/city's design of individual EV fleet investment projects is still undefined. The country-specific e-mobility ecosystem constraints and respective solutions are not formulated (not even broadly), nor are broad terms agreed with the respective governments on which the modality of public or public-private or mostly private sector led implementation of the programme. This lack of country/city-specific design – even at a broader level – makes it difficult to assess the needs of the recipient.

24. The iTAP understands that due to the programmatic approach of this proposal, there is a degree of generality which is by design. However, it is difficult to assess the needs of the recipient or even whether concessionality is needed without understanding the specific needs and market bottlenecks of the cities where the GCF capital will flow.

25. The iTAP, the AEs (AfD) and EEs (Proparco and GiZ) had an in-depth conversation on the questions-and -answers call, on the relative importance of e-mobility policies, enabling environment, governance framework, policy reforms, and the sequencing of policies/enabling conditions versus investment projects. The AE argued that the intervention strategy is to use concrete implementation projects to foster policies, regulations, and frameworks conducive to the widespread adoption of e-buses. The AE argued that this can be done by having concrete workable implementation cases, for example, of concession contracts or business models such as the separation of bus ownership and bus operations conducive to e-buses. Such approaches, together with high quality e-bus system design, enable e-buses to be commercially competitive to fossil units. Only once it is apparent that e-buses can operate in a reliable manner and with comparable total costs to diesel buses will governments, city authorities, transport operators, and others be willing to establish and actually implement policy targets such as a ban on fossil-fuel buses. Stakeholders consulted by the programme team have repeatedly stressed the point that investment projects involving significant fleets (not pilot projects) need to be implemented prior to taking critical regulatory decisions, which affect public transport services. The proof of

concept is thus critical and the sequencing of starting with investment projects, which are then used as a concept approach, accompanied, and followed by policy advice is therefore core to this programme.

26. The iTAP would have preferred to have seen some specific examples of what *regulatory enabling environment is needed in each of the target cities*, even if they are to be implemented in parallel or after the investment in EV fleets has been carried out. This would have provided more assurance to the iTAP that the AE knows what needs to be done, even if the sequencing is performed – unusually – in parallel or after the investment. The iTAP is also encouraged by the fact that the programme has detailed investment criteria to ensure that GCF funds are used in the appropriate area and with the largest possible impact.

1.6 Country ownership

Scale: High

27. The programme is well-aligned with the programme countries' policy objectives on low-emission transport, urban development, climate-change commitments in NDCs, and sectoral plans, which include concrete e-mobility targets. All seven countries have also established/are formulating EV policies and legislation. Some of the countries have established specific targets for the EV sector, to which this proposed programme will contribute.

28. The design of the programme, from the concept note stage until funding proposal development, was based on a detailed stakeholder assessment, which included multiple meetings with the national designated authorities and stakeholders in the public and private sectors relevant to the programme, as well as the programme countries' diagnostics and market studies.

29. The EV sector in the programme countries has (i) limited access to commercial funding; (ii) early-mover costs and disadvantages, including high-performance risks; and (iii) a lack of sufficient commercial viability, including profitability and acceptable risk levels. Therefore, financiers are not currently able to fund such activities with commercial loans. Governments and transport operators also lack the capacity and know-how on the appropriate technology choices, and the most appropriate and conducive policies and business models to foster cost-effective EV deployment.

1.7 Efficiency and effectiveness

Scale: Medium to high

30. Projects of the proposed programme, target a high co-finance ratio of 61 per cent (leverage factor of 1.6). Future investments are not expected to require further GCF finance due to cost decreases and learning curve experience and the same approach as in approved GCF e-mobility projects for public transit (Inter-American Development Bank, Corporación Andina de Fomento – Andean Development Corporation and Asian Development Bank programmes).

31. The mitigation cost is EUR 27/t CO₂ which is low for a transport programme.

32. The required concessionality for jump-starting the EV public transport mobility has been clearly measured, thought through, and justified. Most impressive is the ability of the AE to draw on other successful country experiences in showing the path to potential commercial viability if the right country-specific projects are designed and implemented. The benchmark report (annex 22 to the funding proposal) clearly shows that all countries with large fleets of commercial EVs (>100 units) have used, at least initially, financial-support instruments in the form of grants to kick-start EV deployment. The market studies realized in all countries also show that EVs are currently not commercially viable (with different degrees of non-viability). The market studies, however, also reveal that in the medium term due to price decreases, a commercial uptake of EVs is viable.

33. The blending of GCF capital with the capital of AFD and Proparco will lower the cost of capital for each project which invests in EV buses and related infrastructure. The lower cost of capital would enable the upfront higher costs of the EV buses – vis-à-vis the cost of diesel buses – to be tariff neutral for the ultimate beneficiaries. As the expected cost of EV buses decreases, the need for concessionality will be removed – this is one of the few programmes which clearly delineates a path for concessionality to accelerate climate investment and a path for GCF concessionality to become unnecessary with scale.

34. One item which was discussed on the call, is the fact that both the AE and its EE (Proparco) have not considered hedging any portion of their hard-currency investment to their borrowers, which surprised the iTAP. Given the current macroenvironment and the high indebtedness of many of their target countries (most of target countries' external debt is also in hard currency), this may negatively impact the effectiveness of their programmes and the potential climate impact.

II. Overall remarks from the independent Technical Advisory Panel

35. The programme is trying to create scale in a sector which requires it for two reasons: (i) in order for the cost of capital (related to perception of risk) to come down and (ii) the cost of the EV fleets to decrease. The programme aims to affect both those important levers and is well-designed from an impact, country ownership, and efficiency and effectiveness point of view. The main underlying assumption of the AfD programme is that these changes will happen inevitably but by creating scale, the programme will accelerate these positive transformations for the EV sector in the pilot cities and countries.

36. The programme is by design vague on details on which mode of implementation it will use to carry out the EV bus fleet investment and necessary infrastructure (localized grid upgrades, depots, etc.), and whether it will be public-private partnership, public-led, or private-led. However, it has selected suitable partners in AFD, Proparco and Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ), with the necessary respective competences to assess, evaluate, design, and implement whichever design makes most sense for each locality.

37. The main gap that the iTAP finds in this programme is a **lack of** focus on mobilizing the billions of local savings and local capital into EV bus investments, either as co-investors in the current programme or as future investors in the space. Latin America has deep capital markets: the region's financial markets have substantial liquidity, a broad range of financial instruments, and significant participation from various types of investors. Moreover, local capital investors are the right partners for long-term financing sustainability of the targeted sector. In addition, if they were to be involved from the beginning, local denominated co-investors could also act also as a risk mitigant against currency depreciations and foreign shocks. Large scale infrastructure in Latin America, such as this, need not only be invested in hard currency. Development Financial Institutions (DFIs) must do better.

38. The iTAP recommends that the Board approve this funding proposal with the following recommendation:

- (a) AFD, Proparco and GIZ to co-Design with local stakeholders of an investment strategy to attract local investors in this space either through, inter alia, co-investment, exits, refinancing, or as sole investors in follow-on opportunities. We strongly recommend that they allocate a significant portion of the technical assistance AND the GCF capital to enable, design, co-pilot and implement co-investments into the target projects with local capital – the iTAP suggests for the private sector investments of this programme, that e.g., *inter alia*, local financial institutions are crowded in by being allowed to invest in more risk averse senior positions in the capital stack – in order to ensure that after the

AFD programme is over, the local financial markets take up the baton of financing and scaling up this important sector for their economies, the health of their people, and the planet.

Response from the accredited entity to the independent Technical Advisory Panel's assessment of FP237

Proposal name:	E-Motion: E-Mobility and Low Carbon Transportation
Accredited entity:	Agence Française de Développement (AFD) (French Development Agency)
Countries:	Argentina, Brazil, Colombia, Costa Rica, Dominican Republic, Mexico, Peru
Programme size:	Large

Impact potential
<i>iTAP Scale: High</i> The AE (AFD) thanks the iTAP for its positive assessment of this key criterion, and agrees with such assessment.
Paradigm shift potential
<i>iTAP Scale: Medium</i> In relation to para. 3 of the iTAP assessment, and E-Motion being a programme, we concur with the uncertainty regarding the implementation arrangements and models that will be set up for each future subproject within the programme, as presented in the Funding Proposal. The AE and the EEs will pay close attention during implementation to ensure that selected subprojects maximize the modal shift towards sustainable mobility. Regarding iTAP's recommendation (paras. 4. and 5.) to leverage co-investments from the local financial sector, please refer to our response to "Overall remarks".
Sustainable development potential
<i>iTAP Scale: High</i> The AE (AFD) thanks the iTAP for its positive assessment of this criterion, and agrees with such assessment.
Needs of the recipient
<i>iTAP Scale: Medium</i> Regarding the needs of the countries (para. 4.), and as noticed by iTAP (para. 5.), E-Motion follows a programmatic approach and the detailed design of each subproject cannot be known at this stage. The AE would only point that the results of the Market Study conducted in each country of the Programme contributes to give insights on their e-mobility policies and needs. The AE thanks the iTAP in pointing out the importance of sequencing the activities in each city / country (para 6.) and is being encouraged by the fact that the programme has

detailed investment criteria to ensure that GCF funds are used in the appropriate areas and with the largest possible impacts (para. 7.).

Country ownership

iTAP Scale: High

The AE (AFD) thanks the iTAP for its positive assessment of this criterion, and agrees with such assessment.

Efficiency and effectiveness

iTAP Scale: Medium-High

The AE (AFD) thanks the iTAP for its positive assessment of this criterion, and takes note of iTAP's comment in para. 5.

Overall remarks from the independent Technical Advisory Panel:

The AE thanks iTAP for its recommendation (para. 4.) that GCF Board approves the E-Motion Funding Proposal.

The AE welcomes the iTAP's overall remark (para. 3.) regarding the Funding Proposal's lack of focus on attracting and mobilizing "local investors", and acknowledges iTAP's recommendation (para. 4a.) for the AE (AFD) and the EEs (Proparco and GIZ) to "co-Design with local stakeholders [of] an investment strategy to attract local investors" and to "allocate a significant portion of the technical assistance AND the GCF capital to enable, design, co-pilot and implement co-investments into the target projects with local capital".

This recommendation is in line with the discussion during the iTAP meeting, and the AE fully concurs. Over the past and coming years, new business models (such as the separation of bus ownership and bus operation) to develop E-mobility will attract new stakeholders. Those stakeholders may be public, such as (i) electricity operators who are part of E-mobility financial schemes in some countries, and (ii) public development banks, such as FDN in Colombia with whom AFD is already working with on E-mobility. Those stakeholders might also be private. The E-Motion Program will indeed support those new developments by including a solid technical assistance on E-mobility to make it more attractive over the years, thanks to the sequencing of activities as mentioned above. The 7-years implementation period will enable such sequencing.

Gender Assessment E-Mobility Program



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Acronyms

AFD French Development Agency

CAF Latin American Development Bank (former Corporación Andina de Fomento)

CEPAL Comisión Económica para América Latina y el Caribe

EV Electric Vehicles

GBV Gender-Based Violence

GHG Green House Emissions

GIZ German Corporation for International Cooperation GmbH

IADB Inter-American Development Bank

ICRW International Center for Research on Women

IFC International Finance Corporation

ILO International Labor Organization

PISA Program for International Student Assessment

PMU Project Management Unit

TUMI Transformative Urban Mobility Initiative

UN United Nations

WB World Bank

WHO World Health Organization

Introduction

The Green Climate Fund acknowledges in its Gender policy that its initiatives to combat climate change are more sustainable, equitable and have more probabilities of reaching their objectives when gender equality and women empowerment questions are included in designing and implementing its programs and projects.

The role of the E-mobility and low carbon transportation multi-country facility to support innovation and implement measures to combat climate change from a gender perspective, to contribute to reducing the gender gap, and to prevent and minimize harm is three-folded: First, the transport sector is a catalyst to change social norms and this facility will do it in many ways, like by raising awareness on sexual harassment and violence against women and girls, by providing visibility to women working in non-traditional roles in the EV (electric vehicle) markets, and by considering the mobility of care when designing transport operations. Second, this new e-transport paradigm will lever women's access to jobs in traditionally sex-segregated occupations or improve access to investment loans for female entrepreneurs. Third, the capacity building, training, and sharing of know-how to women and men designing the investment plans in the program will systematize gender mainstreaming in the transport sector.

The E-mobility and low carbon transportation multi-country facility seeks to both directly and indirectly reduce Green House Gas (GHG) emissions from the transport sector in Latin America through electrifying intensively used vehicle fleets and generating structural changes in the sector that will lead to further EV adoption (and therefore higher GHG reductions and social co-benefits). The program is split in three components:

- Component 1: Establishment of a sustainable mobility ecosystem encouraging electrification and shift to public transport (TA)
- Component 2: Investive measures for PT mode shift (FA)
- Component 3: Deployment of EV fleets (FA)

This gender assessment provides information about gender equality and the empowerment of women as well as its links with the transport sector in the following countries: Argentina, Brazil, Colombia, Costa Rica, Dominican Republic, Mexico, and Peru. The assessment is organized into two sections that feed the recommendations that will design the proposed Gender Action Plan.

The first part covers data on relevant indicators to offer a snapshot of women's unique situation in each of the seven countries. The large amount of available data related to gender equality and women's empowerment prompted a selection of those indicators with available and updated data that were reliable across different sources. This section summarizes the most recent data and analyses found in ad hoc country studies on gender or data sources from reliable international entities. The indicators selected reveal gender gaps in education, labor force, and political participation and furnish data on maternal mortality, adolescent pregnancy, and gender-based violence. For this assessment, most data are the national average or referred to urban women. The unique multidimensional and overlapping inequalities of each country has been included slightly because of the complexity of an assessment for seven countries, and only when a particular report provided strong data. This section finishes with a sub-section that describe laws and regulation in each country that can foster or limit economic opportunities for women analyzing seven indicators: mobility, workplace, pay, marriage, parenthood, entrepreneurship, assets and pension.

The emphasis on displaying the most recent country data available derives from the current COVID-19 pandemic situation. The numbers are changing and proposing the most recent data will allow us to better know these countries' situation in terms of gender equality before the health crisis. Nevertheless, the first section begins with a small summary that offers available information about the pandemic's impact on the situation of women in the Latin American and the Caribbean region.

The second section presents initiatives and best practices carried out on these countries and the region on gender equality and urban transportation. Given the large huge number of initiatives on gender and urban transportation going on in these countries and in the region, this report is presenting best practices grouped by topic and activities to facilitate the reading. This second section is structured in three interrelated topics: mobility, safety, and jobs. After the sub-section about jobs, there is a specific subsection on the impact of the pandemic on women's employment, which has used the available data at the time of drafting the assessment.

Following, a third section describes the capacity of the AFD, PROPARCO and GIZ to implement, monitor and evaluate the gender actions proposed in the GAP

Finally, the assessment concludes with the recommendations drawn to guide the Gender Action Plan.

Section 1: Gender assessment by country

Summary

The last two decades brought significant advances for women in Latin American Countries. All of the countries in this program ratified the Convention Belem do Pará, whose main achievement was to incorporate specific legislation to include “femicide” as a specific aggravated crime. Since its ratification in 1994, countries have been advancing to prevent, sanction, and eradicate violence against women. In general, for the Latin American and the Caribbean region, Gender Based Violence (GBV) is endemic and underreported, and before 2006 no country in LAC established femicide as a crime, while by 2015 16 countries already did. Likewise, in many parts of the World, civil society has raised strongly against GBV and Argentina, Chile, México, Perú, and Uruguay have marched under the motto “Ni una menos” (Not one woman less).

The seven countries have included quotas of around 30 percent for political representation, and many countries have reached it at the national level yet at the local level, most lag behind. These countries have advanced enormously in maternal health, but inside every country, disparity persists geographically and by ethnic group or race. Parity in education is mostly reached in all countries at the primary and secondary levels, and in many cases, girls are outperforming boys; whereas in tertiary education attainment varies by country, and careers in STEM (Science, Technology, Engineering and Maths) remain predominantly masculine. Occupational segregation persists across countries and women’s careers concentrates in low paid jobs and informality, which moreover, have little and low benefits, resulting in lower earnings and well-being. Despite many laws, gender wage gaps persist across sectors and gender roles impede women’s access to better economic opportunities since domestic responsibilities are mostly in their hands.

Education in girls is affected by adolescent pregnancy, which is still high and despite the downward regional tendency in fertility rates, it is only exceeded by Sub-Saharan Africa. Marriage and sex are illegal for minors in all these countries, yet teenage pregnancies and minors living in union occur. Plans, strategies, and entities (Ministries or agencies) have been given more power or have been recently created. These entities are not funded enough to carry on, enforce, and follow up with all the gender equality agenda.

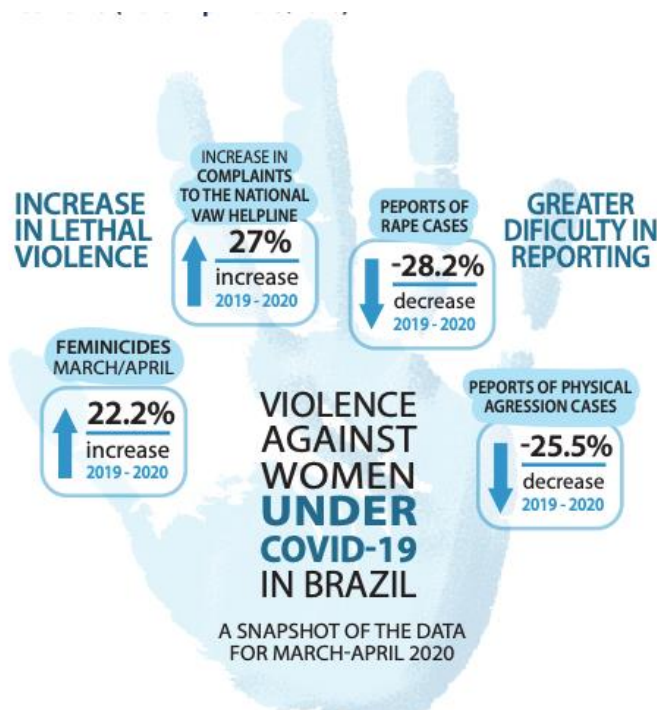
The outlook to both keeping the progress and continue reducing the gaps in gender equality is challenging after the COVID-19 pandemic. UN Women and other international organizations fear that recent gains in gender equality might be lost. The COVID-19 pandemic has worsened women’s situation and has aggravated differences amongst groups. In Brazil, the maternal death rate due to COVID-19 is twice as higher for black women than for white ones¹. Gender based violence has increased and women in lockdowns are living with their aggressor under the same roof and can’t access support networks. The figure 1 below shows how only in Brazil, from March to April 2020 compared to the same period in 2019, there has been an increase of 22 percent in femicides and a decrease of 28.2 percent in reporting cases of rape. Complaints increased 27 percent and reports for physical aggression cases decreased by 25.5 percent². In Colombia the calls to the domestic violence helpline increased by 91 percent, in Mexico by 36 percent, in Argentina by 25 percent (7 out of 10 murders happen at home). It is estimated that less than 40 percent of domestic violence survivors report it; adding to that, figures of actual gender-based violence

¹ UN Women 2020. From insight to action: Gender equality in the wake of Covid-19

² WB Ag 2020 Addressing Violence against women under COVID-19 in Brazil

are worse during COVID-19 because the fear of getting infected prevents women from seeking help in health services. Lack of critical resources and access to key reproductive and health services due to stay-at-home orders might have increased teenage pregnancies and maternal mortality. There is evidence that other forms of violence, such as sexual exploitation and abuse have increased in others health crises³. Many women have abandoned the labor market to lead their household's dynamics during the pandemic, with severe lockdowns in most countries in the region and schools and workplaces closed, while access to decent work has deteriorated for all. More information about the impact on women's jobs is included latter at the end of section two.

Figure 1. Snapshot of violence against women in Brazil during Covid-19 (March-April 2019/2020)⁴



Following, the report presents women's unique situation by country to allow the reader to understand the realities of women in each of the seven countries. As mentioned before, analyses from country gender profiles and ad hoc reports have been put together with the most up-to-date indicators found from reliable statistical data sources. It is not intended to be exhaustive, but given the number of countries and difficulties to compare them, this part of the report seeks to provide the panorama of each country presenting the same indicators to facilitate the organization of the information. To finish this section, there is analysis on the laws and regulations that limit or foster economic opportunities for women in these countries.

³ WB April 2020 Gender dimensions of the COVID-19 Pandemic

⁴ WB Ag 2020 Addressing Violence against women under COVID-19 in Brazil

Argentina⁵

Maternal mortality rate is low: 39 per 100.000 births in 2017 (CEPAL), and all Argentinian pregnancies received professional care.

13 percent was the adolescent pregnancy ratio⁶ in 2010. In 2015, adolescent pregnancy was 63 per 1,000 women. This rate can be related to the rate of young adults (1 of 5) who are not in employment, education and training, which almost 25 percent of them are women. (IADB 2015)

252 women died for gender reasons, 179 women of them died by their intimate partner in 2019 (0.8 per 100.000 women) (CEPAL). In 2016, 100.000 cases of gender-based violence were registered, where the perpetrator was mostly an intimate partner. Data is scarce, often underestimated and data collection presents challenges. Despite having various institutions collecting data, the protocols differ, variables are not harmonized, victims lack confidence and economic autonomy, mistrust the authorities or are unfamiliar with the processes. In its first two years of operation, the free phone line for women victims of gender-based violence received 60.000 calls (2013-2014) (WB 2018).

Parity is reached in elementary education enrollment and completion and girls slightly surpass boys in secondary enrollment (92 and 87 percent respectively in 2017), but non-indigenous women are four times more likely to finish secondary school than rural indigenous women. Argentinian girls drop out of secondary school rate is 30 percent, and their reasons are pregnancy, motherhood, or an intimate relationship, whereas only 15 percent left school to work (data from PISA 2012). In contrast, 42 percent of boys left school to work, and 24 percent reported they did not like school.

49 percent of women participate in the labor force, compared to 73 percent of men, and for younger adults (15-24), the rate drops to 30 and 45 percent, respectively. Women's domestic responsibilities and high teenage pregnancies contribute to the low integration of women into the Argentinian job market in terms of quantity and quality. Informality represents 30 percent of wages employees, and more than half of women (except in Patagonia) who have not completed secondary education are informal, a proportion that reaches 70 percent in the northern regions of the country (WB 2018). This undermines the acquisition of skills for the less educated people while women's high domestic responsibilities prevent their integration into labor force (IADB 2016).

39 seats are held by women in the national parliament in 2018 and the female rate for ministries is 17 percent. The previous 30 percent quota for women in candidate lists was increased to 50 percent in 2017.

⁵ Data come from "Argentina Group Country Strategy IADB 2016–2019", "Argentina: Escaping Crises, Sustaining Growth, Sharing Prosperity, Systematic Country Diagnostic", 2018 WB, and unless otherwise specified, from CEPAL and The Little Data Book on Gender 2019

⁶ Adolescent pregnancy rate: Total of young women, 15-19 years old, who declare to have had a living baby divided by the total number of women in the same age group, multiplied by 100 (CEPAL)

Brazil⁷

Likewise, in other countries in LAC the Brazilian poorest live in rural areas, but 60 percent of Brazilians poor are urban people. The larger group affected by poverty are Afro-descendants, and extreme poverty hits indigenous people most. Moreover, the multidimensional face of inequality in Brazil is female as well, since women female-headed households are overrepresented below the poverty line. Black women, feminists, lesbians, domestic workers, indigenous women and more women's groups have shaped the women's rights agenda from its many faces and priorities. This gender overview for Brazil comes almost entirely from the WB Snapshot of Brazil in 2016, which was marked by race and geography.

44 women die for 100,000 births in Brazil in 2017. In 2014, 65 percent of maternal deaths were Afro-Brazilian women and 30 percent were white. Almost all Brazilian women receive prenatal care, but access for afro-descendants and indigenous peoples is different than for white women. 75 percent of white women get seven or more prenatal visits compared to roughly 50 percent of afro-descendant women and 25 percent of indigenous peoples' women.

Adolescent pregnancy rate remains high despite the downward trend since the year 2000. In 2017, the rate was 62 births per 1000 women aged 15-19.

Femicide reached 1.941 women in 2019. Rate of violence against women by intimate partner in Brazil is 16.7, and 40 percent of Brazilian women report to have suffered physical, sexual or psychological gender-based violence. Indigenous women almost double non-indigenous gender-based violence average. Domestic and Family Violence's Law in Brazil, known as Maria da Penha's Law, is worldwide known for its comprehensiveness but lacks funding for both implementation mechanisms and enforcement at the sub-national level. Men suffer heavily gang violence and homicide in Brazil.

Parity is achieved in elementary and secondary education while women exceed men in tertiary education enrollment rates in 2017, with 59 percent and 43 percent respectively. Women expected years of schooling in Brazil for 2018 is 15.8, almost 10 years more than in the year 2000 (14.5) compared to 15.1 years for men. In contrast, when analyzing racial and geographic inequalities, 13-16 years old girls in the Southeast and South are more likely to attend school than girls in other regions (poorer), and white men are more likely to attend school than Afro-descendants. At tertiary level less than 50 percent of Afro-descendants attend school compared to whites. Same for women studying Science, Technology, Engineering and Mathematics in tertiary education who represent 36.6 percent graduates compared to 62.4 percent men.

Female labor participation hasn't changed much over the last decade and remains low (59 percent in 2014). Gender wage gaps are exacerbated by race and persist in most educated women. Afro-descendant women wages are less than half than white men's wages.

Female candidates increased 47 percent on the 2014 general election thanks to the 30 percent quota imposed. However, it didn't reflect in the Chamber of Deputies with 9.9 percent of women seats (compared to 8.8 seats secured in the previous 2010 election) and 13.6 percent female senators, which is below the average of 26 percent politicians in LAC parliaments. At the state level only 1 female governor

⁷ WB (2016) A Snapshot of Gender in Brazil Today Institutions, Outcomes, and a Closer Look at Racial and Geographic Differences. The most recent data come from the Little Data Book on Gender 2019

of a total of 27 was elected in 2014. The numbers drop from the previous election in regional parliaments and only 11.8 percent mayors are women.

Colombia⁸

Women's access to health services has brought numerous positive results in maternal health, but adolescent pregnancy rates' improvements are slow. Mothers' mortality at birth or during pregnancy have dropped from 118 per 100,000 births in 1990 to 94 in 2015. In 2015, 90 percent women had access to natal check-ups, and 96 percent of women had access to a skilled professional at delivery.

Teenage pregnancy ratio in 2018 was 11.8 percent. Increase in the use of contraceptives has caused adolescent pregnancy rates (15-19 years old) to fall from 80 birth per 1,000 women in 1990 to 50 in 2016. From 2000 to 2015 the percentage of women who reported having the final decision on their health to fell to their partner or someone else halved (from 12 to 6 percent). This last trend is more obvious among urban and educated women. However, limited women's agency⁹ causes that 20 percent of women refuse to use contraceptive methods. 18 percent of urban female teenagers and 29 percent of rural female teenagers, allege opposition of their partners or someone else. Despite sexual relations with minors is forbidden by law, between 2016 and 2017, the number of children born to girls aged between 10 and 14 years old increased 4 percentage points.

Gender based violence is endemic in Colombia. Data from 2015 shows that almost 48 percent of married women and 27 of women in union have reported physical and emotional violence by their partners. 27 percent reported having suffered physical violence while 18 percent reported being threatened with violence. 128 femicides were registered in 2016, a 12.3 percent more than in 2015. The economic cost of violence in general represents 11 percent of the economy (Londoño and Guerrero 1999)¹⁰ while the economic cost of domestic violence alone amounts to 4 percent of GDP (Rivero and Sánchez 2004)¹¹.

Girls' education enrollment and achievement have reached equality, and Colombian girls are doing better, but when it comes to secondary and tertiary girls' attendance features an adverse pattern. Both boys' and girls' education rates have improved over time; in secondary education 82 percent of girls compared to 76 percent of boys are enrolled. Whereas in tertiary education 63 percent of women are enrolled compared to 54 percent of men. However, patriarchal family dynamics, involvement in household responsibilities and social norms endanger Colombian girls' education as absenteeism and dropout out rates start appearing in secondary and tertiary education. Compared to boys, twice of girls aged 16-24 were not in education, training or employment in 2017.

Economic opportunities have grown for women but the economy of care falls disproportionally in their hands since 37 percent of urban women report to be engaged only in domestic tasks. In 2017, 57.7 percent of Colombian women participated in labor force (more than 18 years old) compared to 80.8 percent Colombian men. Moreover, when Colombian women find jobs, these are usually poorly paid and of low quality. Only one third of women contribute to a pension fund compared to more than fifty percent of men.

⁸ Data from WB June 2019, Colombia Gender Assessment

⁹ Agency is the capacity to make decisions about one's own life and act on them to achieve the desired outcome, free of violence, retribution, or fear.

¹⁰ Rivero, R., y Sánchez, F., (2005) Determinants, effects and costs of domestic violence. Documento CEDE Edición electrónica.

¹¹ Londoño, Juan Luis and Rodrigo Guerrero (1999). "Violencia en América Latina – Epidemiología y Costos." Inter-American Development Bank, Documento de Trabajo R-375

Colombian women's voices are heard in public spheres. The civil conflict affected women's rights and gender inequalities for decades. Colombian women have been actively involved in 2018's peace process highlighting and incorporating gender related issues in the discussions. The peace agreements have declined the likelihood of men suffering from violence, but the risk for women and girls remains high.

Legislation has advanced women's rights but enforcement is lagging behind in many. In politics, despite the 30 per cent quota in the candidate' list only 18 percent members of the parliament were women in 2018. Sex with minors is illegal, but in 2015, 5 percent of women were married before they turned 15 and 23 percent get married before turning 18.

Costa Rica¹²

27 mothers per 100.000 births died in 2019, since the year 2000 (74 deaths per 100.000) the rate is dropping but regional differences persist.

Adolescent pregnancy has been dropping in the last years, from 18.7 percent in 2010 to 12.5 percent births born from women below 19 years old of in 2019. Adolescent pregnancy is associated with the beginning of sexual relationships and despite being illegal, adolescent-adult sexual relationships occur. 56.9 percent of girls whose first sexual relationship happened when they were between 10 and 14 years old, report that their partners were 5 years older. Whereas 8.5 percent of men older than twenty years old report that their partners had between 5 and 10 years old less than them, which means that when sexual intercourses have occurred, they have incurred in a crime¹³. 74.5 percent of girls and teens who report living or having lived under some kind of conjugal bond do not attend school, compared to those who are not (17 percent)¹⁴. Moreover, 88.3 percent of the girls reporting living or have lived under conjugal link declares not to work and 57.3 percent reports working in not paid domestic work (UN Population Fund).

14 women died for gender reasons (femicides) in 2019 (0.6 percent per 100.000 women), 9 of which (0.4 percent per 100.000) were killed by their intimate partner or ex (CEPAL 2020). 57.7 per cent of women surveyed in 2004 were victims of at least one incident of gender-based violence sometime after they turned 16, whereas 24.2 percent reported more than 4 incidents. Older women, between 25 and 49 years old, reported higher rates (62 percent) and 48 percent reported being victims before turning 15 years old (ENVCM, Household Survey Costa Rica 2004).

Only around 50 percent of working age women are included in the Costa Rican labor force, which in 2019 was 54.1 percent. 76 percent of them work in low productivity sectors. The Latinobarómetro 2015 found that 43 percent of Costa Rica's citizens think that women should only work outside the household if their partner is not earning enough (Only 20 percent in Chile's citizens think the same). 32.7 percent of women in Costa Rica (29 percent is the average in LAC countries) don't have their own income compared to 10.5 percent of men. Besides, women spend more than double of their time in unpaid domestic work (33:38h per week vs 13:26h per week)¹⁵

¹² Unless otherwise specified, data come from Instituto Nacional de Estadística de Costa Rica (National Institute of Statistics), INEC 2020

¹³ The Penal Code in Costa Rica under its articles 156, 157, and 158, forbids expressly sexual intercourse with minors of 15 years old, despite victim's consent, and considers sexual relationships with girls under 13 years old a rape.

¹⁴ X Censo de Población y VI de Vivienda del año 2011-INEC (Population and Household Survey 2011, Costa Rica National Statistics Institute)

¹⁵ INEC 2017, Encuesta nacional del uso del tiempo (National survey on time use)

The education gender gap is closed and near all the population is literate. Girls school attendance is similar to boys in elementary and secondary school. While in tertiary education women's attendance rate is superior than men (60.7 percent vs 50 percent), this does not result in more women participating in productive careers or jobs¹⁶.

55.17 percent of ministries are held by women since 2018, and 45.6 percent of the seats in the congress are occupied by women. At the local level, 14.8 percent of majors and were women and 45.6 of city councilwomen in 2018. Costa Rica is one of the best performing countries in gender equality politics not only in the region but in the World.

The Dominican Republic

The poverty gender gap in the Dominican Republic is the largest in the region¹⁷. At the end of 2015, the poverty gap was 4.6 percentage points. In the case of women head of household, the poverty gap raises to 9.1 percentage points, and the highest gaps are for non-educated women (45.6%). People with Haitian origins have worse poverty indicators than the 20 percent poorest of the population; 4.8 percent of the country's population are Haitian women.

95 Dominican mothers died per 100.000 births in 2018 (92 in 2015), while the regional average is 74. Around 100 percent of Dominican women have access to prenatal care and skilled professionals at delivery. Still, the quality of pregnancy and delivery supervision seems to be the leading cause of the high mortality rate. The mortality rate for sexual and reproductive diseases (other than maternal deaths) was 50.25 per 100.000 in 2015.

Dominican Republic ranks first in adolescent pregnancy in the region: 93 birth per 1.000 women, while LAC average is 62, and the average fertility rate for the country is 2.3 children per woman (World Bank 2018). Low educated and poor teens are more affected (50.4 percent women with 1-4 years of education, and 35.1 percent of the lower quintile). Teen pregnancy is related to rape in childhood and teen ages: The probability of getting pregnant before 18 years old for women who have experienced intimate partner violence is 1.86 times higher than women who haven't had this experience. Early pregnancy's probability is 3.26 times higher for women whose abuse happened before or at 18 years old. Dominican women start sexual life early and often with older men, and for women who had two or more intimate partners only 37.8% used condoms in their last intercourse (2013). Around 67 percent of women use a modern contraception method (similar to the regional average in 2016).

148 Dominican Republic women died in 2019 victims of gender-based homicides (CEPAL) and the country ranks first in intimate femicides in the LAC region with 1.5 cases for 100.000 women (CEPAL). Intimate partner violence is related to controlling behaviors and drinking alcohol. 51.3 percent of women who suffered intimate partner violence reported that their partner got drunk regularly compared to 12.5 percent of women whose partner consumes moderately or never. Controlling behaviors produce similar data: 64 percent use 5 or 6 types of control, while only 8 percent don't use controlling behaviors. HIV and GBV are tightly related since prevalence is 2.6 times higher among women who have experienced violence compared to women who haven't (CONAVIHISIDA, 2014). Every day four women died due to HIV/AIDS,

¹⁶ World Economic Forum 2020 Global Gender Gap Report 2020

¹⁷ Unless, otherwise specified, data comes from IADB 2018 Desigualdades de Género en República Dominicana, 2018-2020

six times the regional average, while mobility situates the Dominican Republic the second in the Caribbean. In 2015 there was no gender gap in HIV/AIDS data.

Girls' enrollment rate equals boys for primary education at 93 percent in 2017 (85 and 84 percent respectively in the year 2000), and exceeds boy's (74 percent compared to 66 percent). The elementary completion rate is similar for both sexes, 94 percent of girls and 96 percent for boys. It exceeds slightly for girls in secondary education at 88 percent compared to 82 percent of boys completing secondary school. Women's graduation ratio almost doubled men's with 42 percent and 21 percent, respectively. (WB 2017)

Labor force participation hasn't grown much for women or men in the last two decades. In 2018, the female labor force participation rate was 57 percent women and 82 for men. Vulnerable employment continues being in the hands of women, who holds 57 percent of all vulnerable employment (39 percent is for men's)

Women have doubled their presence in the public sphere in the last years: 27 percent of women occupy the parliament in 2017 (16 in the year 2000). A similar figure, 28 percent of women, have been elected in local government bodies. The female share of women's employment in senior and middle management is 50 percent of the total, compares to 37 percent in 2000. Twenty-three firms have a top female manager and 70 firms have female participation in ownership.

Mexico¹⁸

Poverty in Mexico is feminine, indigenous and rural. Mexico ranks 1st in indigenous people population. 17 percent of population lives in four states with the higher level of indigenous peoples (Chiapas, Oaxaca, Guerrero, and Veracruz) and accounts for 37.5 percent of the population living in poverty (IADB, 2019). For the purpose of this assessment, differences within Regions won't be included and data will include the country's average.

Women access to maternal health has improved in the last decade. Maternal deaths have decreased from 86 deaths per 100,000 live births in 1995 to 38 in 2015. Mexico's social inequalities produces different levels of access to maternal care, being the rural, poor and indigenous women those with lower access. Maternal mortality rates for indigenous women doubles those of non-indigenous women.

Adolescent pregnancy rate remains high. Fertility rate has been decreasing from 157 births per 100,000 adolescents in 1957 to 74 births per 100,000 adolescents in 2015. Knowledge of contraceptive methods is high in teenage girls, compared to adult women. 18 percent of teenage girls who did not used contraceptive methods reported they wanted to get pregnant. Contraceptive use is low on average, but there are differences by regions and by level of education. 14.6 percent of women without education did not know any method compared 0.6 percent with completed secondary. Studies show that teenage pregnancy's impact on girls reduces in 0.8 years school attainment in the short run and they have one year less of education in the long run.

¹⁸ Unless otherwise specified the analysis comes from WB (April 2019) Mexico Gender Assessment and IADB (Nov 2019) Mexico Group Country Strategy 2019-2024

Urgent action needs to be taken to prevent gender-based violence: seven Mexican women per day were killed in 2016¹⁹. In absolute numbers 983 died in 2019 (1.5 per 100.000 women) (CEPAL 2019). Likewise, in other countries in the region advances in legislation haven't been translated into enforcement to protect women. In the case of Mexico, men and women's homicides have followed the same pattern since they started to rise after 2007 war on drug cartels. Since homicides on men are higher Mexico's gender-based violence is overlooked and women don't seek for help probably because of the social acceptability of gender-based violence. Mexican women older than 15 have experienced at least one violent incident (66 percent), intimate partner violence (44 percent), and sexual violence in public spaces (34 percent). Despite intimate partner violence is the most common in Mexico, violence from aggressors' others than intimate partners is increasing.

Prevalence in diabetes in women is related to the rates in female obesity, especially in urban areas. 7 out of 10 Mexican women above 18 years old are considered overweight, and 33 percent of them are obese compared to 24 percent of men. There is evidence of gender discrimination against obese women and a study highlighted that the probability of being called back after an interview is low for obese women compared to men. The low call back rate affects dark-skinned and married woman as well.

Education gender gaps are closed, less girls than boys are out of school but the gap is reversing for boys. On average, Mexican women have 9 years of education compared to 9.3 for men. In 2016, 92 percent of girls graduated from secondary education compared to 86 percent of boys, whereas 28 percent of female students completed tertiary compared to 23 percent of men. Women's enrollment in college for 2017/18 represented 50.3 percent, 55.2 percent of graduates, and 48.8 percent of graduates' in PhD. However, data differs across regions with high risks of girls drops out in some regions such as in Chiapas where girls have almost one year less of education than men. Primary and secondary attainment gap is closed, but at the tertiary level women's attainment is lower than men.

Female and youth labor force participation is low and the care economy is in women's hands. Women's labor force participation in Mexico, 45.5 percent, and there is a gender gap in income of 16.5 percent. In terms of per capita GDP, Mexico is the 5th wealthiest country but according to IDB's Better Jobs Index Mexico ranks 13 out of 17 countries in the region. The gender gaps in labor participation are due to: (i) a greater likelihood of career interruptions, (ii) occupational and sector-based segregation, gender preferences (and/or limitations) in the case of longer or shorter remunerated working hours, (iii) employer discrimination, and (iv) women's responsibilities in the household, which limit the time dedicated to remunerated work. Mexican women do 77 percent of housework and spend six hours daily compared to two of men in unpaid housework. 30 percent of employees in Mexico work 40 hours per week, which constitutes a barrier for women seeking a paid job. Lack of reliable childcare is another barrier for women seeking a job.

Women's public representation almost reach parity at federal level and ranks the 4th country on the world with most women in parliament, but numbers remain low at local levels. A quota requires 50 percent of women on candidate list in elections for national and state congress as well as municipal councils. In 2018²⁰, 48.2 percent of deputies of the federal parliament and 49.2 percent of senators are women.

¹⁹ ONU Mujeres/ SEGOB/ INMUJERES (2017). The long road to justice, prosecuting femicide in Mexico

²⁰ Observatorio de la participación política de las mujeres en Mexico
<https://observatorio.inmujeres.gob.mx/mvc/view/public/index.html?ms=MzE=>

Peru²¹

69 women die during pregnancy or 42 days after delivery in Peru (PAHO, 2017). Maternal mortality is now similar to the regional average (74). All pregnancies receive follow up care in Peru and almost all deliveries (94 percent) are attended by professional staff. Data for rural, mostly indigenous people women, are worse. However, the Government have many policies including and respecting indigenous people's preferences in order to increase maternal care access among this group.

48.5 births occurred over 1000 women age 15-19 in Peru (WB 2018). That represents around 10 percent of adolescent girls in Peru (Cepal 2017). Even though is lower than the regional average, it differs among groups; the poorest quintile had 24.5 percent of teen pregnancies while rural women reached 22.5 (compared to 10.8 percent in urban areas) in 2015.

128 women died in femicides in 2019 which represents a 0.8 per 100,000 women, 98 are perpetrated by intimate partners or ex-partners (0.6 per 100,000 women) (CEPAL 2019). It shows a decreasing tendency over the last ten years, together with a decreasing tendency in intimate partner violence. In 2016 a 64 percent of victims were between 18-34 years and half of the cases occurred inside the house (between 2009-2016) (Instituto Nacional de Estadística del Perú). In 2014, 70.8 percent of women reported having suffered violence from their husband or partner, mostly being psychological (67.4 percent), 32 percent physical and 7.9 percent sexual violence. Even though reporting has increased, most victims don't search assistance because they don't trust the institutions and social barriers impede them to leave abusive relationships. (WB 2018)

Parity is reached in primary and secondary education enrollment, although women in the poorest quintile are less likely to enroll in secondary education. Completion is higher for girls who outperform boys in Peru. The gross tertiary graduation ratio is 61 for women and 43 for men, this gap has been increasing since 2017 at the advantage of women (WB 2018)

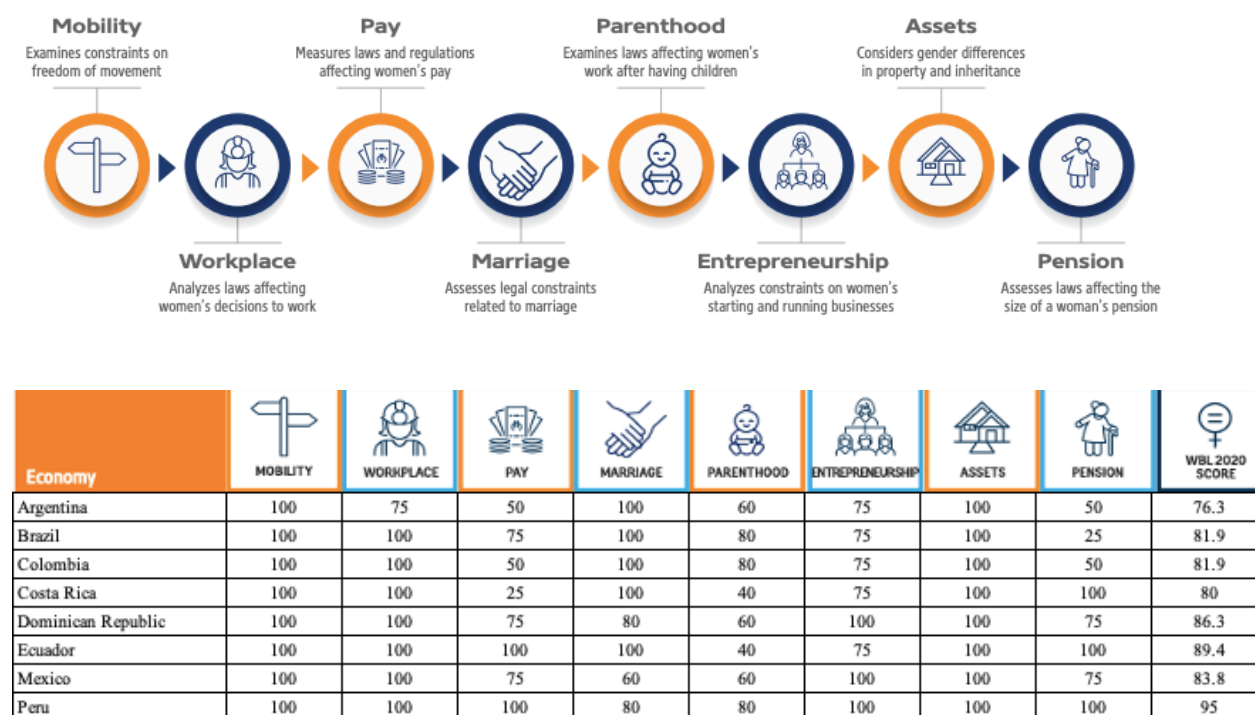
Female labor force participation rate was 70 percent for women and 85 percent for men in 2018 (Little data book on gender 2019). Larger gaps in labor force participation are found in married women, or women with kids, afro-Peruvian women and indigenous people women. Peru's ranks among high in female labor force participation in the region. Between 1993 and 2007 out of the total increase in the women's employment rate, 27 percent can be attributed to the drop in Peruvian's women fertility rates, which went from 3 births per woman in 1996 to less than 2.5 in 2015. However, over twice as many women (23.6 percent) in the group 15-29 years old are neither working or studying (ni-nis) compared to men, but the share is lower than the regional average. This share seems related to gender roles and the use of the time.

26.2 percent of women are represented in the Low Chamber (Cepal 2019). The 30 percent gender quota was established in 1997 and in 2017 women occupied 31.6 percent of ministries and 37 in 2019, although most occupy ministries in social areas. Only 2.9 % of mayors are women, this percentage has not changed in Peru since 1998 and is low compared to the regional average. The national political fragmentation impacts the local government which many don't have representation from the larger political parties, and is likely responsible of the lack of trust by the population in institutions.

²¹ Data come from "Gender Gaps in Peru" WB 2018, CEPAL and the Little Data Book on Gender 2019

Laws and Regulations on Economic Opportunities for women

Countries have laws and regulations that foster or impede women's access to economic opportunities. The report "Women, Business and the Law 2020" illustrates with seven indicators the legal differences between men and women as they transition through different stages of working life²². The table below collects the scores of our seven countries in LAC. A score of 100 means that there are no legal constraints on women's freedom in a particular indicator, whereas a score less than 100 indicates at least one legal constraint. Norms and practices are not captured by the indicators; only laws and regulations are considered.



The table reveals that in the 7 Program countries, women do not have legal constraints for movement. The mobility indicator is based on four questions: Can a woman choose where to live in the same way as a man? Can a woman travel outside her home in the same way as a man? Can a woman apply for a passport in the same way as a man? Can a woman travel outside the country in the same way as a man? Laws and regulations in property and inheritance are not discriminative for women in any of these countries either.

The best score as a country is for Peru, with the only restrictive laws are in marriage and parenthood, while the worse score is for Argentina. In Argentina, a law states that women are not allowed to work at night (not applicable to nurses, domestic workers and public shows) neither are allowed to work in cleaning of moving machinery, nor can they be an engine driver stoker. The former is in line with the differential in the pay indicator: women in Argentina cannot work in specific tasks (deemed dangerous) thus, they can be excluded from some higher paid jobs. Similarly, in Costa Rica, laws do not mandate equal remuneration for work of equal value and women are not allowed to work in jobs deemed hazardous,

²² Women, Business and the Law 2020, WB. All the data elaborated in this section comes from each country's analysis for 2020.

arduous or morally inappropriate, which affect its score in “pay”, which is the lowest of the seven countries.

Equal wages and equal rights to a pension (such as retirement or childcare that accounts for pension benefits) are only protected in Costa Rica, and Peru. Whereas, in Brazil, the country with the lowest score, women are not legally allowed to retire with full or partial pension benefits as men, and the mandatory retirement age is different (65 for women and 70 for men). Parenthood laws in Costa Rica are worse than in the other countries. In Costa Rica, parental leave is not paid, there is no paid leave to fathers and the government doesn’t administer 100% of maternity leave benefits.

Analyzing entrepreneurship, Argentina, Brazil, and Colombia do not have laws that expressly prohibit discrimination in access to credit based on gender. Whereas in the other countries discrimination against women is prohibited explicitly in the credit or financial institutions’ laws or general laws like those on Equality of men and women (ex: Mexico).

Most gaps in education (enrolment and attainment) in the seven countries have been closed, but it has not translated into more and better economic opportunities for women. In Colombia, estimates on the potential economic losses over the long term due to gender gaps in the labor market amount to 17.6 percent in gross income per capita. In education, not only women outnumber men, but their skills and preparation are now as high as those of men, if not higher²³. Colombia’s population is aging, as in many countries in LAC region, and to keep with the economic growth, the untapped potential of unemployed and inactive women should be considered. Adequate investing in women’s human capital can increase women’s productivity as well. The former will be particularly important after the COVID-19 pandemic, as we will see later, because women have stepped back and have taken over the care economy, leaving behind the pursue of a career.

Section 2: Overview of Gender Equality and Urban Transport links in these countries

According to UN Habitat, two-thirds of the population will be living in cities by 2050 (UN Habitat, 2018). Together with pollution, overcrowding, lack of modal integration, are some of the problems that the increase of urban population is exacerbating. 46.2% of surveyed people in Peru responded that Transportation is the second most important problem affecting their quality of life (first is insecurity) (IADB, 2019). As this report revealed at the end of section one, women from these countries do not have limitations of mobility in the legal structures, but when we look at their agency freedom of movements differ. Agency in mobility is the ability to make and act upon decisions related to one’s mobility and make full use of public transport systems (WB, 2020). Lack of access to transportation and safety issues limits women’s participation in the labor market and reduces their participation in the economy by 16.5 percent²⁴. Women turn away working opportunities depending on the traveling conditions to the work station (time spent, return timing, expenses). Besides providing access to education, health and jobs, the sector generates jobs itself but traditionally most opportunities are likely to remain in men’s hands. Hence, when we use gender lens to analyze urban transportation, we discover that the conditions, priorities, roles and beliefs about urban transport infrastructure and services vary between women and men and, consequently, decisions about mobility are constrains. The lack of certain safety features in the design of the transport system can constrain a woman’s decision to choose a certain mode of

²³ WB 2018 Colombia Gender Assessment

²⁴ ILO 2017 World Employment and Social Outlook: Trends for women 2017.

transportation, sometimes in ways that are inoculate to men, such as dark bus stops or overcrowded buses. When it comes to the design of the transport infrastructures, the realities of women, minorities, the elderly and people with disabilities are many times alike and need be taken into account when designing a transport operation. When they serve women, they are adequate for the overall population and enhances quality of the system.

As we see, there is a myriad of links between gender and urban transportation, and these links have been intensely studied in the last years. There are gender differences in the purpose, time, mode of transportation, or perceptions on safety, that affect women's mobility agency and access to economic opportunities, and the characteristics comprised in this section are common to the countries in the region. For the purpose of organizing the information the initiatives are grouped in three dimensions that are interrelated: (i) mobility, (ii) safety, and (iii) jobs. Each dimension reveals the unique situation for these countries and groups activities and best practices; initiatives for other countries in the Latin America and the Caribbean region are included occasionally. The first sub-section about mobility patterns includes an analysis of gender-segregated ride-hailing. Whereas, the third sub-section on jobs, covers the impact of the pandemic on women's and men's jobs.

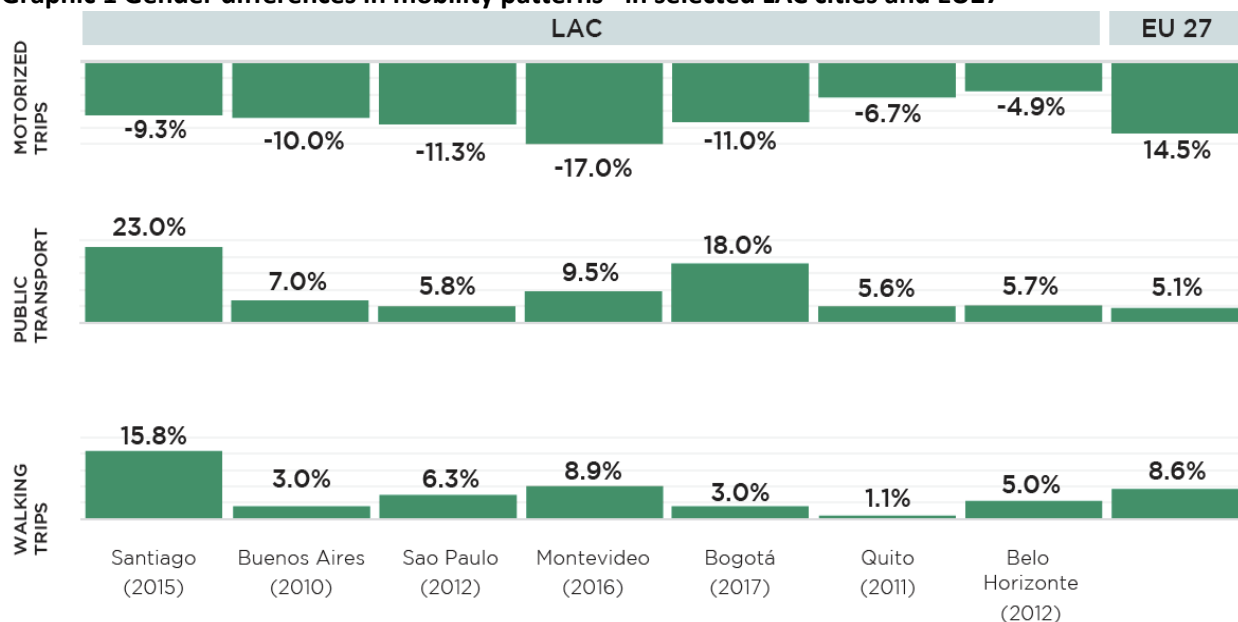
Mobility in Urban Transport

Female urban transport ridership characteristics differ from that of men. Most women are captives of public transportation because either they don't own a private vehicle or they don't have the means to use one (gas, insurance). For long distances, they become dependents on public transportation (AFD 2015). Women rely more on public transportation than men, especially buses; women chain different destinations in a single trip (Ex: drop kids at school, buy food, visit relatives in need, or get to work), and many times they travel off-peak hours. They are often accompanied by others (Ex: children, elderly) and carry bags. Their commuter times are often longer and more expensive. This is because women combine multiple wage jobs and care responsibilities (family errands, care for the elderly, the ill, the kids, and their community) in one trip. Sometimes, the nature of the wage work (Ex. As part-time nannies, providers of cleaning services) requires getting on and off in multiple destinations, and when multimodal fare is not in place, they have to buy a new ticket every time, or just walk more. If there is not public transportation available, they just walk or decide not to travel. Women in these countries, use occasionally official taxis when traveling with heavy bags or children, but most times they use informal taxis or taxi-buses ("colectivos" in Buenos Aires and Lima) which have a number of benefits for women: they fill route gaps, their price is shared by passengers, they provide comfort and, reduce stress from the trip. Some taxi drivers discriminate against women with bags, and some cities have subsidies to reduce taxis' fare for certain passengers (women, students, or elderly). In Lima, women who live in the high hills feel discriminated by moto-taxi drivers when asked to pay higher fares. Ride hailing is being frequently used by women and sometimes it brings cheaper fares to women (WB 2020)

The graph 1 below shows the differences in percentage female-male of their mobility patterns (motorized trips, public transportation, and walking trips) in seven LAC countries compared to the EU27. Common mobility patterns are found in the region and they are similar to developed countries in the EU for women: Women do more trips by foot than men, they use more public transportation and they do less motorized trips compare to men. However, when women are in higher income groups, they show similar mobility patterns than men, especially when they can buy a private car. Women are more susceptible of a modal change to individual modes of transportation if perceived safer, or they can choose not to travel if

perceived unsafe. Additionally, women can transfer their fears and perceptions of public transportation to other generations (CAF 2018)

Graphic 1 Gender differences in mobility patterns* in selected LAC cities and EU27²⁵



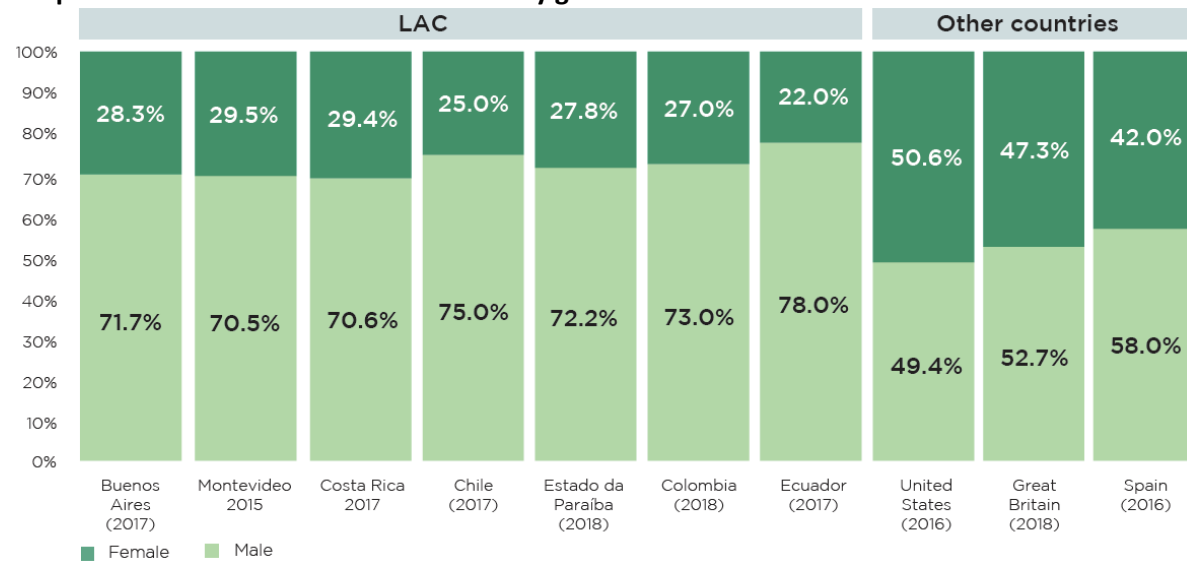
Note: *Differences in percentages (female - male)

The growth of motorization rates in the LAC region and its negative consequences on public transportation have a direct impact on women's mobility. Women use significantly more public transportation and walk more, whereas men drive more cars. This is observed in the distribution of drivers' licenses by gender, where female drivers' licenses represent merely 30 percent of the total licenses in the region. In contrast, some developed countries are in the process of closing or have closed the gender gap among drivers²⁶. In Costa Rica, driving licenses for men double those of women, but when we analyze the type of license the gender gap is even bigger for productive activities. Only 10 percent are authorized bus drivers, and between 7 and 12 percent of taxi concessions of airports and regular taxis are owned by women (it might not reflect women taxi drivers, only ownership)²⁷.

²⁵ Stylized Urban Transportation Facts in Latin America and the Caribbean, Technical Note N ° IDB-TN-1640, IADB 2019

²⁶ Stylized Urban Transportation Facts in Latin America and the Caribbean, Technical Note N ° IDB-TN-1640, IADB 2019

²⁷ COSEVI 2020, (Consejo de Seguridad Vial de Costa Rica)

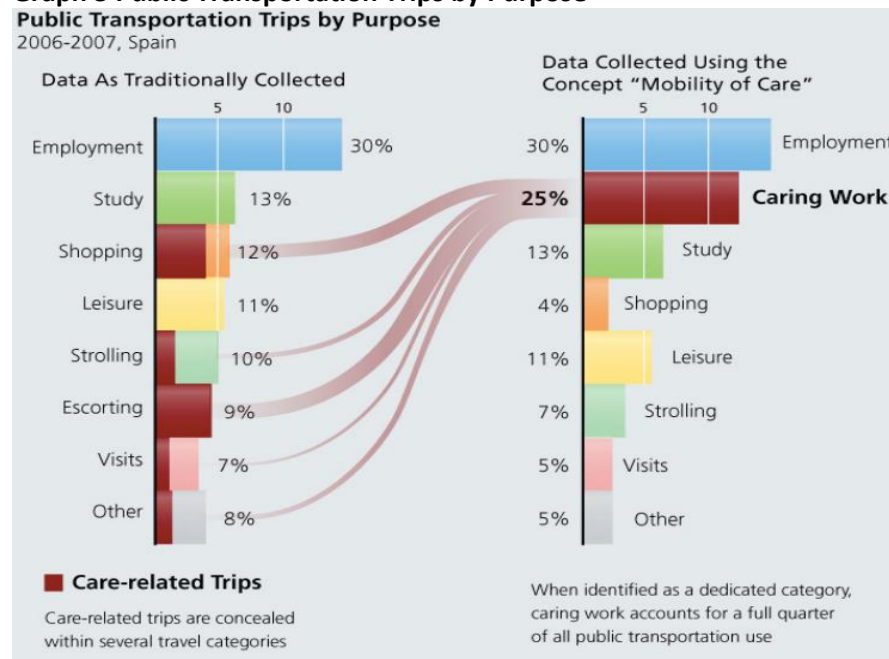
Graphic 2 Distribution of driver's licenses by gender in LAC and selected countries²⁸

The mobility of care²⁹ is unnoticed by data collection methods. For the purpose of designing and calculating ridership, data collection methods focus traditionally in one trip pattern (Ex: travel to and from work or to and from study) while the mobility related to care responsibilities is overlooked because it is fractured in multitude purposes and fall in different categories (Ex: shopping, or escorting). The graph 1 below, visualize the traditional way on the left and the new concept which groups trips for activities related to care on the right (maintenance of the household and care for others). The number of trips for care work and those for paid work are similar for this example's data. Traditionally, surveys have undervalued the "reproductive" trips, considering only the "productive" trips for the design of transport systems. This gender bias underestimates others ridership than commuting, and a poor urban transport design has negative consequences for the agency of women.

²⁸ Stylized Urban Transportation Facts in Latin America and the Caribbean, Technical Note N ° IDB-TN-1640, IADB 2019

²⁹ Inés Sánchez de Madariaga <https://unhabitat.org/mobility-of-care-ines-sanchez-de-madariaga>

Graph 3 Public Transportation Trips by Purpose³⁰



To give visibility to mobility patterns, the IADB has a Client Map Methodology³¹ which breaks into 12 segments a trip's experience in Bogotá, Santiago and Medellín. Then different profiles of riders (women, elderly, people with disabilities) are assessed and the results help to design better urban transport.

Fare schemes can foster women's use of public transportation. Women have more price sensitivity than men because of their unique mobility pattern (chain trips, with family members, off-peak hours) and the poor suffer the most with unregulated tariffs (CAF 2020). Urban transportation must promote a fare structure that subsidize short trips or propose an integrated fare system. Women's value of time differs from men's as they plan their trips to accommodate a busy agenda of care activities and are very concerned about the time of their return home, so besides safety, waiting time and fare are key to their decisions. Further, because most women's trips are not related to income generating activities, women might give up convenient modes of transportation and walk to save money³². Buenos Aires decided to increase transport tariff in 2016 and la Defensoría del Pueblo³³ (Office of the Ombudsman) received considerable claims from vulnerable riders, which 90 percent are women with small kids, and specifically domestic workers.

Ride-hailing has become an alternative for women to increase their urban mobility in safer conditions than other modes. The main characteristics of ride-hailing, many times match women's unique needs. Its on-demand services are convenient for women's off-peak mobility, the service can cover gaps in public transportation given that public transportation mostly covers masculine trips (from home to work/study centers), and women's perception of security augments when ride-hailing at night, especially when the

³⁰ Sanchez de Madariaga research 2009, 2010, 2013a, 2013b, Image by Eric Steiner
<https://genderedinnovations.stanford.edu/case-studies/transportation.html#tabs-2>

³¹ Mapas de Experiencia de Viaje para medir la accesibilidad del transporte público, IADB 2011

³² Approaches for gender responsive urban Transport Mobility, 2018 GIZ, TUMI, SUTP

³³ The Defensoría del Pueblo was created in the city of Buenos Aires to protect human rights from administration or public services actions or omissions. Its work on gender and its Observatorio de Género were promoted by the movement "Ni una menos".

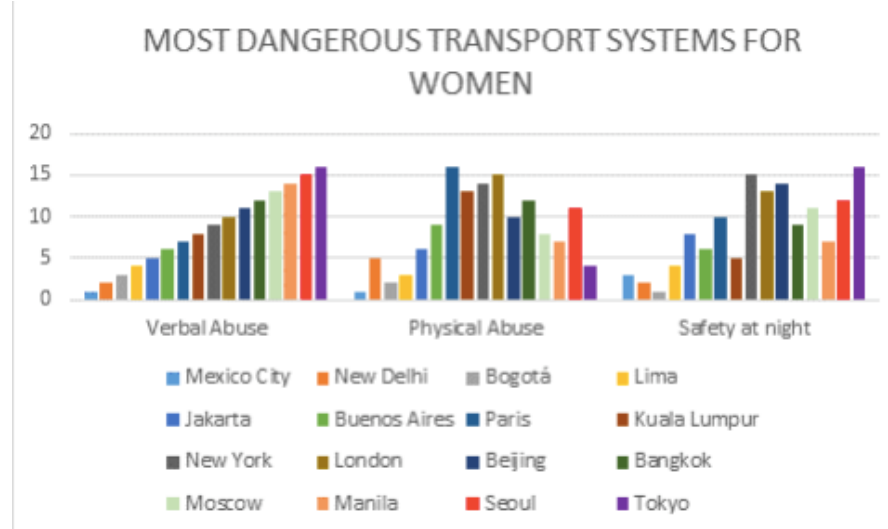
supplier of the services has an app with safety features to monitor and share the ride and to know about the driver's reputation.

Gender-segregated ride-hailing benefits female passengers and drivers. Safety and security are the most limiting, or facilitating, factor for female's agency in mobility and access to jobs. In 2018, a survey carried out in 6 countries (Mexico being one of them) for the IFC³⁴ found that 20 percent of women riders reported that the lack of women drivers reduces the trip they would make, whereas 44 percent said that they would be more likely to use a ride-hailing service if they could select a woman as driver. Ride-hailing is an alternative for women when there are gaps in the public transport system since women's trips are chained, because it has become a safer mode of transportation thanks to technology. At Uber Brazil, more than half of women started driving more when gender segregated transport was offered, which manifests how women can get jobs as drivers in taxis when they perceive a safe environment at work.

Safety in Urban Transport

Six out of 10 women reported to have been harassed in public transportation in Latin American countries. In 2014, a survey³⁵ in 15 out of the 20 largest cities in the World, asked 6550 women and experts in urban planning, about their perceptions when traveling in public transportation. Women's perceptions were collected around the following concepts: How safe women felt travelling alone at night, the risk of being verbally or physically harassed, the likelihood that other passengers would come to their assistance, the trust that authorities would investigate reports of harassment or violence, and the availability of safe public transport. The graph 4 below reveals the ranking of the most dangerous transport systems for women found by this study:

Graph 4 The Most Dangerous Transport Systems for Women³⁶



³⁴ Driving Toward Equality: Women, Ride-Hailing, and the Sharing Economy, IFC 2018

³⁵ The survey was carried out by YouGov and The Thomson Reuters Foundation. Not all the cities in the world were surveyed, so the results ranked the cities surveyed. <https://news.trust.org//spotlight/most-dangerous-transport-systems-for-women/>

³⁶ Observatorio de la participación política de las mujeres en México [https://observatorio.inmujeres.gob.mx/mvc/view/public/index.html?ms=MzE=Violence against women&girls resource guide](https://observatorio.inmujeres.gob.mx/mvc/view/public/index.html?ms=MzE=Violence%20against%20women&girls%20resource%20guide). Transport Brief, WB et al.(Oct 2015)

Bogota, the world's 16th largest capital, was ranked as the most unsafe city for women to travel alone at night. More than six out of 10 women in Mexico City, 64 percent, polled online by YouGov said they had been groped or experienced physical harassment on public transport. Lima ranked 3rd capital perceived as dangerous by women riders, and Buenos Aires came as 6th. The 2018 study "Ella se mueve segura" ("she's moving safely") from CAF presented results of a survey carried out to women in different transport systems of Latin América. In Guadalajara 62 percent of women reported being afraid of suffering a sexual attack, and 60 percent and 64 percent of women reported being victims of sexual harassment in Lima and Bogota, respectively. Most of these events are not reported to authorities nor shared with relatives or friends, and many times happens traveling from and to the transport systems³⁷. In Costa Rica, 80 percent of women report being victims of violent situation in public spaces³⁸.

Perception of transport systems safety shapes women's agency in mobility in such a way that they rather pay higher fares and spend longer journeys than face exposure to risks. In Rio de Janeiro, minivans are more expensive, their trips are more delayed and queues are longer than trains for long distance commutes. Nonetheless, young women choose to travel in minivans because they feel protected from harassment thanks to the seats for passengers that minivans have installed³⁹.

Adequate urban transport infrastructure can boost women's participation in the sector, both as passengers and providers. Good lighting and vigilance cameras can transform women's perception of safety and discourage perpetrators from committing crimes. Both simple solutions prevent crimes, not only crimes against women. Cameras are common in subways and intermodal and main stations, and recently they are installed in buses to protect both passengers and drivers. Installation, operation, and maintenance are costly and can't detect harassments, only aggressions. Combining safety agents and cameras is the most effective solution⁴⁰. Since 2014, Buenos Aires installed cameras and GPS in public transportation units to monitor traffic registered inside and outside the bus. These kinds of tools can potentially be used for safety as well⁴¹. Particularly at dawn and dusk, isolated bus stops, empty or too crowded buses, poorly lighted park-and-ride lots represent stressful settings for women that limit their mobility agency. The surroundings of the transport infrastructure play a role in creating traveling stress for women (IFC, 2020). Further, measures easy to implement, such as allowing women to get off the bus between bus stops at night. These random bus stop locations might be better lit or be closer to women's destinations, which can radically change their perception of safety at night. In Montreal and Toronto (Canada) these stops are permitted (CAF 2020).

Women adopt specific strategies, which can work or not, but make women feel safer in the public transport system. Sexual harassment in public transport includes leering, winking, offensive gestures, unwanted touching, groping, pressing against the body, indecent exposure, and sexual assault (WBG, IADB, ICRW, 2015). Women change clothing, take long detours to avoid unsafe places, avoid boarding crowded buses or stay near the walls, and even carry non-lethal weapons. They feel safer when traveling in a group, during off-peak hours, in well-known and familiar environments and when they develop friendly relationships with drivers. In Manizales (Colombia), a medium-sized city where trips usually need only one mode of transportation, 73 percent of women have suffered at least one incident of harassment, sexual violence, or street abuse, and 47 percent changed their routines because of them. These women

³⁷ Ella se mueve segura. Un estudio sobre la seguridad personal en las mujeres, CAF & Fia Foundation. (2018)

³⁸ Sexual and Reproductive Health Survey 2016 Costa Rica (2016).

³⁹ WB 2020, Why does she move?

⁴⁰ CAF 2020, Ella se mueve segura.

⁴¹ IADB May 2020, Género y Transporte, Buenos Aires. Transport Gender Lab

chose longer and safer routes changed transport modes (to taxi or motorcycle) and changed the way they dress, travel with others, or select a determined seat in the bus⁴².

Campaigns to raise awareness about harassment together with safety protocols to report abuses can protect women and reduce incidents. Sex-segregation in massive urban transport is a temporary measure that increases mobility and protects women from harassment. Women-only buses or carriages on trains have been operating for decades in many cities worldwide (Brazil, Mexico, Japan, India, Indonesia, Nepal, United Arab Emirates, Nepal, Thailand) and have received praises and critics. Women feel safer when using exclusive spaces, but segregating women does not solve the problem. Moreover, women passengers on non-segregated modes are perceived as open to sexual advances, despite the fact that the initiative reduces incidences (WB, 2019). In 2008, Mexico DF launched Viajemos Seguras (“Let’s we women travel safe”) sex-segregating units in its transport system and was implemented in other estates later. But the program did more than that, it located booths in transport stations to file complaints, transportation staff received training on violence against women and girls, vehicles and stations displayed prevention campaigns, and bus systems and trains had women only cars. In 2015, Mexico DF responded to women’s groups claims and implemented a more ambitious program, which installed emergency buttons in 90 percent of the Atenea buses (drivers can make a silence alarm call to an operative) and 271 buses had cameras installed.

Mobile applications facilitate women protection and contribute to gathering data, which supply information to design safer transport infrastructure and raise awareness on harassment. Digital technology helps women track and share their location with family and friends. Ride hailing companies have developed applications for passengers to share additional information on their trip and identify both the driver and the car’s plates. Passengers can leave comments after their journey and quantify a driver’s professionalism, cleanliness of the vehicle, and quality of their trip. In Brazil, “Chega de FiuFiu” (Enough whistling) is an initiative created by an NGO to geo-localize and foster reporting of harassment. The video documented the women’s regular interaction when walking and using public transportation to raise awareness of the harassment that women suffer routinely. In 2018, Bogota under the “Me nuevo Segura” study, adapted the Indian application Safetipin to measure the index of safety perception at night of different routes and places by women. Safetipin is based on nine variables to obtain a safety score: Level of lighting, Openness (sightlines), Visibility (overhanging trees, high walls etc), Level of crowding or emptiness, Security (agents), Walk/ footpath, Availability of public transport, Usage of the public space (for instance is it male-dominated or mixed) and, Feeling/ emotional response to the space (sometimes also called “eyes on the street”). While analyzing personal data of respondents, sixty-five percent of the study participants were below 35 years old and 78 percent work or study, and 50 percent finished tertiary education⁴³. Age, sex, education, and poverty are variables to consider when implementing technology to protect women in public transport since affordability (costs of telephones or internet) and user knowledge can be barriers for many women⁴⁴, especially the poorest.

The transport sector can serve as a catalyst to teach women their rights. The Via Lilas⁴⁵ program was launched in 2015 in Rio de Janeiro as part of the implementation of the Maria da Penha’s law on gender-based violence. Electronic kiosks, “totens” were installed in the Supervia Rio’s suburban rail lines, which

⁴² IADB mobiliblog (Ag 2020): Violencia de género: enemiga de la movilidad sostenible

⁴³ CAF, TUMI, Bogota, 2019 “Me Nuevo Segura” Mediciones nocturnas en materia de seguridad para las mujeres

⁴⁴ IADB 2020, ¿Desigualdades en el mundo digital?: Brechas de género en el uso de las TIC. Washington, D.C

⁴⁵ WB 2015. Vias Lilas <https://blogs.worldbank.org/transport/rio-lilas-initiative-using-transport-infrastructure-help-reduce-gender-based-violence>

serve around 700,000 daily users. Anonymously, women seek data on the health network, women's rights laws, and gender-based violence protocols.

Sensibilization and prevention campaigns on Gender-Based Violence can equip public transport workers with tools to understand it. In the Dominican Republic, a gender-based violence campaign delivered workshops to drivers' trade unions working in buses and "motoconchistas" (drivers of private cars that operate as public transport). The workshops included topics such as new masculinities and gender-based violence in the domestic spaces and with passengers in the transport environment. (IADB Gender Lab, 2019)

Feedback and grievance mechanisms located in the transport systems help women and girls to inform authorities when they are victims of violence. Capacity building and coordination among different bodies (police, judiciary system, staff) is essential to provide useful and appropriate assistance to victims. Often, survivors of abuse perceive authorities in charge of taking care of them as lacking credibility (WB 2020).

Lack of trust in authorities is common prevent women from reporting abuses. In a survey in 2016 in Lima, 35 percent of public transport users don't trust in police arriving at the scene when reporting the rape of a woman in the public transport system. For 41.5 percent their trust is low, 18.9 percent reported some trust, and 3.9 percent trust a lot that police will arrive⁴⁶.

Jobs and Urban Transport

The transport sector provides men and women access to education, health and economic opportunities. In this section, we are focusing only on the jobs that the sector generates by itself. Jobs can be generated during the design and at the construction of the transport infrastructure, and during its operations and maintenance. In this program, most jobs will be generated during the operation and maintenance of the e-buses, e-taxis and e-vans and at the charging station's maintenance. Investments in infrastructure are common to boost the economy and it will be the case after this health crisis; thus, this is a unique opportunity to lever jobs for women.

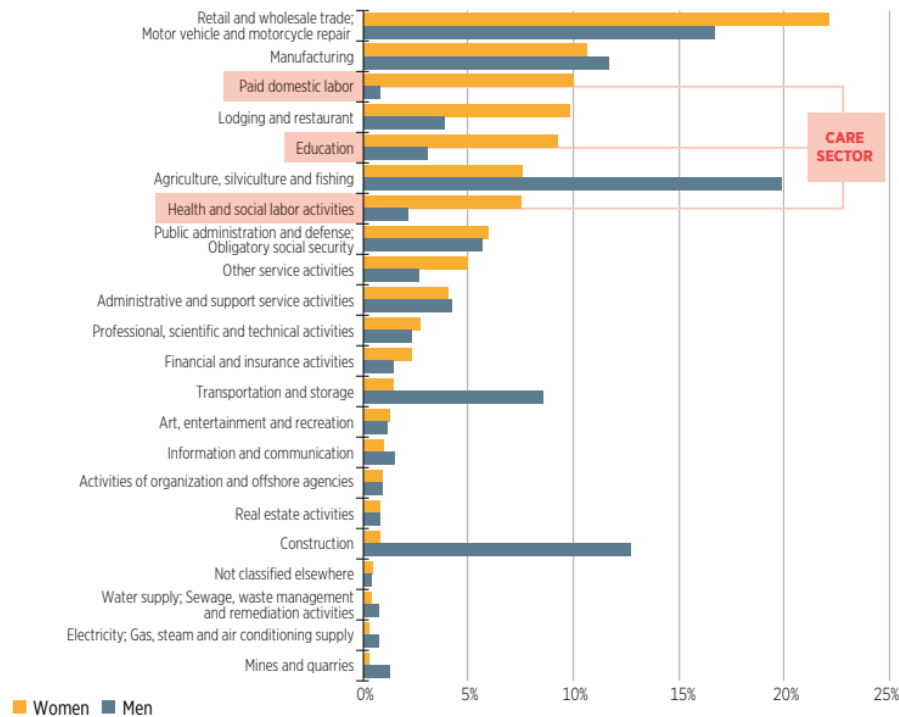
The gender gap in jobs in the transport sector is enormous. Typically, women work in feminized and low-paid tasks such as cleaning or administration, while men work as drivers, mechanics, or supervisors. Thus, both the number of women hired or trained and the quality of their contracts are low. Specific efforts need to be made to reach greater diversity in the sector. To bridge the job's gap, it is crucial to attract, secure and maintain jobs with more vacancies (ex: commercial drivers), with a higher paycheck (ex: mechanics), or that are involved in decision making (investors).

The graph 5 below, describes that around 30 percent of women work in the care sector (traditionally dominated by women), compared to 6 percent of men. This reflects the regional wage gap in which women earn between 5 and 30 percent less than men with similar educational level and demographic characteristics⁴⁷.

⁴⁶ IADB 2016, Transporte Público desde una perspectiva de género: percepción de inseguridad y victimización en Asunción y Lima.

⁴⁷ The Future of Work in LAC (2019), IADB. Graph designed by IADB with ILO data 2017

Graph 5 Occupational Segregation: women care, men build⁴⁸



Barriers to entry in the transport sector remain the same, despite women's employment benefits the sector. Stereotypes, lack of family support, low rates of girls studying STEM, are some of the traditional barriers for women to work in sex-segregated jobs, sometimes perceived as unsafe. Reshaping women's tasks when pregnant for their protection, expensive professional driving licenses, and stigma hinder more equality workforce in the sector. The power of labor unions can both ease or impede women's access to jobs in the sector. Advertise that all the vacancies are open for women, adapt working schedules to women's care needs at home, building dressing rooms and provide childcare, offer scholarships to train women are incentives for the sector to attract them. Working with trade unions is vital in masculinized occupations to allow women to enter. Still in 2017, in Argentina, the train labor unions do not allow women to be drivers (CAF 2020).

In Cali, 25 women out of 2000 workers are drivers in the integrated urban transport system, MIO. 13 percent of women are urban traffic police, 9 percent are supervisors and 14 work in operations. In private companies working for the massive transport system (GIT) there are 33 women working in administration, seven are bus operators and two are assistants of private operators' general services (IADB 2020).

In Transantiago (Chile), women drivers said that the visibility of other women carrying a diversity of tasks and positions incentive other women to work as drivers. From 2012 on, in Santiago, new regulation prompted the private companies to collaborate with municipalities to help B1 drivers to obtain professional driving licenses. Most women got to take these driving courses thanks to scholarships (72 percent of women surveyed, 161 women), and then, they got the jobs. The price of professional licenses is a barrier as well for women willing to work as drivers. It is difficult for women head of households to attend these courses since they don't have the time to leave their jobs or can't leave their family responsibilities. Women drivers in Santiago were perceived as more compromised with the job, more responsible, have less sinistrality than men, or the accidents are less severe. Most women drivers were

⁴⁸ Ibid

perceived as more kind, empathic, calm, they treat the passengers better and resolved conflict situations without escalating problems. After women started working in Transantiago men colleagues changed their behaviors, become more respectful and watched their languages. (IADB, Transport GenderLab 2019). A growing body of evidence shows how companies can benefit from investing in women as employees, entrepreneurs, customers, and community partners. Such investments benefit companies in several ways, from broadening the talent pool to increasing productivity, providing an opportunity to transform local and global markets⁴⁹. Even though the data is scarce⁵⁰, women drivers tend to have fewer traffic accidents than men. The savings derived from the reduction in traffic accidents can benefit transport operators when hiring women drivers.

Following recent legislation to grant and enforce equal opportunities, most countries in the region have carried out initiatives to extend women's employment opportunities in the transport sector.

The IADB Transport Gender Lab has published several recent transport initiatives that create jobs for women. In Mexico DF, in 2008, the program Atenea was created to reduce traffic to and from schools by hiring women bus drivers, who would offer more confidence to parents in the school routes. The program got extended to women, the elderly and kids only buses and school routes. In 2017 there were 136 buses for the program, and 75 drivers were women.

In the state of Jalisco where the police were linked to corruption and high rates of alcoholism were considered the leading cause of traffic deaths, the government renovated the traffic police with a call for women named "Buscamos a las mejores" (We are seeking the best women) to attract women to work as traffic officers. Moreover, another call searched for women for the "Salvando Vidas" (Saving lives) program whose focus was on improving people's perception of the police as a transparent department that applies law to everyone.

In 2018, the city of Guadalajara hired 64 women as urban drivers and the objective is to train, certify and hire 600.

In Buenos Aires, 54 percent of 2.750 traffic agents were women. Moreover, 43 percent of other positions in the urban transportation system were occupied by women such as supervisors, shift coordinators, and chiefs of operative buses. Additionally, as of today, 641 women are working in the control and supervision of traffic in Bogota.

In the Dominican Republic, the metropolitan agency for bus services, selected women from its staff who worked as cashiers and were familiar with the system were trained to drive buses. Sixteen women are currently in charge of driving in the "Corredor de la Mujer" (Lady's Corridor) and a new corridor is expected to have 100 percent women drivers. These drivers recorded videos and shared them in media on Women's Day. Four women workers of the subway (50 percent of drivers in the subway are women) shared their career experience in the media on Women's Day as well, from their start as drivers until they got to positions with more responsibility.

⁴⁹ IFC, March 2017. Investing in Women, new evidence from the business case

⁵⁰ World Health Organization Factsheet, 2002. Gender and Road Traffic Injuries

Women, Jobs and COVID-19

Women's labor force participation⁵¹ has been steadily rising in the last decade for each of the seven countries until the pandemic. The devastating impact of COVID-19 has affected disproportionately women's labor force participation. The large gender gaps in employment favor men, but these gaps were gradually narrowing in the last years for all countries. The COVID-19 pandemic has left women in a worse situation than before.

The sex-desegregated table 1 below provides a glimpse of the labor supply for the last years, which in the second quarter of 2020 has dropped around 10 points in Argentina, Colombia, Dominica Republic, and Mexico, 7 in Brazil, and 27 points in Peru. For Argentina, Brazil, and Colombia, the difference between men and women's labor force participation was narrowing from around 20-25 difference points to 20 in the first quarter of 2019, but in Dominican Republic the women-men gap narrowed from around 30 to 20 points. In Costa Rica women's rate in the 2nd quarter of 2020 is almost the same that that of 2011. In contrast with these dropping numbers globally, 70⁵² percent of nurses, front-line care workers, midwives, and community health workers are women, whose jobs have been vital for controlling the health crisis. Women working in the health sector are more affected by COVID-19, and studies from the USA and European countries have revealed that positive COVID-19 cases are two to three times higher for women than for their male colleagues.

⁵¹ The labour force participation rate is a measure of the proportion of a country's working-age population that engages actively in the labour market, either by working or looking for work; it provides an indication of the size of the supply of labour available to engage in the production of goods and services, relative to the population at working age. In contrast, employment comprise all persons above a specified age who during a specified brief period, either one week or one day, were in the following categories: paid employment or self-employment.

⁵² Solidarity Center 2020. "Domestic workers left out in the cold". Accessed 10 August 2020.

Table 1 Annual average national labour force participation⁵³ rates by sex, 2011–2020

Annual average national labour force participation rates by sex, 2011–2020 (M: Men, W: Women)										1st quarte r	1st quarte r	2nd quarte r	2nd quarte r
Country	201 1	201 2	201 3	201 4	201 5	201 6	201 7	201 8	2019 a	2019a	2020a	2019a	2020a
Argentina	59.5	59.3	58.9	58.3	57.7	57.5	57.8	58.5	59.1	58.9	58.6	59.5	49.2
M	72.9	72.2	72.0	70.9	70.1	69.4	69.7	69.6	69.9	69.8	68.7	70.2	58.0
W	47.4	47.6	47.1	46.9	46.4	46.9	47.6	48.7	49.4	49.0	49.5	49.9	41.2
Brazil	60.0	61.4	61.3	61.0	61.3	61.4	61.7	61.6	62.0	61.7	61.0	62.1	55.3
M	70.8	73.1	72.9	72.5	72.4	72.3	72.0	71.7	71.7	71.6	70.8	71.7	65.5
W	50.1	50.8	50.7	50.6	51.2	51.4	52.3	52.5	53.2	52.8	52.1	53.4	46.3
Colombia	63.7	64.5	64.2	64.2	64.7	64.5	64.4	64.0	63.3	63.5	61.6	62.9	54.8
M	75.1	75.4	74.9	74.9	75.2	74.9	74.8	74.6	73.9	74.2	72.7	73.4	66.2
W	52.8	54.1	53.9	54.0	54.8	54.5	54.5	53.8	53.1	53.3	50.9	52.9	43.9
Costa Rica	58.4	62.5	62.2	62.6	61.2	58.4	58.8	60.7	62.5	62.4	63.4	63	57.6
M	73.6	76.2	75.5	75.9	74.3	72.4	73	74.3	74.4	74.4	74.7	75.1	70.5
W	44.2	48.4	48.6	49.2	48.1	44.3	44.5	46.9	50.6	50.3	52.1	50.8	44.6
Dominican Republic	57.8	59.0	58.7	59.1	61.8	62.3	62.2	63.6	65.1	64.9	63.4	65.2	56.7
M	73.1	74.4	74.1	74.6	76.3	76.6	76.1	77.8	78.4	78.8	76.3	78.5	70.6
W	43.7	44.0	43.7	44.0	48.1	48.9	49.0	50.4	52.7	52.0	51.5	52.7	43.9
Mexico	59.8	60.4	60.3	59.8	59.8	59.7	59.3	59.6	60.1	59.5	59.9	60.2	49.4
M	78.5	78.8	78.5	78.3	78.0	77.7	77.6	77.4	77.2	76.9	76.4	77.1	63.5
W	42.8	43.9	43.9	43.1	43.4	43.4	43.0	43.5	44.7	43.7	44.9	44.9	36.7
Peru	73.9	73.6	73.2	72.3	71.6	72.2	72.4	72.3	72.7	72.9	70.2	72.0	45.3
M	82.7	82.4	82.0	81.4	81.0	81.2	81.0	80.7	81.1	81.5	78.3	79.6	54.7
W	65.2	64.8	64.5	63.3	62.3	63.3	64.0	64.0	64.5	64.4	62.2	64.5	36.1

In the second quarter of 2020, there have been significant drops in employment rates that have affected both women and men, and the progress of the pandemic's devastation will worsen the figures. ILO says that women's employment dropped by 18.1% compared to men's, which dropped by 15.1%. Women's workforce is more extensive in the third sector, which is most affected by the pandemic (domestic services, restaurants and hotels, commercial activities). Forty percent of all employed women in the world (estimated at 510 million women) work in hard-hit sectors, compared to 36.6 percent of employed men. Women's paid labor and women-run businesses are being hit hardest. One of the occupational categories more affected is that of wage earners employed by households, most of them are women of relatively low-income households, who lack essential work benefits (health leave, notice period, severance pay). In Latin America, 93 percent of the 11 to 18 million domestic workers are women (UN WOMEN 2020). This pandemic affects women, young people, and immigrants the most for a variety of reasons (For instance: they mostly work in the informal market, no new positions being created for young people, or

⁵³ Own elaboration from data from: ILO & ECLAC (Nov 10, 2020). "Employment Situation in LAC (Number 23): Employment trends in an unprecedented crisis: Policy challenges". Data comes from household surveys. Some data are preliminary figures or new series, please visit the document to know more about these figures.

lack of legal papers to work). The global pandemic impacted employment rates in many ways; many companies have ceased operations or remain on stand-by due to country restrictions. Absenteeism has been a consequence of the pandemic since workers have not been able to attend their jobs' locations. Companies and countries have been designing ways to adapt to the temporary situation without firing workers (Ex: income support mechanisms or early leave) (ILO, Nov 2020).

The open unemployment rate⁵⁴ is larger in women because the household care responsibilities have increased with the pandemic. The table 2 below clearly shows tendencies. According to ILO, Dominican Republic's open unemployment rate drastically decreased to 3.2 in the second quarter of 2020 from 6.4 in the same period of 2019. Both people who were unemployed before the pandemic and those who lost their jobs because of COVID-19 are not looking for jobs. Figures are similar for both sexes in the Dominican Republic. The number of people in open unemployment⁵⁵ in this health crisis increased less among women (by 7.7%) than among men (29.4%); the contrary is expected in an economic crisis. Female open employment has increased less than expected, probably due not only to the loss of jobs but also for the increase in care work, resulting in women not looking for paid work. While for men, who are still considered the breadwinners of the family, looking for jobs is easier after they lost them during the pandemic (ILO 2020). Women hold their selves responsible for the family care burden, while men are still seen as "help" in the region (WB, 2020). The fact that open unemployment has rose more for men (in all the seven countries) does not mean that the crisis has more hit men compared to women; it means that further research on open employment is required.

⁵⁴ Open unemployment refers to those looking for a job that cannot find one because of the economic conditions.

Table 2 Average annual national open unemployment rate by sex, 2011–2020⁵⁶

Average annual national open unemployment rate by sex, 2011–2020 (M: Men, W: Women)										1st quart er	1st quart er	2nd quart er	2nd quart er
Country	2011	2012	2013	2014	2015	2016	2017	2018	2019	2019	2020	2019	2020
Argentina	7.2	7.2	7.1	7.3	6.5	8.5	8.4	9.2	9.8	10.1	10.4	10.6	13.1
M	6.3	6.1	6.1	6.5	5.7	7.8	7.5	8.2	9.2	9.2	9.7	10.2	12.8
W	8.5	8.8	8.5	8.4	7.6	9.4	9.5	10.5	10.7	11.2	11.2	11.2	13.5
Brazil	6.7	7.3	7.1	6.8	8.5	11.5	12.7	12.3	11.9	12.7	12.2	12.0	13.3
M	4.9	6.0	5.8	5.7	7.3	10.1	11.3	10.8	10.1	10.9	10.4	10.3	12.0
W	9.1	9.2	8.9	8.2	10.1	13.3	14.6	14.1	14.0	14.9	14.5	14.1	14.9
Colombia	10.0	9.7	9.0	8.5	8.3	8.6	8.8	9.0	9.9	11.2	11.8	9.5	18.3
M	7.8	7.5	7.0	6.7	6.4	6.7	6.8	7.1	7.8	8.8	9.4	7.7	16.0
W	12.9	12.5	11.6	10.9	10.7	11.0	11.3	11.6	12.6	14.4	15.1	11.8	21.6
Costa Rica	7.7	10.2	9.4	9.6	9.6	9.5	9.1	10.3	11.8	11.3	12.5	11.9	24
M	6	8.9	8.3	8.1	8	8	7.5	8.4	9.3	9.3	8.6	9.9	20
W	10.3	12.2	11.1	11.9	12.2	12.1	11.6	13.2	15.3	14.2	18	15	30.4
Dominican Rep	6.1	6.7	7.4	6.7	7.3	7.1	5.5	5.7	6.2	5.8	5.7	6.4	3.2
M	4.7	5.1	5.3	4.8	5.2	4.8	4.0	3.5	3.9	3.7	3.6	4.2	2.4
W	8.3	9.2	10.5	9.7	10.5	10.5	7.8	8.8	9.3	8.9	8.6	9.5	4.3
Mexico	5.2	4.9	4.9	4.8	4.3	3.9	3.4	3.3	3.5	3.4	3.4	3.5	4.8
M	5.2	4.9	4.9	4.8	4.3	3.9	3.3	3.2	3.5	3.3	3.5	3.5	5.3
W	5.2	4.9	5.0	4.9	4.5	3.9	3.6	3.4	3.5	3.5	3.4	3.6	3.9
Peru	4.0	3.7	4.0	3.7	3.5	4.2	4.1	3.9	3.9	5.2	5.1	3.6	8.8
M	3.7	3.2	3.4	3.4	3.4	3.9	3.8	3.5	3.5	4.4	4.5	3.3	9.7
W	4.4	4.4	4.7	4.0	3.6	4.6	4.4	4.4	4.5	6.2	5.8	4.0	7.5

This situation negatively influences younger women, particularly younger mothers who, when facing long periods of inactivity, experience shows that their career paths are drastically altered. The more time absent from school or work, the greater the risks of exclusion from the labor market, the precariousness of jobs, or the informality. This latter happens to young men as well.

Cleaning, cooking, and serving meals, teaching children and shopping for the family are the major workloads that have been intensified recently. The pandemic has made a point in the care work, which traditionally governments have ignored and the economic markets have devaluated. In all of Latin American and the Caribbean region's countries, and globally, the burden of the economy of care falls entirely into women's responsibility. Despite the well-timed and record measures taken by governments to tackle COVID-19's economic outcomes, the little social protection measures were lacking gender lens. Reconciliation of paid and un-paid work and family's care must be addressed to allow women to go back to paid work (UN Women 2020).

⁵⁶Own elaboration from data from: Employment Situation in Latin America and the Caribbean (Number 23): Employment trends in an unprecedented crisis: ECLAC&ILO, Nov 2020. Data comes from household surveys. Some data are preliminary figures or new series, please visit the document to know more about these figures.

Gender Capacity in AFD, PROPARCO and GIZ

The following paragraph explain with detail the capacity of AFD to manage and monitor the gender-related activities at project level and the particular tools and experiences of PROPARCO and GIZ that will generate important synergies to succeed during implementation.

AFD has vast experience with large portfolios of projects with the mandate to include gender equality and women's empowerment. Since 2013, AFD defined its strategy on Gender and the Reduction of Gender Inequalities based on three priorities: prevent gender inequality in its operations, promote gender and women's empowerment and support in societies on these issues. Today, its objectives are more ambitious and since 2018 it has set up a rating system to mainstream gender in all its activities, based on the classification of OCDE (DAC markers: DAC 0: the project does not target gender equality, DAC 1: gender equality is an important and deliberate objective of the project, DAC 2: gender equality is the main objective of the project). In 2020, 67.1% of projects financed by AFD integrate gender; thus, it has largely achieved the objective of 50 percent of the volume of commitments to be gender-responsive and rated 1 or 2 and 749MM of Euros will fund projects whose main objective is to reduce gender equality (DAC 2). Moreover, AFD manages a Support Fund for Feminist Civil Organizations aiming at building the organizational, financial and technical capacities of feminist organizations in partner countries and promote exchanges and networking between them to give them the means to take action at the local, regional and international levels. Together with 26 banks, AFD, signed a Paris development bank Statement on gender equality and women's empowerment which defines four areas of action: (i) Strengthen the commitment of Public Development Banks to gender equality through a dedicated institutional strategy; (ii) Increase the share of financial flows for gender-responsive operations; (iii) Contribute to gender-responsive climate policies; and (iv) Improve dialogue and collaboration with all stakeholders on gender issues, including feminist civil society organizations. AFD experience in the region and long term collaboration with regional and local agencies will facilitate the dialogue with public transportations during the implementation of gender activities at project level.

PROPARCO belongs to the 2X Challenge and brings gender-lens investing experience to support the project's opportunities to increase women's economic agency. PROPARCO (société de Promotion et Participation à la Coopération économique) is a subsidiary of AFD and as part of AFD Group follows its gender strategy. Its work focuses on private sector development, provides funding (loans, guarantees, equity and quasi-equity) and support (technical assistance and expertise) to both businesses and financial intermediaries across Africa, Asia, Latin America and the Middle-East. The financing of infrastructure projects represents more than 30% of PROPARCO's activities in these countries. PROPARCO mobilizes its funds to increase women's participation in development countries through promoting women's financial inclusion, empowerment and access to essential goods and services following the 2X Challenge initiative, which is widely adopted by other development institutions. PROPARCO's criteria to achieve gender equality are four: 1) Entrepreneurship, access to financing for women-owned and operated business; 2) Leadership, access to management positions; 3) Employment, access to quality jobs; 4) Essential good and services, access to goods and services benefiting women in particular. PROPARCO will implement the financing activities and TA of the Program where Project Owners (PO) belong to private sector and its mandate and know-how to mainstream gender in its activities will facilitate women's inclusion.

GIZ is a pioneer organization working on gender and transportation. In 2007 GIZ published the toolkit "Gender and Urban Transport", which has been a must have tool for development practitioners for more

than a decade and has been updated in 2018. TUMI (Transformative Urban Mobility Initiative, implemented by GIZ) launched the international conference “Women Mobilize Women” which since 2018 shaped a new narrative around gender and transportation and triggered follow up initiatives and networks of female change makers from all over the world. One of the regional networks on gender and mobility generated in the conference is located in Latin America and encourages the inclusion of women in decision making, planning and operation of transport. GIZ experience in transport will accelerate the current tendency to include gender and vulnerable groups and its work in this facility will multiply the synergies among local and regional actors.

Recommendations

In these seven countries, gender equality has been improving in recent years, but the COVID-19 pandemic has worsened it. The country snapshots on gender and the review of the numerous initiatives are enough to recommend that all the investment plans in this facility include gender strategies and gender action plans to prevent violence against women and girls and guarantee women access to direct jobs and economic opportunities by the projects receiving loans.

The second section of the assessment reveals that these countries are already implementing programs and actions in their urban transport sector seeking more gender equality. The E-mobility and low carbon Transportation facility is a timely opportunity to build from the many initiatives already in place. The TA will house all the activities related to gender equality and women’s empowerment. The efforts to establish a sustainable mobility ecosystem at regional, national and local levels plan to embed gender inclusion which will multiply the effects of the gender-related activities in the new paradigm. The outcomes and activities of the Gender Action Plan are shaped in line with the facility’s.

The technical assistance will support the adopters implement a people centered design. Adding to the differentiated risks and vulnerabilities of men and women detailed in the gender assessment, the adopters will be guided on how to apply intersectionality enablers. Adding to the afro descendants and indigenous peoples present in many of these countries, intersecting identities of age, social economic status, social cultural context, education, disability among others⁵⁷ need to be considered. Together with universal design approach, accessibility and reasonable accommodations⁵⁸ will be factored in when electrifying transport systems. The technical assistance will guide the adopters to use tools that guarantee that women and vulnerable groups are represented during multi-stakeholders consultations so that their perspectives can be factored into policies, road maps, trainings and campaigns. The adopters should use tools such as accessibility and reasonable accommodations in a way that maintain confidentiality and privacy to allow women and vulnerable groups to share their perspectives. An environment where women and vulnerable groups feel safe will allow the adopters to include perspectives during design and implementation that might not be considered otherwise. Consultations in a safe environment will be

⁵⁷ UN Women, UNRPD, and more (2021). Intersectionality resource guide and toolkit, An Intersectional approach to leave no one behind

⁵⁸ Universal design means the design of products, environments, programmes and services to be usable by all people, to the greatest extent possible, without the need for adaptation or specialised design. Universal design shall not exclude assistive devices for particular groups of persons with disabilities where this is needed. Reasonable accommodation means necessary and appropriate modification and adjustments not imposing a disproportionate or undue burden, where needed in a particular case, to ensure to persons with disabilities the enjoyment or exercise on an equal basis with others of all human rights and fundamental freedoms.

particularly emphasized when designing and implementing sexual harassment programs and prevention campaigns.

The review of public policies at country level will integrate gender lens, this means that the technical assistance will help find obstacles and risks in the policies being reviewed that may impede women from benefiting from the EV mass adoption. The themes covered by the topics in section two of this assessment (mobility, safety, and jobs) constitute a guide that the technical assistance will use when providing support to the investment plans. The public and private sector investment plans will receive technical assistance to systematically include a gender assessment and gender action plan with concrete measures, activities and products that will ensure that women benefit equally from the massive adoption of EV through direct jobs. The technical assistance will help identify the potential sexual harassment risks to consider when designing and implementing e-mobility transport systems. The adopters will receive guidance to develop mechanisms that respond, address and monitor sexual harassment in the new e-mobility systems. The objective of these technical assistance is that the adopters include Sexual Harassment Programs that contain safe, confidential and ethical grievance mechanisms for survivors of sexual harassment and GBV. These programs are to be context-specific, build from and improve existing mechanisms or design. Moreover, these programs must prevent further risks such as victim blaming and discrimination and mitigate potential impacts on indigenous people and afro descendants and other intersecting identities.

Specific actions to guarantee women feel safe when using the EV, as workers and as users, are imperative in the region. Gender-based violence is endemic in these countries and underreported. Women in the region are victims in the transport sector, and the COVID-19 pandemic is exacerbating the numbers. The urban transport system represents an environment where harassment and sexual abuses occur quotidianly in the regular trips women do every day. Incidents will happen in the e-buses, in the e-taxis, while waiting, walking to and from the stops or using charging stations. Incidents increase at night, in isolated vehicles and unattended places, but in crowded units as well. Safety measures not only benefit women but all the users. The recent study “Ella se mueve segura” (She’s moving safely) represents a timely and adequate guide from the region that will be used by the program to build capacity and learn at project level, thanks to an online training. The program will include communication campaigns to raise awareness and prevent sexual harassment in the public transport sector. The purpose of the South-to-South learning visit is to have adopters visualize a sexual harassment response program that includes grievance mechanisms. The technical assistance will bring adopters together to discuss and acquire the know-how to carry out these programs in their transport systems.

Women are not discriminated in access to credit in any of the seven countries. The program promotes women entrepreneurs to prepare investments loans to entry in the e-transport market in order to prevent leaving them behind. Women will benefit from the capacity building and trainings on EV and financing that will be delivered by the facility.

Women are not represented in the jobs generated by the transport sector in the region. The pandemic complicates the return to work for those women who lost their jobs and those whose care responsibilities have increased. This facility is an opportunity for women in these seven countries to access the jobs generated by the projects to be designed with the investment loans. For instance, the public sector projects will help women obtain professional driving licenses to be able to opt for the direct jobs that the e-buses, e-taxis and e-vans will create. This can be done via scholarships for women to receive trainings on professional driving. The technical assistance will also support investment projects for e-taxis adoption on how to include segregated ride-hailing from the market that is already present and growing in the

region. The technical assistance will help the adopters to consider vulnerable groups who do not have access to technology.

To prevent female turnover and retain women workers in the projects, an activity is focused on the promotion of work-life balance for the projects. The activity will include workshops that cover practical examples to learn how GBV, Sexual harassment and other barriers affect women's participation in the work place, with concrete examples on the case of intersectionalities. The aim is for the adopters to understand gender inequalities, at a family and work level, and to appoint a person in charge in the organization to monitor and report advances in gender equality and inclusive environments in the organization. Paternity and maternity leaves, availability of day care, lactation rooms, wage gaps, recruitment and promotion criteria hindering women's advancement in the sector and benefits of diversity working environments will be covered within these workshops. A gender focal point will be appointed to advance the work-live balance agenda of the adopters, to monitor and report gender equality in the human resources policies and to act as a sexual harassment ombudsperson.

Data collection is crucial to learn, share and improve transport systems, an activity will help transport systems to sex-disaggregate data systematically in order to inform public transport agencies on how to improve gender equality. Baselines will be determined before specific projects are financed in order to monitor and measure the indicators.

The number of expected households (male and female in each country) with improved access to low emission energy sources depends on each country's energy policy, and the number of people with access to e-mobile systems will depend on each bus line. Likewise, the estimated additional male and female passengers who will use low-carbon transport as a result of funding support in the different countries will depend on both the lines to be electrified and the personal choice that determine a modal change.

The Gender Action Plan guarantees that these general recommendations are transformed into specific activities by the facility, and that they have a budget and are measured and monitored, and adjusted when needed.

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Gender Action Plan

E-Mobility and Low Carbon Transportation

	Indicators and Targets	Timeline	Responsibilities	Costs in EUR
Impact Statement: Women experiences and benefits from the Transport Sector are improved				
Outcome Statement: The targeted transportation sub-sectors are more gender-responsive and provide equal opportunities for men and women				
Output 1: Gender considerations to be included during the revision of roadmaps and policies (National level) ¹				
Activity 1.1. Usage of a gender perspectives including mainstreaming of the same when designing and reviewing roadmaps and policies for different EV segments ²	Sector specific and country-level gender assessments undertaken as part of the development of low carbon vehicle roadmaps. GAPs to be developed Target: 1 per country	During year 1-3, on a rolling basis	GIZ Gender Specialist	116,800
	% of the policies and roadmaps supported that include a gender perspective³ - Users' satisfaction of new PT environment, disaggregated by sex (% very satisfied, satisfied, dissatisfied) Target: 100% of policies and roadmaps supported by the Program are reviewed and gender equality considerations adopted	Year 5		

¹ Sex-disaggregated data will be reported for all the activities in the program.

² Multistakeholder consultations will be taking place during the implementation of the program. Based on those, information on contextualized gendered needs of different countries and diverse needs of the population in terms of e-mobility in the public transport sector will be provided.

³ The program will support the design of new roadmaps and policies and will review current policies to recommend how to adapt them to create a e-mobility conducive ecosystem. This activity seeks to include a gender perspective in the process to provide recommendations to policies to be inclusive. In particular, policies will be reviewed following the differences found in the assessment in terms of mobility, safety and jobs. Moreover, these reviews will identify sexual harassment risk, propose response, grieving and monitoring mechanisms, and consider intersectionality.

	Indicators and Targets	Timeline	Responsibilities	Costs in EUR
Output 2: Support a knowledge exchange and collaboration on gender aspects in transportation (Regional level)				
Activity 2.1. To increase awareness about gender equality gaps and opportunities in EV sector	• % of capacity building activities (i.e: webinars, workshops, publications) of the facility that include awareness on gender equality and urban transport⁴ Target: 80% • % of participants of the exchange platform attending trainings and capacity building activities on EV that are women Target: 30%⁵ women, 70%men • Participants perceiving the content on gender of the capacity building activities relevant (Very Relevant, Moderately relevant, not relevant)⁶ • Regional network on women in e-mobility supported and strengthened - to share and exchange experiences, promote mentorship, and produce knowledge Target: 1 network	Year 1 - 5	GIZ	110,400

⁴ This activity guarantees that gender equality is learned and integrated into the activities to be carried out by the facility related to capacity building. It is expected that in some capacity building activities it will not be feasible to integrate gender aspects. Besides, this activity will foster and monitor women's participation on the capacity building activities, making it mandatory to reach a minimum of 30 percent women attending. These activities will consider accessibility and reasonable accommodations to allow intersecting identities of age, social economic status, social cultural context, education, disability among others to benefit from all of the capacity building activities.

⁵ A target between 30% -40 % aims at ensuring that women at least constitute a «critical minority ». Targeting a 30% of women to participate is the critical mass needed to guarantee that they are members of this platform in a sector which is traditionally masculine. A parity of 50% is very difficult to achieve and it is not realistic. in a traditionally masculine sector.

⁶ The satisfaction surveys in the activities will include a question to measure the quality of the content on gender and also seeks to find out if the content needs to be adjusted.

	Indicators and Targets	Timeline	Responsibilities	Costs in EUR
Activity 2.2. To increase knowledge on how to include gender perspective in Urban Transport projects via capacity building online	Online module⁷ on Gender and Urban Transport developed Target: 1 Participants of the trainings Target: 1 person minimum from a representative from the local public authority in each city where an investment of the program is foreseen ⁸	Year 2	GIZ	97,400
Activity 2.3. Implement South-to-South learning from best practices in the region, e.g. “Bajale al acoso” Quito	South-to-South learning visit to best practices in the region, e.g. “Bajale al acoso” Quito Target: 1	Year 4	GIZ	109,200
Activity 2.4 Design of a communication campaign about sexual harassment and protocols of attention in public transport	- Public communication campaign about sexual harassment in public transportation and protocol of attention⁹ in the new EV fleet and system Target: 1 - Women’s feelings about reporting sexual harassment in public transportation (Will report, Not sure whether they will report, Will not report)	Year 3	GIZ	143,300
Output 3: Integrate gender components in the projects financed by the Program				

⁷ The online module will have verification means of attendance. The training will be practical with initiatives on the region (no theoretical), 3 hours max by day. 1 Module on mobility (1 day), 1 Module on Jobs (2 days), 1 module on Safety (3 days), 1 Module on how to prepare a gender assessment in a real project and preparing a gender action plan (1 day). The content will cover tools to include intersectional identities in all the modules.

⁸ The facilitated training, with accessibility features to support inclusion, will require one person of each city/agency/investment to receive it. At this stage it is not possible to know the number of participants. The online format allows the content to be used multiple times.

⁹ The public communication campaigns will be designed with consultations addressing specific perceptions and risks those different identities face, with significant attention to mitigate indigenous peoples and afro descendants’ potential risks. The protocols of attention will develop/enhance safe, confidential, and ethical grievance reporting mechanisms for survivors and monitoring mechanisms to identify, respond and address risks of sexual harassment. This technical assistance will guide the adopters to develop protocols of attention that are inclusive and safe for different identities present in each country/city of the new e-transport system and will benefit from the implementation of technologies, such as panic buttons or telephone applications.

	Indicators and Targets	Timeline	Responsibilities	Costs in EUR
Activity 3.1. Gender perspective is included in the investment plans	% of projects that include a Gender Equality Strategy with a GAP which contains specific and measurable activities ¹⁰ Target:100%	Year 1-5	Project owners with support of gender consultant	0 (financed by the Project owner)
Activity 3.2. Improve sex disaggregation of data collection on urban transport	% of urban passenger transport projects whose user's data systems collects and presents data disaggregated to inform the PT Target: 100%	Year 1--5	Public transport authority with support of gender consultants	108,000
Activity 3.3. Improved women's access to jobs	% projects with direct finance by the Program guaranteeing at least 30% of skilled jobs related directly to the project are reserved to women¹¹, with special attention to measures that can attract and retain women in the industry Target: 80%	Year 5	Projects' owners	0 (financed by the Project owner)
Activity 3.5 Raise awareness on sexual harassment in public transport through capacity building	% of public transport projects with activities (communication campaign or workshops) aiming at preventing sexual harassment Target: 100% % of projects with reporting systems designed for victims of sexual harassment in collaboration with local	Year 1-5	AFD/ Gender Specialist	92,800

¹⁰ The project owners will learn on how to draft a gender assessment and GAP in the online module, in this activity they will get support from consultants to implement gender in their own investment plans. In particular, the adopters will include sexual harassment programs (risks assessments, design and grievance mechanisms), and activities to improve women's mobility and employment. The adopters will be guided on how to include intersectional identities in their assessments.

¹¹ This is monitoring that the GAP of the investment's plans are reserving a critical minority of jobs to women. The target is not set to 80% because is expected that some investments plans might not generate jobs.

	Indicators and Targets	Timeline	Responsibilities	Costs in EUR
	authorities (such as police, judiciary systems)¹² Target:100%			
Activity 3.6. Promotion of gender equality at the workplace	- Capacity building for the staff of public transport projects on gender equality and work-life balance issues via workshop¹³ Target: one workshop for each public transport project % of adopters including annual surveys on work-life balance to their staff and sharing the results Target: 100% - Participants satisfied of the work-life balance workshops (Very satisfied, somewhat satisfied, not satisfied) % of adopters with one focal point in charge of gender equality Target: 100%	Year 5	Project owners with support of gender consultant	72,800

¹² The adopters will receive guidance on developing clear mechanisms of identifying, responding, and addressing risks of sexual harassment. These reporting systems will be built from existing mechanisms already in the cities when available and plan to engage different local actors such as the police. The adopters will receive guidance to enhance safe, confidential, and ethical grievance reporting mechanisms for survivors and monitoring mechanisms.

¹³ Target learners are workers of the system (drivers, cashiers, etc.) as well as managerial and administrative staff. The activity will include workshops that cover practical examples to learn how GBV, Sexual harassment and other barriers affect women's participation in the work place, with concrete examples on the case of intersectionality. The aim is for the adopters to understand gender inequalities, at a family and work level, and to appoint a person in charge in the organization to monitor and report advances in gender equality and inclusive environments in the organization. Paternity and maternity leaves, availability of day care, lactation rooms, wage gaps, recruitment and promotion criteria hindering women's advancement in the sector and benefits of diversity working environments will be covered by the curriculum. A gender focal point will be appointed to advance the work-life balance agenda of the adopters, to monitor and report gender equality in the human resources policies and to act as a sexual harassment ombudsperson. A model of survey will be shared in the workshop to be used by the agency/company adopter of the technology to allow adopters to use it regularly with their workforce.